MATHEMATICS
ANNOTATED READING NOTES

PDE

Partial Differential Equations

英文精读札记与中文边注

Classical PDE and Sobolev Spaces

Reading companion to Lawrence C. Evans
Partial Differential Equations, Second Edition

基于 Evans 原著的学习整理

$$\partial_t u - \Delta u = 0$$

LUMENBOOK

2026-11-09

澄明讲义 / 数学阅读札记

How to Use These Notes

These notes follow Lawrence C. Evans's *Partial Differential Equations*, second edition. The English PDF supplies the chapter order and mathematical source. The supplied Chinese translation is a reference for terminology and reading support.

English text and margin notes

Read the main argument first. Each Chinese margin note addresses a specific mathematical issue: the role of a hypothesis, the structure of an operator, a counterexample, a limiting case, or the connection to a later method. Headings name the mathematical issue. Longer calculations and proofs belong in the main text and are marked as editorial supplements.

边注的数学任务

边注应回答可检验的问题：某条件在哪里使用？删去后有什么反例？方程的主部如何决定方法？不以句式分析、常用词翻译或正文复述占据边栏。

Expanded proofs

An expanded proof will state the assumptions, explain the strategy, and fill in omitted calculations. Each use of approximation, integration by parts, compactness, or passage to a limit must identify the applicable hypotheses. Source theorem numbers will be recorded explicitly; the notes' own numbering does not automatically coincide with the original book.

Current status

The prefaces and Chapters 1–6 are annotated. Chapter 2 emphasizes harmonic and heat-ball means; Chapter 3 develops characteristics, Hopf–Lax and entropy solutions. Chapter 4 treats representation methods, asymptotics and analytic Cauchy theory. Chapter 5 develops Sobolev theory; Chapter 6 applies it to weak elliptic solutions, regularity, maximum principles and eigenvalues. Chapters 7–12 and Appendices A–E remain outlines. Source comparison, editorial checking, and independent review are recorded separately in the project documentation.

The introduction precedes Part I, as in the source. The three parts contain Chapters 2–4, 5–7, and 8–12. Appendices A–E supply background material. The bilingual glossary is an added study aid. Detailed subsection headings will be added when their source pages are studied.

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Preface to the second edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xi–xii; PDF pages 12–13. Chinese reference: PDF page 9.

原著序言全文；中文边注为整理者补充，英文印次信息按所提供的英文 PDF 保留。

Let me thank everyone who over the past decade has provided me with suggestions and corrections for improving the first edition of this book.

I am extraordinarily grateful. Although I have not always followed these many pieces of advice and criticism, I have thought carefully about them all. So many people have helped me out that it is unfortunately no longer feasible to list all their names. I have also received great help from everyone at the AMS, especially Sergei Gelfand, Stephen Moye and Arlene O'Sean. The NSF has generously supported my research during the writing of both the original edition of the book and this revision. I will continue to maintain lists of errors on my homepage, accessible through the `math.berkeley.edu` website.

When you write a big book on a big subject, the temptation is to include everything.

A critic famously once imagined Tolstoy during the writing of *War and Peace*: "The book is long, but even if it were twice as long, if it were three times as long, there would always be scenes that have been omitted, and these Tolstoy, waking up in the middle of the night, must have regretted. There must have been a night when it occurred to him that he had not included a yacht race…" (G. Moore, *Avowals*).

This image notwithstanding, I have tried to pack into this second edition as many fascinating new topics in partial differential equations (PDE) as I could manage, most notably in the new Chapter 12 on nonlinear wave equations. There are new sections on Noether's Theorem and on local minimizers in the calculus of variations, on the Radon transform, on Turing instabilities for reaction-diffusion systems, etc. I have rewritten and expanded the previous discussions on blow-up of solutions, on group and phase velocities, and on several further subjects. I have also updated and greatly increased citations to books in the bibliography and have moved

爆破与延拓判据

有限时间爆破必须说明失控的量：可能是解的幅度、梯度，或某个 Sobolev 范数。局部存在与全局存在之间的关键，是控制延拓定理所要求的范数。

references to research articles to within the text. There are countless further minor modifications in notation and wording. Most importantly, I have added about 80 new exercises, most quite interesting and some rather elaborate. There are now over 200 in total.

And there is a yacht race among the problems for Chapter 10.

LCE
January, 2010
Berkeley

This third printing of the second edition corrects all known mistakes, as of February, 2022.

印次差异

英文此处为 2022 年；中译本对应写 2015 年，并附译者注说明参考了 2020 年勘误。此处按英文 PDF 保留，不能混作同一印次。

Preface to the first edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xiii–xv; PDF pages 14–16. Chinese reference: PDF pages 11–12.

原著序言全文；中文边注为整理者补充。保留作者的写作原则、致谢和署名。

I present in this book a wide-ranging survey of many important topics in the theory of partial differential equations (PDE), with particular emphasis on various modern approaches.

I have made a huge number of editorial decisions about what to keep and what to toss out, and can only claim that this selection seems to me about right. I of course include the usual formulas for solutions of the usual linear PDE, but also devote large amounts of exposition to energy methods within Sobolev spaces, to the calculus of variations, to conservation laws, etc.

My general working principles in the writing have been these:

a. PDE theory is (mostly) not restricted to two independent variables. Many texts describe PDE as if functions of the two variables (x, y) or (x, t) were all that matter. This emphasis seems to me misleading, as modern discoveries concerning many types of equations, both linear and nonlinear, have allowed for the rigorous treatment of these in any number of dimensions. I also find it unsatisfactory to "classify" partial differential equations: this is possible in two variables, but creates the false impression that there is some kind of general and useful classification scheme available in general.

高维中的主部

二阶线性算子的主部由矩阵 $A = (a^{ij})$ 描述，其二次型 $\xi^T A \xi$ 决定椭圆性等结构。二维判别式只能覆盖这一框架的特殊情形，不能据此统一处理任意阶数、方程组和非线性。

b. Many interesting equations are nonlinear.

My view is that overall we know too much about linear PDE and too little about nonlinear PDE. I have accordingly introduced nonlinear concepts early in the text and have tried hard to emphasize everywhere nonlinear analogues of the linear theory.

c. Understanding generalized solutions is fundamental. Many of the partial differential equations we study, especially nonlinear first-order equations, do not in general possess smooth solutions. It is therefore essential to devise some kind of proper notion of generalized or weak solution. This is an important but subtle undertaking, and much of the hardest material in this book concerns the uniqueness of appropriately defined weak solutions.

广义解仍需选择原则

守恒律的分布弱形式容许不连续解，但单凭积分恒等式一般无法保证唯一性，还需熵条件。Hamilton–Jacobi 方程则采用黏性解；解的定义必须配合方程结构。

d. PDE theory is not a branch of functional analysis. Whereas certain classes of equations can profitably be viewed as generating abstract operators between Banach spaces, the insistence on an overly abstract viewpoint, and consequent ignoring of deep calculus and measure theoretic estimates, is ultimately limiting.

估计与抽象框架的分工

把 PDE 写成算子方程后，仍须证明适用条件：例如椭圆问题的强制性来自能量估计，非线性极限常需紧性或单调性。抽象存在定理不会自动提供这些针对具体方程的估计。

e. Notation is a nightmare.

I have really tried to introduce consistent notation, which works for all the important classes of equations studied. This attempt is sometimes at variance with notational conventions within a given subarea.

f. Good theory is (almost) as useful as exact formulas. I incorporate this principle into the overall organization of the text, which is subdivided into three parts, roughly mimicking the historical development of PDE theory itself. Part I concerns the search for explicit formulas for solutions, and Part II the abandoning of this quest in favor of general theory asserting the existence and other properties of solutions for linear equations. Part III is the mostly modern endeavor of fashioning general theory for important classes of nonlinear PDE.

估计提供可检验的结论

若能建立 $\|u-\tilde{u}\|_Y \le C\|g-\tilde{g}\|_X$，就同时得到唯一性与定量稳定性。这类结论即使没有显式公式，也能说明数据误差如何传到解。

Let me also explicitly comment here that I intend the development within each section to be rigorous and complete (exceptions being the frankly heuristic treatment of asymptotics in §4.5 and an occasional reference to a research paper). This means that even locally within each chapter the topics do not necessarily progress logically from "easy" to "hard" concepts. There are many difficult proofs and computations early on, but as compensation many easier ideas later. The student should certainly omit on first reading some of the more arcane proofs.

渐近展开的证明责任

形式上把幂级数代入 PDE 只能给出系数关系。要证明渐近公式有效，还须在指定区域和范数中估计余项；没有余项控制的形式展开不能作为解存在或误差界的证明。

I wish next to emphasize that this is a textbook, and not a reference book. I have tried everywhere to present the essential ideas in the clearest possible settings, and therefore have almost never established sharp versions of any of the theorems. Research articles and advanced monographs, many of them listed in the Bibliography, provide such precision and generality. My goal has rather been to explain, as best I can, the many fundamental ideas of the subject within fairly simple contexts.

I have greatly profited from the comments and thoughtful suggestions of many of my colleagues, friends and students, in particular: S. Antman,

J. Bang, X. Chen, A. Chorin, M. Christ, J. Cima, P. Colella, J. Cooper, M. Crandall, B. Driver, M. Feldman, M. Fitzpatrick, R. Gariepy, J. Goldstein, D. Gomes, O. Hald, W. Han, W. Hrusa, T. Ilmanen, I. Ishii, I. Israel, R. Jerrard, C. Jones, B. Kawohl, S. Koike, J. Lewis, T.-P. Liu, H. Lopes, J. McLaughlin, K. Miller, J. Morford, J. Neu, M. Portilheiro, J. Ralston, F. Rezakhanlou, W. Schlag, D. Serre, P. Souganidis, J. Strain, W. Strauss, M. Struwe, R. Temam, B. Tvedt, J.-L. Vazquez, M. Weinstein, P. Wolfe, and Y. Zheng.

I especially thank Tai-Ping Liu for many years ago writing out for me the first draft of what is now Chapter 11.

I am extremely grateful for the suggestions and lists of mistakes from earlier drafts of this book sent to me by many readers, and I encourage others to send me their comments, at `evans@math.berkeley.edu`. I have come to realize that I must be more than slightly mad to try to write a book of this length and complexity, but I am not yet crazy enough to think that I have made no mistakes. I will therefore maintain a listing of errors which come to light, and will make this accessible through the `math.berkeley.edu` homepage.

Faye Yeager at UC Berkeley has done a really magnificent job typing and updating these notes, and Jaya Nagendra heroically typed an earlier version at the University of Maryland. My deepest thanks to both.

I have been supported by the NSF during much of the writing, most recently under grant DMS-9424342.

LCE
August, 1997
Berkeley

Chapter 1

Introduction

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 1; printed pp. 1–13; PDF pages 18–30.

Chinese reference: 中译本：引言，书内第 1–9 页；PDF 第 19–27 页。

Editorial convention. The source exposition and all five source problems are retained below. Chinese margin notes and passages headed "Editorial supplement" are added by the editor. Only the two equations numbered (1) and (2) in the source carry those source numbers here; supplementary displays are unnumbered.

This chapter surveys the principal theoretical issues concerning the solving of partial differential equations.

To follow the subsequent discussion, the reader should first of all turn to Appendix A and look over the notation presented there, particularly the multiindex notation for partial derivatives.

1.1 Partial Differential Equations

A *partial differential equation* (PDE) is an equation involving an unknown function of two or more variables and certain of its partial derivatives.

Using the notation explained in Appendix A, we can write out symbolically a typical PDE, as follows. Fix an integer $k \geq 1$ and let U denote an open subset of $\mathbb{R}^n$.

Definition (source, §1.1). An expression of the form

$$F(D^k u(x), D^{k-1}u(x), \dots, Du(x), u(x), x) = 0 \quad (x \in U) \tag{1}$$

is called a *kth-order partial differential equation*, where

$$F : \mathbb{R}^{n^k} \times \mathbb{R}^{n^{k-1}} \times \cdots \times \mathbb{R}^n \times \mathbb{R} \times U \to \mathbb{R}$$

is given and $u : U \to \mathbb{R}$ is the unknown.

We solve the PDE if we find all u verifying (1), possibly only among those functions satisfying certain auxiliary boundary conditions on some part Γ of ∂U. By finding the solutions we mean, ideally, obtaining simple, explicit solutions, or, failing that, deducing the existence and other properties of solutions.

Definitions (source, §1.1).

(i) The partial differential equation (1) is called *linear* if it has the form

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

for given functions a_α $(|\alpha| \le k), f$. This linear PDE is *homogeneous* if $f \equiv 0$.

(ii) The PDE (1) is *semilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iii) The PDE (1) is *quasilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(D^{k-1}u, \dots, Du, u, x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iv) The PDE (1) is *fully nonlinear* if it depends nonlinearly upon the highest order derivatives.

导数数组与对称性

原书把 $D^k u$ 写成含 n^k 个分量的数组。若 $u \in C^k$，混合偏导可交换，独立分量只有 $\binom{n+k-1}{k}$ 个。前者按求导次序储存，后者按多重指标计数；两种记号并不矛盾。

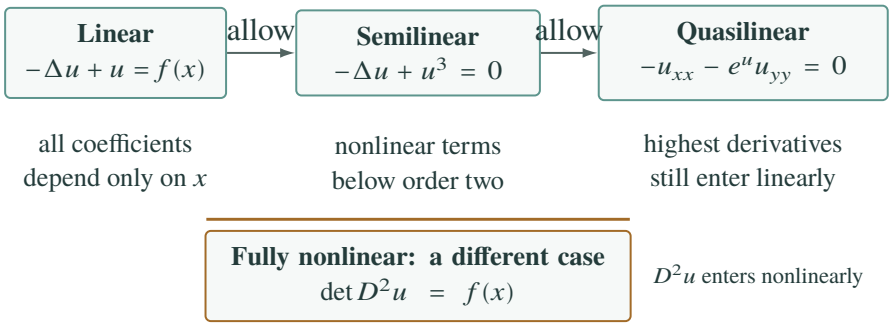


Diagram T1.1. Classify by the highest derivatives. The first three classes allow successively more dependence on lower-order variables. In the last example the highest derivatives themselves enter nonlinearly. All displayed examples are second order.

分类看最高阶导数

先找阶数 k，再看 $D^k u$ 如何出现。半线性：最高阶系数只依赖 x；拟线性：系数可依赖低阶导数。例如 $u_t + uu_x = 0$ 是一阶拟线性，而 $u_t + uu_x + u_{xxx} = 0$ 是三阶半线性。

A *system of partial differential equations* is, informally speaking, a collection of several PDE for several unknown functions.

Definition (source, §1.1). Let k be a positive integer. An expression of the form

$$\mathbf{F}(D^k\mathbf{u}(x), \dots, D\mathbf{u}(x), \mathbf{u}(x), x) = \mathbf{0} \quad (x \in U) \tag{2}$$

is called a *kth-order system of partial differential equations*, where

$$\mathbf{F} : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \cdots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times U \to \mathbb{R}^m$$

is given and

$$\mathbf{u} : U \to \mathbb{R}^m, \qquad \mathbf{u} = (u^1, \dots, u^m)$$

is the unknown.

Here we are supposing that the system comprises the same number m of scalar equations as unknowns $(u^1, \dots, u^m)$. This is the most common circumstance, although other systems may have fewer or more equations than unknowns. Systems are classified in the obvious way as being linear, semilinear, etc.

Notation. We write "PDE" as an abbreviation for both the singular "partial differential equation" and the plural "partial differential equations".

Editorial supplement: the multiindex convention

For $\alpha = (\alpha_1, \dots, \alpha_n) \in \{0, 1, 2, \dots\}^n$, set

$$|\alpha| = \sum_{i=1}^{n} \alpha_i, \qquad D^\alpha u = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} u.$$

The *principal part* of a linear operator of order k is $\sum_{|\alpha|=k} a_\alpha(x) D^\alpha$; the terms of smaller order are its lower-order terms. Here $D^0 u = u$. For example, when $n = 2$, $D^{(2,1)}u = u_{x_1 x_1 x_2}$ and $|(2,1)| = 3$.

The words linear, semilinear and quasilinear describe nested forms. When a single classification is requested, we report the most restrictive applicable form: linear first, then semilinear, then quasilinear. These definitions concern the displayed differential operator. An algebraic rewriting that changes its form need not preserve its classification.

等价零点不保结构

把 $Lu = 0$ 改写成 $(Lu)^3 = 0$，经典解集不变，但新表达式对最高阶导数非线性，且在线性化时可能退化。因此分类必须说明采用哪个算子表达式，不能只看方程的解集。

1.2 Examples

There is no general theory known concerning the solvability of all partial differential equations. Such a theory is extremely unlikely to exist, given the rich variety of physical, geometric, and probabilistic phenomena which can be modeled by PDE. Instead, research focuses on various particular partial differential equations that are important for applications within and outside of mathematics, with the hope that insight from the origins of these PDE can give clues as to their solutions.

Following is a list of many specific partial differential equations of interest in current research. This listing is intended merely to familiarize the reader with the names and forms of various famous PDE. To display most clearly the mathematical structure of these equations, we have mostly set relevant physical constants to unity. We will later discuss the origin and interpretation of many of these PDE.

Throughout $x \in U$, where U is an open subset of $\mathbb{R}^n$, and $t \geq 0$. Also $Du = D_x u = (u_{x_1}, \dots, u_{x_n})$ denotes the gradient of u with respect to the spatial variable $x = (x_1, \dots, x_n)$. The variable t always denotes time.

1.2.1 Single partial differential equations

a. Linear equations

1. Laplace's equation
$$\Delta u = \sum_{i=1}^{n} u_{x_i x_i} = 0.$$
2. Helmholtz's (or eigenvalue) equation
$$-\Delta u = \lambda u.$$
3. Linear transport equation
$$u_t + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$
4. Liouville's equation
$$u_t + \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

5. Heat (or diffusion) equation

$$u_t - \Delta u = 0.$$

6. Schrödinger's equation

$$\mathrm{i}u_t + \Delta u = 0.$$

7. Kolmogorov's equation

$$u_t - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

8. Fokker–Planck equation

$$u_t - \sum_{i,j=1}^{n} (a^{ij} u)_{x_i x_j} - \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

9. Wave equation

$$u_{tt} - \Delta u = 0.$$

10. Klein–Gordon equation

$$u_{tt} - \Delta u + m^2 u = 0.$$

11. Telegraph equation

$$u_{tt} + 2du_t - u_{xx} = 0.$$

12. General wave equation

$$u_{tt} - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

13. Airy's equation

$$u_t + u_{xxx} = 0.$$

14. Beam equation

$$u_{tt} + u_{xxxx} = 0.$$

b. Nonlinear equations

1. Eikonal equation

$$|Du| = 1.$$

2. Nonlinear Poisson equation

$$-\Delta u = f(u).$$

随流不变与质量守恒

输运方程沿 $\dot{X} = b(X,t)$ 保持 $u(X(t),t)$ 不变。Liouville 方程展开后多出 $(\operatorname{div} b)u$，描述密度随体积伸缩而变。只有 $\operatorname{div} b = 0$ 时两式一致。

算子与形式伴随

令 $L = a^{ij}\partial_{ij} - b^i\partial_i$（重复指标求和）。分部积分给出 $L^*\rho = \partial_{ij}(a^{ij}\rho) + \partial_i(b^i\rho)$。两式分别为 $u_t = Lu$ 与 $\rho_t = L^*\rho$，漂移项符号因此不同。

阶数与时间初值

热方程、波方程都是二阶 PDE，但前者时间一阶，后者时间二阶，通常分别需给 $u(0)$ 与 $(u(0), u_t(0))$。梁方程总阶数为四、时间阶数为二；总阶数不能直接决定初值个数。

3. p-Laplacian equation

$$\operatorname{div}(|Du|^{p-2}Du) = 0.$$

4. Minimal surface equation

$$\operatorname{div}\left(\frac{Du}{(1+|Du|^2)^{1/2}}\right) = 0.$$

5. Monge–Ampère equation

$$\det(D^2u) = f.$$

6. Hamilton–Jacobi equation

$$u_t + H(Du, x) = 0.$$

7. Scalar conservation law

$$u_t + \operatorname{div}\mathbf{F}(u) = 0.$$

8. Inviscid Burgers' equation

$$u_t + uu_x = 0.$$

9. Scalar reaction-diffusion equation

$$u_t - \Delta u = f(u).$$

10. Porous medium equation

$$u_t - \Delta(u^\gamma) = 0.$$

11. Nonlinear wave equation

$$u_{tt} - \Delta u + f(u) = 0.$$

12. Korteweg–deVries (KdV) equation

$$u_t + uu_x + u_{xxx} = 0.$$

13. Nonlinear Schrödinger equation

$$\mathrm{i}u_t + \Delta u = f(|u|^2)u.$$

退化与奇异

对 p-Laplacian，$p > 2$ 时主部在 $Du = 0$ 退化；$1 < p < 2$ 时其线性化系数在那里奇异。散度形式及相应弱形式仍有意义，不能把只在 $Du \neq 0$ 合法的展开直接用于临界点。

多孔介质的扩散系数

在 $u > 0$ 的光滑区，$\Delta(u^\gamma) = \gamma u^{\gamma-1}\Delta u + \gamma(\gamma-1)u^{\gamma-2}|Du|^2$。通常取 $\gamma > 1, u \geq 0$，主部在 $u = 0$ 退化；$0 < \gamma < 1$ 对应快扩散情形。

1.2.2 Systems of partial differential equations

a. Linear systems

1. Equilibrium equations of linear elasticity

$$\mu\Delta\mathbf{u} + (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

2. Evolution equations of linear elasticity

$$\mathbf{u}_{tt} - \mu\Delta\mathbf{u} - (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

3. Maxwell's equations

$$\begin{cases}\mathbf{E}_t = \operatorname{curl}\mathbf{B},\\ \mathbf{B}_t = -\operatorname{curl}\mathbf{E},\\ \operatorname{div}\mathbf{B} = \operatorname{div}\mathbf{E} = 0.\end{cases}$$

b. Nonlinear systems

1. System of conservation laws

$$\mathbf{u}_t + \operatorname{div}\mathbf{F}(\mathbf{u}) = \mathbf{0}.$$

2. Reaction-diffusion system

$$\mathbf{u}_t - \Delta\mathbf{u} = \mathbf{f}(\mathbf{u}).$$

3. Euler's equations for incompressible, inviscid flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

4. Navier–Stokes equations for incompressible, viscous flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} - \Delta\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

约束的传播

真空 Maxwell 系统中，$\partial_t \operatorname{div}\mathbf{E} = 0$、$\partial_t \operatorname{div}\mathbf{B} = 0$，因为散度作用于旋度为零。散度条件是初值必须满足且由演化保持的约束，不能视作任意额外数据。

See Zwillinger [Zw] for a much more extensive listing of interesting PDE.

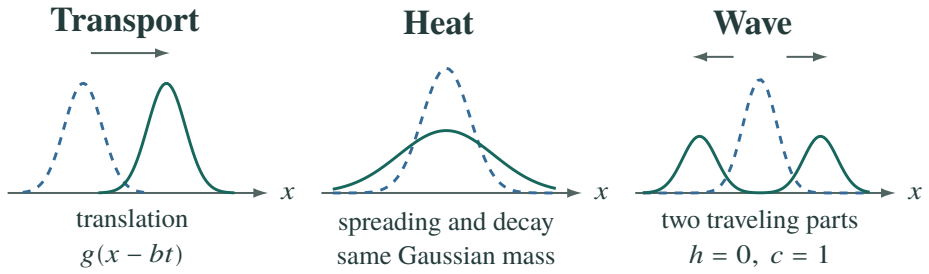


Diagram T1.2. Three evolution mechanisms. Dashed curves are initial profiles and solid curves are later profiles. Transport translates a shape; heat spreads a Gaussian while preserving its integral; the one-dimensional wave equation with zero initial velocity splits a profile into two half-amplitude translates. The panel axes are schematic and use separate scales.

Editorial supplement: two principal-part calculations

For $1 < p < \infty$ and a C^2 function at a point where $Du \neq 0$, the chain rule gives

$$\operatorname{div}(|Du|^{p-2}Du) = \sum_{i,j=1}^{n} A_{ij}(Du)u_{x_i x_j},$$

$$A(q) = |q|^{p-2}\left(I + (p-2)\frac{q \otimes q}{|q|^2}\right).$$

The eigenvalue in the direction of q is $(p-1)|q|^{p-2}$; in every orthogonal direction it is $|q|^{p-2}$. This proves ellipticity at $q \neq 0$ and explains its failure to be uniform as $|q| \to 0$ or $|q| \to \infty$ when $p \neq 2$. At $p = 2$ the equation is exactly Laplace's equation.

Similarly, the minimal surface equation expands as

$$\frac{1}{\sqrt{1 + |Du|^2}} \sum_{i,j=1}^{n} \left(\delta_{ij} - \frac{u_{x_i}u_{x_j}}{1 + |Du|^2}\right) u_{x_i x_j} = 0.$$

The matrix in parentheses has eigenvalues 1 transverse to Du and $(1 + |Du|^2)^{-1}$ parallel to Du. Thus it is positive definite for each finite gradient, and a bound on $|Du|$ supplies a uniform ellipticity bound. Both calculations exhibit quasilinear second-order equations.

压力是约束乘子

对光滑不可压流取散度，得到 $-\Delta p = \sum_{i,j}(\partial_i u^j)(\partial_j u^i)$。压力由速度及边界条件耦合确定；没有独立的压力演化方程。仅看速度的扩散项会漏掉这一非局部约束。

1.3 Strategies for Studying PDE

As explained in §1.1 our goal is the discovery of ways to solve partial differential equations of various sorts, but—as should now be clear in view of the many diverse examples set forth in §1.2—this is no easy task. And indeed the very question of what it means to "solve" a given PDE can be subtle, depending in large part on the particular structure of the problem at hand.

1.3.1 Well-posed problems, classical solutions

The informal notion of a well-posed problem captures many of the desirable features of what it means to solve a PDE. We say that a given problem for a partial differential equation is well-posed if

(i) the problem in fact has a solution;
(ii) this solution is unique; and
(iii) the solution depends continuously on the data given in the problem.

The last condition is particularly important for problems arising from physical applications: we would prefer that our (unique) solution changes only a little when the conditions specifying the problem change a little. (For many problems, on the other hand, uniqueness is not to be expected. In these cases the primary mathematical tasks are to classify and to characterize the solutions.)

Now clearly it would be desirable to "solve" PDE in such a way that (i)–(iii) hold. But notice that we still have not carefully defined what we mean by a "solution". Should we ask, for example, that a "solution" u must be real analytic or at least infinitely differentiable? This might be desirable, but perhaps we are asking too much. Maybe it would be wiser to require a solution of a PDE of order k to be at least k times continuously differentiable. Then at least all the derivatives which appear in the statement of the PDE will exist and be continuous, although maybe certain higher derivatives will not exist. Let us informally call a solution with this much smoothness a classical solution of the PDE: this is certainly the most obvious notion of solution.

So by solving a partial differential equation in the classical sense we mean if possible to write down a formula for a classical solution satisfying (i)–(iii) above, or at least to show such a solution exists, and to deduce various of its properties.

Monge–Ampère 的分支

当 $n \geq 2$，行列式非线性地依赖 Hessian。在 D^2u 正定的凸解分支上，线性化系数为正定的余子式矩阵。离开该分支，椭圆性没有保证；仅写 $\det D^2u = f$ 还不足以指定椭圆问题。

适定性依赖拓扑

连续依赖必须指定数据空间 X 和解空间 Y。常见目标是 $\|u-\tilde{u}\|_Y \leq C\|g-\tilde{g}\|_X$。同一方程更换范数、时间方向或边界条件后，适定性可能改变。

1.3.2 Weak solutions and regularity

But can we achieve this? The answer is that certain specific partial differential equations (e.g. Laplace's equation) can be solved in the classical sense, but many others, if not most others, cannot. Consider for instance the scalar conservation law

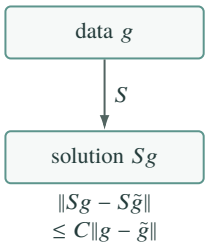
$$u_t + F(u)_x = 0.$$

We will see in §3.4 that this PDE governs various one-dimensional phenomena involving fluid dynamics, and in particular models the formation and propagation of shock waves. Now a shock wave is a curve of discontinuity of the solution u; and so if we wish to study conservation laws, and recover the underlying physics, we must surely allow for solutions u which are not continuously differentiable or even continuous. In general, as we shall see, the conservation law has no classical solutions but is well-posed if we allow for properly defined generalized or weak solutions.

This is all to say that we may be forced by the structure of the particular equation to abandon the search for smooth, classical solutions. We must instead, while still hoping to achieve the well-posedness conditions (i)–(iii), investigate a wider class of candidates for solutions. And in fact, even for those PDE which turn out to be classically solvable, it is often most expedient initially to search for some appropriate kind of weak solution.

The point is this: if from the outset we demand that our solutions be very regular, say k-times continuously differentiable, then we are usually going to have a really hard time finding them, as our proofs must then necessarily include possibly intricate demonstrations that the functions we are building are in fact smooth enough. A far more reasonable strategy is to consider as separate the existence and the smoothness (or regularity) problems. The idea is to define for a given PDE a reasonably wide notion of a weak solution, with the expectation that since we are not asking too much by way of smoothness of this weak solution, it may be easier to establish its existence, uniqueness, and continuous dependence on the given data. Thus, to repeat, it is often wise to aim at proving well-posedness in some appropriate class of weak or generalized solutions.

图 T1.M：连续依赖的方向



先说明数据与解各用什么范数，再检验解映射是否连续。存在、唯一和连续依赖是三个不同问题。

经典解可局部存在

“没有经典解”不能理解成任意初值、任意短时间都无解。光滑 Burgers 初值通常先产生局部经典解，再因特征线相交形成激波。失效的是一般情形下全局保持光滑的要求。

Now, as noted above, for various partial differential equations this is the best that can be done. For other equations we can hope that our weak solution may turn out after all to be smooth enough to qualify as a classical solution. This leads to the question of regularity of weak solutions. As we will see, it is often the case that the existence of weak solutions depends upon rather simple estimates plus ideas of functional analysis, whereas the regularity of the weak solutions, when true, usually rests upon many intricate calculus estimates.

弱形式与唯一性

分部积分把导数转移给光滑试验函数，降低对 u 的正则性要求。但守恒律的分布弱解通常不唯一；还需熵条件筛选物理解。“弱解存在”本身不等于问题已经适定。

Let me explicitly note here that once we are past Part I (Chapters 2–4), our efforts will be largely devoted to proving mathematically the existence of solutions to various sorts of partial differential equations, and not so much to deriving formulas for these solutions. This may seem wasted or misguided effort, but in fact mathematicians are like theologians: we regard existence as the prime attribute of what we study. But unlike theologians, we need not always rely upon faith alone.

1.3.3 Typical difficulties

Following are some vague but general principles, which may be useful to keep in mind:

(i) Nonlinear equations are more difficult than linear equations; and, indeed, the more the nonlinearity affects the higher derivatives, the more difficult the PDE is.
(ii) Higher-order PDE are more difficult than lower-order PDE.
(iii) Systems are harder than single equations.
(iv) Partial differential equations entailing many independent variables are harder than PDE entailing few independent variables.
(v) For most partial differential equations it is not possible to write out explicit formulas for solutions.

None of these assertions is without important exceptions.

Editorial supplement: estimates, instability, and shocks

The following three calculations make the preceding distinctions precise. They are editorial additions, not proofs stated in the source introduction.

1. A forward heat estimate. Let U be a bounded smooth domain, and let u, v be sufficiently smooth solutions of $u_t - \Delta u = 0$ with the same homogeneous Dirichlet boundary condition. For $w = u - v$, multiplication

by w and integration by parts give

$$\frac{1}{2}\frac{d}{dt}\int_U w^2\,dx + \int_U |Dw|^2\,dx = \int_{\partial U} w\,\partial_\nu w\,dS = 0.$$

Consequently, for every time within their common interval of existence,

$$\|u(t) - v(t)\|_{L^2(U)} \le \|u(0) - v(0)\|_{L^2(U)}.$$

This proves uniqueness in this smooth class and an L^2 continuous-dependence estimate. It does not prove existence. Extending the calculation to weak solutions requires a justification for the time derivative and the test function w; the formal identity alone is not that justification.

边界项决定估计

这里消失的是 $w\partial_\nu w$，依据为相同的齐次 Dirichlet 条件。若边值不同，就不能直接令边界项为零。同样的能量计算必须随边界条件调整，不能只从方程形式照抄结论。

2. Reversing time destroys this stability. On $U = (0, \pi)$ fix $T > 0$, with zero boundary values, and define

$$u_j(x,t) = e^{j^2(T-t)}\sin(jx), \qquad 0 \le t \le T.$$

Every u_j is a smooth forward heat solution. Its terminal and initial norms satisfy

$$\|u_j(T)\|_2 = \sqrt{\pi/2}, \qquad \|u_j(0)\|_2 = e^{j^2T}\sqrt{\pi/2}.$$

More decisively, $v_j = e^{-j^2T/2}u_j$ satisfies

$$\|v_j(T)\|_2 \longrightarrow 0, \qquad \|v_j(0)\|_2 \longrightarrow \infty.$$

Thus the map from terminal data to the initial solution cannot be continuous at zero in L^2, even when restricted to these smooth solvable data. No claim of nonexistence is needed for this instability argument.

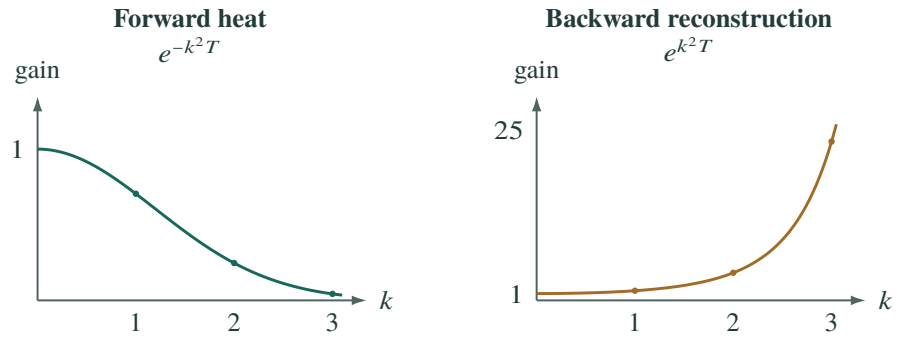


Diagram T1.3. High frequencies distinguish stability from solvability. For the Dirichlet mode $\sin(kx)$ on $(0, \pi)$, forward heat evolution multiplies amplitude by e^{-k^2T}; reconstruction multiplies by its reciprocal. Here $T = 0.35$ and vertical scales differ. Arbitrarily small terminal errors in high modes can therefore give large reconstructed initial errors.

3. Smooth data can create a Burgers singularity. For $u_t + uu_x = 0$ on the line, let $u(x,0) = g(x)$, where $g \in C^1(\mathbb{R})$ and both g and g' are bounded. Along a characteristic starting at ξ,

$$X(\xi,t) = \xi + tg(\xi), \qquad u(X(\xi,t),t) = g(\xi).$$

Before $\partial_\xi X = 1 + tg'(\xi)$ vanishes, inversion and the chain rule give

$$u_x(X(\xi,t),t) = \frac{g'(\xi)}{1+tg'(\xi)}.$$

If $g'(\xi_0) < 0$, this denominator vanishes at $t = -1/g'(\xi_0) > 0$, so the formula cannot continue as a C^1 solution through that event. If the negative minimum $m = \min_{\mathbb{R}} g'$ is attained, the first such time is $T_* = -1/m$. A bounded amplitude therefore does not prevent derivative blow-up.

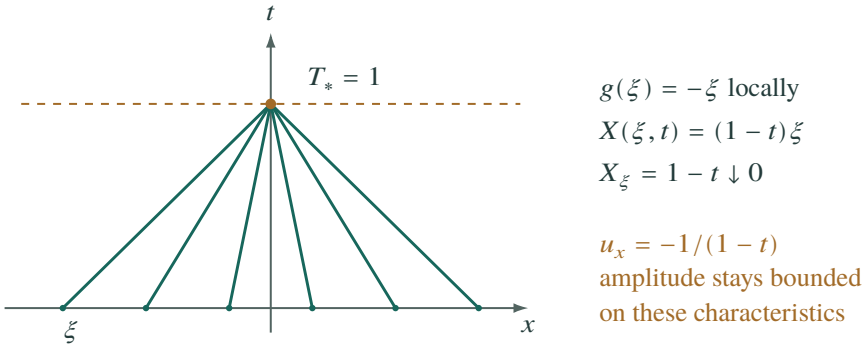


Diagram T1.4. Loss of invertibility precedes a shock. For the local initial profile $g(\xi) = -\xi$, the characteristics meet at $t = 1$. The characteristic formula is valid for $t < 1$; the meeting point depicts failure of a classical continuation, rather than several values of an admissible solution at one point.

To include a discontinuity, write $F(u) = u^2/2$. A locally bounded weak solution with initial data g must satisfy

$$\int_0^\infty \int_{\mathbb{R}} (u\varphi_t + F(u)\varphi_x)\,dx\,dt + \int_{\mathbb{R}} g(x)\varphi(x,0)\,dx = 0$$

for every smooth test function compactly supported in $\mathbb{R} \times [0,\infty)$. The initial term follows from time integration by parts; compact support removes the spatial boundary term.

For a piecewise constant state u_L to the left of $x = st$ and u_R to the right, the delta coefficient in $u_t + F(u)_x$ is $-s(u_R - u_L) + F(u_R) - F(u_L)$. It vanishes precisely when

$$s(u_R - u_L) = F(u_R) - F(u_L).$$

This is the Rankine–Hugoniot condition. For Burgers with $u_L \neq u_R$, it gives $s = (u_L + u_R)/2$. Yet it permits both compressive and expansive

爆破的量必须说明

Burgers 特征线上 $u = g(\xi)$ 始终有界，失控的是 u_x。这叫梯度爆破，不能误写成 $|u| \to \infty$。局部存在定理的延拓判据通常取决于某个特定范数，而不是笼统的“解变大”。

jumps. The entropy criterion selects a compressive shock, $u_L > u_R$, for which $u_R < s < u_L$; a jump with $u_L < u_R$ is not the entropy solution. This calculation explains why merely imposing the weak identity is insufficient. The general entropy theory belongs to the later chapters.

1.4 Overview

This textbook is divided into three major Parts.

Part I: Representation Formulas for Solutions

Here we identify those important partial differential equations for which in certain circumstances explicit or more-or-less explicit formulas can be had for solutions. The general progression of the exposition is from direct formulas for certain linear equations to far less concrete representation formulas, of a sort, for various nonlinear PDE.

Chapter 2 is a detailed study of four exactly solvable partial differential equations: the linear transport equation, Laplace's equation, the heat equation, and the wave equation. These PDE, which serve as archetypes for the more complicated equations introduced later, admit directly computable solutions, at least in the case that there is no domain whose boundary geometry complicates matters. The explicit formulas are augmented by various indirect, but easy and attractive, "energy"-type arguments, which serve as motivation for the developments in Chapters 6, 7 and thereafter.

Chapter 3 continues the theme of searching for explicit formulas, now for general first-order nonlinear PDE. The key insight is that such PDE can, locally at least, be transformed into systems of ordinary differential equations (ODE), the characteristic equations. We stipulate that once the problem becomes "only" the question of integrating a system of ODE, it is in principle solved, sometimes quite explicitly. The derivation of the characteristic equations given in the text is very simple and does not require any geometric insights. It is in truth so easy to derive the characteristic equations that no real purpose is had by dealing with the quasilinear case first.

We introduce also the Hopf–Lax formula for Hamilton–Jacobi equations (§3.3) and the Lax–Oleinik formula for scalar conservation laws (§3.4). (Some knowledge of measure theory is useful here but is not essential.) These sections provide an early acquaintance with the global theory of these important nonlinear PDE and so motivate the later Chapters 10 and 11.

Chapter 4 is a grab bag of techniques for explicitly (or kind of explicitly) solving various linear and nonlinear partial differential equations, and the reader should study only whatever seems interesting. The section on the Fourier transform is, however, essential. The Cauchy–Kovalevskaya Theorem appears at the very end. Although this is basically the only general existence theorem in the subject, and thus logically should perhaps be regarded as central, in practice these power series methods are not so prevalent.

局部存在不等于稳定

Cauchy–Kovalevskaya 定理要求解析系数、解析初值及非特征初始面，给出局部解析解。它不直接给出光滑或 Sobolev 数据下的连续依赖，也不保证全局解。

Part II: Theory for Linear Partial Differential Equations

Next we abandon the search for explicit formulas and instead rely on functional analysis and relatively easy "energy" estimates to prove the existence of weak solutions to various linear PDE. We investigate also the uniqueness and regularity of such solutions and deduce various other properties.

Chapter 5 is an introduction to Sobolev spaces, the proper setting for the study of many linear and nonlinear partial differential equations via energy methods. This is a hard chapter, the real worth of which is only later revealed, and requires some basic knowledge of Lebesgue measure theory. However, the requirements are not really so great, and the review in Appendix E should suffice. In my opinion there is no particular advantage in considering only the Sobolev spaces with exponent $p = 2$, and indeed insisting upon this obscures the two central inequalities, those of Gagliardo–Nirenberg–Sobolev (§5.6.1) and of Morrey (§5.6.2).

指数与维数

Sobolev 空间的指数不能随意固定。当 $1 \le p < n$，临界指数 $p^* = np/(n-p)$ 控制可积性增益；当 $p > n$，Morrey 估计给出 Hölder 连续性。维数在这里直接决定可得的正则性。

In Chapter 6 we vastly generalize our knowledge of Laplace's equation to other second-order elliptic equations. Here we work through a rather complete treatment of existence, uniqueness and regularity theory for solutions, including the maximum principle, and also a reasonable introduction to the study of eigenvalues, including a discussion of the principal eigenvalue for nonselfadjoint operators.

Chapter 7 expands the energy methods to a variety of linear partial differential equations characterizing evolutions in time. We broaden our earlier investigation of the heat equation to general second-order parabolic PDE and of the wave equation to general second-order hyperbolic PDE. We study as well linear first-order hyperbolic systems, with the aim of motivating the developments concerning nonlinear systems of conservation laws in Chapter 11. The concluding Section 7.4 presents the alternative functional analytic method of semigroups for building solutions.

(Missing from this long Part II on linear partial differential equations is any discussion of distribution theory or potential theory. These are

important topics, but for our purposes seem dispensable, even in a book of such length. These omissions do not slow us up much and make room for more nonlinear theory.)

Part III: Theory for Nonlinear Partial Differential Equations

This section parallels for nonlinear PDE the development in Part II but is far less unified in its approach, as the various types of nonlinearity must be treated in quite different ways.

Chapter 8 commences the general study of nonlinear partial differential equations with an extensive discussion of the calculus of variations. Here we set forth a careful derivation of the direct method for deducing the existence of minimizers and discuss also a variety of variational systems and constrained problems, as well as minimax methods. Variational theory is the most useful and accessible of the methods for nonlinear PDE, and so this chapter is fundamental.

Chapter 9 is, rather like Chapter 4 earlier, a gathering of assorted other techniques of use for nonlinear elliptic and parabolic partial differential equations. We encounter here monotonicity and fixed point methods and a variety of other devices, mostly involving the maximum principle. We study as well certain nice aspects of nonlinear semigroup theory, to complement the linear semigroup theory from Chapter 7.

不同的广义解机制

守恒律用积分恒等式表达守恒，再加熵不等式；Hamilton–Jacobi 方程的黏性解用接触试验函数表达微分不等式。两者的稳定性和唯一性机制不同，不能仅以“弱导数存在”统一定义。

Chapter 10 is an introduction to the modern theory of Hamilton–Jacobi PDE and in particular to the notion of “viscosity solutions”. We encounter also the connections with the optimal control of ODE, through dynamic programming.

Chapter 11 picks up from Chapter 3 the discussion of conservation laws, now systems of conservation laws. Unlike the general theoretical developments in Chapters 5–9, for which Sobolev spaces provide the proper abstract framework, we are forced to employ here direct linear algebra and calculus computations. We pay particular attention to the solution of Riemann’s problem and to entropy criteria.

Chapter 12, an introduction to nonlinear wave equations, is new with this edition. We provide long time and short time existence theorems for certain quasilinear wave equations and an in-depth examination of semilinear wave equations, especially for subcritical and critical power nonlinearities in three space dimensions. To complement these existence theorems, the final section identifies various criteria ensuring nonexistence of solutions.

Appendices A–E provide for the reader’s convenience some background material, with selected proofs, on inequalities, linear functional

analysis, measure theory, etc.

The Bibliography is an updated and extensive listing of interesting PDE books to consult for further information. Since this is a textbook and not a reference monograph, I have mostly not attempted to track down and document the original sources for the myriads of ideas and methods we will encounter. The mathematical literature for partial differential equations is truly vast, but the books cited in the Bibliography should at least provide a starting point for locating the primary sources. (Citations to selected research papers appear throughout the text.)

1.5 Problems

Source problems (Evans, Chapter 1, Problems 1–5)

The numbering in this list is the source problem numbering. Solutions in the next subsection are editorial additions.

1. Classify each of the partial differential equations in §1.2 as follows:
   (a) Is the PDE linear, semilinear, quasilinear or fully nonlinear?
   (b) What is the order of the PDE?
2. Let k be a positive integer. Show that a smooth function defined on $\mathbb{R}^n$ has in general
$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$$
distinct partial derivatives of order k.
(Hint: This is the number of ways of inserting $n-1$ dividers | within a row of k symbols $\circ$: for example, $\circ\circ \,||\, \circ\circ\circ \,|\, \circ \,|\, \circ\circ\circ \,||\, \circ\circ\circ\circ|$. Explain why each such pattern corresponds to precisely one of the partial derivatives of order k.)

The next exercises provide some practice with the multiindex notation introduced in Appendix A.

3. Prove the Multinomial Theorem:
$$(x_1 + \cdots + x_n)^k = \sum_{|\alpha|=k} \binom{|\alpha|}{\alpha} x^\alpha,$$
where $\binom{|\alpha|}{\alpha} := |\alpha|!/\alpha!$, $\alpha! = \alpha_1! \cdots \alpha_n!$, and $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. The sum is taken over all multiindices $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $|\alpha| = k$.
4. Prove Leibniz's formula:
$$D^\alpha(uv) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v,$$

where $u, v : \mathbb{R}^n \to \mathbb{R}$ are smooth, $\binom{\alpha}{\beta} := \alpha!/[\beta!(\alpha-\beta)!]$, and $\beta \le \alpha$ means $\beta_i \le \alpha_i$ $(i = 1, \dots, n)$.

5. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is smooth. Prove

$$f(x) = \sum_{|\alpha| \le k} \frac{1}{\alpha!} D^\alpha f(0) x^\alpha + O(|x|^{k+1}) \qquad \text{as } x \to 0$$

for each $k = 1, 2, \dots$. This is Taylor's formula in multiindex notation. (Hint: Fix $x \in \mathbb{R}^n$ and consider the function of one variable $g(t) := f(tx)$.)

Editorial supplement: solutions to Problems 1–5

Problem 1: classification. The following tables use the most restrictive applicable class, for generic nonlinearities. The order is the largest total derivative order in space and time. Coefficients and forcing terms are prescribed, not additional unknowns, unless explicitly stated otherwise.

Scalar linear equation (§1.2.1a)	Class	Order
1. Laplace	Linear	2
2. Helmholtz	Linear	2
3. Transport	Linear	1
4. Liouville	Linear	1
5. Heat	Linear	2
6. Schrödinger	Linear	2
7. Kolmogorov	Linear	2
8. Fokker–Planck	Linear	2
9. Wave	Linear	2
10. Klein–Gordon	Linear	2
11. Telegraph	Linear	2
12. General wave	Linear	2
13. Airy	Linear	3
14. Beam	Linear	4

Scalar nonlinear equation (§1.2.1b)	Class	Order
1. Eikonal	Fully nonlinear	1
2. Nonlinear Poisson	Semilinear	2
3. p-Laplacian	Quasilinear	2
4. Minimal surface	Quasilinear	2
5. Monge–Ampère	Fully nonlinear	2
6. Hamilton–Jacobi	Fully nonlinear	1
7. Scalar conservation law	Quasilinear	1
8. Inviscid Burgers	Quasilinear	1
9. Reaction-diffusion	Semilinear	2
10. Porous medium	Quasilinear	2
11. Nonlinear wave	Semilinear	2
12. KdV	Semilinear	3
13. Nonlinear Schrödinger	Semilinear	2

The p-Laplacian entry assumes $1 < p < \infty$, $p \neq 2$, using its classical expansion away from $Du = 0$ as in §1.2. The porous medium entry assumes $\gamma > 1$ and uses the expansion where $u > 0$. The Monge–Ampère entry assumes $n \geq 2$ and given f. If $H(q, x)$ is affine in q, Hamilton–Jacobi is quasilinear (in fact linear in this particular displayed form). If $f(u)$ is affine, the nonlinear Poisson, reaction-diffusion and nonlinear wave equations become linear. For the Schrödinger equations the unknown is complex-valued; one may equivalently use its real and imaginary parts as a real system.

先约定参数与分支

表中按一般非线性列出。$p = 2$ 时 p-Laplacian 变为线性；$\gamma = 1$ 时多孔介质式变为热方程；$n = 1$ 且 f 已知时 Monge–Ampère 式是 $u_{xx} = f$。忽略这些例外会把分类误当成方程名称的固定属性。

System (§1.2.2)	Class	Order
a1. Elastic equilibrium	Linear	2
a2. Elastic evolution	Linear	2
a3. Maxwell	Linear	1
b1. Conservation laws	Quasilinear	1
b2. Reaction-diffusion	Semilinear	2
b3. Incompressible Euler	Quasilinear	1
b4. Incompressible Navier–Stokes	Semilinear	2

For a smooth scalar flux, the chain rule gives

$$\operatorname{div} \mathbf{F}(u) = \sum_i (F^i)'(u) u_{x_i},$$

hence its quasilinear form. For a system with m components, $\mathbf{F} : \mathbb{R}^m \to \mathbb{R}^{m \times n}$ and $(\operatorname{div} \mathbf{F}(\mathbf{u}))^a = \sum_i \partial_{x_i} F^{ai}(\mathbf{u})$. Affine fluxes give linear equations or systems.

For the incompressible flow entries, the unknowns are $(\mathbf{u}, p)$. Navier–Stokes has constant coefficients on the second derivatives of $\mathbf{u}$; its nonlinear convection term has order one. This is the sense of the semilinear classification in the table. The divergence equation has order one, and there

are no second derivatives of p in the displayed system. It is a system of mixed differential orders, not an unconstrained uniformly parabolic system for all $n+1$ unknowns. Euler has first-order derivatives with coefficients depending on $\mathbf{u}$, hence is quasilinear.

Problem 2: counting independent derivative operators.

Expanded proof (editorial solution to source Problem 2).

For a C^k function, the order of differentiation does not matter. Thus a distinct derivative operator of order k is determined by a multiindex $\alpha \in \{0,1,2,\ldots\}^n$ satisfying $\alpha_1 + \cdots + \alpha_n = k$. Represent α_i by α_i circles in the ith compartment between $n-1$ dividers. Empty compartments are allowed: they encode $\alpha_i = 0$. This is a bijection.

Among the $k+n-1$ positions, choose the $n-1$ divider positions, giving $\binom{k+n-1}{n-1} = \binom{k+n-1}{k}$. For $n = 1$ there are no dividers and exactly one derivative, as the same formula predicts. This counts operators; special functions may have several derivatives with equal values. □

Problem 3: the multinomial theorem.

Expanded proof (editorial solution to source Problem 3).

Expand the product of k copies of $(x_1 + \cdots + x_n)$ by distributivity. A term x^α is obtained by selecting x_i from exactly α_i of the factors. First choose α_1 of the k factors, then α_2 of the remaining factors, and so on. The number of such choices is

$$\binom{k}{\alpha_1}\binom{k-\alpha_1}{\alpha_2}\cdots\binom{\alpha_n}{\alpha_n} = \frac{k!}{\alpha_1!\cdots\alpha_n!}.$$

Grouping all terms with the same multiindex proves the formula. The identity also holds for $k = 0$, with the single multiindex $\alpha = 0$. □

两类组合系数

多项式展开中 $\binom{k}{\alpha} = k!/\alpha!$，要求 $|\alpha| = k$。乘积求导中 $\binom{\alpha}{\beta} = \prod_i \binom{\alpha_i}{\beta_i}$，逐坐标分配导数。后者一般不等于 $\binom{|\alpha|}{|\beta|}$。

Problem 4: the multiindex product rule.

Expanded proof (editorial solution to source Problem 4).

Apply the one-variable product rule α_i times in coordinate x_i, for each $i = 1,\ldots,n$. If exactly β_i of these derivatives fall on u, the remaining $\alpha_i - \beta_i$ fall on v, and the coefficient is $\binom{\alpha_i}{\beta_i}$. Since mixed derivatives commute under the smoothness assumption, the resulting term is $D^\beta u\, D^{\alpha-\beta} v$, with coefficient

$$\prod_{i=1}^{n}\binom{\alpha_i}{\beta_i} = \frac{\alpha!}{\beta!(\alpha-\beta)!}.$$

Summing over all coordinatewise choices $0 \le \beta_i \le \alpha_i$ proves the identity. For $\alpha = 0$ it reduces to $uv = uv$. □

Problem 5: Taylor's formula with a uniform remainder.

Expanded proof (editorial solution to source Problem 5).

It suffices that $f \in C^{k+1}$ on a neighborhood of 0. Choose $r > 0$ so the closed ball $\overline{B}_r(0)$ lies inside this neighborhood. For $|x| < r$ set $g_x(t) = f(tx)$, $0 \le t \le 1$. Repeated differentiation and the multinomial theorem give, for $0 \le j \le k+1$,

$$g_x^{(j)}(t) = \left(\sum_{i=1}^{n} x_i \partial_{x_i}\right)^j f(tx) = \sum_{|\alpha|=j} \frac{j!}{\alpha!} x^\alpha D^\alpha f(tx).$$

The one-variable Taylor formula with integral remainder gives

$$f(x) = \sum_{j=0}^{k} \frac{g_x^{(j)}(0)}{j!} + R_k(x),$$
$$R_k(x) = \frac{1}{k!} \int_0^1 (1-t)^k g_x^{(k+1)}(t)\, dt.$$

Substituting the derivative formula converts the finite sum into

$$\sum_{|\alpha|\le k} \frac{D^\alpha f(0)}{\alpha!} x^\alpha.$$

Continuity and compactness give a finite common bound

$$M = \max_{\substack{|\alpha|=k+1 \\ y\in\overline{B}_r(0)}} |D^\alpha f(y)|.$$

Using $|x^\alpha| \le |x|^{k+1}$ and $\int_0^1 (1-t)^k\, dt = 1/(k+1)$, we obtain

$$|R_k(x)| \le M \left(\sum_{|\alpha|=k+1} \frac{1}{\alpha!}\right) |x|^{k+1}.$$

The constant depends only on n, k and the indicated local derivative bound, not on the direction of x. This proves the required $O(|x|^{k+1})$. □

1.6 References

Klainerman's article [Kl] is a nice modern overview of the field of partial differential equations.

Good general texts and monographs on PDE include Arnold [Ar2], Courant–Hilbert [C-H], Craig [Cr], DiBenedetto [DB1], Folland [F1], Friedman [Fr2], Garabedian [G], John [J2], Jost [Jo], Levy–Shearer [L-S] McOwen [MO], Mikhailov [M], Olver [O2] Petrovsky [Py], Rauch [R1], Renardy–Rogers [R-R], Showalter [Sh], Smirnov [Sm], Smoller [S], Strauss [St2], Taylor [Ta], Thoe–Zachmanoglou [T-Z], Vasy [Vy], Zauderer [Za], and many others. The prefaces to Arnold [Ar2] and to Bernstein [Bt] are particularly interesting reading. Zwillinger's handbook [Zw] on differential equations is a useful compendium of methods for PDE.

Editorial source note. The bracketed keys above are Evans's bibliography identifiers, retained as source identifiers rather than local citation numbers. Their full entries are in the supplied English source, printed p. 689 onward (PDF page 706 onward). This list records the supplied printing; it is not an updated literature survey. The supplied Chinese printing has a shorter list, omitting, for example, Craig [Cr], Jost [Jo], Levy–Shearer [L-S], Olver [O2], and Vasy [Vy]. The English printing controls the transcription here.

余项须对方向一致

逐条射线使用一元 Taylor 公式还不够：余项常数可能依赖方向。在固定闭球上统一控制所有 $k+1$ 阶导数，才得到真正的多元 $O(|x|^{k+1})$。仅假设 C^k 通常只能得到 $o(|x|^k)$。

Part I

Representation Formulas for Solutions

Chapter 2

Four Important Linear Partial Differential Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 2; printed pp. 17–87; PDF pages 34–104.

Chinese reference: 中译本：书内第 13 页起；球均值见第 18–19 页，热球均值见第 35–37 页。

Editorial study edition. The exposition and proofs are reconstructed in the order of Evans, Chapter 2; they are not a verbatim transcription. Source theorem numbers are identified separately from our chapter-wide numbering. Chinese mathematical notes, worked examples, figures, and exercise solutions are editorial additions.

Our main route is

$$\text{fundamental solutions} \longrightarrow \text{representation formulas} \longrightarrow \text{estimates and uniqueness.}$$

The central comparison is between harmonic means (§2.2.2) and the heat-ball mean (§2.3.2). The latter is a positive weighted average over past space-time, not a uniform average over an ordinary ball.

四个模型四种机制

传输沿特征线搬运；Laplace 方程描述静态平衡；热方程耗散并向未来扩散；波动方程保存能量并有限速传播。相同的“平均”符号背后可能是完全不同的测度。

Conventions and prerequisites. $\omega_n = |B_1|$ and $|\partial B_1| = n\omega_n$. The elliptic operator is $-\Delta$; the heat operator is $L = \partial_t - \Delta_x$; the wave speed is one unless stated otherwise. $C^{2,1}$ means two continuous spatial derivatives and one continuous time derivative. Basic multivariable calculus, Green's identities, polar integration, dominated convergence, convolution, and smooth cutoffs are prerequisites. Boundary correctors on general domains are explicitly conditional on their existence; ball and half-space kernels are constructed here.

2.1 Transport Equation

2.1.1 Characteristics and the initial-value problem (Evans §2.1.1)

For fixed $b \in \mathbb{R}^n$, consider $u_t + b \cdot Du = 0$, $u(x,0) = g(x)$. Through (x,t) trace the line

$$X(s) = x + (s-t)b, \qquad 0 \le s \le t.$$

For a C^1 solution, the chain rule gives

$$\frac{d}{ds} u(X(s),s) = Du(X(s),s) \cdot b + u_t(X(s),s) = 0.$$

Integrating from 0 to t proves necessity of

$$\boxed{u(x,t) = g(x-tb).}$$

Conversely, if $g \in C^1$, differentiation of this formula yields $u_t = -b \cdot Dg(x-tb)$ and $Du = Dg(x-tb)$, proving the PDE and initial condition. This establishes both existence and uniqueness in the classical class, rather than merely guessing a candidate.

回溯特征线

计算当前点 (x,t) 的值，要沿速度 b 回到初始平面，落点是 $x-tb$。正负号可用一维脉冲向右移动检验：$g(x-bt)$ 的峰位于 $x = bt$。

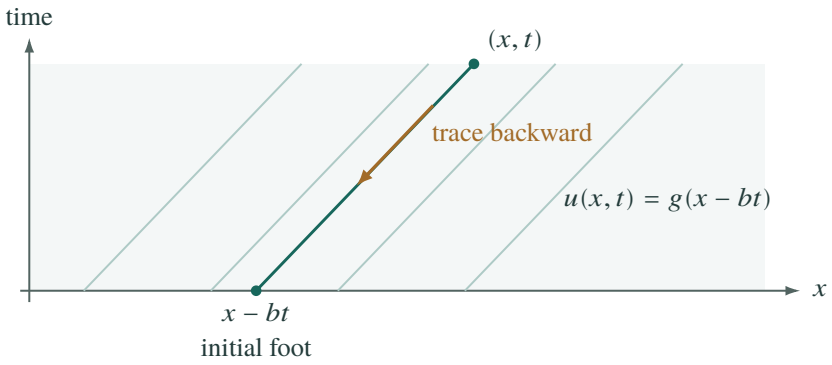


Diagram T2.1. Read a transport solution backward. The solution is constant on each characteristic when the source is zero. With $b > 0$, the forward line tilts right, while evaluation at a target requires its leftward initial foot $x - bt$. The orange arrow indicates evaluation backward in time, not a negative physical velocity.

Nonsmooth data and weak solutions. For $g \in L^1_{\mathrm{loc}}$, the same formula is locally integrable. For a compactly supported smooth test function on $\mathbb{R}^n \times [0,\infty)$, change variables $x = z + tb$:

$$\begin{aligned}\int_0^\infty \int u(\phi_t + b \cdot D\phi)\,dx\,dt &= \int_{\mathbb{R}^n} g(z) \int_0^\infty \frac{d}{dt}\phi(z+tb,t)\,dt\,dz \\ &= -\int_{\mathbb{R}^n} g(z)\phi(z,0)\,dz.\end{aligned}$$

Fubini is justified by compact test support and local integrability. This is the distributional formulation with the initial datum included. There is no need to assign a classical derivative at a moving jump. Translations preserve L^p norms: $\|u(\cdot,t)\|_p = \|g\|_p$ whenever $g \in L^p$.

2.1.2 Forcing and Duhamel's principle (Evans §2.1.2)

If $u_t + b \cdot Du = f$ and $u(x,0) = g(x)$, then

$$\frac{d}{ds}u(X(s),s) = f(X(s),s).$$

Integration along the same characteristic gives

$$\boxed{u(x,t) = g(x - tb) + \int_0^t f(x - (t-s)b, s)\, ds.}$$

For C^1 data differentiate under the integral, with the upper endpoint contributing $f(x,t)$; the remaining integrals cancel between u_t and $b \cdot Du$. This directly verifies the formula.

For the additional constant term cu, multiply along characteristics by e^{cs}:

$$u(x,t) = e^{-ct}g(x - tb) + \int_0^t e^{-c(t-s)}f(x - (t-s)b, s)\, ds.$$

The damping factor belongs to the elapsed time $t - s$, not to s alone.

源项沿轨迹累积

在时间 s 注入的源，之后还要平移 $(t-s)b$ 才到达当前点。Duhamel 的共同思想是把每个时刻的小冲量，按齐次方程传播后相加。

2.2 Laplace's Equation

2.2.1 Fundamental solution (Evans §2.2.1, Theorem 1)

A radial function $v(x) = F(r)$, $r = |x| > 0$, satisfies

$$\Delta v = F''(r) + \frac{n-1}{r}F'(r).$$

Thus harmonicity off zero gives $(r^{n-1}F')' = 0$. Integration yields a constant plus r^{2-n} if $n \geq 3$, or a constant plus $\log r$ if $n = 2$. Choose the constants so that the inward flux is one:

$$\Phi(x) = \begin{cases} [n(n-2)\omega_n]^{-1}|x|^{2-n}, & n \geq 3, \\ -(2\pi)^{-1}\log|x|, & n = 2. \end{cases} \qquad -\Delta\Phi = \delta_0.$$

符号由通量固定

我们求解 $-\Delta u = f$，故要求 $-\int_{\partial B_r}\partial_\nu\Phi = 1$。若改用 Δ 作基本算子，基本解及 Green 公式的符号也要一起改。

Expanded proof (the distributional singularity).

The displayed function is locally integrable and harmonic away from zero, and $\partial_r\Phi = -1/(n\omega_n r^{n-1})$. For $\phi \in C_c^\infty$, apply Green's identity on a large ball containing its support, with $\overline{B}_\varepsilon$ removed:

$$\int_{|x|>\varepsilon} \Phi(-\Delta\phi) = \int_{\partial B_\varepsilon} (\Phi\partial_r\phi - \phi\partial_r\Phi)\, dS.$$

The first term tends to zero: it is $O(\varepsilon)$ for $n \geq 3$, and $O(\varepsilon|\log\varepsilon|)$ for $n = 2$. The second is the normalized spherical mean of ϕ, tending to $\phi(0)$. Local integrability permits passage in the volume integral. Hence $-\Delta\Phi = \delta_0$ as a distribution. □

For $f \in C_c^2(\mathbb{R}^n)$ put $u = \Phi * f$. Write $u(x) = \int \Phi(y) f(x-y)\,dy$ and differentiate f, not the singular kernel. On each compact set of x, all these integrals have common bounded support in y and are dominated by an integrable multiple of $|\Phi|$. Thus $u \in C^2$ and

$$-\Delta u(x) = \int \Phi(y)\,(-\Delta_y f(x-y))\,dy = f(x)$$

by the distribution identity just proved. This avoids an illegal unqualified differentiation of $D^2\Phi$ at zero. For $n = 1$, $\Phi(x) = -|x|/2$ has the same normalization.

2.2.2 Mean-value formulas (Evans §2.2.2, Theorems 2–3)

Write $\omega_n = |B_1|$ and $\sigma_{n-1} = |\partial B_1| = n\omega_n$. For $\overline{B}_r(x) \subset U$ define the *normalized* means

$$S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u\,dS, \qquad A_u(x,r) = \frac{1}{\omega_n r^n} \int_{B_r(x)} u\,dy.$$

The sphere is a boundary of dimension $n-1$; the ball has dimension n. These are two different averaging measures.

平均不是积分本身

球面平均除以面积，球体平均除以体积。对常数 $u = 1$，两者都必须等于 1；漏掉 r^{n-1} 或 r^n 就会破坏这一检验。

Theorem 2.1 (Mean-value identities)
If $u \in C^2(U)$ and $\Delta u = 0$, then

$$u(x) = S_u(x,r) = A_u(x,r) \quad \text{whenever } \overline{B}_r(x) \subset U.$$

More generally, for any C^2 function,

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u(y)\,dy. \tag{M}$$

Expanded proof (all radius and surface factors).

1. Remove the moving surface before differentiating. Set $y = x + r\theta$ on the unit sphere. Its surface Jacobian is r^{n-1}, so

$$S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} u(x + r\theta)\,dS_\theta.$$

On a fixed compact range of radii, Du is continuous and bounded. Differentiation under this fixed finite integral is legitimate:

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} Du(x + r\theta)\cdot\theta\,dS_\theta.$$

Transforming back, $\theta = \nu(y)$ is the outward unit normal, giving

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} Du\cdot\nu\,dS.$$

The divergence theorem on the smooth ball changes the flux into $\int_{B_r(x)} \Delta u$. This proves (M).

2. Identify the constant. If $\Delta u = 0$, the derivative in (M) vanishes for every admissible r. Hence the spherical mean is independent of radius. Continuity gives

$$|S_u(x,r) - u(x)| \le \sup_{|y-x|\le r} |u(y) - u(x)| \longrightarrow 0.$$

The constant is therefore exactly $u(x)$, not an unspecified constant.

3. Recover the ball average by adding spherical shells. Polar integration yields

$$\int_{B_r(x)} u = \int_0^r \int_{\partial B_\rho(x)} u \, dS \, d\rho = \int_0^r n\omega_n \rho^{n-1} u(x) \, d\rho = \omega_n r^n u(x).$$

Divide by the volume. This also proves, for general continuous u,

$$A_u(x,r) = \frac{n}{r^n} \int_0^r \rho^{n-1} S_u(x,\rho) \, d\rho. \tag{SB}$$

□

Can one recover spheres from balls? Yes. Differentiating $r^n A_u(x,r)$ in (SB) gives

$$S_u(x,r) = A_u(x,r) + \frac{r}{n} \partial_r A_u(x,r).$$

Consequently a continuous function with the ball-mean identity for every small radius also has the sphere-mean identity. A single radius at a single center is not sufficient.

Theorem 2.2 (Converse, including a continuous version)
If $u \in C^2(U)$ has either mean identity for every sufficiently small ball at each center, then $\Delta u = 0$. If $u \in C(U)$ has this property at every center and all locally admissible small radii, it is smooth and harmonic.

Expanded proof.

For C^2 functions Taylor-expand at x with a uniform $o(r^2)$ error. Symmetry of the sphere cancels linear and mixed terms; its normalized second moments are $\mathbb{E}(\theta_i\theta_j) = \delta_{ij}/n$. The ball has moments $\mathbb{E}(z_i z_j) = r^2\delta_{ij}/(n+2)$. Thus

$$S_u(x,r) = u(x) + \frac{r^2}{2n}\Delta u(x) + o(r^2),$$

$$A_u(x,r) = u(x) + \frac{r^2}{2(n+2)}\Delta u(x) + o(r^2). \tag{T}$$

图 T2.M：球体由球壳相加



shell volume
$n\omega_n \rho^{n-1}\, d\rho$

先对每个半径做球面平均，再乘球壳体积并积分。体积平均的径向权重不是均匀的。

方程如何变成平均

关键中介是通量：$\Delta u = 0$ 令每个球的总法向通量为零，通量又恰好是球面平均对半径的导数。故平均不随尺度变化。

An exact identity for arbitrarily small r forces $\Delta u(x) = 0$.

For a merely continuous u, first turn ball means into sphere means using (SB), if necessary. Take a radial smooth mollifier of mass one, supported in B_ε. In polar coordinates,

$$(\rho_\varepsilon * u)(x) = \int_0^\varepsilon \rho_\varepsilon(r) \int_{\partial B_r(x)} u \, dS \, dr = u(x).$$

The convolution is smooth on the smaller domain; hence u is smooth there and throughout U. Now apply the C^2 converse. This proves the smoothness assertion without differentiating an initially nonsmooth function. □

Subharmonic and superharmonic functions. If $\Delta v \geq 0$, (M) implies $S_v(x,r) \geq v(x)$ and then $A_v(x,r) \geq v(x)$. If $\Delta v \leq 0$, reverse both inequalities. For $-\Delta u = f$, integrating (M) and changing the order of integration gives, for $n \geq 3$,

$$u(x) = S_u(x,r) + \int_{B_r(x)} \frac{|y-x|^{2-n} - r^{2-n}}{n(n-2)\omega_n} f(y)\, dy. \tag{P}$$

Indeed $\int_{|y-x|}^r \rho^{1-n}\, d\rho = (|y-x|^{2-n} - r^{2-n})/(n-2)$. For $n = 2$ replace the kernel by $(2\pi)^{-1}\log(r/|y-x|)$. The kernel is nonnegative: a positive source raises $u(x)$ above its boundary average under the convention $-\Delta u = f$.

Concrete checks. In two dimensions, $u(y) = y_1^2 - y_2^2$ and $u(y) = y_1 y_2$ have $\Delta u = 0$, so both means reproduce $u(x)$ at every center. For $v(y) = |y|^2$,

$$S_v(0,r) = r^2, \qquad A_v(0,r) = \frac{n}{n+2} r^2, \qquad v(0) = 0.$$

For $n = 1$, the sphere consists of two endpoints: $S_u(x,r) = [u(x-r) + u(x+r)]/2$. Harmonic functions are affine, and the same identities remain valid with $\omega_1 = 2$.

2.2.3 Consequences of averaging (Evans §2.2.3)

Theorem 2.3 (Strong maximum principle; source Theorems 4–5)
If U is bounded and $u \in C^2(U) \cap C(\overline{U})$ is harmonic, its maximum and minimum are attained on ∂U. If U is connected and a maximum is attained inside, u is constant. Consequently the classical Dirichlet problem $-\Delta u = f$, $u = g$ has at most one solution in this class.

二阶误差读出 Laplace

普通函数的平均一般不等于中心值；首个对称性允许的误差就是 $r^2\Delta u$。调和性让这一误差及全部更高阶平均误差精确抵消。

别混同一元中值定理

这里是积分平均精确重现中心值，不是声称“存在某个中间点”。后面的热球公式也保持中心值，但必须改变平均的几何区域和权重。

Expanded proof.

Let $M = \max_{\overline{U}} u$ and suppose $u(x) = M$ at an interior point. The ball identity gives $\int_{B_r(x)} (M - u) = 0$. The integrand is nonnegative and continuous. If it were positive at one point, continuity would give a positive integral on a smaller ball, a contradiction. Thus $u = M$ throughout $B_r(x)$. The set $\{u = M\}$ is open and relatively closed in U. On a connected component containing an interior maximum it is the whole component, and continuity transfers this value to that component's boundary, which lies in ∂U. This proves the boundary maximum statement even if U has several components. Apply the argument to $-u$ for minima. The difference of two Dirichlet solutions is harmonic with zero boundary values, so its maximum and minimum are both zero. □

正权平均的等号条件

平均值达到所有被平均数的上界，意味着它们几乎处处都达到该上界；连续性再把“几乎处处”提升为处处。热球的强最大值原理使用同一机制。

Smoothness and derivative estimates (source Theorems 6–7). Radial convolution equals u by the mean identity, as proved in §2.2.2, so harmonic functions are C^∞. Fix a radial mollifier ρ with support in B_1. For $B_r(x) \Subset U$, use a convolution radius $r/2$ near x:

$$|D^\alpha u(x)| = \left| \int D^\alpha \rho_{r/2}(x - y) u(y)\, dy \right| \le C_{\alpha,n} r^{-n-|\alpha|} \|u\|_{L^1(B_r(x))}. \tag{D}$$

All derivatives land on the smooth kernel; its scaling explains the power of r. With a first derivative the same argument gives $\|Du\|_{L^\infty(B_{R-\delta})} \le C_n \delta^{-1} \|u\|_{L^\infty(B_R)}$.

> **Theorem 2.4** (Liouville and bounded Poisson representation; source 8–9)
> *A bounded harmonic function on $\mathbb{R}^n$ is constant. For $n \ge 3$ and $f \in C_c^2$, every bounded solution of $-\Delta u = f$ is $\Phi * f + C$.*

Expanded proof.

Apply (D) for $|\alpha| = 1$ on $B_R(x)$: $|Du(x)| \le C_n R^{-1} \|u\|_\infty \to 0$ as $R \to \infty$. Thus $Du = 0$ and connectedness makes u constant. For $n \ge 3$, $\Phi * f$ is continuous and tends to zero at infinity, hence bounded. Subtract it from u and apply Liouville. For $n = 2$, the logarithmic potential need not be bounded at infinity, so that final step cannot be asserted with the same hypotheses. □

> **Theorem 2.5** (Analyticity; source Theorem 10)
> *Harmonic functions are real analytic in the interior.*

Expanded proof (quantitative iteration rather than smoothness alone).

Fix $B_R(x_0) \Subset U$, with $M = \|u\|_{L^\infty(B_R)}$. Every derivative of u is harmonic, because derivatives commute with Δ. Iterate the first-derivative

estimate k times on concentric balls whose successive radii differ by $R/(2k)$. For any $|\alpha| = k$ this gives

$$\|D^\alpha u\|_{L^\infty(B_{R/2})} \le M(2C_n k/R)^k \le M(2eC_n/R)^k k!.$$

The last inequality follows from $\log(k!) = \sum_{j=1}^{k} \log j \ge \int_1^k \log s\, ds \ge k\log k - k$. For h small enough that the segment from x_0 to $x_0 + h$ stays in $B_{R/2}$, apply one-dimensional Taylor's theorem to $q(t) = u(x_0 + th)$. Its kth derivative is

$$q^{(k)}(t) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} h^\alpha D^\alpha u(x_0 + th).$$

Dividing the remainder by $k!$ and using $\sum_{|\alpha|=k} |h^\alpha|/\alpha! = |h|_1^k/k!$ bounds it by $M(2eC_n|h|_1/R)^k$, tending to zero for sufficiently small h. Thus the multivariable Taylor series equals u near x_0. □

Theorem 2.6 (Harnack; source Theorem 11)
For connected $V \Subset U$ and nonnegative harmonic u, $\sup_V u \le C_{U,V,n} \inf_V u$.

Expanded proof.

Choose $\rho > 0$ with every $B_{2\rho}(z)$, $z \in V$, inside U. If $x, y \in V$ and $|x - y| \le \rho$, then $B_\rho(y) \subset B_{2\rho}(x)$ and positivity gives

$$u(x) = \frac{1}{|B_{2\rho}|} \int_{B_{2\rho}(x)} u \ge 2^{-n} \frac{1}{|B_\rho|} \int_{B_\rho(y)} u = 2^{-n} u(y).$$

Exchange x, y for the other inequality. Compactness gives a finite $\rho/4$ net of V. Connectedness permits a chain between any two net points with successive centers separated by at most ρ; one may obtain this by covering a path in V and deleting loops in the finite cover graph. There is a uniform finite bound on the chain length. Multiplying the local factors proves the global comparison. If one value is zero, the same inequalities force every value in V to be zero, so the statement includes that case. □

光滑不等于解析

C^∞ 只说明每阶导数存在；解析还需控制高阶导数的增长。这里的 $k!$ 型估计让 Taylor 余项趋零，不能省略这一环。

2.2.4 Green and Poisson kernels (Evans §2.2.4)

For sufficiently smooth u, v, the two Green identities are

$$\int_U Du \cdot Dv = -\int_U v\Delta u + \int_{\partial U} v\partial_\nu u,$$

$$\int_U (u\Delta v - v\Delta u) = \int_{\partial U} (u\partial_\nu v - v\partial_\nu u).$$

The second is the difference of the first identity in the two orders.

Boundary correction. For fixed $x \in U$, suppose a harmonic corrector h_x with $h_x(y) = \Phi(y - x)$ on ∂U has been constructed. Set $G(x, y) = \Phi(y - x) - h_x(y)$. Then

$$-\Delta_y G(x, y) = \delta_x, \qquad G(x, y) = 0 \quad (y \in \partial U).$$

General-domain existence of this corrector is not proved by the notation; it belongs to Dirichlet existence theory later in the book. The following representation is conditional on such a sufficiently regular Green function. Explicit examples below avoid that existence issue.

Green 函数是修正后的点源

基本解有正确的点源奇性，却通常不满足边界条件。减去调和修正项保留奇性、消掉边界值；“定义修正项”本身不等于证明它存在。

Theorem 2.7 (Representation and symmetry; source Theorems 12–13)
For $-\Delta u = f$, $u = g$ on the boundary,

$$u(x) = \int_U G(x, y) f(y)\, dy - \int_{\partial U} g(y)\, \partial_{\nu_y} G(x, y)\, dS_y. \qquad \text{(G)}$$

Furthermore $G(x, y) = G(y, x)$.

Expanded proof.

Apply the second Green identity on $U \setminus \overline{B}_\varepsilon(x)$ to u and $G(x, \cdot)$. Away from x, $\Delta G = 0$. On the outer boundary $G = 0$, leaving $\int g \partial_\nu G$. On the inner boundary the singular normal derivative has total flux one, with normal directed into the removed ball; the resulting term tends to $u(x)$. Terms involving $G \partial_\nu u$ tend to zero at rates $O(\varepsilon)$ or $O(\varepsilon |\log \varepsilon|)$. Thus $\int_U Gf = \int_{\partial U} g \partial_\nu G + u(x)$, giving (G).

For symmetry, apply the same second identity to $G(x, \cdot), G(z, \cdot)$ with small disjoint balls about x, z removed. Both are harmonic on the remaining domain and zero on the outer boundary. At the sphere around x, the term with the singular normal derivative tends to $G(z, x)$ and the opposite product tends to zero; the sphere around z gives $-G(x, z)$. Their sum is zero, proving symmetry. □

The upper half-space (source Theorem 14). For $x_n > 0$ let $x^* = (x', -x_n)$. Reflection gives $G(x, y) = \Phi(y - x) - \Phi(y - x^*)$. The singularity of the image lies outside the domain, and the two distances agree on $y_n = 0$. Its boundary normal derivative gives

$$P(x, z) = \frac{2x_n}{n\omega_n (|x' - z|^2 + x_n^2)^{n/2}}, \qquad z \in \mathbb{R}^{n-1}.$$

$$u(x) = \int_{\mathbb{R}^{n-1}} P(x, z) g(z)\, dz.$$

The kernel is harmonic in x, positive, and has mass one. To check the mass without imposing boundary decay, put $z = x' + x_n w$; the remaining integral is $[2/(n\omega_n)] \int (1 + |w|^2)^{-n/2}\, dw = 1$. Its value follows by

polar coordinates, or the Gaussian integral and Gamma substitution. For bounded continuous g, split the boundary integral into $|z - x_0'| < \delta$ and its complement. Uniform continuity near x_0' controls the first error; the latter kernel mass tends to zero as $x \to (x_0', 0)$. This proves boundary recovery and, by differentiation on compact interior sets, harmonicity. Uniqueness here is among bounded solutions: a bounded zero-boundary harmonic function extends oddly across the plane, is harmonic by the reflection argument in Problem 10, and is constant by Liouville, hence zero.

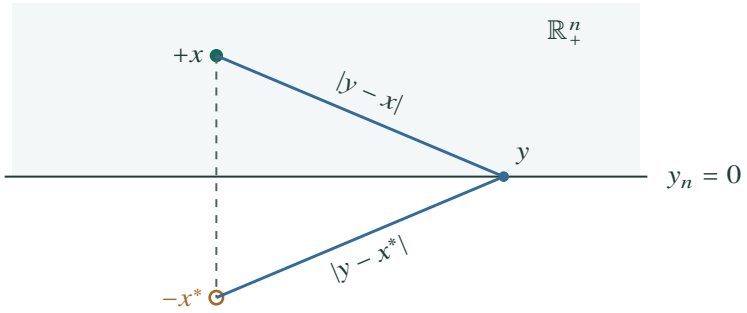


Diagram T2.2. A reflected point source enforces the boundary value. The solid source at x is inside the upper half-space; the image at x^* is outside and has the opposite sign. Every boundary point y is equidistant from the pair, so their potentials cancel there. The labels indicate source signs, not signed coordinate locations.

The ball (source Theorem 15). For $n \geq 3$, $0 < |x| < R$, put $x^* = R^2 x/|x|^2$. An explicit Green function is

$$G(x, y) = \Phi(y - x) - (R/|x|)^{n-2} \Phi(y - x^*).$$

On $|y| = R$, $|y - x^*| = (R/|x|)|y - x|$, so the boundary value is zero. At $x = 0$ the formula has the limit $\Phi(y) - \Phi(R)$. In two dimensions use instead

$$G(x, y) = \frac{1}{2\pi} \log \frac{|x|\,|y - x^*|}{R|y - x|},$$

with the corresponding center limit $(2\pi)^{-1} \log(R/|y|)$. Differentiating in the boundary normal gives, for every $n \geq 2$,

$$P_R(x, y) = \frac{R^2 - |x|^2}{n\omega_n R|x - y|^n}, \qquad u(x) = \int_{\partial B_R} P_R(x, y) g(y)\, dS_y. \tag{PB}$$

Expanded proof (why the boundary formula solves the problem).

The constructed Green function and (G) applied to $u = 1$ give $\int_{\partial B_R} P_R = 1$. This uses a known constant harmonic function, so does

not assume existence for arbitrary data. For $|y| = R$, direct differentiation yields

$$\Delta_x\big[(R^2 - |x|^2)|x-y|^{-n}\big] = 2n(R^2 - |y|^2)|x-y|^{-n-2} = 0.$$

Thus the integral is harmonic; all differentiations are dominated on compact subsets separated from the sphere. For $x \to y_0 \in \partial B_R$ split $g(y) - g(y_0)$ into a small spherical neighborhood and its complement. Positivity and mass one control the near part by the modulus of continuity. On the complement the denominator stays bounded below while $R^2 - |x|^2 \to 0$, so the far part tends to zero. This proves the boundary data. The maximum principle gives uniqueness. At $x = 0$, P_R is constant and (PB) reduces exactly to the spherical mean identity. □

2.2.5 Energy and Dirichlet's principle (Evans §2.2.5, Theorems 16–17)

Let U be bounded and smooth enough for the following integrations. For two $C^2(\overline{U})$ solutions of the same Dirichlet problem, their difference w has $\Delta w = 0$ and $w|_{\partial U} = 0$. Green's first identity gives $\int_U |Dw|^2 = -\int_U w\Delta w + \int_{\partial U} w\partial_\nu w = 0$. Thus w is constant on each component, and its boundary values make it zero.

For $-\Delta u = f$, $u = g$, define

$$J(v) = \int_U \left(\tfrac{1}{2}|Dv|^2 - fv\right) dx.$$

For $v = u + w$ with $w|_{\partial U} = 0$,

$$J(v) - J(u) = \int_U (Du\cdot Dw - fw) + \tfrac{1}{2}\int_U |Dw|^2 = \tfrac{1}{2}\int_U |Dw|^2 \geq 0.$$

Here the linear term vanishes by Green's identity and the PDE. Conversely, if a smooth u minimizes J for its boundary data, test $v = u + \varepsilon\phi$ with $\phi \in C_c^\infty(U)$ and both signs of ε. Differentiating the quadratic polynomial at zero gives $\int Du\cdot D\phi = \int f\phi$, hence $-\Delta u = f$. This proves the equivalence in the smooth class; existence of a minimizer in a Sobolev class uses the later functional-analytic theory.

中心平均与偏心平均

从球心看，边界各方向权重相同；从偏心点看，靠近该点的边界权重更大，变成 Poisson 核。球面均值是 Poisson 表示的球心特例。

2.3 Heat Equation

2.3.1 The Gaussian kernel and Duhamel (Evans §2.3.1)

Heat balance with unit diffusivity is $u_t = \Delta u$. The scaling $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$ suggests a mass-preserving kernel $t^{-n/2}F(x/\sqrt{t})$. Substitution gives $\Delta F + \frac{1}{2}z\cdot DF + \frac{n}{2}F = 0$. Taking $F(z) = Ce^{-|z|^2/4}$ satisfies this equation;

从第二章走向第五章

能量只含一阶导数，所以弱形式 $\int Du\cdot D\phi = \int f\phi$ 自然进入 H^1。第二章的显式公式与第五章的函数空间方法在这里接上。

the product of one-dimensional Gaussian integrals fixes $C = (4\pi)^{-n/2}$. Thus

$$K(x,t) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)} \quad (t>0), \qquad K = 0 \quad (t \le 0).$$

The value at the single point $(0,0)$ is interpreted distributionally. Direct differentiation verifies

$$K_t = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K, \qquad \Delta K = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K.$$

Also $K > 0$, $\int K(x,t)\,dx = 1$, and $\int_{|x|>\delta} K(x,t)\,dx \to 0$ as $t \downarrow 0$. The last assertion follows from $x = \sqrt{t}\,z$ and the integrable Gaussian tail.

Theorem 2.8 (Source Theorem 1)
For bounded continuous g,

$$u(x,t) = \int_{\mathbb{R}^n} K(x-y,t) g(y)\,dy$$

is smooth for $t > 0$, solves the heat equation, and tends to $g(x_0)$ as $(x,t) \to (x_0,0)$.

Expanded proof.

On compact subsets of $t > 0$, every kernel derivative is a polynomial times a Gaussian with an integrable uniform majorant. Differentiate under the integral and use $K_t = \Delta K$. For initial convergence use mass one:

$$u(x,t) - g(x_0) = \int K(x-y,t)\,[g(y) - g(x_0)]\,dy.$$

Given $\varepsilon > 0$, continuity makes the bracket smaller than ε for $|y - x_0| < \delta$. For $|x - x_0| < \delta/2$, the far part is bounded by $2\|g\|_\infty$ times a Gaussian tail outside $B_{\delta/2}(x)$, which tends to zero. This proves the joint limit. Completing the square in a product of two Gaussians gives $K_t * K_s = K_{t+s}$, the semigroup law. □

For a time-dependent source with zero initial data, Duhamel gives

$$v(x,t) = \int_0^t (K_{t-s} * f(\cdot,s))(x)\,ds.$$

For example assume f is smooth on the time interval, with uniformly compact spatial support and bounded indicated derivatives. Change to $\tau = t - s$ if needed to keep differentiations away from the singular upper endpoint. Moving spatial derivatives to f gives

$$v_t = f(x,t) + \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds, \qquad \Delta v = \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds.$$

抛物缩放为何是平方

空间二阶导数缩放为 λ^2，时间一阶导数也必须缩放为 λ^2 才保持方程。热球的高度因此正比 r^2，不是 r。

瞬时传播不等于大量传播

$K(x,t) > 0$ 对任意 x、$t > 0$ 成立，所以非零非负初值立即在全空间产生正值；远处的影响可极小，但不是严格为零。

The endpoint term uses $K_\varepsilon * f(\cdot,t) \to f(\cdot,t)$. The bounds on $f, D^2 f$ provide domination, proving $Lv = f$ and $v(\cdot,0) = 0$. Adding $K_t * g$ solves the full initial-value problem (source Theorem 2). We use these explicit smoothness assumptions rather than claim that arbitrary continuous forcing gives classical second derivatives without further justification.

2.3.2 Heat-ball mean values (Evans §2.3.2, Theorem 3)

We now expand the connection with harmonic means. The region, the time orientation, and the weight are all essential.

1. The kernel determines the geometry

Let $K(z,\tau) = (4\pi\tau)^{-n/2} e^{-|z|^2/(4\tau)}$, $\tau > 0$. For a target (x,t) define

$$E_r(x,t) = \{(y,s) : s < t,\ K(x-y, t-s) > r^{-n}\}.$$

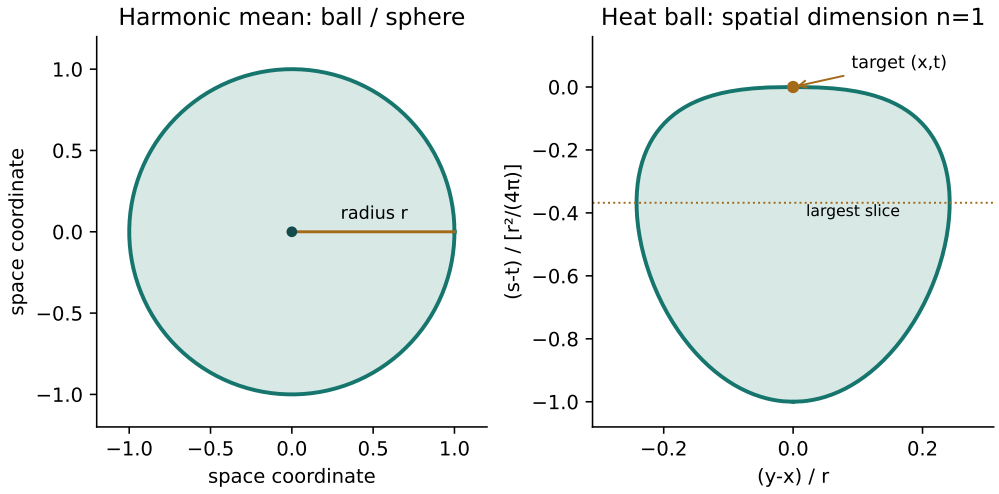
The open or closed choice on the ordinary boundary does not affect the integral. Put $\tau = t - s$ and $a = r^2/(4\pi)$. Taking logarithms gives

$$0 < \tau < a, \qquad |y - x| < R_r(\tau), \qquad R_r(\tau)^2 = 2n\tau \log(a/\tau). \quad \text{(HB)}$$

Each past slice is an ordinary spatial ball. Its radius vanishes at both tips and is largest at $\tau = a/e$:

$$\max R_r(\tau) = r\sqrt{\frac{n}{2\pi e}}.$$

Thus r is a kernel-level scale, not the largest spatial radius.



Editorial figure. Left: an elliptic ball and its boundary. Right: a one-dimensional spatial section of a heat ball, with time increasing toward the target. Spatial and temporal axes use different scales.

热球为何朝向过去

核 $K(x-y, t-s)$ 只在 $s < t$ 有意义。热球下端在 $t - r^2/(4\pi)$，上端是目标时刻 t；它不是以 (x,t) 为对称中心的时空球。

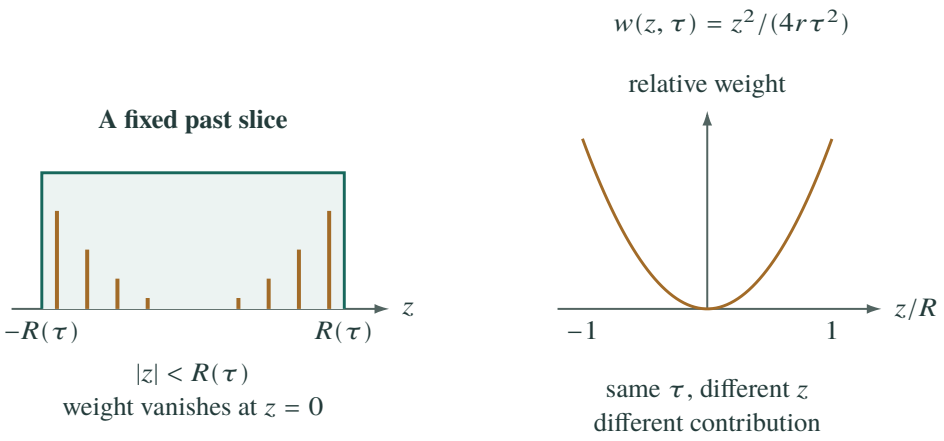


Diagram T2.3. A heat ball carries a nonuniform averaging measure. In one spatial dimension, fix a past time $\tau = t - s$. The blue band marks the allowed interval, while orange heights illustrate the factor $z^2/(4r\tau^2)$. The right panel shows its quadratic spatial dependence up to a fixed positive factor. This is a slice of the full space–time probability measure, not a separately normalized conditional density.

2. Statement and full proof

Theorem 2.9 (Watson–Evans mean-value formula)
If $u \in C^{2,1}$ solves $Lu = u_t - \Delta u = 0$ on a neighborhood of $\overline{E}_r(x,t)$, then

$$\boxed{u(x,t) = \frac{1}{4r^n} \iint_{E_r(x,t)} u(y,s) \frac{|x-y|^2}{(t-s)^2}\, dy\, ds.} \tag{H}$$

The nonnegative measure

$$d\mu_r(y,s) = \frac{\mathbf{1}_{E_r(x,t)}}{4r^n} \frac{|x-y|^2}{(t-s)^2}\, dy\, ds$$

has mass one, so this is $u(x,t) = \mathbb{E}_{\mu_r}[u(Y,S)]$.

Expanded proof (including the source's omitted normalization).

A. Compute the mass exactly. At fixed τ, $\int_{B_R} |z|^2\, dz = n\omega_n R^{n+2}/(n+2)$. With $b = n/2$ and $a = r^2/(4\pi)$, polar integration gives

$$\begin{aligned} I_r &:= \iint_{E_r(0,0)} \frac{|y|^2}{s^2}\, dy\, ds \\ &= \frac{n\omega_n}{n+2} (2n)^{b+1} \int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau. \end{aligned}$$

Set $q = \log(a/\tau)$, so $\tau = ae^{-q}$. Then

$$\int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau = a^b \int_0^\infty e^{-bq} q^{b+1}\, dq = a^b \frac{\Gamma(b+2)}{b^{b+2}}.$$

区域和权重缺一不可

均匀热球抽样使用 $dy\, ds/|E_r|$；这里使用 $|x-y|^2/[4r^n(t-s)^2]$。即使区域画对了，丢掉权重仍然不是均值定理。

Here $\Gamma(c) = \int_0^\infty z^{c-1}e^{-z}\,dz$; the last equality uses $z = bq$. Substitute $\omega_n = \pi^b/\Gamma(b+1)$, $a^b = r^n/(4\pi)^b$, and $\Gamma(b+2) = (b+1)\Gamma(b+1)$. Cancellation yields

$$I_r = 4r^n, \qquad \mu_r(E_r) = 1.$$

This also proves integrability of the seemingly singular top-tip weight for every $n \geq 1$.

B. Differentiate after fixing the domain. Translate the target to $(0,0)$; thus integration times s are negative. Scaling of K gives $E_r = \{(ry, r^2 s) : (y,s) \in E_1\}$. Define

$$\phi(r) = r^{-n} \iint_{E_r} u(y,s)\frac{|y|^2}{s^2}\,dy\,ds = \iint_{E_1} u(ry, r^2 s)\frac{|y|^2}{s^2}\,dy\,ds.$$

The Jacobian is r^{n+2} and the weight contributes r^{-2}, explaining the cancellation. First suppose u smooth. Differentiate on E_1, then scale back:

$$\phi'(r) = A + B,$$

$$A = r^{-n-1}\iint_{E_r}(y\cdot Du)\frac{|y|^2}{s^2}, \qquad B = r^{-n-1}\iint_{E_r} 2u_s\frac{|y|^2}{s}. \tag{AB}$$

Bounded derivatives on compact sets and the integrable weights justify this operation for radii in a compact admissible interval.

C. Find a boundary-vanishing function. Set

$$\psi_r = \log\left(r^n K(y,-s)\right) = n\log r - \frac{n}{2}\log(-4\pi s) + \frac{|y|^2}{4s}.$$

It is positive inside E_r, zero on its lateral boundary, and

$$D_y\psi_r = \frac{y}{2s}, \qquad (\psi_r)_s = -\frac{n}{2s} - \frac{|y|^2}{4s^2}. \tag{L}$$

Since $2|y|^2/s = 4y\cdot D\psi_r$, spatial integration by parts gives

$$B = -4r^{-n-1}\iint_{E_r}\psi_r(nu_s + y\cdot Du_s).$$

Integrate the last term in time and insert (L):

$$\begin{aligned} B &= r^{-n-1}\iint_{E_r}[-4n\psi_r u_s + 4(\psi_r)_s\, y\cdot Du] \\ &= -4nr^{-n-1}\iint_{E_r}\psi_r u_s - 2nr^{-n-1}\iint_{E_r}\frac{y\cdot Du}{s} - A. \end{aligned}$$

The A terms cancel. Use $y/s = 2D\psi_r$ and integrate spatially once more:

$$\boxed{\phi'(r) = 4nr^{-n-1}\iint_{E_r}\psi_r(\Delta u - u_s).} \tag{R}$$

For a caloric function the right side is zero, so ϕ is constant.

D. Check both tips and the regularity assumption. Zero lateral values alone do not justify ignoring the singular tip. First truncate to $\varepsilon < \tau < a - \eta$. The lateral terms vanish because $\psi_r = 0$. At the top cap, $R = O(\sqrt{\tau \log(a/\tau)})$ and

$$\int_{B_R} |y| \psi_r \, dy \le C R^{n+1} \log(a/\tau) \longrightarrow 0.$$

This controls the time-boundary term containing $y \cdot Du$. At the lower cap, $R \to 0$ and $\psi_r \to 0$, so that term also vanishes. The weight remaining near the upper tip is bounded by a constant times

$$\int_0^{\varepsilon} \tau^{n/2-1} [\log(a/\tau)]^{n/2+1} \, d\tau \longrightarrow 0.$$

The differentiated terms in (AB) and those with ψ_r have the same integrable powers or better ones. Let $\varepsilon, \eta \downarrow 0$.

For only $C^{2,1}$ regularity, convolve in space-time on a slightly larger neighborhood. The constant-coefficient operator L commutes with convolution, so smooth approximations still solve the equation where needed. They converge uniformly on the compact region; the integrable weight passes the established mean identity to u. This avoids requiring mixed third derivatives in the original function. At a terminal cylinder time, take limits from interior target times.

E. Identify the constant by shrinking. The heat ball lies in $|y| \le C_n r$, $|s| \le r^2/(4\pi)$. Uniform continuity and the mass computation imply $\phi(r) \to 4u(0,0)$ as $r \downarrow 0$. Therefore $\phi(r) = 4u(0,0)$ for every admissible radius. Divide by four and restore the target (x,t). □

3. The precise connection to elliptic averages

For general smooth u let $M_r[u] = \phi(r)/4$. The preceding computation, before imposing the PDE, gives

$$\frac{d}{dr} M_r[u](x,t) = -\frac{n}{r^{n+1}} \iint_{E_r(x,t)} \log\left(r^n K(x-y, t-s)\right) Lu(y,s) \, dy \, ds. \tag{HR}$$

Thus $Lu \le 0$ implies $M_r[u]$ nondecreasing, and shrinking r gives $u(x,t) \le M_r[u](x,t)$. Reverse the signs for supersolutions. This parallels $\Delta v \ge 0$ and $\partial_r S_v \ge 0$ in (M).

对数核为何出现

ψ_r 的零边界值消掉侧边界项；空间梯度产生平均权重；时间导数让缩放导数的两项抵消。这三种作用共同决定热球的定义。

A counterexample to unweighted replacements. The polynomial $u(y,s) = |y-x|^2 + 2n(s-t)$ has $Lu = 0$ and $u(x,t) = 0$. At time t its uniform spatial ball average is $nr^2/(n+2) > 0$. On $B_r(x) \times (t-r^2, t)$ its uniform space-time average is $nr^2/(n+2) - nr^2 \neq 0$. A symmetric Euclidean space-time ball cancels the time term but still leaves a positive spatial square. None of these is the required mean.

Stationary solutions. If $u(y,s) = h(y)$, then $Lu = 0$ is exactly $\Delta h = 0$. At fixed τ, the heat-ball weight is radial in $y - x$. Integrate over spherical shells first: every spherical mean of h equals $h(x)$, and the remaining scalar integral is the total mass one. Thus heat-ball averaging of a stationary solution is an aggregate of elliptic sphere averages. A time-dependent caloric solution needs the past slices as well as the weight.

Moments and a converse: editorial supplement. There is an exact probability description of μ_r. Put $b = n/2$, $a = r^2/(4\pi)$. Let Q have Gamma density proportional to $q^{b+1}e^{-bq}$, $q > 0$, and set $\tau = ae^{-Q}$. Conditional on τ, choose a uniform direction and radius $\varrho = R_r(\tau)U^{1/(n+2)}$, with U uniform on $(0,1)$. Step A gives the time density after the logarithmic substitution. The conditional spatial radial density is proportional to ϱ^{n+1}, because the volume Jacobian contributes ϱ^{n-1} and the weight contributes ϱ^2. Hence this construction samples the actual weighted measure.

Gamma integration and symmetry yield

$$\mathbb{E}(Y_i - x_i) = 0, \qquad \mathbb{E}[(Y_i - x_i)(Y_j - x_j)] = 2\delta_{ij}\mathbb{E}\tau,$$

$$\mathbb{E}\tau = c_n r^2, \qquad c_n = \frac{1}{4\pi}\left(\frac{n}{n+2}\right)^{n/2+2} > 0.$$

For the second moment, conditional radial integration gives $\mathbb{E}(|Y-x|^2 \mid \tau) = [(n+2)/(n+4)]\,2n\tau Q$. Integrating the Gamma density then gives $\mathbb{E}|Y-x|^2 = 2n\mathbb{E}\tau$. Taylor expansion for $C^{2,1}$ functions therefore implies

$$M_r[u](x,t) = u(x,t) + c_n r^2(\Delta u - u_t)(x,t) + o(r^2). \qquad \text{(HT)}$$

The remainder is uniform on the shrinking region, and $\mathbb{E}(|Y-x|^2 + \tau) = O(r^2)$ controls its integral. An exact identity for all small radii at every center forces $Lu = 0$. Compare this with the elliptic expansion (T).

两个次解的符号

椭圆次调和是 $\Delta v \geq 0$，即 $-\Delta v \leq 0$；抛物次解是 $v_t - \Delta v \leq 0$。统一算子符号后，两者都是“中心不超过正权平均”。

最简单的公式自检

$|y-x|^2 + 2n(s-t)$ 的空间曲率与时间变化恰好平衡。正确权重必须把二者抵消；普通空间平均看不到时间补偿。

几何平均在检测算子

球面平均的二阶误差是 $r^2\Delta u/(2n)$；热球平均的二阶误差是 $c_n r^2(\Delta u - u_t)$。两种几何平均分别检测各自的 PDE。

Do not confuse this with Gaussian averaging. At a fixed earlier time s, the whole-space solution can be written $u(x,t) = \int K(x-y,t-s)u(y,s)\,dy$ in a class where the Cauchy problem is unique. That is a global spatial Gaussian average on one time slice. Heat-ball averaging is local in space-time and uses many past slices. The fundamental solution is the same; the two averaging measures differ.

2.3.3 Maximum principles and regularity (Evans §2.3.3)

Write $U_T = U \times (0,T]$ and $\Gamma_T = (\overline{U} \times \{0\}) \cup (\partial U \times [0,T])$. The top $U \times \{T\}$ is not part of this parabolic boundary.

Theorem 2.10 (Source Theorems 4–5)
On bounded U, a continuous classical solution satisfies $\max_{\overline{U}_T} u = \max_{\Gamma_T} u$. If U is connected and the global maximum occurs at (x_0,t_0) with $x_0 \in U$, $t_0 > 0$, then u is constant on $\overline{U} \times [0,t_0]$. The initial-boundary problem has at most one classical solution for given forcing and parabolic boundary data.

Expanded proof.

For the weak maximum assertion set $v = u - \varepsilon t$. At a maximum in U_T, including the terminal time, the left time derivative is nonnegative and the spatial Laplacian is nonpositive. Thus $Lv \geq 0$, contradicting $Lv = -\varepsilon$. A maximum must lie on Γ_T. Let $\varepsilon \downarrow 0$.

For the strong assertion put $M = \max u$. The heat-ball mean at an interior maximum averages $u \leq M$ against a probability density strictly positive except on a negligible axis. Equality forces $u = M$ almost everywhere in the heat ball; continuity makes it equal everywhere. This propagates equality backward locally.

To reach a distant past point, use a polygonal spatial path in U and assign strictly decreasing times to its finitely many vertices. A small backward portion of each space-time line lies in a heat ball about its later endpoint: the spatial displacement is $O(\tau)$, whereas the heat-ball radius is of order $\sqrt{\tau \log(1/\tau)}$, which is larger for small τ. On each segment the earliest time of an already-constant terminal portion is attained by continuity. If it were not the segment's earliest endpoint, a small heat ball would extend it, a contradiction. Propagate through the finite chain. Every earlier interior point therefore has value M; continuity reaches time zero and the boundary. Finally apply the weak principle to both signs of the difference of two solutions for uniqueness. □

强最大值结论只向过去

内部时刻 t_0 达到全局最大值，会强迫此前恒定；不自动强迫此后恒定。边界条件在未来改变时，解也可以随后改变。

Whole-space uniqueness needs a growth class (source 6–7). Suppose $Lu \le 0$, $u(\cdot,0) \le 0$ and $u(x,t) \le Ae^{a|x|^2}$ for $0 \le t \le T$. Choose $b > a$ and the positive caloric barrier

$$V(x,t) = (1-4bt)^{-n/2} \exp\left(\frac{b|x|^2}{1-4bt}\right) \quad (0 \le t < 1/(4b)).$$

Its PDE follows from direct Gaussian-type differentiation. For fixed $\varepsilon > 0$, $u - \varepsilon V < 0$ on the lateral boundary of a sufficiently large ball, uniformly on a shorter interval $[0,\delta]$, $\delta < 1/(4b)$; the exponent $b > a$ ensures domination. Its initial values are nonpositive. The bounded-cylinder principle gives $u \le \varepsilon V$. Let the radius tend to infinity, then $\varepsilon \downarrow 0$. Repeat on consecutive time intervals of length δ, using the same growth bound, to cover $[0,T]$. This proves the whole-space maximum principle and, for $|u - \tilde{u}| \le Ae^{a|x|^2}$, uniqueness. Without a growth restriction the source discusses nonphysical zero-initial-data solutions; this draft does not use their existence as a substitute for specifying a uniqueness class.

Interior smoothing and derivative estimates (source 8–9). For a backward cylinder $Q_r = B_r(x_0) \times (t_0 - r^2, t_0]$ inside the domain, choose a smooth cutoff ζ equal to one on $Q_{3r/4}$, vanishing near the bottom and lateral boundary of Q_r. Then

$$L(\zeta u) = (\zeta_s - \Delta\zeta)u - 2D\zeta \cdot Du.$$

Duhamel and spatial integration by parts in the last term yield, on $Q_{r/2}$,

$$\begin{aligned} u(x,t) = \iint_{Q_r} \big[& K(x-y,t-s)(\zeta_s + \Delta\zeta)(y,s) \\ & + 2D_y K(x-y,t-s) \cdot D\zeta(y,s) \big] u(y,s)\, dy\, ds. \end{aligned} \tag{C}$$

Set the kernel to zero for $s \ge t$. Where cutoff derivatives do not vanish, either $t - s$ stays separated from zero or $|x - y|$ stays separated from zero. In the latter case every derivative of the Gaussian tends to zero faster than any power as $s \uparrow t$. Thus the combined kernel in (C) is smooth with bounded derivatives on the relevant compact sets, despite the original fundamental solution's singularity. A preliminary local convolution proves (C) for merely $C^{2,1}$ functions, then passes to the limit. Differentiating (C) now gives $u \in C^\infty$ locally.

On a unit cylinder, its kernel derivatives are bounded by constants $C_{\alpha,l}$. Rescale $v(z,\tau) = u(x_0 + rz, t_0 + r^2\tau)$ to obtain

$$\sup_{Q_{r/2}} |D_x^\alpha \partial_t^l u| \le C_{\alpha,l,n} r^{-n-2-|\alpha|-2l} \|u\|_{L^1(Q_r)}.$$

One time derivative costs two powers of spatial scale. For completeness, spatial analyticity can also be read from (C), using the elementary complex-analytic parameter-integral theorem and Cauchy estimates as additional prerequisites. Fix a time and replace the spatial variable by $z = x + iv$ in the Gaussian, using the polynomial $\sum_j (z_j - y_j)^2$ in its exponent. Where $t - s$ is small and cutoff derivatives are nonzero, $|x - y|$ is bounded below by some $d > 0$. For $|v| < d/2$ the real part of that polynomial remains positive, bounded below by a fixed multiple of d^2. The resulting Gaussian and its spatial derivatives have an integrable bound uniform on a small complex neighborhood. Where $t - s$ is bounded below there is no singularity at all. Thus (C) extends holomorphically in z near every interior spatial point. Cauchy's formula supplies factorial derivative bounds, proving real analyticity in space. This argument says nothing comparable about analyticity in time; smoothness alone does not establish that.

2.3.4 Energy and backward uniqueness (Evans §2.3.4, source 10–11)

Let w be the difference of two sufficiently smooth solutions with the same lateral boundary values. Then $w = 0$ on ∂U and

$$\frac{1}{2}\frac{d}{dt}\int_U w^2 = \int_U w\Delta w = -\int_U |Dw|^2.$$

Zero initial difference gives zero energy at every later time. This proves forward uniqueness and displays dissipation.

Backward uniqueness requires more than this decreasing-energy identity. Assume enough boundary and mixed derivative regularity for the next identities (smooth solutions up to the closed cylinder suffice). Set $e(t) = \int_U w^2$. Since $w_t = 0$ on the fixed lateral boundary,

$$e' = -2\int |Dw|^2, \qquad e'' = -4\int Dw \cdot Dw_t = 4\int (\Delta w)^2.$$

Cauchy–Schwarz in $\int |Dw|^2 = -\int w\Delta w$ gives $(e')^2 \le ee''$. Wherever $e > 0$, this means $(\log e)'' \ge 0$. If e were positive at t_1 and first vanished at $t_2 \le T$, convexity on $[t_1, t_2)$ would give, for $0 < \theta < 1$ and $b < t_2$,

$$e((1-\theta)t_1 + \theta b) \le e(t_1)^{1-\theta} e(b)^{\theta}.$$

Let $b \uparrow t_2$. The right side vanishes, contradicting positivity at the earlier interior time. Thus $e(T) = 0$ forces $e \equiv 0$.

切断函数移走奇点

把导数转移给核后，核奇点附近的切断导数全为零；远离空间中心时高斯又快速衰减。因此才可任意多次求导，而不是直接对奇异积分硬求导。

2.4 Wave Equation

2.4.1 Spherical means and explicit solutions (Evans §2.4.1)

The wave equation $u_{tt} - \Delta u = 0$ needs two initial data: $u(\cdot,0) = g$, $u_t(\cdot,0) = h$. Unlike the heat equation, it scales with $t \mapsto \lambda t$ when $x \mapsto \lambda x$.

One dimension: d'Alembert (source Theorem 1). Write $\xi = x + t$, $\eta = x - t$. Then $\partial_t^2 - \partial_x^2 = -4\partial_\xi \partial_\eta$. Thus $u_{\xi\eta} = 0$, and integration gives $u = F(\xi) + G(\eta)$. At $t = 0$, $F + G = g$ and $F' - G' = h$, whereas $F' + G' = g'$. Solving for F', G' and integrating yields

$$u(x,t) = \frac{g(x+t) + g(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy. \tag{DA}$$

For $g \in C^2$, $h \in C^1$, direct differentiation verifies the equation, $u(\cdot,0) = g$, and $u_t(\cdot,0) = h$. Only initial points in $[x-t, x+t]$ matter.

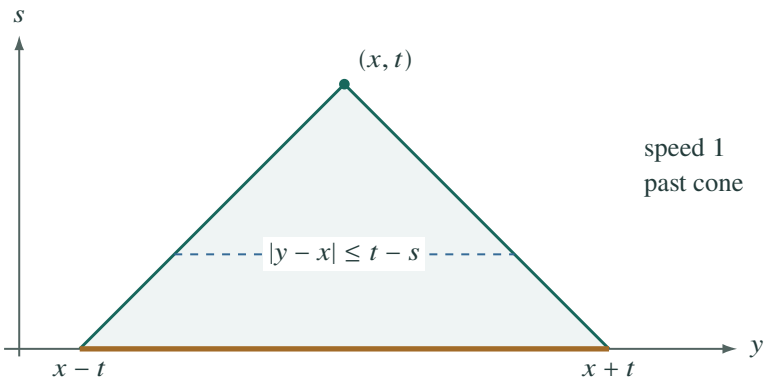


Diagram T2.4. The wave equation has a finite domain of dependence. For speed one in one dimension, the solution at (x,t) uses initial data only on $[x-t, x+t]$. At an intermediate time s, the relevant interval is $|y - x| \le t - s$. In higher dimensions this interval becomes a ball. A forced equation also accumulates sources from the past cone.

The common spherical-mean identity. Fix a spatial center x and set $M(r,t) = |\partial B_r|^{-1} \int_{\partial B_r(x)} u(y,t)\,dS$. The radius-flux computation used for harmonic means now gives

$$M_r = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u, \qquad (r^{n-1}M_r)_r = r^{n-1} M_{tt}.$$

Thus M satisfies the Euler–Poisson–Darboux equation

$$M_{tt} = M_{rr} + \frac{n-1}{r} M_r, \qquad M(0,t) = u(x,t), \quad M_r(0,t) = 0. \tag{EPD}$$

逆向唯一不等于逆向稳定

知道最终状态可唯一确定过去，不代表微小最终误差只造成微小过去误差。热方程高频逆向增长仍会导致严重不稳定，第一章的例子依然适用。

两个初值各管什么

初始位移 g 生成两列反向平移的形状；初始速度 h 在区间内累积。热方程只需一个初值，而波动方程的二阶时间导数要求两个。

The second radius derivative differentiates the ball integral by shells; the wave equation changes Δu into u_{tt}.

Three dimensions: Kirchhoff. For $n = 3$, $W(r,t) = rM(r,t)$ satisfies $W_{tt} = W_{rr}$. Extend M evenly in r, hence W oddly. Its initial data are $rS_g(x,r)$ and $rS_h(x,r)$. Use (DA) for W and differentiate in r at zero. Since $W_r(0,t) = u(x,t)$, this gives

$$u(x,t) = \partial_t\big[tS_g(x,t)\big] + tS_h(x,t). \tag{KIR}$$

Equivalently

$$u(x,t) = \frac{1}{4\pi}\int_{\partial B_1}\left[g(x+t\theta) + tDg(x+t\theta)\cdot\theta + th(x+t\theta)\right]dS_\theta.$$

The latter fixed-sphere expression makes initial limits transparent. The formula solves the wave equation: by (EPD) applied to spatial spherical means of g or h, tS_h satisfies the wave equation in the variables (x,t); time differentiation handles the g term. At zero, $tS_h = th(x) + O(t^3)$ and $\partial_t(tS_g) = g(x) + O(t^2)$. These observations prove existence as well as the derivation from a hypothetical solution, for smooth data.

Two dimensions: descent from three dimensions. Make g, h independent of a third coordinate. Project the upper and lower hemispheres onto the disk. Each contributes surface Jacobian $t/\sqrt{t^2 - |y-x|^2}$. Formula (KIR) becomes

$$u(x,t) = \partial_t\left[\frac{1}{2\pi}\int_{B_t(x)}\frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,dy\right] + \frac{1}{2\pi}\int_{B_t(x)}\frac{h(y)}{\sqrt{t^2-|y-x|^2}}\,dy. \tag{W2}$$

Use $y = x + tz$ before differentiating: the singularity $(1-|z|^2)^{-1/2}$ is integrable on the fixed unit disk. This justifies differentiation for sufficiently smooth data. Descent preserves the PDE because the third-coordinate derivative is zero; the three-dimensional initial limits give the desired two data.

All odd dimensions (source Theorem 2). For $n = 2k+1 \geq 3$, put $D_r = r^{-1}\partial_r$ and $\gamma_n = 1\cdot 3\cdots(n-2)$. Define

$$S_n(t)f = \frac{1}{\gamma_n}D_t^{k-1}\big[t^{n-2}S_f(x,t)\big].$$

Then

$$u(x,t) = \partial_t S_n(t)g + S_n(t)h. \tag{WO}$$

Here $S_n(t)$ denotes a solution operator, whereas $S_f(x,t)$ is the normalized spherical average defined earlier.

同一个通量公式三次出现

Laplace 方程令球面平均对半径不变；波动方程把同一半径变化联系到二阶时间导数；热方程则需要带方向的时空平均。不要因都出现球面而混为一谈。

降维为何出现权重

投影把球面压到圆盘，靠近圆周时球面近乎竖直，面积 Jacobian 变大，产生 $(t^2-|y-x|^2)^{-1/2}$。它是几何投影权重，不是热球的权重。

Expanded proof (the radial intertwining step).

Let $E = r\partial_r$ and $A_k F = D_r^{k-1}(r^{2k-1}F) = rP(E)F$, where $P(E) = \prod_{j=1}^{k-1}(E+2j+1)$; for $k=1$ the product is one. This factorization follows successively from $D_r(r^a F) = r^{a-2}(E+a)F$. The radial Laplacian is $r^{-2}E(E+2k-1)$. The polynomial identity

$$(E+1)P(E) = (E+2k-1)P(E-2)$$

therefore proves

$$(A_k F)'' = A_k\left(F'' + \frac{2k}{r}F'\right).$$

For $F(r) = S_f(x,r)$ the bracket equals $\Delta_x F$ by the divergence identity behind (EPD). Hence $A_k S_f$, now with $r = t$, solves the wave equation. It commutes with Δ_x since the operator acts only on r.

Spherical symmetry gives $S_f(x,t) = f(x) + O(t^2)$ with an even smooth extension at zero. The coefficient of $tf(x)$ in $A_k S_f$ is $P(0) = 3\cdot 5\cdots(2k-1) = \gamma_n$. All other terms start at t^3. Thus $S_n(t)f = tf + O(t^3)$, giving initial value zero and initial velocity f. Its time derivative has initial value f and initial velocity zero. This proves (WO) for smooth data and explains its constants. Finite differentiability follows by differentiating the fixed-sphere integrals; the source sufficient assumptions are $g \in C^{m+1}$, $h \in C^m$, $m = (n+1)/2$. □

All even dimensions (source Theorem 3). For even $n \geq 2$, put $\gamma_n = 2\cdot 4\cdots n$ and

$$B_f(x,t) = \frac{t^n}{\omega_n t^n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy = \frac{1}{\omega_n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy.$$

Set $S_n(t)f = \gamma_n^{-1} D_t^{(n-2)/2} B_f(x,t)$. The same formula $u = \partial_t S_n(t)g + S_n(t)h$ holds. To derive it, extend the data constantly in one extra coordinate, apply (WO) in dimension $n+1$, and project both hemispheres. Their Jacobian gives $2t/\sqrt{t^2-|y-x|^2}$; the sphere and ball area constants simplify to the displayed γ_n. The inherited PDE and initial limits prove it. For smooth data there is no endpoint issue; for the source finite regularity statement take $m = (n+2)/2$, $g \in C^{m+1}$, $h \in C^m$. When $n = 2$, this reduces to (W2).

For odd $n \geq 3$, only data on the sphere at distance t enter the formula, the sharp Huygens phenomenon. For even n, the whole ball enters and a trailing influence remains. Dimension one is an exception to the odd-dimensional sharp-front statement: (DA) includes an interval integral of h.

2.4.2 Forcing (Evans §2.4.2, Theorem 4)

The operator $S_n(t)$ above solves the homogeneous equation with initial displacement zero and velocity f. For smooth forcing set

$$v(t) = \int_0^t S_n(t-s)f(s)\,ds.$$

Since $S_n(0) = 0$ and $\partial_t S_n(0) = I$, differentiate twice:

$$v_t = \int_0^t \partial_t S_n(t-s)f(s)\,ds, \qquad v_{tt} = f(t) + \int_0^t \partial_t^2 S_n(t-s)f(s)\,ds.$$

The integrals in v_{tt} and Δv cancel. Thus $v_{tt} - \Delta v = f$, with both initial data zero. Add the homogeneous solution for general initial data.

2.4.3 Energy and finite propagation (Evans §2.4.3, Theorems 5–6)

For a smooth solution with fixed zero Dirichlet boundary values,

$$\mathscr{E}(t) = \frac{1}{2}\int_U (u_t^2 + |Du|^2), \qquad \mathscr{E}'(t) = \int_{\partial U} u_t \partial_\nu u = 0.$$

The boundary term vanishes because $u = 0$ there for all times implies $u_t = 0$ there. Apply this to the difference for uniqueness; zero initial energy gives $Du = u_t = 0$, and zero initial displacement removes constants.

For local propagation, consider the shrinking ball $B_{R-t}(x_0)$ and the cone energy

$$e(t) = \frac{1}{2}\int_{B_{R-t}(x_0)} (u_t^2 + |Du|^2).$$

Differentiating the moving domain contributes a negative surface term. The PDE and spatial integration by parts yield

$$e'(t) = \int_{\partial B_{R-t}} \left[u_t \partial_\nu u - \tfrac{1}{2}(u_t^2 + |Du|^2)\right] dS \le 0.$$

The inequality follows from $2u_t\partial_\nu u \le u_t^2 + |\partial_\nu u|^2 \le u_t^2 + |Du|^2$. If both initial data vanish on $B_R(x_0)$, $e(0) = 0$ and hence $e(t) = 0$ for $t < R$. Along each vertical segment inside the cone $u_t = 0$ and its initial value is zero, so $u = 0$ throughout the cone. Continuity gives the apex value when applicable.

传播速度来自能量通量

球半径以速度 1 收缩时，移边界损失足以压住波的能量流入。热核虽处处为正，波的锥外影响却严格为零；这是两个方程的本质区别。

2.5 Problems

Source: Evans §2.5, Problems 1–24, printed pp. 81–86 (PDF pp. 98–103). Statements are concisely restated, and solutions are editorial. Smoothness is assumed wherever a displayed differentiation requires it. Problems 3–5 and 17 are especially useful checks on the mean-value arguments.

2.5.1 Problems 1–8: transport and elliptic averages

1. Transport with a zeroth-order term. Solve $u_t + b \cdot Du + cu = 0$, $u(\cdot, 0) = g$.

Solution. Along $X(s) = x + (s-t)b$, the quantity $e^{cs}u(X(s), s)$ has derivative zero. Evaluate at $s = 0, t$ to obtain $u(x,t) = e^{-ct}g(x - tb)$. □

2. Rotation invariance of the Laplacian.

Solution. For an orthogonal matrix O, the Hessian of $v(x) = u(Ox)$ is $D^2v = O^T(D^2u)(Ox)O$. Its trace equals $\operatorname{tr}((D^2u)(Ox)OO^T) = \Delta u(Ox)$. Thus a harmonic u gives a harmonic v. □

3. The center value for Poisson's equation. Derive the center formula for $-\Delta u = f$ on B_r, $u = g$ on ∂B_r.

Solution. Integrate (M) from zero to r:

$$S_u(0,r) - u(0) = -\frac{1}{n\omega_n}\int_0^r \rho^{1-n}\int_{B_\rho} f(y)\,dy\,d\rho.$$

Changing the order gives (P), with the nonnegative kernel $[|y|^{2-n} - r^{2-n}]/[n(n-2)\omega_n]$ when $n \geq 3$. Replace $S_u(0,r)$ by the boundary average of g. The kernel's singularity is locally integrable, justifying the change for bounded f; signed f is treated by absolute integrability. □

4. A direct maximum-principle proof.

Solution. For $\varepsilon > 0$, $v = u + \varepsilon|x|^2$ satisfies $\Delta v = 2n\varepsilon > 0$. At an interior maximum its Hessian would be negative semidefinite and $\Delta v \leq 0$, impossible. Thus v reaches its maximum on the boundary. Bound the added quadratic uniformly on bounded U and let $\varepsilon \downarrow 0$. □

5. Subharmonic functions. Show the mean inequality and boundary maximum principle for $\Delta v \geq 0$; prove that $\phi(u)$ and $|Du|^2$ are subharmonic for harmonic u and smooth convex ϕ.

Solution. Formula (M) makes spherical means nondecreasing, and (SB) gives $v(x) \leq A_v(x, r)$. For the boundary maximum argument use $v + \varepsilon|x|^2$, as in Problem 4. The chain rule gives $\Delta\phi(u) = \phi''(u)|Du|^2 + \phi'(u)\Delta u \geq 0$. Since harmonic functions are smooth, differentiating the PDE is valid:

$$\Delta|Du|^2 = 2|D^2u|^2 + 2Du \cdot D\Delta u = 2|D^2u|^2 \geq 0.$$

□

凸性与平均相容

调和函数经凸函数变换后不一定仍调和，却成为次调和。Jensen 不等式给直觉，链式法则中的 $\phi''(u)|Du|^2$ 给出可核验的原因。

6. A uniform Poisson bound.

Solution. Enclose U in $B_R(0)$, put $F = \|f\|_\infty$, $G = \|g\|_\infty$, and set $b(x) = G + F(R^2 - |x|^2)/(2n)$. Then $b \geq G$ on ∂U and $-\Delta b = F$. The function $u - b$ is subharmonic because $-\Delta(u - b) = f - F \leq 0$; its boundary values are nonpositive. Problem 5 yields $u \leq b$. Repeat for $-u$, giving $\|u\|_\infty \leq G + R^2F/(2n)$. □

7. Explicit Harnack bounds on a ball.

Solution. For positive harmonic u on B_R with continuous boundary values, use (PB), $R - |x| \leq |x - y| \leq R + |x|$, and $\int_{\partial B_R} u = n\omega_n R^{n-1} u(0)$:

$$R^{n-2}\frac{R - |x|}{(R + |x|)^{n-1}}u(0) \leq u(x) \leq R^{n-2}\frac{R + |x|}{(R - |x|)^{n-1}}u(0).$$

If u is only given on the open ball, first use radii $\rho < R$ with $|x| < \rho$, then let $\rho \uparrow R$. □

8. Verify the ball Poisson formula.

Solution. The full verification following (PB) establishes mass one using the known solution 1, harmonicity by direct differentiation, and boundary convergence by a near/far split. Positivity controls the near error, and the factor $R^2 - |x|^2$ kills the far error. This proves existence; the maximum principle supplies uniqueness. □

2.5.2 Problems 9–17: boundaries and the heat equation

9. Lipschitz boundary data can give an unbounded normal derivative. Let bounded continuous half-space data satisfy $g(z) = |z|$ for $|z| \le 1$, $n \ge 2$. Show the Poisson solution has unbounded gradient near zero.

Solution. The boundary value at zero is zero. For $\lambda > 0$,

$$\frac{u(\lambda e_n)}{\lambda} = \frac{2}{n\omega_n} \int_{\mathbb{R}^{n-1}} \frac{g(z)}{(|z|^2 + \lambda^2)^{n/2}}\, dz.$$

The portion $|z| > 1$ is uniformly bounded in absolute value, by boundedness of g and the integrable radial tail. On $\lambda < |z| < 1$, polar coordinates bound the remaining positive integral below by $c_n \int_\lambda^1 d\rho/\rho$, diverging. A bounded gradient would bound this difference quotient, a contradiction. □

10. Odd reflection across a flat zero boundary.

Solution. First assume u is C^2 up to the flat boundary of a half-ball, harmonic inside, and zero on the flat portion. Tangential first and second derivatives vanish there. The PDE then makes $u_{nn} = 0$ there. For the odd extension, tangential derivatives change sign below the plane, normal first derivatives match, mixed derivatives match, and normal second derivatives match because their boundary values are zero. Thus the extension is C^2 and harmonic.

For only continuous boundary regularity, take a small full ball centered on the flat plane. Prescribe its continuous odd-reflected spherical data and solve by (PB). Uniqueness makes this harmonic extension odd, hence zero on the plane. On the upper half-ball it and u have the same full boundary data, so the maximum principle makes them equal. Its lower half is the desired odd reflection. Such balls cover the interface. □

反射要检查到二阶

奇延拓的函数值匹配只是第一步。要直接证明 C^2，还需用零边界值消去切向二阶导数，再用方程推出法向二阶导数为零。

11. Kelvin transform. Show $\overline{u}(x) = |x|^{2-n} u(x/|x|^2)$ is harmonic when u is harmonic, on the inverted domain away from zero.

Solution. Put $r = |x|$, $z = x/r^2$ and $e = x/r$. Then $Dz = r^{-2}(I - 2ee^T)$, so $Dz(Dz)^T = r^{-4}I$, and $\Delta z_i = -2(n-2)x_i/r^4$. Also $\Delta r^{2-n} = 0$ off zero. In the Laplacian product and chain rule, the first-derivative terms cancel: the contribution from $r^{2-n}\Delta z_i$ is opposite to that from $2D(r^{2-n})\cdot Dz_i$. The remaining Hessian term is

$$\Delta_x\big[r^{2-n}u(z)\big] = r^{-n-2}(\Delta u)(z).$$

This proves the claim, including $n = 2$ with prefactor one. □

12. Parabolic scaling and its infinitesimal generator.

Solution. For $\lambda > 0$, define $u_\lambda(x,t) = u(\lambda x, \lambda^2 t)$. The chain rule gives

$$Lu_\lambda = \lambda^2 (Lu)(\lambda x, \lambda^2 t) = 0.$$

Differentiate this identity in λ at one. The result is $L(x \cdot Du + 2tu_t) = 0$. Nonzero negative λ also gives a solution by spatial reflection; $\lambda = 0$ requires data at time zero and is not needed for the argument. □

13. A self-similar profile. For $u(x,t) = v(x/\sqrt{t})$ in one space dimension, derive the kernel.

Solution. For $z = x/\sqrt{t}$, $u_t = -zv'/(2t)$ and $u_{xx} = v''/t$. Thus $v'' + (z/2)v' = 0$, so $v' = ce^{-z^2/4}$ and $v(z) = c\int_0^z e^{-q^2/4}\,dq + d$. Choose $c = (4\pi)^{-1/2}$ and $d = 1/2$ to obtain initial step data. Differentiating in x gives $u_x = (4\pi t)^{-1/2} e^{-x^2/(4t)} = K(x,t)$. The distributional derivative of the initial step is δ_0, so this construction indeed produces the fundamental solution. □

14. Heat flow with reaction and forcing.

Solution. For $u_t - \Delta u + cu = f$, let $v = e^{ct}u$. Then $Lv = e^{ct}f$ and $v(\cdot,0) = g$. Duhamel gives

$$u(x,t) = e^{-ct}(K_t * g)(x) + \int_0^t e^{-c(t-s)}(K_{t-s} * f(\cdot,s))(x)\,ds.$$

□

15. Heat equation on a half-line with boundary forcing. For zero initial data and boundary value $g(t)$, $g(0) = 0$, derive

$$u(x,t) = \int_0^t \frac{x\,e^{-x^2/[4(t-s)]}}{\sqrt{4\pi}(t-s)^{3/2}} g(s)\,ds, \qquad x > 0.$$

Solution. The zero-boundary half-line heat Green kernel is $K(x-y,\tau) - K(x+y,\tau)$. Its inward boundary flux gives the Poisson kernel $P(x,\tau) = -2K_x(x,\tau) = xe^{-x^2/(4\tau)}/(\sqrt{4\pi}\tau^{3/2})$. It satisfies the heat equation for $x, \tau > 0$. The substitution $z = x/(2\sqrt{\tau})$ shows $\int_0^\infty P(x,\tau)\,d\tau = 1$. As $x \downarrow 0$, the mass on $\tau > \delta$ tends to zero. Thus $P(x,\cdot)$ is an approximate identity at time zero. After extending g by zero to negative times, its convolution tends to $g(t)$; compatibility $g(0) = 0$ makes the corner continuous. For fixed $x > 0$, the exponential kills all singular powers at $\tau = 0$, justifying differentiation and the PDE. The time interval shrinks to zero at the initial limit, proving the zero initial datum. □

16. Direct parabolic maximum principle.

Solution. The argument $v = u - \varepsilon t$ in §2.3.3 supplies the proof. At a top-time maximum the left derivative is nonnegative; one does not need to pretend that the terminal point is a two-sided time maximum. □

17. Subsolutions and convex transformations.

Solution. For $Lv \le 0$, (HR) and its limit at zero imply $v(x,t) \le M_r[v]$. The perturbed maximum argument from Problem 16 applies to subsolutions, giving the boundary maximum principle on bounded cylinders. If ϕ is smooth and convex and $Lu = 0$, then

$$L\phi(u) = \phi'(u)Lu - \phi''(u)|Du|^2 \le 0.$$

For a smooth caloric u, spatial and time derivatives are caloric, and $L(w^2) = -2|Dw|^2$ whenever $Lw = 0$. Sum for $w = \partial_i u$ and $w = u_t$:

$$L(|Du|^2 + u_t^2) = -2|D^2u|^2 - 2|Du_t|^2 \le 0.$$

□

椭圆与热的凸性公式

椭圆侧是 $\Delta\phi(u) = \phi''(u)|Du|^2 \ge 0$；热算子侧是 $L\phi(u) = -\phi''(u)|Du|^2 \le 0$。符号不同，得到的平均不等式方向一致。

2.5.3 Problems 18–24: waves, systems, and energy

18. Stokes' rule. If u has wave data $(0,h)$, show $v = u_t$ has data $(h,0)$.

Solution. Time differentiation commutes with $\partial_t^2 - \Delta$, so v solves the same PDE. At zero, $v = h$ and $v_t = u_{tt} = \Delta u = \Delta 0 = 0$. This is also the reason the displacement part of the solution is a time derivative of the velocity solution operator. □

19. Null coordinates and one-way waves.

Solution. The equation $u_{xy} = 0$ first makes u_x independent of y; integrating gives $u = F(x) + G(y)$. For $\xi = x + t$, $\eta = x - t$, the wave operator is $-4\partial_\xi\partial_\eta$, so $u = F(x+t) + G(x-t)$. The initial conditions give $F' = (g' + h)/2$, $G' = (g' - h)/2$, producing (DA). A purely right-moving profile $G(x-t)$ requires $h = -g'$. A purely left-moving profile $F(x+t)$ requires $h = g'$. Additive constants can be allocated to either profile without changing these derivative conditions. □

20. Undistorted radial profiles. For nonzero attenuation $\alpha(r)$ and nonnegative delay $\beta(r)$, $\beta(0) = 0$, suppose $\alpha(r)\phi(t - \beta(r))$ solves the wave equation for every smooth profile ϕ.

Solution. Insert the form into $u_{tt} - u_{rr} - (n-1)u_r/r$. Independence of ϕ, ϕ', ϕ'' gives

$$\alpha(1 - \beta'^2) = 0, \qquad 2\alpha'\beta' + \alpha\beta'' + \frac{n-1}{r}\alpha\beta' = 0,$$

$$\alpha'' + \frac{n-1}{r}\alpha' = 0.$$

For a nontrivial smooth attenuation, the first and the nonnegative-delay condition give $\beta(r) = r$; the middle equation gives $\alpha = Cr^{-(n-1)/2}$. Substitution into the last equation yields $(n-1)(n-3)C = 0$. Thus $n = 1$, $\alpha = C$, or $n = 3$, $\alpha = C/r$. The identically zero attenuation is a trivial exception and has to be excluded to make the dimension claim true. □

量词“任意波形”很强

把 ϕ, ϕ', ϕ'' 的系数分别置零，依赖于要求对所有光滑 ϕ 成立。若只要求某一个特定波形，不能推出相同的维数限制。

21. Maxwell and elastic waves.

Solution. Differentiate $E_t = \operatorname{curl} B$ and use $B_t = -\operatorname{curl} E$:

$$E_{tt} = -\operatorname{curl}\operatorname{curl} E = -\nabla \operatorname{div} E + \Delta E = \Delta E.$$

Likewise $B_{tt} = \Delta B$, using the divergence constraints. For elasticity, $u_{tt} - \mu\Delta u - (\lambda + \mu)\nabla \operatorname{div} u = 0$. Taking divergence and curl respectively gives

$$w_{tt} - (\lambda + 2\mu)\Delta w = 0, \qquad \mathbf{w}_{tt} - \mu\Delta \mathbf{w} = 0,$$

where $w = \operatorname{div} u$ and $\mathbf{w} = \operatorname{curl} u$. When $\mu > 0$ and $\lambda + 2\mu > 0$, the longitudinal and transverse speeds are $\sqrt{\lambda + 2\mu}$ and $\sqrt{\mu}$. □

22. Telegraph equation from velocity switching. Given $u_t + u_x = d(v - u)$ and $v_t - v_x = d(u - v)$, show each component satisfies $w_{tt} + 2dw_t - w_{xx} = 0$.

Solution. Let $a = u+v$, $b = u-v$. Adding and subtracting gives $a_t + b_x = 0$ and $b_t + a_x = -2db$. Differentiate the first in time: $a_{tt} = -b_{xt} = a_{xx} + 2db_x = a_{xx} - 2da_t$. Differentiate the second in time and use $a_{xt} = -b_{xx}$: $b_{tt} = b_{xx} - 2db_t$. Linear combinations give the same telegraph equation for u, v. □

23. A compact space-time source and its outgoing shape. In the diamond $S = \{|x| + |t-2| < 1\}$, take $f = 1$ below $t = x+2$ and $f = -1$ above it, zero elsewhere. Solve with zero wave data and describe u for $t > 3$.

Solution. The one-dimensional Duhamel formula is

$$u(x,t) = \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} f(y,s)\,dy\,ds.$$

Set $\alpha = s + y$, $\beta = s - y$. Then S becomes $1 < \alpha < 3$, $1 < \beta < 3$, and $dy\,ds = \frac{1}{2}d\alpha\,d\beta$. The sign depends only on β: it is $+1$ for $\beta < 2$, -1 for $\beta > 2$. The backward cone imposes $\alpha \le t + x$, $\beta \le t - x$. Define

$$F(a) = \min\{2, \max\{0, a-1\}\},$$

$$H(b) = \begin{cases} 0, & b \le 1, \\ b-1, & 1 < b \le 2, \\ 3-b, & 2 < b < 3, \\ 0, & b \ge 3. \end{cases}$$

The integrals factor, giving $u(x,t) = \frac{1}{4}F(t+x)H(t-x)$. If $t > 3$ and $H(t-x) \neq 0$, then $t + x = 2t - (t-x) > 3$, so $F(t+x) = 2$. Therefore

$$u(x,t) = \tfrac{1}{2}H(t-x) \quad (t > 3).$$

It is a right-moving triangular pulse of height $1/2$, supported on $t - 3 < x < t - 1$, with peak at $x = t - 2$. The sign-balanced source cancels the other outgoing pulse. This piecewise forcing is interpreted through the integral formula; across its jump lines the PDE is read distributionally. □

特征坐标把菱形变方形

$\alpha = s + y$, $\beta = s - y$ 同时把波锥不等式变成坐标上界。二重积分才能分解成两个一维截断积分；漏掉 Jacobian 的 $1/2$ 会让最终脉冲高度错一倍。

24. Equipartition in one dimension. For compactly supported smooth g, h, show total wave energy is constant and kinetic and potential energies eventually coincide.

Solution. Write $u = F(x+t) + G(x-t)$ as in Problem 19. Although the primitives F, G need not be compactly supported, their derivatives $F' = (g' + h)/2$, $G' = (g' - h)/2$ are. Since $u_t = F' - G'$ and $u_x = F' + G'$,

$$k(t) + p(t) = \int_{\mathbb{R}} \left(F'(x+t)^2 + G'(x-t)^2\right) dx$$

is independent of t by translation invariance. Also

$$k(t) - p(t) = -2\int_{\mathbb{R}} F'(x+t)G'(x-t)\,dx.$$

For sufficiently large t the translated compact supports are disjoint, so this integral is zero. If the initial supports lie in $[-R, R]$, every $t > R$ suffices. □

2.6 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 2, printed pp. 17–87 (local PDF pp. 34–104). The exposition here is reconstructed for study; source theorem numbering is retained only as an explicit source identifier.

Evans's §2.6 attributes the heat-ball theorem and the associated strong maximum principle to N. Watson, *Proceedings of the London Mathematical Society* 26 (1973), pp. 385–417. The proof above follows the source's scaling/logarithm method and adds the full mass computation, tip estimates, scale-defect identity, and moment interpretation. The latter supplements are editorial deductions; no separate Watson-paper comparison is claimed. The English source omits the normalization calculation; the Chinese translator's footnote on printed p. 37 suggests polar coordinates and a Gamma substitution. The present proof spells out those constants.

The source points to Gilbarg–Trudinger, Chapters 2–4, for elliptic theory; John and Friedman for the heat equation; Mikhailov for regularity; Payne for backward uniqueness; and Folland and Strauss for wave representation formulas. These are source reading directions, not claims that those books were independently consulted for this draft.

The Chinese chapter opening (PDF p. 31) provides section starts at printed pp. 13, 14, 30, 43, 56, and 60 for §2.1–§2.6. Its mean-value and heat-ball pages were visually checked for terminology and formulas as recorded in the chapter log. This is not a claim of a line-by-line comparison of the entire Chinese chapter.

Scope and review. Elementary calculus, measure-theoretic limit theorems, the Gaussian integral and Gamma integral, and the explicitly invoked complex-analytic parameter-integral/Cauchy facts are prerequisites. General-domain existence of harmonic boundary correctors is left to later Dirichlet theory. All four model equations, their representation methods, and the 24 source exercises have study explanations here; mathematical self-checks and numerical moment checks are not independent mathematical review.

Chapter 3

Nonlinear First-Order PDE

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 3; printed pp. 89–158; PDF pages 106–175.

Chinese reference: 中译本：第三章，书内第 61 页起；章首页为 PDF 第 79 页。

Editorial study edition. This chapter reconstructs Evans's exposition and proofs in the source order; it is not a verbatim transcription. Source theorem identifiers are distinguished from our chapter-wide numbering. Expanded calculations, Chinese mathematical annotations, figures, and exercise solutions are editorial additions.

For $F = F(p, z, x)$ we study $F(Du, u, x) = 0$, with p denoting the gradient variable and z the function-value variable. The main route is

$$\begin{aligned} &\text{envelopes and characteristic ODE} \\ &\longrightarrow \text{ local smooth solutions} \\ \longrightarrow &\text{ variational formulas and admissible weak solutions.} \end{aligned}$$

从局部光滑到全局弱解

特征线在相空间中仍可光滑演化，但投影到物理空间后可能相交。单值光滑解因此失效；Hamilton–Jacobi 选择半凹分支，守恒律选择满足熵条件的跳跃。

Conventions and prerequisites. Spatial gradients are column vectors. Time is a distinguished extra coordinate for evolution equations. Hamiltonians and fluxes are real valued; convexity and superlinearity are stated wherever used. We use local ODE existence/uniqueness and smooth dependence, the implicit and inverse function theorems, standard integration and weak derivative calculus, Rademacher's theorem, and Gronwall's inequality as explicit prerequisites. Supporting hyperplanes for finite convex functions are invoked for the nonsmooth convex-duality statements.

3.1 Complete Integrals, Envelopes

3.1.1 Complete integrals (Evans §3.1.1)

An n-parameter family $u(x;a)$ is a complete integral for $F(Du,u,x)=0$ in n independent variables if every member solves the PDE and the $n\times(n+1)$ matrix

$$(D_a u,\ D^2_{ax}u)$$

has rank n. The condition concerns variation of the whole first jet, not only the gradient.

If this family factored through $n-1$ parameters, $u(x;a)=v(x;\psi(a))$, the chain rule would factor the displayed matrix through $D_a\psi$, whose rank is at most $n-1$. Thus full rank rules out that redundancy.

完全积分不是所有解

它是含足够独立参数的一族解，不能把“完全”理解成已经穷尽全部解。边界值通常是一整个函数，有限多个常数不能直接编码任意边界数据。

Three useful families. For Clairaut's equation $x\cdot Du+f(Du)=u$,

$$u(x;a)=a\cdot x+f(a)$$

solves the PDE and $D^2_{ax}u=I$, proving full rank. For the eikonal equation $|Du|=1$, use $u=a\cdot x+b$ with $|a|=1$. A local chart of the sphere has $n-1$ independent parameters and b adds one; the independent tangential gradient variations and value variation give rank n. This family is locally parametrized by an open subset of $\mathbb{R}^n$, rather than treating the sphere as an open set.

For $u_t+H(D_x u)=0$ in $n+1$ independent variables,

$$u(x,t;a,b)=a\cdot x-tH(a)+b.$$

There are $n+1$ parameters. The a derivatives change $D_x u$ by the standard basis, and the b derivative changes the value by one without changing the gradient. The corresponding jet matrix has rank $n+1$, as required.

3.1.2 Envelopes and singular integrals (Evans §3.1.2, Theorem 1)

Theorem 3.1
Let $U(x;a)$ be a smooth family of solutions and suppose $D_a U(x;\alpha(x))=0$ with α continuously differentiable. Then $v(x)=U(x;\alpha(x))$ is also a solution wherever this construction is defined.

Expanded proof.

For each coordinate, the full chain rule is

$$\partial_i v=\partial_{x_i}U(x;\alpha(x))+\sum_j \partial_{a_j}U(x;\alpha(x))\,\partial_i\alpha_j.$$

The sum vanishes by the stationary-parameter condition. Thus v and the selected family member have the same value and the same spatial gradient at x. Substitution into F proves the PDE. To obtain a smooth stationary parameter locally, one sufficient condition is invertibility of $D^2_{aa}U$, by the implicit function theorem. That condition is sufficient, not part of the abstract envelope identity. □

For $U(x,t;a) = a \cdot x - t|a|^2$, stationarity gives $a = x/(2t)$, and the envelope is $v(x,t) = |x|^2/(4t)$ for $t > 0$. Indeed $v_t = -|x|^2/(4t^2)$ and $|D_x v|^2 = |x|^2/(4t^2)$. This example uses $H(p) = |p|^2$; later the common choice $H = |p|^2/2$ has a different factor of two.

For $U(x;a) = \pm\sqrt{1-|x-a|^2}$, the PDE is $U^2(1+|DU|^2) = 1$. Stationarity yields $a = x$ and the envelopes $v = \pm 1$. The derivative with respect to a of the upper branch is $(x-a)/\sqrt{1-|x-a|^2}$; changing the branch changes the sign.

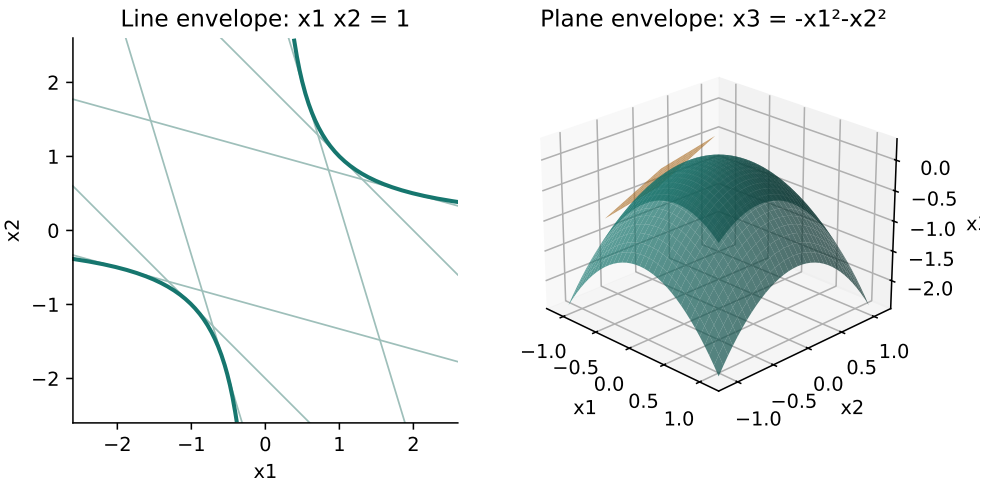
One can restrict the parameters to $a = (a', h(a'))$ and then form an envelope over a'. The stationary equations become $U_{a_i} + U_{a_n} h_{a_i} = 0$. The resulting general integral depends on the arbitrary function h, whenever a smooth envelope exists. It still need not describe every PDE solution: a factorized equation $F_1(Du,u,x)F_2(Du,u,x) = 0$ may have a branch belonging only to F_2.

为何参数导数消失

对 $U(x,a(x))$ 求导时，不能忘掉参数链式项。正是 $D_a U = 0$ 使它消失，包络与被选中的曲面才在该点相切。

二次 Hamiltonian 的系数

若 $H(p) = |p|^2$，其对偶为 $L(v) = |v|^2/4$；若 $H(p) = |p|^2/2$，则 $L(v) = |v|^2/2$。检查公式时，把 PDE、对偶与包络中的系数一起核对。



Editorial figure for Problem 2. A hyperbola is tangent to a family of lines; a negative paraboloid is tangent to the associated family of planes. Elimination and parameter-domain restrictions are worked out in the exercise solution.

3.2 Characteristics

3.2.1 Deriving the full characteristic system (Evans §3.2.1)

Suppose first that a C^2 solution exists. Along a curve $X(s)$ set $Z(s) = u(X(s))$ and $P(s) = Du(X(s))$. Then

$$\dot{P}_i = \sum_j u_{x_i x_j}(X)\dot{X}_j.$$

Differentiate the PDE itself in x_i:

$$\sum_j F_{p_j} u_{x_j x_i} + F_z u_{x_i} + F_{x_i} = 0.$$

Choosing $\dot{X}_j = F_{p_j}$ makes the unknown Hessian sum match the first expression. Hence

$$\boxed{\begin{aligned} \dot{X} &= F_p(P, Z, X), \\ \dot{P} &= -F_x(P, Z, X) - F_z(P, Z, X)P, \\ \dot{Z} &= P \cdot F_p(P, Z, X). \end{aligned}} \tag{CH}$$

All partial derivatives of F here hold its other arguments fixed. The third equation is the chain rule for $u(X(s))$.

选择路径是为了消去 Hessian

任意路径的梯度变化含 D^2u，并不封闭。取速度 F_p 后，恰好能用微分后的 PDE 替换这组二阶导数，得到只含 X, P, Z 的 ODE。

An invariant of the system. Even before knowing $P = Du$, solutions of (CH) satisfy

$$\frac{d}{ds}F(P, Z, X) = F_p \cdot (-F_x - F_z P) + F_z(P \cdot F_p) + F_x \cdot F_p = 0.$$

Initial data with $F(P(0), Z(0), X(0)) = 0$ therefore remain on the PDE constraint surface. This invariant alone does not yet prove that the transported vector P is the gradient of a single-valued function.

3.2.2 Reduced systems and explicit examples (Evans §3.2.2)

For $b(x) \cdot Du + c(x)u = f(x)$, only

$$\dot{X} = b(X), \qquad \dot{Z} = f(X) - c(X)Z$$

are needed to compute values. If $b = b(x, u)$, the value equation remains $\dot{Z} = -c(X, Z)$ for $b(x, u) \cdot Du + c(x, u) = 0$; the path now depends on the unknown value.

Rotation. For $x_1 u_{x_2} - x_2 u_{x_1} = u$ in the first quadrant with $u(r, 0) = g(r)$,

$$X(s) = (r\cos s, r\sin s), \qquad Z(s) = e^s g(r).$$

Thus $u(x_1, x_2) = e^{\arctan(x_2/x_1)} g(\sqrt{x_1^2 + x_2^2})$. This is also immediate from $u_\theta = u$ in polar coordinates.

Value blow-up without crossing. For $u_{x_1} + u_{x_2} = u^2$, $u(x_1, 0) = g(x_1)$,

$$X(s) = (r + s, s), \qquad Z(s) = \frac{g(r)}{1 - sg(r)},$$

so

$$u(x_1, x_2) = \frac{g(x_1 - x_2)}{1 - x_2 g(x_1 - x_2)}$$

on the connected region reached before a denominator vanishes. If $g(r) > 0$, the characteristic value blows up at $s = 1/g(r)$. Continuing the rational expression beyond its pole is not continuing a finite classical solution through the blow-up time.

两种失效机制

半线性例子可在路径不相交时因 Z 爆破失效；守恒律常见的是值仍有界、空间梯度先爆破。只检查特征线位置还不够。

A fully nonlinear example. For $u_{x_1} u_{x_2} = u$, $u(0, r) = r^2$, compatibility gives $P_2(0) = 2r$ and $P_1(0) = r/2$ when $r \neq 0$. The system is $\dot{P}_i = P_i$, $\dot{X}_1 = P_2$, $\dot{X}_2 = P_1$, $\dot{Z} = 2P_1 P_2$. It integrates to

$$X_1 = 2r(e^s - 1), \quad X_2 = \tfrac{r}{2}(e^s + 1), \quad Z = r^2 e^{2s}.$$

Since $4X_2 - X_1 = 4r$ and $4X_2 + X_1 = 4re^s$,

$$u(x_1, x_2) = \frac{(x_1 + 4x_2)^2}{16}.$$

Direct differentiation verifies the PDE and boundary data everywhere. The characteristic construction just given is locally invertible only where $r \neq 0$; the polynomial formula extends across the excluded locus. An explicit extension does not mean the local existence theorem was applicable at the characteristic boundary point $r = 0$.

3.2.3 Admissible and noncharacteristic data (Evans §3.2.3)

Flatten the initial hypersurface to $x_n = 0$ and write its coordinates as $y \in \mathbb{R}^{n-1}$. Prescribing $u(y, 0) = g(y)$ fixes

$$Z(0) = g(y), \quad P_i(0) = g_{y_i}(y) \ (i < n), \quad F(P(0), g(y), (y, 0)) = 0. \tag{IC}$$

Only the normal derivative $P_n(0)$ remains to be found. At an admissible root p^0, the sufficient condition $F_{p_n}(p^0, g(y_0), (y_0, 0)) \neq 0$ lets the implicit function theorem solve for a smooth branch $q_n(y)$. Different roots may produce different local solutions; uniqueness is relative to the chosen admissible branch.

For a curved surface, choose a smooth extension of g and write $p = D_{\tan} g + \lambda \nu$, where ν is its unit normal. Differentiating the scalar constraint in λ gives

$$\frac{d}{d\lambda} F(D_{\tan} g + \lambda \nu, g, x) = F_p \cdot \nu.$$

Thus the invariant geometric noncharacteristic condition is $F_p \cdot \nu \neq 0$: the characteristic velocity crosses the surface. Changing the normal's orientation changes the sign, not this condition.

For example $u_{x_2} = 0$ with data on $x_2 = 0$ is noncharacteristic. The equation $u_{x_1} = 0$ on the same boundary is characteristic: nonconstant data in x_1 are incompatible, while constant data leave the x_2 dependence undetermined. Failure of the hypothesis can mean either nonexistence or nonuniqueness.

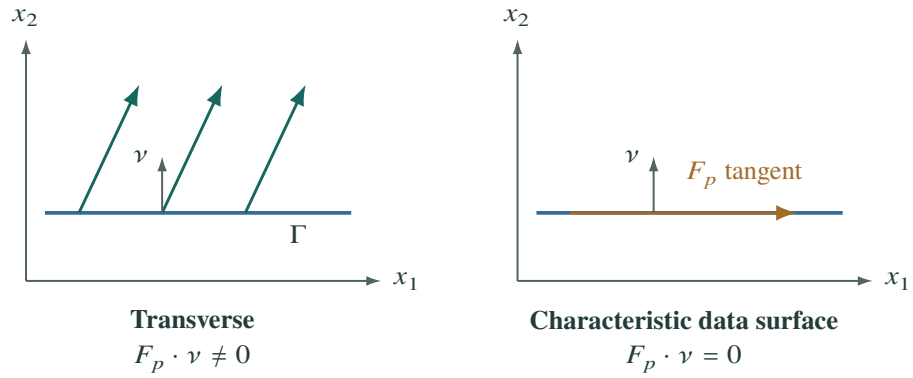


Diagram T3.1. Noncharacteristic data permit local reconstruction. The projected characteristic direction is F_p. Transversality makes the initial-surface parameters together with the flow parameter independent local coordinates. A tangent direction makes this coordinate construction singular; it does not by itself prove that no solution can exist.

3.2.4 Reconstructing a local solution (Evans §3.2.4, Theorem 2)

Theorem 3.2

For smooth F, g and a chosen admissible noncharacteristic triple, (CH) with data (IC) reconstructs a smooth local solution. The construction is unique locally among smooth solutions sharing that branch of boundary gradient.

Expanded proof (including the gradient compatibility step).

1. Solve the ODE with boundary parameters. Use the boundary-parameterized solution

$$(X, P, Z)(y, 0) = ((y, 0), q(y), g(y)).$$

边界值没有给出完整梯度

沿边界求导只能得到切向分量，法向分量要由 PDE 求出。代数方程可能无根或多根；非特征条件保证的是已选根附近的局部唯一分支。

非特征条件是局部条件

$F_p \cdot \nu \neq 0$ 让特征穿过初始曲面，从而给出局部坐标。它不保证以后投影仍可逆；法向速度的正负另用于区分流入、流出方向。

Local ODE theory with smooth dependence supplies a common parameter neighborhood. At $s = 0$, the columns X_{y_i} are e_i, $i < n$, and $X_s = F_p$. Hence

$$\det D_{(y,s)} X(y_0, 0) = F_{p_n}(p^0, z^0, x^0) \neq 0.$$

The inverse function theorem gives local coordinates $(y, s) = (y(x), s(x))$. Define $u(x) = Z(y(x), s(x))$.

2. Preserve the algebraic PDE constraint. The invariant above and initial compatibility give $F(P(y,s), Z(y,s), X(y,s)) = 0$. We still have to prove $P(y,s) = Du(X(y,s))$.

3. Propagate the contact identities. The s identity $Z_s = P \cdot X_s$ is in (CH). For every boundary parameter set $R_i = Z_{y_i} - P \cdot X_{y_i}$. Initially $R_i = 0$, by the tangential compatibility in (IC). Differentiate:

$$\begin{aligned}(R_i)_s &= P_{y_i} \cdot X_s - P_s \cdot X_{y_i} \\ &= F_p \cdot P_{y_i} + F_x \cdot X_{y_i} + F_z P \cdot X_{y_i}.\end{aligned}$$

Differentiating the preserved identity $F(P, Z, X) = 0$ in y_i yields $F_p \cdot P_{y_i} + F_x \cdot X_{y_i} = -F_z Z_{y_i}$. Therefore $(R_i)_s = -F_z R_i$ with zero initial value. Uniqueness for this scalar linear ODE gives $R_i = 0$.

4. Return to physical coordinates. Together the identities say $dZ = P \cdot dX$ for all parameter differentials. Since $D_{(y,s)}X$ is invertible, every physical differential occurs, so $Du = P$. The preserved constraint now becomes $F(Du, u, x) = 0$. At $s = 0$ we recover $u = g$. Any other smooth solution with the same boundary gradient branch generates the same characteristic ODE data, so ODE uniqueness and coordinate invertibility identify the solutions. □

3.2.5 Evolution equations and breakdown (Evans §3.2.5)

For $u_t + \mathbf{f}'(u) \cdot D_x u = 0$ with initial data g,

$$X(a,t) = a + t\mathbf{f}'(g(a)), \qquad u(X(a,t),t) = g(a).$$

Eliminating a gives $u = g(x - t\mathbf{f}'(u))$ while the map is invertible. Its Jacobian is $I + t\,\mathbf{f}''(g(a)) \otimes Dg(a)$. The rank-one determinant identity yields

$$\det D_a X = 1 + t\, Dg(a) \cdot \mathbf{f}''(g(a)).$$

In one dimension,

$$u_x(X(a,t),t) = \frac{g'(a)}{1 + tf''(g(a))g'(a)}.$$

不能提前把 P 叫作梯度

从 ODE 构造出来的 P 起初只是向量场。必须证明 $dZ = P \cdot dX$，再利用坐标可逆性，才能认定 $P = Du$；否则只保留 $F(P, Z, X) = 0$ 还没有解出 PDE。

If the denominator first vanishes with $g'(a) < 0$ and $f'' > 0$, gradients blow up although $u = g(a)$ remains finite. For Burgers flux $f(u) = u^2/2$ and bounded C^1 data with $m = \inf g' < 0$, the first loss of a uniform positive Jacobian is at $T_* = -1/m$. If the infimum is attained, that characteristic gives an actual zero Jacobian at T_*.

For $u_t + H(D_x u, x) = 0$, the full system becomes Hamilton's equations and an action equation:

$$\dot{X} = H_p(P, X), \qquad \dot{P} = -H_x(P, X), \qquad \dot{Z} = P \cdot H_p - H.$$

When $H = H(p)$, $P = Dg(a)$ stays constant and

$$X = a + tDH(Dg(a)), \qquad Z = g(a) + t[Dg(a) \cdot DH(Dg(a)) - H(Dg(a))].$$

This formula is local until the projection ceases to be invertible. The next section gives a variational continuation with a selection rule.

相空间不交不保证投影不交

扩展状态 (X, P, Z) 的 ODE 可保持唯一性，但只看 X 时不同状态可落在同一位置。物理空间的逆映射失效，正是特征构造中光滑解失效的关键。

3.3 Introduction to Hamilton–Jacobi Equations

3.3.1 Variations and Hamilton's ODE (Evans §3.3.1)

For a path $X : [0, t] \to \mathbb{R}^n$ with fixed endpoints consider

$$\mathcal{A}[X] = \int_0^t L(\dot{X}(s), X(s))\, ds.$$

Suppose a smooth minimizer exists. For every smooth variation V vanishing at both endpoints, differentiate the action at $X + \varepsilon V$:

$$0 = \int_0^t [L_v(\dot{X}, X) \cdot \dot{V} + L_x(\dot{X}, X) \cdot V]\, ds.$$

Integration by parts, with the endpoint terms zero, gives $\frac{d}{ds} L_v(\dot{X}, X) = L_x(\dot{X}, X)$ by the fundamental lemma of the calculus of variations. These are the Euler–Lagrange equations. They are necessary conditions; their derivation does not prove a minimizer exists or that every stationary path minimizes.

Assume now the velocity–momentum relation $P = L_v(v, x)$ can be inverted, and define $H(P, x) = P \cdot v(P, x) - L(v(P, x), x)$. In differentiating this identity the terms containing derivatives of v cancel because $P = L_v$. Thus $H_p = v$ and $H_x = -L_x$. The Euler–Lagrange equations become

$$\dot{X} = H_p(P, X), \qquad \dot{P} = -H_x(P, X).$$

For $L(v, x) = |v|^2/2 - V(x)$ this gives $P = v$, $H = |P|^2/2 + V(x)$ and $\ddot{X} = -DV(X)$. For an autonomous Hamiltonian, $\frac{d}{ds} H = H_p \cdot (-H_x) + H_x \cdot H_p = 0$.

3.3.2 Convex duality and Hopf–Lax (Evans §3.3.2)

A function A is superlinear if $A(v)/|v| \to +\infty$ as $|v| \to \infty$. For a finite convex function define

$$A^*(p) = \sup_v \{p \cdot v - A(v)\}.$$

Superlinearity of A makes the objective tend to $-\infty$, so the maximum is attained and A^* is finite. It is convex as a supremum of affine functions. For every $R > 0$,

$$A^*(p) \geq R|p| - \max_{|v|=R} A(v);$$

hence A^* is also superlinear when A is finite everywhere.

Theorem 3.3 (Convex duality; source Theorem 3)
For finite convex superlinear A, $A^{**} = A$ and

$$A(v) + A^*(p) \geq p \cdot v.$$

Equality holds precisely when p is a supporting slope of A at v. For differentiable A, this means $p = DA(v)$.

Expanded proof.

The inequality is the definition of the supremum. A supporting slope satisfies $A(w) \geq A(v) + p \cdot (w - v)$ for every w. Rearranging shows the supremum defining $A^*(p)$ is attained at v, giving equality. Conversely equality rearranges to that support inequality. Always $A^{**}(v) \leq A(v)$. Choose a supporting slope at v; then the equality just established gives the reverse inequality. Existence of supporting hyperplanes for finite convex functions is the invoked convex-analysis prerequisite. For a differentiable convex function the tangent-plane inequality provides this slope directly as $DA(v)$. □

The supporting-plane argument for $A^{**} = A$ also holds for every finite convex A without superlinearity; in that case A^* may take the value $+\infty$. Subdifferentials of such extended-valued functions are taken only at points where the function is finite.

两个来源得到同一组 ODE

变分法从作用量的驻值推出 Hamilton 方程；特征线法从 Hamilton–Jacobi PDE 推出同一方程。前者给出力学意义，后者解释它如何搬运函数与梯度。

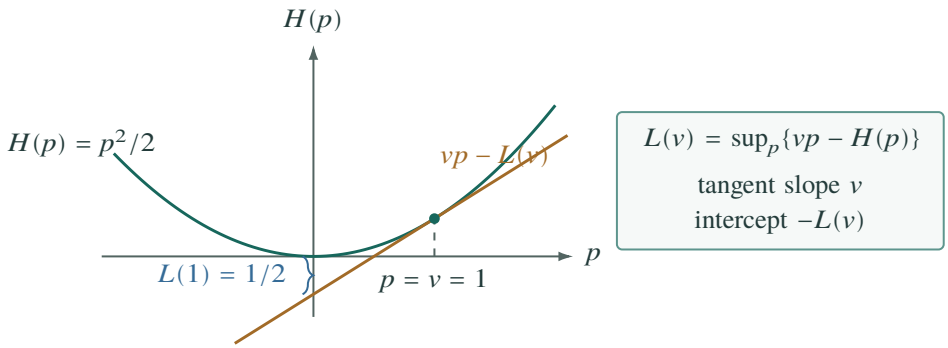


Diagram T3.2. Legendre duality is a supporting-line construction. For $H(p) = p^2/2$ and $v = 1$, maximizing $vp - H(p)$ finds the tangent point $p = 1$. The tangent's intercept is $-L(1) = -1/2$. Convexity makes the whole graph lie above this line, which is exactly the Fenchel inequality.

For the rest of this subsection assume H is finite, convex, smooth and superlinear on $\mathbb{R}^n$, set $L = H^*$, and let g be globally Lipschitz with constant K. Then L is finite, convex and superlinear, and $H = L^*$.

Why a straight path gives the formula (source Theorem 4). Since L no longer depends on position, Jensen's inequality gives for any absolutely continuous path with $X(0) = y$, $X(t) = x$,

$$\int_0^t L(\dot X(s))\,ds \geq tL\left(\frac{1}{t}\int_0^t \dot X(s)\,ds\right) = tL((x-y)/t).$$

The straight path $X(s) = y + s(x-y)/t$ attains equality. Minimizing over its starting point gives

$$\boxed{u(x,t) = \min_{y\in\mathbb{R}^n}\{tL((x-y)/t) + g(y)\}.} \qquad \text{(HL)}$$

The minimum exists: with $v = (x-y)/t$, its objective is at least $g(x) + t[L(v) - K|v|]$, coercive in v. Comparing with $y = x$ also proves that every minimizer satisfies $L(v) - K|v| \leq L(0)$, hence $|v| \leq R_K$ for a fixed finite R_K. This is the first finite-range localization of the minimization.

对偶变换中的极大值

求 $v \cdot p - H(p)$ 的驻点只是候选；凸性让该目标关于 p 凹，超线性增长使它在无穷远趋负无穷，才保证极大值确实取得。

超线性排除逃向远方的起点

g 的下降至多是线性的，而 $tL((x-y)/t)$ 在位移很大时增长更快。这不仅给出最小值存在，还把最优起点限制在 $|x-y| \leq Rt$ 内。

直线最优依赖哪些条件

这里用到 L 的凸性和无位置依赖。若 $L = L(v,x)$，不同路径经过不同势场，不能只用端点位移除以时间就替代整条作用量。

Lemma 3.4 (Dynamic programming; source Lemma 1)
For $0 < s < t$,

$$u(x,t) = \min_z\{(t-s)L((x-z)/(t-s)) + u(z,s)\}. \qquad \text{(DP)}$$

Expanded proof.

For any z and a minimizing starting point y for $u(z,s)$, convexity applied to $(x-y)/t = [(t-s)/t](x-z)/(t-s) + (s/t)(z-y)/s$ gives $u(x,t) \le (t-s)L((x-z)/(t-s)) + u(z,s)$. This proves one inequality for the infimum over z. Conversely take a minimizer y in (HL) and choose $z = y + (s/t)(x-y)$. Both subinterval velocities then equal $(x-y)/t$. Use this same y as a competitor for $u(z,s)$ to obtain the opposite inequality. Coercivity ensures the intermediate minimum exists. □

Lemma 3.5 (Lipschitz bounds and initial values; source Lemma 2)
The function in (HL) is globally Lipschitz in (x,t) *for* $t \ge 0$, *when defined by* $u(x,0) = g(x)$.

Expanded proof.

If y minimizes at x, use $y + h$ as competitor at $x + h$. The velocity is unchanged, so $u(x+h,t) - u(x,t) \le K|h|$. Exchange $x, x+h$ for the reverse inequality.

For time zero, the competitor $y = x$ gives $u(x,t) \le g(x) + tL(0)$, while duality gives

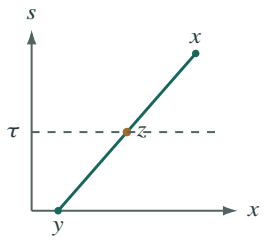
$$u(x,t) \ge g(x) + t \inf_v [L(v) - K|v|] = g(x) - t \max_{|p|\le K} H(p).$$

Thus $|u(x,t) - g(x)| \le Ct$, uniformly in x, with

$$C = \max\{|L(0)|, \max_{|p|\le K} |H(p)|\}.$$

Apply the same argument to (DP), using the spatial Lipschitz bound for $u(\cdot,s)$, to get $|u(x,t) - u(x,s)| \le C|t-s|$. □

图 T3.M：把路径在中途切开



中间点 z 也要优化。两段代价相加后再取下确界，才得到动态规划恒等式。

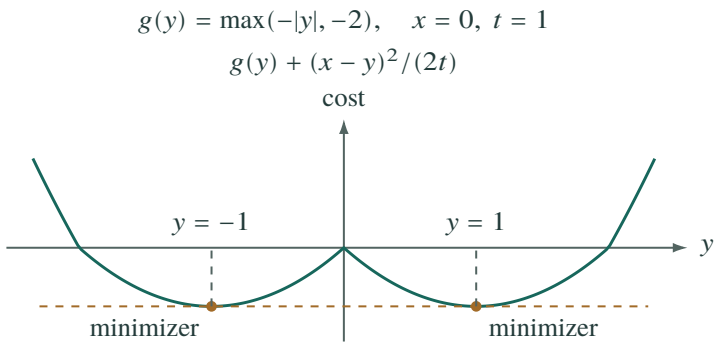


Diagram T3.3. Equal minimizing costs can produce a corner in the value. With $H(p) = p^2/2$, $t = 1$ and $g(y) = \max(-|y|, -2)$, the Hopf–Lax cost at $x = 0$ has two minimizers $y = \pm 1$. Their candidate endpoint slopes $(x-y)/t$ differ. Indeed near $x = 0$ the value is $u(x,1) = -1/2 - |x|$, so it is not differentiable there.

Theorem 3.6 (PDE at differentiability points; source Theorems 5–6)
The Hopf–Lax function satisfies $u_t + H(Du) = 0$ at every point where it is differentiable, hence almost everywhere by Rademacher's theorem.

Expanded proof.

At a differentiability point (x,t), (DP) with the competitor x over a short forward time gives $u(x+hv, t+h) \le hL(v) + u(x,t)$ for every fixed v. Divide by $h > 0$ and pass to zero: $u_t + v \cdot Du \le L(v)$. Supremizing in v yields $u_t + H(Du) \le 0$.

For the other sign choose a minimizing y in (HL) at (x,t) and put $v_* = (x-y)/t$. At time $t-h$, $z = x - hv_*$ lies on the straight minimizing path. Using y as competitor there yields $u(x,t) - u(z, t-h) \ge hL(v_*)$. Differentiability gives $u_t + v_* \cdot Du \ge L(v_*)$. Since $H(Du) \ge v_* \cdot Du - L(v_*)$, the reverse sign follows. Neither argument differentiates a potentially nonsmooth minimizing point $y(x,t)$. □

For $H(p) = |p|^2/2$, $L(v) = |v|^2/2$ and $u(x,t) = \min_y\{|x-y|^2/(2t) + g(y)\}$. At a smooth minimizing branch, $(x-y)/t = Dg(y)$, so $x = y + tDg(y)$ and $Du = Dg(y)$, recovering the local characteristic formulas. When branches compete, the minimum remains meaningful even though their smooth projection has broken down.

最小值不一定可微

多个最优起点可能同时出现，不能直接对 $y(x,t)$ 求导。动态规划用固定竞争者给一侧不等式，再沿一条最优路径给另一侧，避开了这个问题。

3.3.3 Semiconcavity and uniqueness (Evans §3.3.3)

An a.e. PDE alone is insufficient for uniqueness. For $H(p) = p^2$ and $g = 0$ in one dimension, both $u = 0$ and

$$v(x,t) = \min\{0, |x| - t\}$$

are Lipschitz and satisfy $v_t + v_x^2 = 0$ off the three lines $x = 0, \pm t$. At the central upward corner, $v(h,t) + v(-h,t) - 2v(0,t) = 2|h|$ for $|h| < t$. This cannot be bounded by a constant times h^2.

Definition 3.7

A function f is semiconcave with constant C if

$$f(x+z) - 2f(x) + f(x-z) \le C|z|^2.$$

For continuous f this is equivalent to concavity of $f(x) - C|x|^2/2$, by the midpoint criterion and continuity. For smooth f it is equivalent to $D^2 f \le CI$.

半凹只限制向上弯曲

允许向下尖角，禁止二阶差分过大的向上尖角。它不是要求 Hessian 的绝对值有界，也不是要求函数本身凹；先减去一个二次函数后才凹。

Theorem 3.8 (Source Lemmas 3–4)

If g is semiconcave with constant C, the Hopf–Lax function has the same spatial constant for all t. If instead $D^2 H \ge \theta I$, $\theta > 0$, it satisfies

$$u(x+z,t) - 2u(x,t) + u(x-z,t) \le \frac{|z|^2}{\theta t},$$

even for initial g that is merely Lipschitz.

Expanded proof (including the dual estimate of Problem 15).

For the first assertion choose a minimizer y at (x,t). Use $y+z$ and $y-z$ as competitors at $x+z$ and $x-z$. All three velocities are equal, so their L terms cancel. The remaining second difference is that of g at y.

For the second assertion, uniform convexity makes the maximizer $p(v)$ in $L(v) = \max_p(v \cdot p - H(p))$ unique, with $v = DH(p(v))$. Strong monotonicity gives

$$\theta |p(v) - p(w)|^2 \le (v - w) \cdot (p(v) - p(w)),$$

so $|p(v) - p(w)| \le |v - w|/\theta$. Comparing the optimizing expressions at v and $v + h$ shows $DL(v) = p(v)$; equivalently, apply the implicit function theorem to $DH(p) = v$. Hence DL is $1/\theta$-Lipschitz. Integrating its increment along a segment gives

$$L(v+h) \le L(v) + DL(v) \cdot h + \frac{|h|^2}{2\theta}.$$

Add the estimates for h and $-h$: $L(v+h)+L(v-h)-2L(v)\le |h|^2/\theta$. Now use the same minimizing y at all three target points. The initial terms cancel and $h=z/t$ gives the claimed constant.

In particular, taking $v=(v_1+v_2)/2$, $h=(v_1-v_2)/2$ proves

$$\tfrac12 L(v_1)+\tfrac12 L(v_2)\le L((v_1+v_2)/2)+\frac{|v_1-v_2|^2}{8\theta}.$$

This is the dual inequality omitted from the source lemma's proof. □

Theorem 3.9 (Uniqueness in the source weak class; source Theorems 7–8)
Let H be smooth, convex and superlinear. Among globally Lipschitz functions with the same initial datum, satisfying the HJ equation a.e. and the spatial semiconcavity bound

$$u(x+z,t)-2u(x,t)+u(x-z,t)\le C(1+t^{-1})|z|^2,$$

there is at most one solution. The Hopf–Lax function belongs to this class if g is semiconcave or H is uniformly convex.

Expanded proof (the source's regularized cone argument).

Let $w=u-\tilde u$. Subtract the a.e. PDE and integrate DH on the gradient segment:

$$w_t+b\cdot Dw=0,\qquad b=\int_0^1 DH(\lambda Du+(1-\lambda)D\tilde u)\,d\lambda.$$

Both gradients lie in a fixed ball, so $|b|\le R$ for a finite R.

1. Smooth the coefficients, preserving a one-sided bound. Locally convolve $u,\tilde u$ in space-time and define b_ε by the same expression with their smoothed spatial gradients. For $t>2\varepsilon$, semiconcavity implies

$$D^2u_\varepsilon,\ D^2\tilde u_\varepsilon\le C'(1+t^{-1})I.$$

To check this, integrate their second-difference inequalities against the nonnegative smoothing kernel, bound $1/s$ by $2/t$ on its time support, then divide by the squared increment and let it tend to zero. The gradients remain bounded and converge a.e. to the original ones.

Writing $A=D^2H\ge 0$ and B for the convex combination of smoothed Hessians gives $B\le C'(1+t^{-1})I$. Thus $\operatorname{tr}(AB)\le C'(1+t^{-1})\operatorname{tr}A$, by conjugating with $A^{1/2}$ and taking traces. Since A is bounded on the gradient ball, $\operatorname{div} b_\varepsilon\le C''(1+t^{-1})$. Also $|b_\varepsilon|\le R$ and $b_\varepsilon\to b$ a.e.

2. Integrate a nonnegative transform in a shrinking cone. Take a smooth nonnegative function ϕ, and put $v = \phi(w)$. Weak Lipschitz calculus gives $v_t + b \cdot Dv = 0$ a.e. On $B_{\rho(t)}(x_0)$, $\rho(t) = r + R(T - t)$ with $r > 0$, set $e(t) = \int v$. The moving-boundary term and $|b_\varepsilon| \le R$ make its boundary flux nonpositive. For a.e. t this yields

$$e'(t) \le C''(1 + t^{-1})e(t) + \int_{B_{\rho(t)}} (b_\varepsilon - b) \cdot Dv.$$

These integrations hold for Lipschitz v by approximation, or its Sobolev integration formula on the smooth ball. For each fixed $\delta > 0$, the error tends to zero in $L^1(\delta, T)$ by dominated convergence on the compact cone. Integrate against a nonnegative time test function and let $\varepsilon \to 0$, obtaining the same differential inequality without the error on $[\delta, T]$.

3. Start Gronwall away from the singular time zero. Let M be the sum of the time Lipschitz constants. Since $w(\cdot, 0) = 0$, $|w(x, \delta)| \le M\delta$. Choose $\phi = \phi_\delta$ zero on $[-M\delta, M\delta]$ and strictly positive outside. Then $e(\delta) = 0$. Gronwall on $[\delta, T]$ has a finite coefficient integral, and gives $e(T) = 0$. Hence $|w| \le M\delta$ on $B_r(x_0)$ at time T, using continuity to pass from a.e. points. Let $\delta \downarrow 0$. The ball and T were arbitrary, so $w = 0$. Existence in the stated class follows from the two semiconcavity estimates and the preceding Hopf–Lax theorem. □

This is the semiconcavity-based weak formulation used in this chapter. Viscosity solutions provide a broader comparison framework in Chapter 10; the a.e. equation and the present selection condition must not be silently interchanged.

凸性把单边 Hessian 界传给速度

两个半凹函数的 Hessian 只需有上界。因 D^2H 半正定，迹运算仍给出 $\operatorname{div} b$ 的上界；若失去凸性，这一步的符号便没有保证。

不能从零时刻硬套 Gronwall

$1/t$ 在零附近不可积。证明先用时间 Lipschitz 界把差值截掉，在正时刻 δ 得到零初始能量，再应用 Gronwall，最后令 $\delta \to 0$。

3.4 Introduction to Conservation Laws

3.4.1 Shocks and entropy (Evans §3.4.1)

We now work in one space dimension:

$$u_t + f(u)_x = 0, \qquad u(x, 0) = g(x).$$

The conservative form must be retained after smoothness breaks down. An integral solution is a locally bounded function satisfying

$$\int_0^\infty \int_{\mathbb{R}} [u\phi_t + f(u)\phi_x]\, dx\, dt + \int_{\mathbb{R}} g(x)\phi(x, 0)\, dx = 0 \qquad \text{(CL)}$$

for every smooth compactly supported test function on $\mathbb{R} \times [0, \infty)$. For smooth u, integration by parts proves equivalence with the PDE. For discontinuous u, the expression $f'(u)u_x$ is not an ordinary product of functions and cannot replace (CL) without additional theory.

Theorem 3.10 (Rankine–Hugoniot)
Across a smooth shock curve $x = \gamma(t)$ with traces u_l, u_r, an otherwise smooth integral solution must satisfy

$$\gamma'(t)(u_r - u_l) = f(u_r) - f(u_l). \tag{RH}$$

For unequal states, the speed is the secant slope of f.

Expanded proof.

Write locally $u = u_l + (u_r - u_l)\mathbf{1}_{\{x>\gamma(t)\}}$, with the smooth extensions on each side understood. The singular part of u_t is $-\gamma'[u]\delta_{x=\gamma(t)}$; that of $f(u)_x$ is $[f]\delta_{x=\gamma(t)}$. Here $[u] = u_r - u_l$ and $[f] = f(u_r) - f(u_l)$. The regular parts vanish by the PDE on each side. The remaining distribution is zero precisely when (RH) holds. Equivalently, integrate each smooth side of (CL) and add their oppositely oriented interface fluxes. □

For strictly convex f, an admissible shock has $u_l > u_r$. Its speed satisfies $f'(u_r) < \gamma' < f'(u_l)$, since secant slopes lie strictly between endpoint tangent slopes. Characteristics enter the shock from both sides. An upward jump also satisfies (RH) with its secant speed, but would send characteristics outward and is rejected by entropy.

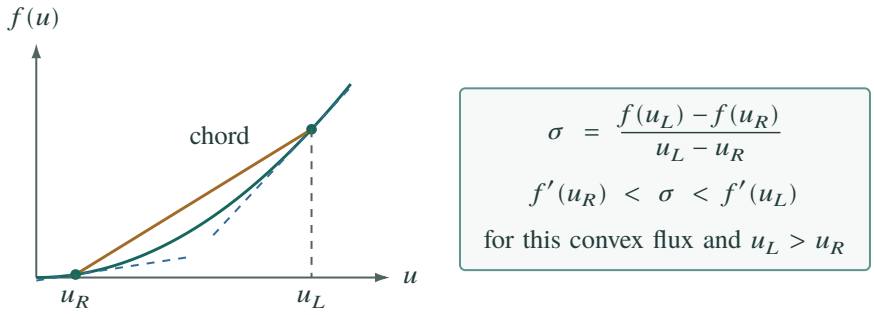


Diagram T3.4. Shock speed is a secant slope. For $f(u) = u^2/2$, the secant between $u_R = 0.2$ and $u_L = 1.4$ has slope $\sigma = 0.8$. The tangent slopes satisfy $0.2 < 0.8 < 1.4$, so characteristics from both sides enter the shock. Reversing the spatial states keeps the secant slope but violates the compression direction.

Why weak conservation alone is not unique. For Burgers flux $f(u) = u^2/2$ and data $g = 0$ on $x < 0$, $g = 1$ on $x > 0$, both the upward step

先守恒后求导

光滑时 $f(u)_x = f'(u)u_x$；跳跃时 u_x 含集中测度。守恒形式能直接对 $f(u)$ 作分布求导，避免给不连续函数与 Dirac 的乘积随意赋值。

moving at speed $1/2$ and

$$u(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t \end{cases}$$

satisfy (CL). In the fan the PDE is $-x/t^2 + (x/t)(1/t) = 0$, and both interfaces are continuous. Each candidate attains the same initial data in local L^1. Only the fan satisfies the one-sided bound derived below.

跳跃条件不负责筛选方向

Rankine–Hugoniot 只保证接口两侧通量守恒，上跳和下跳都可能满足。熵条件额外保留压缩型跳跃，排除向外散开的非物理解。

Convex entropy pairs: an editorial explanation. For smooth convex η, define q by $q' = \eta' f'$. A smooth solution satisfies $\eta(u)_t + q(u)_x = 0$. For the viscous equation $u_t + f(u)_x = \varepsilon u_{xx}$,

$$\eta(u)_t + q(u)_x = \varepsilon \partial_{xx}\eta(u) - \varepsilon \eta''(u)|u_x|^2 \le \varepsilon \partial_{xx}\eta(u).$$

If bounded viscous solutions converge strongly locally, their limit therefore satisfies the distributional entropy inequality. This is a conditional explanation of the selection rule, not a proof of viscous existence or convergence.

For a downward shock $l = u_l > u_r = r$ with secant speed s, let $C(v) = f(v) - f(r) - s(v-r) \le 0$ on $[r,l]$. The singular entropy coefficient is

$$[q] - s[\eta] = -\int_r^l \eta'(v)C'(v)\,dv = \int_r^l \eta''(v)C(v)\,dv \le 0.$$

Thus the convex entropy inequality selects the same direction. For general nonsmooth solutions this chapter uses the source's one-sided derivative condition as its explicit uniqueness class.

3.4.2 The Lax–Oleinik construction (Evans §3.4.2)

Assume f smooth and $f'' \ge \theta > 0$ on $\mathbb{R}$, and $g \in L^\infty$. An additive constant in f does not affect the PDE; take $f(0) = 0$. Let

$$P(x) = \int_0^x g(y)\,dy, \qquad L = f^*, \qquad G = (f')^{-1} = L'.$$

The potential

$$W(x,t) = \min_y \{tL((x-y)/t) + P(y)\} \tag{LOP}$$

is a Hopf–Lax solution of $W_t + f(W_x) = 0$. The candidate conservation-law solution is $u = W_x$ a.e.

Since f' is strictly increasing and onto, G exists and is $1/\theta$-Lipschitz. Differentiating the convex dual at its unique maximizer gives $L' = G$ and $L'' = 1/f''(G) > 0$.

Theorem 3.11 (Source Theorem 1)
For each $t > 0$, minimizing starting points in (LOP) are ordered with x. The minimizer is unique except at at most countably many x. At differentiability points of W,

$$\boxed{u(x,t) = G((x - y(x,t))/t).} \tag{LO}$$

Expanded proof.

1. Ordering and countability. Put $c(x,y) = tL((x-y)/t)$. Its mixed derivative is $c_{xy} = -L''((x-y)/t)/t < 0$. If $x_1 < x_2$ and proposed minimizers had $y_1 > y_2$, integration of this mixed derivative on the rectangle gives

$$c(x_1,y_2) + c(x_2,y_1) < c(x_1,y_1) + c(x_2,y_2).$$

Adding $P(y_1) + P(y_2)$ contradicts the sum of the two minimizing inequalities. Hence every minimizer at x_1 is at most every minimizer at x_2. The minimum sets are nonempty compact sets by coercivity. If a set has two distinct endpoints $y_- < y_+$, its interval (y_-, y_+) is disjoint from the corresponding intervals for other x. Assigning a rational point to each nonempty interval proves there are at most countably many such x.

2. Differentiate the value without differentiating the optimizer. For a fixed minimizing y, use that same point as a competitor at $x + h$. Then $W(x+h,t) - W(x,t) \le c(x+h,y) - c(x,y)$. For positive and negative h, differentiability of W gives $W_x = L'((x-y)/t) = G((x-y)/t)$. This works even if P is not differentiable at the minimizing y. Strict monotonicity of G also forces uniqueness of y at such a point. □

Theorem 3.12 (Integral solution and initial trace; source Theorem 2)
The construction gives a bounded integral solution of (CL). It satisfies $u(\cdot,t) \to g$ in L^1_{loc} as $t \downarrow 0$.

Expanded proof.

First W is spatially Lipschitz with constant $\|g\|_\infty$, hence $|u| \le \|g\|_\infty$ a.e. The a.e. HJ equation $W_t + f(W_x) = 0$ is an equality of locally integrable

HJ 解是守恒律解的原函数

$W_t + f(W_x) = 0$ 对空间作分布求导，就得到 $u_t + f(u)_x = 0$。给初始原函数 P 加常数只会给 W 加同一常数，不改变 u 或最优起点。

可数例外指固定的时间切片

每个多重最优点对应一个互不相交的起点开区间，再各选一个有理数，才得到可数性。这不表示整个时空只存在可数个激波点，也不等于全局光滑。

最优起点的跳跃产生激波

$y(x,t)$ 保持单调，但可发生跳跃。跨过竞争分支时，$(x-y)/t$ 随之下降，因此 $u = G((x-y)/t)$ 出现向下跳跃；这把变分选择和熵方向联系起来。

functions. Multiply it by ϕ_x. Integration by parts first in time and then in space gives

$$\iint W_t \phi_x = -\iint W \phi_{xt} - \int P\phi_x(\cdot,0) = \iint W_x \phi_t + \int g\phi(\cdot,0).$$

All supports are compact and W is Lipschitz; thus these are valid weak integration identities. Substituting $u = W_x$ proves (CL).

To strengthen initial recovery, fix a Lebesgue point x of g, with value g_0. Then $P(x+h) = P(x) + g_0 h + o(|h|)$. The coercive minimizer bound gives $|x-y|/t \le R$ uniformly. Writing $y = x - tv$, the minimizing objective becomes

$$P(x) + t[L(v) - g_0 v] + o(t)$$

uniformly for $|v| \le R$. The strictly convex function $L(v) - g_0 v$ has unique minimizer $v_* = f'(g_0)$. Compactness implies every minimizing velocity tends to v_*: otherwise a convergent subsequence away from v_* would minimize the limiting strictly convex objective. Formula (LO) therefore gives $u(x,t) \to G(v_*) = g_0$. The uniform L^∞ bound and dominated convergence on compact intervals give the asserted local L^1 convergence. □

3.4.3 A one-sided estimate and uniqueness (Evans §3.4.3)

For the Lax–Oleinik solution, choose a monotone minimizing representative $y(x,t)$ (for example the largest minimizer). For $z > 0$, $y(x+z,t) \ge y(x,t)$ and therefore

$$u(x+z,t) - u(x,t) \le \frac{z}{\theta t}. \tag{OS}$$

Indeed the corresponding velocity can increase by at most z/t; G is increasing and $1/\theta$-Lipschitz. If the velocity decreases, the displayed upper bound is automatic. Thus $u(x,t) - x/(\theta t)$ is nonincreasing at fixed t. It has one-sided limits and only downward jumps, although its decrease may be arbitrarily steep.

The source's entropy class consists of bounded integral solutions with a representative satisfying $u(x+z,t) - u(x,t) \le C(1+t^{-1})z$ for $z > 0$ a.e. The term 1 permits a more general bound than (OS).

Theorem 3.13 (Source Theorem 3)
For smooth convex f, an entropy solution in this one-sided class with a prescribed bounded initial datum is unique up to null sets. Uniform convexity is used for the explicit Lax–Oleinik construction, but plain convexity suffices for this uniqueness statement.

单边界允许激波

约束是 $u(x+z) - u(x) \le Cz/t$，不是绝对值的 Lipschitz 界。向下跳跃完全允许，向上跳跃则被禁止；这正适合包含压缩激波的解空间。

Expanded proof (dual transport, with two separate small-time scales).

Let $w = u - \tilde{u}$ and

$$b(x,t) = \int_0^1 f'(\lambda u + (1-\lambda)\tilde{u})\, d\lambda.$$

Then $f(u) - f(\tilde{u}) = bw$. Subtracting the two integral identities with the same initial datum gives

$$\iint w(\phi_t + b\phi_x) = 0 \tag{DU}$$

for tests compactly supported in $\mathbb{R} \times [0,\infty)$. The common data cancel exactly; no pointwise initial derivative is assumed.

1. Regularize the velocity and control its upper derivative. Smooth bounded extensions of $u, \tilde{u}$ in space-time by nonnegative kernels and set $b_\varepsilon = \int_0^1 f'(\lambda u_\varepsilon + (1-\lambda)\tilde{u}_\varepsilon)\, d\lambda$. The coefficients are uniformly bounded, say $|b_\varepsilon| \le R$, and converge a.e. to b. For $t > 2\varepsilon$, the one-sided bound, averaged against the kernel and divided by z, gives $(u_\varepsilon)_x, (\tilde{u}_\varepsilon)_x \le C'(1 + t^{-1})$. Because $f'' \ge 0$ and is bounded on the common bounded value range,

$$(b_\varepsilon)_x = \int_0^1 f''(\lambda u_\varepsilon + (1-\lambda)\tilde{u}_\varepsilon)[\lambda (u_\varepsilon)_x + (1-\lambda)(\tilde{u}_\varepsilon)_x]\, d\lambda \le C''(1+t^{-1}).$$

Negative derivatives need no lower bound.

2. Solve a smooth backward test equation. Fix $\psi \in C_c^\infty(\mathbb{R} \times (0,T))$. Let v_ε solve $(v_\varepsilon)_t + b_\varepsilon (v_\varepsilon)_x = \psi$ with $v_\varepsilon(\cdot, T) = 0$. The characteristic formula is

$$v_\varepsilon(x,t) = -\int_t^T \psi(X_\varepsilon(s;x,t), s)\, ds, \qquad \dot{X}_\varepsilon = b_\varepsilon(X_\varepsilon, s).$$

Bounded speed makes all its supports lie in a fixed spatial interval. It vanishes near T, so it is an admissible smooth test in (DU).

For any fixed $\delta > 0$, $\varepsilon < \delta/2$, differentiate the flow in x:

$$\partial_x X_\varepsilon(s;x,t) = \exp\left(\int_t^s (b_\varepsilon)_x(X_\varepsilon(\tau), \tau)\, d\tau \right).$$

The upper derivative bound for $t \ge \delta$ gives a uniform finite bound on this positive factor. Differentiating the test formula therefore yields $|(v_\varepsilon)_x| \le C_\delta$ for $\delta \le t \le T$.

3. A fixed early interval controls total variation. Choose a fixed $t_* > 0$ so that $\psi = 0$ for $0 < t < 2t_*$. For $t \le t_*$, v_ε is transported unchanged along characteristics to time t_*. A one-dimensional smooth flow preserves order and is a diffeomorphism of $\mathbb{R}$ (its speed is bounded). Hence composition with this flow preserves total variation:

$$\int_{\mathbb{R}} |(v_\varepsilon)_x(x,t)|\,dx = \operatorname{Var} v_\varepsilon(\cdot,t) = \operatorname{Var} v_\varepsilon(\cdot,t_*) \le D.$$

The final bound follows from the derivative bound at the fixed t_* and the fixed compact support. Crucially, D is independent of both ε and a later splitting time $\delta < t_*$.

4. Pass the dual identity to the limit. Use v_ε in (DU):

$$\iint w\psi = \iint w(b_\varepsilon - b)(v_\varepsilon)_x.$$

Split at $t = \delta$, with $0 < \delta < t_*$. On $[\delta, T]$ the integrand tends to zero by dominated convergence, using C_δ and common compact support. On $(0,\delta)$, boundedness of w, b_ε, b and the total-variation estimate bound its absolute integral by $CD\delta$. First let $\varepsilon \downarrow 0$ for this fixed δ, then let $\delta \downarrow 0$. Thus $\iint w\psi = 0$ for every interior test ψ, so $w = 0$ a.e. □

一维保序带来变差守恒

光滑流关于初始位置严格递增，因此复合流只是重新排列同一条曲线上的取值次序，不增加总变差。这里不需要流的导数在零时刻附近有统一上界。

两个小时间不能混用

先固定源项消失的时段 t_*，用它得到统一的变差界；再另取 $\delta < t_*$ 切开误差。若让估计常数也随 δ 发散，就不能仅凭区间长度趋零断言误差消失。

3.4.4 Riemann's problem (Evans §3.4.4, Theorem 4)

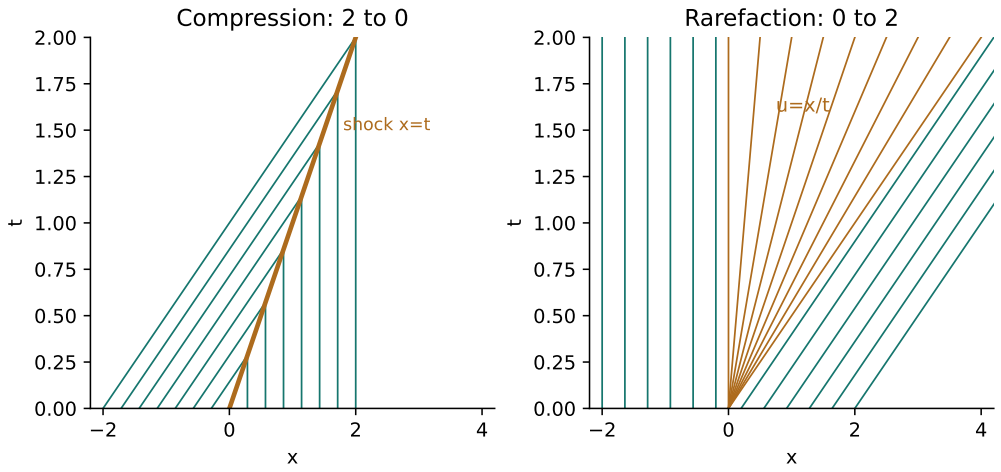
For $g = u_l$ on $x < 0$ and $g = u_r$ on $x > 0$, the entropy solution for a strictly convex flux is as follows. If $u_l > u_r$, it is a shock with speed $s = [f(u_l) - f(u_r)]/(u_l - u_r)$. If $u_l < u_r$, it is the rarefaction

$$u(x,t) = \begin{cases} u_l, & x/t \le f'(u_l), \\ G(x/t), & f'(u_l) < x/t < f'(u_r), \\ u_r, & x/t \ge f'(u_r). \end{cases} \tag{RF}$$

Equal states give a constant solution.

Expanded proof.

The shock satisfies the PDE off its interface, (RH) on it, and the downward entropy direction. In the fan, $u_t = -(x/t^2)G'(x/t)$ and $f'(u)u_x = (x/t)G'(x/t)/t$, which cancel. The fan joins continuously to both constant states and $u_x = G'(x/t)/t \le 1/(\theta t)$. Both candidates are integral solutions and approach their Riemann data locally in L^1; the transition region has width $O(t)$. The entropy uniqueness theorem just proved identifies them as the selected solutions. □



Editorial figure. For Burgers flow, a downward step compresses characteristics into a shock; an upward step opens a rarefaction fan. Both geometries use the same flux, but different data.

3.4.5 Long-time decay and N-waves (Evans §3.4.5)

For this subsection assume $f(0) = 0$, f smooth, and $0 < \theta \le f'' \le \Theta < \infty$. These convenient sufficient hypotheses ensure uniform convexity of both f and $L = f^*$, as required in the source asymptotic arguments. Write $\sigma = f'(0)$ and $d = f''(0)$.

Theorem 3.14 (Supremum decay; source Theorem 5)
For bounded integrable initial data, $\|u(\cdot,t)\|_\infty \le Ct^{-1/2}$.

Expanded proof.

The primitive P in (LOP) is bounded by $M = \|g\|_1$. Duality gives $L(\sigma) = 0, L'(\sigma) = 0$. Since $L'' \ge 1/\Theta$, $L(v) \ge (v-\sigma)^2/(2\Theta)$. Compare a minimizing y with $y_0 = x - \sigma t$:

$$\frac{|x - y - \sigma t|^2}{2\Theta t} - M \le tL((x-y)/t) + P(y) \le P(x - \sigma t) \le M.$$

Thus $|(x-y)/t - \sigma| \le C/\sqrt{t}$. As $G(\sigma) = 0$ and G is $1/\theta$-Lipschitz, formula (LO) gives the claim. □

Suppose additionally $\operatorname{supp} g \subset [-R, R]$. Use the equivalent primitive $P(x) = \int_{-\infty}^{x} g$, so $P_- = 0$, $P_+ = m = \int g$. Let $P_* = \min P$,

$$p = -2P_*, \qquad q = 2(m - P_*), \qquad p, q \geq 0.$$

The associated N-wave is

$$N(x,t) = \begin{cases} (x - \sigma t)/(dt), & -\sqrt{pdt} < x - \sigma t < \sqrt{qdt}, \\ 0, & \text{otherwise.} \end{cases} \tag{NW}$$

Its negative and positive signed areas are $-p/2$ and $q/2$, so $\int N = (q - p)/2 = m$.

Theorem 3.15 (Source Theorem 6)
If $p, q > 0$, then $\|u(\cdot,t) - N(\cdot,t)\|_1 \leq Ct^{-1/2}$.

Expanded proof (locating the edges before integrating the error).

Pass to the moving coordinate $\xi = x - \sigma t$ and put $\ell(v) = L(\sigma + v)$. Then $\ell(0) = \ell'(0) = 0$ and $\ell''(0) = 1/d$. The minimizing velocity bound just proved is $|(\xi - y)/t| \leq C/\sqrt{t}$. Taylor expansion on this fixed small-velocity range gives

$$t\ell((\xi - y)/t) = \frac{(\xi - y)^2}{2dt} + O(t^{-1/2}),$$

$$u(x,t) = \frac{\xi - y(x,t)}{dt} + O(t^{-1}). \tag{NT}$$

The constants are uniform for minimizers; the same Taylor bound holds for the bounded starting-point competitors used below when $|\xi| = O(\sqrt{t})$.

Choose $\varepsilon = A/\sqrt{t}$ with a fixed sufficiently large A. We work for large t, so $\varepsilon < 1$.

1. Exterior edges: the solution is zero beyond them. If $\xi < -R - \sqrt{pd(1+\varepsilon)t}$, the competitor $y = \xi$ has objective $P_- = 0$. A minimizing $y \geq -R$ would, by (NT), have objective at least

$$\frac{pd(1+\varepsilon)t}{2dt} + P_* + O(t^{-1/2}) = \frac{pA}{2\sqrt{t}} + O(t^{-1/2}) > 0.$$

This contradicts the competitor when A is large enough. Thus $y < -R$, where $P = 0$, and the strictly positive cost ℓ is minimized only by $y = \xi$. Hence $u = G(\sigma) = 0$. The right-tail argument is the same with the baseline $P_+ = m$ and q. Therefore $u = 0$ outside

$$-R - \sqrt{pd(1+\varepsilon)t} \leq \xi \leq R + \sqrt{qd(1+\varepsilon)t}. \tag{OUT}$$

衰减的是高度，不一定是质量

波的高度可按 $t^{-1/2}$ 降低，同时宽度按 $\sqrt{t}$ 增长。积分质量不一定趋零，所以长期形状应和一个展开的 N 波比较，而非只和零比较。

N 波参数由原函数最低点决定

p, q 不是初值正负质量各乘二。例如先取正脉冲再取等质量负脉冲，原函数始终非负且最终回零，此时 $p = q = 0$，即使初值两种符号都有。

2. Interior edges: the minimizing point stays in the initial support. At $\xi_l = R - \sqrt{pd(1-\varepsilon)t}$, any $y < -R$ has objective at least $P_- = 0$. Choose a point $z \in [-R, R]$ with $P(z) = P_*$. Since $|\xi_l - z| \le \sqrt{pd(1-\varepsilon)t}$ for large t, its objective is at most

$$\frac{p(1-\varepsilon)}{2} + P_* + O(t^{-1/2}) = -\frac{pA}{2\sqrt{t}} + O(t^{-1/2}) < 0.$$

Thus every minimizer at ξ_l is at least $-R$. Similarly at $\xi_r = -R + \sqrt{qd(1-\varepsilon)t}$ it is at most R. Monotonicity of minimizers then gives $|y(x,t)| \le R$ throughout $\xi_l < \xi < \xi_r$. There (NT) implies $|u(x,t) - \xi/(dt)| \le C/t$.

3. Integrate over the middle and the transition strips. The middle interval has length $O(\sqrt{t})$ and lies inside the ideal N-wave interval, so its L^1 error is $O(t^{-1/2})$. Each strip between the interior, ideal, and exterior edges has bounded length, since $\sqrt{(1 \pm A/\sqrt{t})t} - \sqrt{t} = O(1)$. Both $|u|$ and $|N|$ are $O(t^{-1/2})$, giving the same error there. Outside (OUT) both are zero. This proves the bound for large times.

For a bounded time interval the minimizing-velocity bound from (HL) gives finite propagation of the compact support and $|u| \le \|g\|_\infty$. Thus $\|u\|_1$ is uniformly bounded on that interval; $\|N\|_1 = (p+q)/2$. Enlarge C to cover all remaining $t > 0$. □

For Burgers flow with initial data $\mathbf{1}_{(0,1)}$, the fan is $u = x/t$. Until $t = 2$ it meets a height-one plateau ending at shock $x = 1 + t/2$. After $t = 2$ the shock is $x = \sqrt{2t}$ and $u = x/t$ on $0 < x < \sqrt{2t}$, zero elsewhere. This follows from the fan–shock matching equation $\gamma' = \gamma/(2t)$ and $\gamma(2) = 2$. It has height $O(t^{-1/2})$ and constant mass one. Here $p = 0$, $q = 2$; the explicit example illustrates decay, while the general L^1 theorem above was stated with $p, q > 0$.

边缘位置也要估计

只有波内 $O(1/t)$ 点态误差还不够做全空间积分。还要把两侧支集误差压进宽度 $O(1)$ 的条带，再乘以高度 $O(t^{-1/2})$，才能得到总误差。

3.5 Problems

Source: Evans §3.5, Problems 1–20, printed pp. 154–158 (PDF pp. 171–175). Statements below are concisely restated. All solutions are editorial, with smoothness assumed where required; extensions beyond the smooth class are identified explicitly.

3.5.1 Problems 1–8: envelopes and characteristics

1. A complete Hamilton–Jacobi integral.

Solution. For $U = a \cdot x - tH(a) + b$, $D_x U = a$ and $U_t = -H(a)$, proving the PDE. There are $n+1$ parameters for $n+1$ independent variables. The derivatives with respect to a_i independently vary $D_x U$; the derivative with respect to b changes the value by one without changing any gradient component. These $n+1$ jet variations are linearly independent, proving completeness. □

2. Envelopes of lines and planes.

Solution. For $G = x_1 + a^2 x_2 - 2a$, the stationary condition is $G_a = 2ax_2 - 2 = 0$. Thus $a \neq 0$, $x_2 = 1/a$, and $G = 0$ then gives $x_1 = a$. The envelope is $x_1 x_2 = 1$, with both branches represented. The line with $a = 0$ has no finite stationary contact. For $G = 2a_1 x_1 + 2a_2 x_2 - x_3 + a_1^2 + a_2^2$, stationarity yields $a_i = -x_i$ and elimination gives $x_3 = -x_1^2 - x_2^2$. The derivatives of this paraboloid equal the slopes $2a_i$ of the selected plane, directly verifying tangency. The figure in §3.1 shows both constructions. □

3. Eliminate parameters to obtain a PDE.

Solution. (a) For $\sum_i a_i x_i^2 + u^3 = 0$, implicit differentiation gives $2a_i x_i + 3u^2 u_{x_i} = 0$. Multiply by x_i, sum, and eliminate $\sum_i a_i x_i^2 = -u^3$:

$$u^2(3x \cdot Du - 2u) = 0.$$

On a nonzero branch this is $3x \cdot Du = 2u$. For a C^1 graph it extends to the zero set by continuity from nonzero points, while on the interior of $\{u = 0\}$ the gradient is zero. This avoids division by a possibly zero coordinate x_i.

(b) A unit sphere centered at $(a, 0)$ has $|x-a|^2+u^2 = 1$. Differentiation yields $x - a = -uDu$. Substitution eliminates a and gives $u^2(1 + |Du|^2) = 1$, on regions where the sphere is a differentiable graph. □

参数消元要记录分支

除以 u、x_i 或某个行列式前，先说明它不为零。得到的简化 PDE 可能丢掉零解或特殊分支；保留未约去的形式有助于检查。

4. Forced transport.

Solution. For $u_t + b \cdot Du = f(x,t)$, use time as the characteristic parameter: $\dot{X} = b, \dot{Z} = f(X, s)$. Starting from the target, $X(s) = x - (t-s)b$. Therefore

$$u(x,t) = g(x - tb) + \int_0^t f(x - (t-s)b, s)\, ds.$$

This is the same formula as in Chapter 2. In the full characteristic system the spatial gradient satisfies $\dot{P} = D_x f$ and the time-gradient component satisfies $\dot{Q} = f_t$. □

5. Three characteristic calculations.

Solution. (a) For $x_1 u_{x_1} + x_2 u_{x_2} = 2u$, start at $(a, 1)$. Then $X = (ae^s, e^s)$, $Z = e^{2s} g(a)$. For $x_2 > 0$, $s = \log x_2$ and $a = x_1/x_2$, so $u = x_2^2 g(x_1/x_2)$.

(b) For $x_1 u_{x_1} + 2x_2 u_{x_2} + u_{x_3} = 3u$, start at $(a, b, 0)$. Then $X = (ae^s, be^{2s}, s)$, $Z = e^{3s} g(a, b)$, giving $u(x) = e^{3x_3} g(x_1 e^{-x_3}, x_2 e^{-2x_3})$.

(c) For $u u_{x_1} + u_{x_2} = 1$, start at (a, a) with $Z(0) = a/2$. The equations integrate to

$$X_2 = a + s, \qquad Z = a/2 + s, \qquad X_1 = a + (a/2)s + s^2/2.$$

Eliminate $s = X_2 - a$: $X_1 = a(1 - X_2/2) + X_2^2/2$. Thus

$$a = \frac{2x_1 - x_2^2}{2 - x_2}, \qquad u(x_1, x_2) = x_2 - a/2$$

locally near boundary points (a, a) with $a \neq 2$. The parameter Jacobian is $1 - x_2/2$. At $(2, 2)$ the boundary conditions themselves are incompatible: Tangential differentiation gives $u_{x_1} + u_{x_2} = 1/2$, whereas the PDE and $u = 1$ give the same left side equal to one. There is no C^1 solution through that boundary point. □

6. Flow Jacobians and a continuity equation.

Solution. Let $X(s; x, t)$ solve $\dot{X} = b(X)$ with $X(t; x, t) = x$. On any interval where the smooth flow exists, its derivative matrix $M = D_x X$ satisfies $M_s = Db(X)M$ and $M(t) = I$. Jacobi's determinant formula gives

$$J_s = \operatorname{tr}(M^{-1} Db(X) M) J = (\operatorname{div} b)(X) J, \quad J(t) = 1.$$

In particular $J > 0$ and is the exponential of the divergence integral. For the forward map $Y(t; a)$, the desired density satisfies

$$\frac{d}{dt}\big[u(Y(t; a), t) \det D_a Y(t; a)\big] = 0.$$

Set the product equal to $g(a)$; differentiating proves $u_t + b \cdot Du + u \operatorname{div} b = 0$. Inverting the flow then gives $u(x, t) = g(X(0; x, t)) J(0; x, t)$. A smooth vector field need not have a global complete flow; the formula is asserted on its actual existence interval unless completeness is assumed. □

密度与被搬运的数值不同

连续性方程保持的是 $u(Y,t)\det D_a Y$，因为体积也在变化；纯传输保持 $u(Y,t)$。当速度无散时 Jacobian 为一，两者才在这一点上重合。

7. A curved noncharacteristic surface.

Solution. The tangential boundary gradient is fixed. Write admissible gradients as $D_{\tan} g + \lambda \nu$. The derivative of their scalar PDE constraint in λ is $F_p \cdot \nu$. Nonvanishing permits local implicit solution for the normal component. Equivalently, the characteristic column F_p and the boundary tangent columns form a basis exactly when this scalar product is nonzero. □

8. Verify the implicit conservation formula.

Solution. Let $a = x - t\mathbf{f}'(u)$ and $u = g(a)$. With $D = 1 + tDg(a)\cdot\mathbf{f}''(u) \neq 0$, differentiate:

$$Du = \frac{Dg(a)}{D}, \qquad u_t = -\frac{Dg(a)\cdot \mathbf{f}'(u)}{D}.$$

Hence $u_t + \mathbf{f}'(u)\cdot Du = 0$. The nonzero denominator is exactly the rank-one determinant from the characteristic projection. Its loss is not repaired by the implicit formula itself. □

3.5.2 Problems 9–16: variations and convex duality

9. A free initial endpoint.

Solution. Variations have $V(t) = 0$ but arbitrary $V(0)$. After integration by parts the first variation is

$$\int_0^t \left[L_x - \frac{d}{ds} L_v \right] \cdot V \, ds - L_v(\dot{X}(0), X(0)) \cdot V(0).$$

First choose $V(0) = 0$ to obtain the Euler–Lagrange equation. Then arbitrary $V(0)$ gives $L_v(\dot{X}(0), X(0)) = 0$. Adding the initial cost $g(X(0))$ adds $Dg(X(0))\cdot V(0)$, so the new boundary condition is $L_v(\dot{X}(0), X(0)) = Dg(X(0))$. □

初始代价决定初始动量

自由起点不代表可以忽略端点项。变分边界项的负号与 $g(X(0))$ 的变化抵消后，得到 $P(0) = Dg(X(0))$，恰好接上 HJ 特征初值。

10. Two Legendre transforms.

Solution. (a) For $H(p) = |p|^r/r$, $1 < r < \infty$, the maximizer of $v\cdot p - H(p)$ is parallel to v, with $|p|^{r-1} = |v|$. Writing $1/r + 1/s = 1$ gives $L(v) = |v|^s/s$, also valid at $v = 0$. (b) For $H(p) = \frac{1}{2}p^T A p + b\cdot p$, A symmetric positive definite, stationarity gives $p = A^{-1}(v - b)$. Substitution gives $L(v) = \frac{1}{2}(v-b)^T A^{-1}(v-b)$. Positive definiteness makes this stationary point the unique global maximum of the concave objective. □

11. Subdifferential reciprocity.

Solution. The condition $v \in \partial H(p)$ is $H(q) \geq H(p) + v \cdot (q-p)$ for every q. It is equivalent to p maximizing $v \cdot q - H(q)$, hence to $H(p) + L(v) = p \cdot v$. Using $H = L^*$, the same argument with the roles exchanged is equivalent to $p \in \partial L(v)$. For nonsmooth finite convex functions, this uses the supporting-hyperplane version of the duality established above, rather than an unjustified gradient. □

12. A finite-dimensional dual minimum. For smooth convex superlinear L_i and $H_i = L_i^*$, prove

$$\min_v [L_1(v) + L_2(v)] = \max_p [-H_1(p) - H_2(-p)].$$

Solution. For any v, p, Fenchel's inequalities give $L_1(v) + L_2(v) \geq -H_1(p) - H_2(-p)$. The convex superlinear sum has a minimizing v_*. Smoothness gives $DL_1(v_*) + DL_2(v_*) = 0$. Choose $p_* = DL_1(v_*) = -DL_2(v_*)$. Both Fenchel inequalities are equalities for this pair. Thus the minimum equals the stated dual maximum, which is attained at p_*. □

13. A sharper finite-speed radius.

Solution. Assume here the source's smooth initial datum, with bounded gradient. At a minimizing y in (HL), differentiation in y gives $DL((x-y)/t) = Dg(y)$ when L is smooth. More generally the convex optimality condition is $Dg(y) \in \partial L((x-y)/t)$. Reciprocity and differentiability of H give $(x-y)/t = DH(Dg(y))$. Thus $|x-y| \leq Rt$ for $R = \sup_y |DH(Dg(y))|$, proving the restricted minimum formula on the closed ball $\overline{B}_{Rt}(x)$. For merely Lipschitz g, a minimizing point need not be a differentiability point of g; the coarser coercive radius already proved for (HL) remains available without that assumption. □

14. Distance to a closed set.

Solution. For $H(p) = |p|^2$, $L(v) = |v|^2/4$. If the initial cost is zero on a nonempty closed set E and infinity elsewhere, the extended-value formula reduces to

$$\inf_{y \in E} \frac{|x-y|^2}{4t} = \frac{\operatorname{dist}(x,E)^2}{4t}.$$

Closedness and coercivity give a nearest point. The formula tends to zero on E and to infinity at each point outside E as $t \downarrow 0$. These are extended-value data, outside the globally Lipschitz-data theorem; this is a computation of the proposed formula rather than an invocation of that theorem. For empty E, both sides are infinity by convention. □

不要对不可微的初值硬求导

Lipschitz 函数几乎处处可微，但指定的最优起点仍可能落在不可微集合。习题 13 的驻点等式使用光滑初值；一般情形可先用不依赖求导的强制性估计。

15. Semiconcavity from uniform convexity.

Solution. The proof in §3.3.3 gives all steps: strong monotonicity of DH makes $DL = (DH)^{-1}$ Lipschitz with constant $1/\theta$; integrate its increment to obtain the quadratic upper bound on L; add opposite increments; use the same minimizing starting point and increments $\pm z/t$. The factor is exactly $1/(\theta t)$. □

16. Supremum contraction.

Solution. Let $\delta = \sup|g_1 - g_2| < \infty$. For every competitor y, the objective for u_1 is at most that for u_2 plus δ. Taking infima gives $u_1 \le u_2 + \delta$. Interchanging indices gives the reverse bound, so $\sup|u_1(\cdot,t) - u_2(\cdot,t)| \le \delta$. No differentiation or uniqueness theorem is needed. □

3.5.3 Problems 17–20: entropy and wave interactions

17. An unbounded Burgers entropy example. Consider

$$u(x,t) = \begin{cases} -\frac{2}{3}(t + \sqrt{3x + t^2}), & 4x + t^2 > 0, \\ 0, & 4x + t^2 < 0. \end{cases}$$

Solution. On the right smooth region write $A = \sqrt{3x + t^2}$. Then $u_x = -1/A$ and $u_t = -\frac{2}{3}(1 + t/A)$, whereas $uu_x = \frac{2}{3}(1 + t/A)$. Thus Burgers' equation holds there; it also holds in the zero region. The interface is $\gamma(t) = -t^2/4$. At that point $A = t/2$, so $u_l = 0$, $u_r = -t$. Its speed $\gamma' = -t/2$ equals $(u_l + u_r)/2$, proving (RH). The jump is downward and $u_x < 0$ on the right. Hence at every fixed $t > 0$, u is nonincreasing and satisfies the one-sided entropy bound even with constant zero. The formula has initial trace $g(x) = 0$ for $x < 0$ and $g(x) = -2\sqrt{3x}/3$ for $x > 0$, locally in L^1. It is an unbounded entropy example in this local sense, not an element of the globally bounded uniqueness class stated in §3.4.3. □

例子与定理的解空间不同

这题的熵解无界，不能直接套用要求全局 L^∞ 的唯一性定理。验证 PDE、跳跃条件、单边估计后，还需明确它实际属于哪个局部解类。

18. One-sided bounds survive smoothing.

Solution. If $u(x + h) - u(x) \le Eh$ for $h > 0$, integrate this inequality after translation against a nonnegative mollifier of mass one: $u_\varepsilon(x + h) - u_\varepsilon(x) \le Eh$. Divide by h and pass to $h \downarrow 0$, obtaining $(u_\varepsilon)'(x) \le E$. Only an upper derivative bound is obtained; there is no absolute-value bound. The same argument in each spatial direction gives the regularized Hessian bound used in the HJ uniqueness proof. □

19. Conservation of mass. Assume a continuous integral solution, $f(0) = 0$, and a common compact spatial support on each bounded time interval.

Solution. For a fixed time horizon choose a spatial cutoff equal to one on that common support and use its product with a time test in (CL). The flux term is zero: where the spatial cutoff changes, $u = 0$ and $f(u) = f(0) = 0$. Thus the distributional time derivative of $\int u(x,t)\,dx$ is zero, with initial value $\int g$ supplied by (CL). Continuity of u and the common support make this integral continuous in time, so the equality holds at every time, not only almost everywhere. □

20. Two shocks interacting with one rarefaction. For Burgers flux take $g = 1$ on $x < -1$, 0 on $(-1, 0)$, 2 on $(0, 1)$, and 0 on $x > 1$.

Solution. At -1, a $1 \to 0$ shock starts with speed $1/2$: $\gamma_l = -1 + t/2$. At 0, a $0 \to 2$ fan has $u = x/t$ on $0 < x < 2t$. At 1, a $2 \to 0$ shock starts with speed one: $\gamma_r = 1 + t$.

First stage: $0 < t < 1$**.** The regions from left to right are

x	$(-\infty, \gamma_l)$	$(\gamma_l, 0)$	$(0, 2t)$	$(2t, \gamma_r)$	(γ_r, ∞)
u	1	0	x/t	2	0.

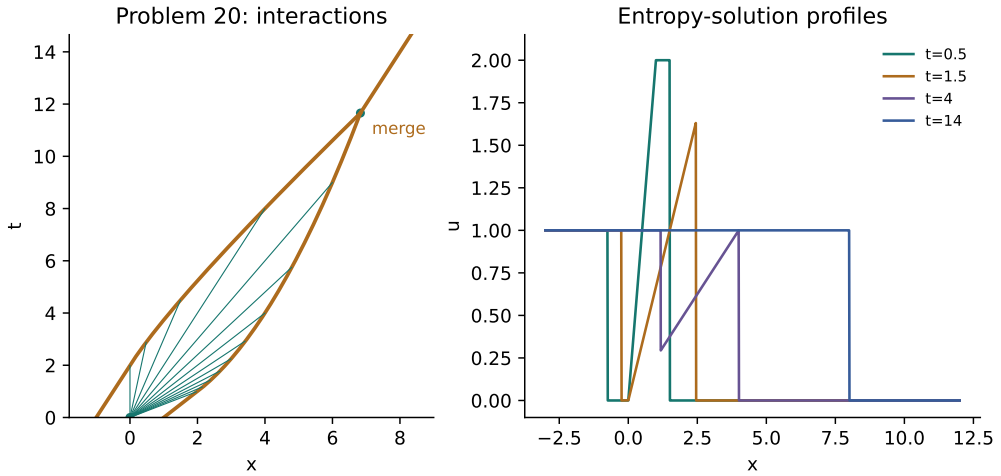
The right shock first meets the fan when $1 + t = 2t$, namely at $t = 1$.

Second stage: $1 \le t < 2$**.** The right shock now separates $u_l = \gamma_r/t$ from zero, so $\gamma_r' = \gamma_r/(2t)$. Matching $\gamma_r(1) = 2$ gives $\gamma_r = 2\sqrt{t}$. The height-two plateau has disappeared. The remaining regions have $u = 1$ for $x < \gamma_l$, $u = 0$ for $\gamma_l < x < 0$, $u = x/t$ for $0 < x < \gamma_r$, and zero to the right. At $t = 2$, the left shock reaches the fan's left edge $x = 0$.

Third stage: $2 \le t < T$**.** The left shock now separates one from γ_l/t, hence $\gamma_l' = (1 + \gamma_l/t)/2$. Multiply by $t^{-1/2}$ to integrate: $(\gamma_l/\sqrt{t})' = 1/(2\sqrt{t})$. Matching $\gamma_l(2) = 0$ gives $\gamma_l = t - \sqrt{2t}$. Between this curve and $\gamma_r = 2\sqrt{t}$, the state is x/t; it is one to the left and zero to the right. The shocks meet at

$$T = (2 + \sqrt{2})^2 = 6 + 4\sqrt{2}, \qquad X = 2\sqrt{T} = 4 + 2\sqrt{2}.$$

Final stage: $t \geq T$**.** Only a $1 \to 0$ shock remains, of speed $1/2$. Continuity of its trajectory at the interaction gives $\gamma(t) = X + (t-T)/2 = 1 + t/2$. Thus $u = 1$ for $x < 1 + t/2$ and $u = 0$ for $x > 1 + t/2$. In all stages each smooth piece solves the PDE, each discontinuity satisfies (RH), and each shock is downward. The fan has derivative $1/t$ and other smooth pieces have zero derivative; therefore the global one-sided bound holds. The initial trace and entropy uniqueness identify this as the required solution. Values assigned exactly on the shock curves do not change the weak solution. □



Editorial figure. The space-time plot records interactions at $t = 1$, $t = 2$ and T; the profiles show all four stages. Independent piecewise Hopf–Lax minimization checks the formulas numerically away from shock ties.

最终激波位置可用质量再检查

相对于 $x = 0$ 处的 $1 \to 0$ 阶跃，初值的积分差为 $-1 + 2 = 1$。最终阶跃应比速度 $1/2$ 的参考激波右移一，故为 $x = t/2 + 1$。

进入稀疏波后速度会变

激波两侧的值不再是两个固定常数，而包含 $\gamma(t)/t$，所以速度要重新由 Rankine–Hugoniot 求出。不能把初始的两条直线一直延长到所有时间。

3.6 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 3, printed pp. 89–158 (local PDF pp. 106–175). The section order and explicit source theorem identifiers follow that edition. The prose, expanded arguments, Chinese notes, plots and exercise solutions are editorial reconstructions.

Evans's §3.6 points to Courant–Hilbert for complete integrals; Carathéodory, Garabedian, John, Chester and Sneddon for characteristics; Lions, Rund and Benton for Hamilton–Jacobi theory; and Lax, Smoller and Whitham for conservation laws. The source attributes the HJ uniqueness argument to Douglis, the entropy uniqueness argument to Oleinik, and the long-time estimates to DiPerna, *Indiana University*

Mathematics Journal 24 (1975), pp. 1047–1071. These are source attributions and reading directions, not claims of independent consultation of each listed work.

The Chinese chapter opening (PDF p. 79) lists section starts at printed pp. 61, 64, 76, 90, 107, and 110 for §3.1–§3.6. Selected Chinese pages for the local characteristic construction, Hopf–Lax and shocks were checked as recorded in the chapter log. A line-by-line comparison of the whole Chinese chapter has not been made.

Review boundary. The chapter invokes ODE existence and smooth dependence, implicit and inverse function theorems, supporting hyperplanes for convex functions, integration and weak calculus, Rademacher, and Gronwall as prerequisites. The local existence proof includes the contact-compatibility step; the two uniqueness arguments explicitly retain their different selection classes. Long-time estimates are stated under convenient sufficient convexity hypotheses, not optimal assumptions. All 20 source exercises have solutions or exact pointers to full earlier arguments. Numerical agreement with selected models and successful compilation do not establish independent mathematical review.

Chapter 4

Other Ways to Represent Solutions

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 4; printed pp. 159–236; PDF pp. 176–253.

Chinese reference: 中译本：第四章，书内第 111 页起；章首页 PDF 第 129 页。

Editorial study edition. This chapter reconstructs the source exposition, not its wording. Source theorem identifiers are stated separately from our numbering. Detailed proofs, Chinese mathematical annotations, exercise solutions and figures are editorial additions.

The organizing question is how a PDE can be represented by simpler objects:

$$\text{modes and profiles} \longrightarrow \text{integral transforms},$$
$$\text{changes of unknown} \longrightarrow \text{asymptotics and power series}.$$

Reading conventions. Fourier transforms use the unitary normalization $(2\pi)^{-n/2}$. The diffusion exponent is denoted by m to distinguish it from multiindices. A formal asymptotic calculation is identified explicitly; its conclusion is not promoted to an unconditional convergence theorem. We invoke standard integration, ODE theory, implicit and inverse function theorems, and elementary complex analysis. Spectral completeness and elliptic solvability are identified where used, rather than proved by writing a series. Sobolev terminology is available in Chapter 5.

表示公式还需要合法性

猜到一个形式不等于已经解决问题。接着要检查初边值、可逆性、级数或积分的收敛，以及能否逐项求导；渐近展开还需另估余项。

4.1 Separation of Variables

4.1.1 Separated modes and the missing convergence question

For $u_t - \Delta u = 0$ with zero Dirichlet data on a bounded domain U, try $u(x,t) = T(t)X(x)$. On a product neighborhood where both factors are nonzero,

$$\frac{T'}{T} = \frac{\Delta X}{X} = -\lambda.$$

Fixing x and then t shows that the common expression is constant. The separated equations are

$$-\Delta X = \lambda X, \quad X|_{\partial U} = 0, \qquad T' = -\lambda T.$$

The identities without division continue across zeros of X. For a nonzero smooth eigenfunction, integration by parts gives

$$\lambda \int_U |X|^2 = \int_U |DX|^2 > 0.$$

Strict positivity follows since a zero-gradient function with zero boundary trace is zero. Thus $e^{-\lambda t}X$ is a heat mode.

分离与叠加是两步

乘积形式只构造单个模态；把多个模态相加依赖方程的线性。非线性方程即使能分离变量，也通常不能把分离解相加来拟合任意初值。

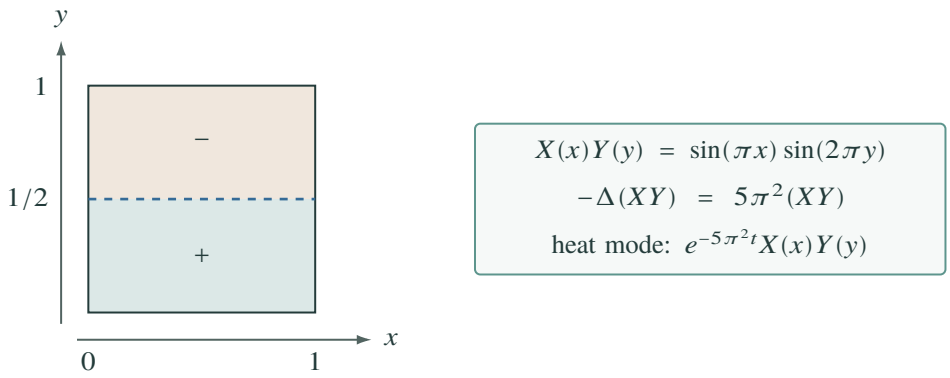


Diagram T4.1. A separated mode and its nodal line. On the unit square, this Dirichlet mode is the product of two one-dimensional modes. The dashed line $y = 1/2$ is a zero set inside the domain. Applying $-\Delta$ adds the two eigenvalues, giving $5\pi^2$; heat evolution then changes only the time factor.

Theorem 4.1 (A justified spectral representation, conditional on the spectral basis)

Assume the Dirichlet eigenfunctions X_j form an orthonormal basis of $L^2(U)$, with eigenvalues $\lambda_j > 0$. For $g \in L^2(U)$, set $g_j = (g, X_j)$ and

$$u(\cdot,t) = \sum_{j=1}^{\infty} e^{-\lambda_j t} g_j X_j. \tag{S4}$$

This converges in L^2, tends to g in L^2 at time zero, and solves the heat

> *equation in the spectral weak sense.*

Expanded proof.

Parseval gives $\sum |g_j|^2 = \|g\|_2^2$, so the tail norm squared is at most $\sum_{j>N} |g_j|^2$, uniformly for $t \geq 0$. Moreover

$$\|u(t) - g\|_2^2 = \sum_j |e^{-\lambda_j t} - 1|^2 |g_j|^2 \longrightarrow 0$$

by dominated convergence for a summable series. For $t \geq \tau > 0$, $\sup_{\lambda \geq 0} \lambda^k e^{-\tau\lambda} < \infty$. Hence time derivatives of every order converge in L^2 away from zero; the spectral Laplacian and time derivative agree termwise. Testing against any finite eigenfunction combination proves the weak equation, and density extends it to the corresponding energy test class. Interior classical smoothness requires the usual elliptic/local regularity facts; it does not follow simply from writing (S4). Existence and completeness of the basis are the later spectral theorem of Chapter 6, not a conclusion of separation itself. □

On $(0, \pi)$ one has $X_j = \sqrt{2/\pi}\sin(jx)$, $\lambda_j = j^2$. The coefficient formula is then an ordinary sine-series expansion. A finite sum is already a classical solution for its own initial datum.

4.1.2 Nonlinear and additive separation

For $u_t - \Delta(u^m) = 0$, $m > 1$, try $u = T(t)X(x) \geq 0$. Homogeneity gives $T'/T^m = \Delta(X^m)/X = \mu$ and therefore

$$T(t) = [c - (m-1)\mu t]^{-1/(m-1)}, \qquad \Delta(X^m) = \mu X.$$

Taking $X = |x|^\alpha$ and using $\Delta r^q = q(q+n-2)r^{q-2}$ for $r > 0$ yields

$$\alpha = \frac{2}{m-1}, \qquad \mu = \alpha m(\alpha m + n - 2) > 0.$$

The positive bracket restricts the solution interval to $t < c/((m-1)\mu)$. At the origin X need not be C^2, but $X^m = r^{2+2/(m-1)}$ is C^2 and its Laplacian extends continuously to zero. Thus the equation holds there in this natural sense. The profile grows at infinity and has infinite total mass.

For the Hamilton–Jacobi equation, additive separation is more natural: $u = X(x) + T(t)$ gives $H(DX) = \mu$ and $T = -\mu t + b$. The affine choice $X = a \cdot x$ recovers the complete family of Chapter 3.

4.1.3 Turing instability: diffusion can destabilize a coupled system

Let $J = D\mathbf{f}(0) = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$, and $D = \operatorname{diag}(d_1, d_2)$, $d_i > 0$. The reaction ODE is linearly asymptotically stable precisely when

$$\operatorname{tr} J = \alpha + \delta < 0, \qquad \det J = \alpha\delta - \beta\gamma > 0.$$

Indeed its two eigenvalues either are real with negative sum and positive product, or are conjugate with negative real part.

For $\mathbf{v}_t - D\Delta\mathbf{v} = J\mathbf{v}$, expand in Laplacian modes. The coefficient of X_j solves

$$s_j' = (J - \lambda_j D)s_j.$$

Its trace is still negative, but its determinant is

$$p(\lambda) = d_1 d_2 \lambda^2 - (d_1\delta + d_2\alpha)\lambda + \det J.$$

Theorem 4.2 (Linear mode instability)
A mode grows exponentially if and only if $p(\lambda_j) < 0$. For a continuous positive instability band it is necessary and sufficient that

$$B = d_1\delta + d_2\alpha > 0, \qquad B^2 > 4d_1 d_2 \det J.$$

Actual instability additionally requires a domain eigenvalue in (λ_-, λ_+), where

$$\lambda_\pm = \frac{B \pm \sqrt{B^2 - 4d_1 d_2 \det J}}{2d_1 d_2}.$$

Expanded proof.

Negative determinant gives two real eigenvalues of opposite signs. Positive determinant and negative trace give negative real parts; zero determinant is a neutral threshold, not strict exponential growth. The quadratic p opens upward and has positive roots exactly under the displayed conditions. Its negative set is the interval between them. The discrete spectrum of the chosen domain must meet this interval. □

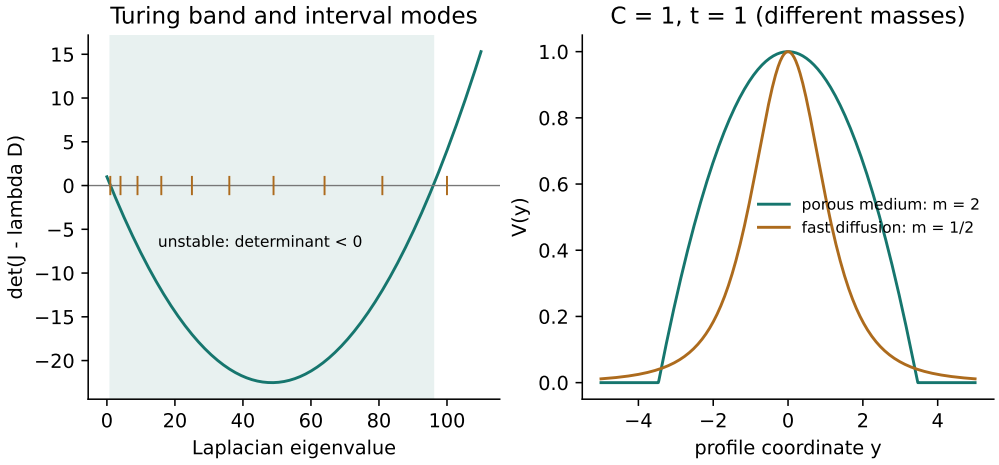
无穷远的条件影响物理解

这个分离解能在有限时间爆破，但它从一开始就有无限质量。不能用它否定有限质量、紧支集数据下的扩散结论；全空间问题也要记录增长类。

For example, take

$$J = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \qquad D = \operatorname{diag}(0.01, 1).$$

Then $\operatorname{tr} J = -2$, $\det J = 1$, while $p(4) = 0.16 - 3.88 + 1 = -2.72$. On $(0, \pi)$ the second sine mode has $\lambda_2 = 4$ and grows. If $D = dI$, by contrast, every eigenvalue is shifted by $-d\lambda_j$, so a stable J remains stable.



Editorial figure. The determinant band is compared with the actual interval spectrum; the other panel compares scaling profiles.

4.2 Similarity Solutions

4.2.1 Plane waves, phase velocity and group velocity

For $u = e^{i(\xi\cdot x - \omega t)}$, direct substitution gives:

equation	dispersion relation
$u_t - \Delta u = 0$	$\omega = -i\lvert\xi\rvert^2$
$u_{tt} - \Delta u = 0$	$\omega = \pm\lvert\xi\rvert$
$u_{tt} - \Delta u + M^2 u = 0$	$\omega = \pm\sqrt{\lvert\xi\rvert^2 + M^2}$
$iu_t + \Delta u = 0$	$\omega = \lvert\xi\rvert^2$
$u_t + u_{xxx} = 0$	$\omega = -\xi^3$

For instance $\partial_{xxx} e^{i\xi x} = -i\xi^3 e^{i\xi x}$, so the last sign is negative with this phase convention. A real ω gives oscillation; a negative imaginary part gives damping. For $\xi \neq 0$, the signed phase speed in direction $\xi/|\xi|$ is $\omega/|\xi|$.

To explain group velocity, set $x = tv$ in a Fourier superposition:

$$u(tv, t) = \int e^{it(\xi\cdot v - \omega(\xi))} a(\xi)\, d\xi.$$

连续波段不等于域内存在模态

代数条件先给出不稳定的 λ 区间；还要看具体区域的离散特征值是否落入其中。$p = 0$ 只是临界中性情形，严格增长需要 $p < 0$。

线性失稳不等于已证明斑图形成

这里证明的是线性化系统存在增长模态。非线性解是否饱和、形成何种稳定空间图案，需要额外的分岔与演化理论，不能由特征值计算直接得出。

Stationary points satisfy $v = D_\xi \omega$. The stationary-phase theorem below makes this observation precise for localized amplitudes and nondegenerate stationary points.

4.2.2 KdV solitary waves

For $u_t + 6uu_x + u_{xxx} = 0$, take $u = v(s)$, $s = x - ct$. The ODE is $-cv' + 6vv' + v''' = 0$.

Theorem 4.3 (The source solitary-wave calculation)
For $c > 0$ and $s_0 \in \mathbb{R}$,

$$u(x,t) = \frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct - s_0)\right)$$

is a decaying traveling wave.

Expanded proof.

One integration gives $v'' - cv + 3v^2 = A$. Multiply by v' and integrate again:

$$\frac{1}{2}(v')^2 = \frac{c}{2}v^2 - v^3 + Av + B.$$

The solitary-wave limits $v, v', v'' \to 0$ at both ends force $A = B = 0$. A positive hump therefore has maximum $c/2$ and obeys $v' = -v\sqrt{c - 2v}$ on its decreasing half. With $v = (c/2)\operatorname{sech}^2\theta$,

$$\frac{dv}{v\sqrt{c - 2v}} = -\frac{2}{\sqrt{c}}\,d\theta \quad (\theta > 0).$$

Thus $\theta = \sqrt{c}(s - s_0)/2$ on that half. Reflection supplies the other half. For a direct check use $(\operatorname{sech}^2 z)'' = 4\operatorname{sech}^2 z - 6\operatorname{sech}^4 z$, which gives $v'' = cv - 3v^2$ everywhere. Differentiating verifies the original PDE, including the crest. □

A solitary-wave formula does not by itself prove the stronger scattering properties often associated with the word soliton. Also the KdV sign in Problems 11–12 is the opposite nonlinear sign; there $u \mapsto -u$ relates the conventions.

4.2.3 Bistable fronts: a detailed shooting argument

Assume $f(0) = f(a) = f(1) = 0$, $0 < a < 1$, $f < 0$ on $(0,a)$, $f > 0$ on $(a,1)$, $f'(0), f'(1) < 0$, and $\int_0^1 f > 0$. An increasing front from zero to one satisfies

$$v' = w, \qquad w' = -cw - f(v),$$
$$(v,w)(-\infty) = (0,0), \quad (v,w)(+\infty) = (1,0).$$

相速不是信息传播速度

Klein–Gordon 的正频相速为 $\sqrt{|\xi|^2 + M^2}/|\xi| > 1$，群速度的大小却小于一。单一相位面的移动不能直接当作信号速度或支集传播速度。

不能把减号分支用在整个波峰两侧

一阶能量式含正负两个分支。上升半支与下降半支要在 $v = c/2$ 处拼合；最后直接核验二阶 ODE，才能排除分支选择造成的假解。

At each endpoint the linear eigenvalues are $(-c \pm \sqrt{c^2 - 4f'(j)})/2, j = 0, 1$. They have opposite signs. We use the standard local stable/unstable manifold theorem for these hyperbolic ODE equilibria, including continuous dependence on c. The following fills in the source's phase-plane shooting outline.

Put $F(v) = \int_0^v f$ and $E = w^2/2 + F(v)$. Then $dE/ds = -cw^2$. Since $F(1) > 0$, a front must have $c < 0$:

$$c \int_{\mathbb{R}} (v')^2 \, ds = -\int_0^1 f(v) \, dv. \tag{BF4}$$

Choose a fixed $b \in (a, 1)$ so close to a that $F(b) < 0$. For $c = 0$ the upper unstable branch from zero and upper stable branch into one meet the section $v = b$ at heights

$$w_L(b; 0) = \sqrt{-2F(b)}, \qquad w_R(b; 0) = \sqrt{2(F(1) - F(b))}.$$

The first is smaller.

Expanded proof (existence and uniqueness of the front speed).

Where $w > 0$, write each trajectory as a graph in v:

$$\frac{dw}{dv} = -c - \frac{f(v)}{w}. \tag{SH4}$$

The endpoint slopes are $w_L/v \to \lambda_0^+(c)$ and $w_R/(1 - v) \to -\lambda_1^-(c)$. As c decreases, the first slope strictly increases and the second strictly decreases. At a first contact between two left branches, their slope difference from (SH4) is the positive difference of $-c$; this prevents crossing. Integrating from the right reverses the comparison for right branches. Thus $w_L(b; c)$ increases strictly as c decreases, while $w_R(b; c)$ decreases strictly.

For $c \le 0$ the left branch stays above its $c = 0$ graph up to b, so cannot stop by hitting zero. The right branch also reaches b: on $(b, 1), f > 0$, and at a hypothetical earlier endpoint with $w = 0$ the equation $w_s = -f < 0$ would give negative w immediately to its right in v, contradicting the positive branch. There is no finite-v upward blowup in these intervals since (SH4) has bounded positive slope whenever w is large. Along the right branch $E(v) \le F(1)$, since energy is increasing in forward time; hence $w_R(b; c) \le \sqrt{2(F(1) - F(b))}$.

To force the left height arbitrarily large, fix $K > 0$. The quantity $|f(v)/v|$ is bounded on $(0, b]$, say by C. On the line $w = Kv$,

$$\frac{dw}{dv} = -c - \frac{f(v)}{Kv} \ge -c - C/K > K$$

when $-c$ is sufficiently large, and $\lambda_0^+(c) > K$ then. The left branch starts above this line and cannot cross downward; so $w_L(b;c) \geq Kb$. The two heights therefore exchange order for sufficiently negative c. Continuous dependence and the intermediate value theorem give a matching speed; strict comparison makes it unique. At the match, ODE uniqueness joins the two trajectories. The hyperbolic endpoint asymptotics take infinite s to reach the endpoints, giving a front on all $\mathbb{R}$, unique up to translation. Integration of the energy identity proves (BF4). □

速度符号取决于连接方向

本稿取 $v(-\infty) = 0$、$v(+\infty) = 1$，故 $\int_0^1 f > 0$ 推出速度为负。若把波形左右反转，速度也反号，不能脱离行波坐标与端态单记正负。

4.2.4 Scaling and the Barenblatt solution

For $u_t - \Delta(u^m) = 0$, try $u = t^{-\alpha}V(y)$, $y = xt^{-\beta}$. The chain rule gives

$$u_t = t^{-\alpha-1}(-\alpha V - \beta y \cdot DV), \qquad \Delta u^m = t^{-\alpha m-2\beta}\Delta_y V^m.$$

Balancing time powers requires $\alpha + 1 = \alpha m + 2\beta$. Finite nonzero time-independent mass requires $\alpha = n\beta$, since $\int u = t^{-\alpha+n\beta}\int V$. Consequently

$$\beta = \frac{1}{n(m-1)+2}, \qquad \alpha = n\beta.$$

For a radial profile $V = W(r)$, the equation integrates to

$$[r^{n-1}(W^m)' + \beta r^n W]' = 0.$$

Regularity and zero flux at $r = 0$ set the integration constant to zero. For $m > 1$, on the positive set,

$$(W^{m-1})' = -\frac{m-1}{m}\beta r, \quad V(y) = (C - k|y|^2)_+^{1/(m-1)}, \quad k = \frac{(m-1)\beta}{2m}.$$

Theorem 4.4 (Barenblatt profile, source §4.2.2)
For $C > 0$, $m > 1$, the above formula defines a nonnegative distributional solution on $\mathbb{R}^n \times (0,\infty)$, with fixed finite mass. Its support radius is $\sqrt{C/k}\,t^\beta$.

Expanded proof (including the free-boundary check).

Inside its positive set the preceding identities prove the PDE. At the moving interface u tends to zero, and so the time derivative of the zero extension has no jump-mass term. Furthermore $V^m = (C - kr^2)_+^{m/(m-1)}$ has zero normal derivative at the interface because $m/(m-1) > 1$. Thus the distributional Laplacian adds no surface delta. The interior time derivative and second spatial derivatives are locally integrable up to the interface: the worst boundary power is $d^{1/(m-1)-1}$, whose exponent is greater than -1. Integration by parts on interior truncated regions, followed by

this integrability and the zero traces, proves the weak PDE across the boundary. Mass conservation follows by the change $x = t^{\beta} y$. Finally for compactly supported continuous ϕ,

$$\int u(x,t)\phi(x)\,dx = \int V(y)\phi(t^{\beta}y)\,dy \longrightarrow \phi(0)\int V.$$

The initial datum is a point mass, not a fixed bounded function. □

The radius grows sublinearly for $m > 1$. Its derivative is finite for each $t > 0$ but diverges as $t \downarrow 0$; finite propagation here means compact support at each positive time, not a uniform speed bound down to the point-source initial time. The fast-diffusion counterpart in Problem 5 has algebraic tails.

自由边界处要检查通量

只在正集内代入公式还不够。这里 $u = 0$ 消去移动边界的时间接口项，$\partial_\nu(u^m) = 0$ 消去空间接口项；扩展后的 PDE 才在分布意义下成立。

4.3 Transform Methods

4.3.1 Fourier transform: normalization and Plancherel

For $f \in L^1(\mathbb{R}^n)$ define

$$\hat{f}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi} f(x)\,dx,$$
$$\mathscr{F}^{-1}h(x) = (2\pi)^{-n/2}\int e^{ix\cdot\xi} h(\xi)\,d\xi.$$

The integrals are absolutely convergent. For Schwartz functions integration by parts and Fubini yield

$$\widehat{\partial_j f} = i\xi_j \hat{f}, \qquad \widehat{f * h} = (2\pi)^{n/2}\hat{f}\,\hat{h}. \tag{F4}$$

For the second identity, set $x = z + y$ in the double integral. Both normalizations must be retained; $\mathscr{F}$ is unitary, whereas ordinary convolution has no normalizing factor.

卷积公式为何多一个常数

每个 Fourier 变换自带 $(2\pi)^{-n/2}$，两个变换相乘就出现两份。卷积只做一次变换，所以要乘回 $(2\pi)^{n/2}$；后面的核常数都由此决定。

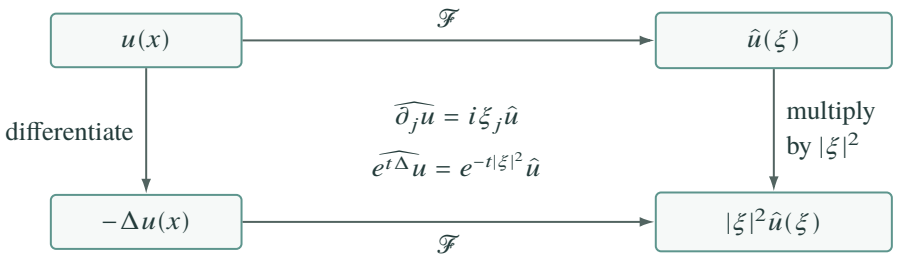


Diagram T4.2. Fourier transformation turns differentiation into multiplication. The square commutes with the unitary normalization used here. There is no 2π in the multiplier because the exponential is $e^{-ix\cdot\xi}$. The heat multiplier is small at large $|\xi|$, connecting this representation with the stability picture in Chapter 1.

Theorem 4.5 (Plancherel and inversion; source Theorems 1–2)
The Fourier transform extends uniquely to a unitary operator on L^2. It preserves inner products, has the displayed inverse, and $\widehat{D^\gamma f} = (i\xi)^\gamma \hat{f}$ whenever the weak derivative is in L^2.

Expanded proof (Gaussian regularization without circular inversion).

1. The Gaussian transform. For $a > 0$, $I(b) = \int_{\mathbb{R}} e^{-ax^2} e^{-ibx}\, dx$. Differentiate under the integral and integrate by parts to get $I'(b) = -bI(b)/(2a)$. Since $I(0) = \sqrt{\pi/a}$, $I(b) = \sqrt{\pi/a}\, e^{-b^2/(4a)}$. Products give the n-dimensional identity. This real-variable proof does not yet require contour deformation.

2. The norm identity. For $f \in L^1 \cap L^2$ and $\eta > 0$, Fubini gives

$$\int e^{-\eta|\xi|^2} |\hat{f}(\xi)|^2\, d\xi = \iint f(x)\overline{f(y)} G_\eta(x-y)\, dx\, dy,$$

where $G_\eta(z) = (4\pi\eta)^{-n/2} e^{-|z|^2/(4\eta)}$. Fubini is legitimate since $\|f\|_1^2 < \infty$ and the frequency Gaussian is integrable. The right side tends to $\|f\|_2^2$ because $G_\eta * f \to f$ in L^2. For completeness,

$$\|G_\eta * f - f\|_2 \le \int G_\eta(z) \|f(\cdot - z) - f\|_2\, dz \longrightarrow 0 :$$

translations are continuous in L^2 by approximation with compact continuous functions; split the last integral into a small ball and its Gaussian tail. Monotone convergence on the left gives the norm identity. Polarization gives the inner-product identity.

3. Inversion and extension. For Schwartz f, (F4) implies rapid decay of $\hat{f}$. Fubini and the Gaussian formula give

$$\mathscr{F}^{-1}(e^{-\eta|\xi|^2}\hat{f}) = G_\eta * f.$$

The right side tends to f; the left tends to $\mathscr{F}^{-1}\hat{f}$ by dominated convergence. Thus inversion holds on Schwartz functions. Approximate arbitrary L^2 data by Schwartz functions. The norm identity makes their transforms Cauchy and makes the limit independent of the approximation. Inversion extends by continuity, so the range is all of L^2. Finally integration by parts against Schwartz tests proves the derivative identity in distributions; if both sides belong to L^2, it is an a.e. equality of those representatives. □

4.3.2 Fourier multipliers and evolution equations

Bessel potential. For $-\Delta u + u = f, f \in L^2$, $\hat{u} = (1 + |\xi|^2)^{-1}\hat{f}$. This gives a unique L^2 distributional solution with $\int (1 + |\xi|^2)^2 |\hat{u}|^2 < \infty$. A rigorous kernel representation is

$$K(x) = \int_0^\infty e^{-s} G_s(x)\, ds, \qquad u = K * f.$$

Tonelli gives $K \geq 0$ and $\|K\|_1 = 1$; its transform is $(2\pi)^{-n/2}/(1 + |\xi|^2)$. The kernel called B in the source is $(2\pi)^{n/2}K$. The singularity of K at zero in some dimensions does not prevent its L^1 integrability or convolution action on L^2.

Heat and Schrödinger equations. Spatial transformation of $u_t - \Delta u = 0$ gives $\hat{u}_t + |\xi|^2 \hat{u} = 0$, hence $\hat{u} = e^{-t|\xi|^2}\hat{g}$. Gaussian inversion recovers $u = G_t * g$ from Chapter 2. For $iu_t + \Delta u = 0$ the sign is instead $\hat{u}_t = -i|\xi|^2\hat{u}$, so

$$\hat{u} = e^{-it|\xi|^2}\hat{g}, \qquad u(x,t) = (4\pi i t)^{-n/2} \int e^{i|x-y|^2/(4t)} g(y)\, dy. \tag{SC4}$$

For $t \neq 0$ use $(it)^{-n/2} = |t|^{-n/2} e^{-i\pi n\, \mathrm{sgn}(t)/4}$. The kernel identity follows by first replacing it with $\eta + it$, $\eta > 0$, and then letting $\eta \downarrow 0$. For $g \in L^1 \cap L^2$, the kernels converge under the spatial integral by domination, while the multipliers converge in L^2. Thus both descriptions agree. For L^2 data this defines a unitary group:

$$\|u(t)\|_2 = \|g\|_2, \qquad \|u(t) - g\|_2^2 = \int |e^{-it|\xi|^2} - 1|^2 |\hat{g}|^2 \longrightarrow 0.$$

If $g, |x|^2 g \in L^1$, differentiation of (SC4) on compact space-time sets away from $t = 0$ verifies the classical PDE. For arbitrary L^2 data the statement is distributional.

Wave and telegraph equations. For $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = h$,

$$\hat{u}(\xi, t) = \cos(t|\xi|)\hat{g} + \frac{\sin(t|\xi|)}{|\xi|}\hat{h}. \tag{WV4}$$

At $\xi = 0$ the quotient is defined as t. It is bounded by t, so the formula has no genuine zero-frequency pole. For $u_{tt} + 2du_t - u_{xx} = 0$ use $\hat{u} = e^{-dt}v$. Then $v'' + (\xi^2 - d^2)v = 0$ and

$$\hat{u} = e^{-dt} \begin{cases} \cosh(qt)\hat{g} + \dfrac{\sinh(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| < d,\ q = \sqrt{d^2 - \xi^2}, \\ \hat{g} + t(\hat{h} + d\hat{g}), & |\xi| = d, \\ \cos(qt)\hat{g} + \dfrac{\sin(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| > d,\ q = \sqrt{\xi^2 - d^2}. \end{cases}$$

L^2 变换不必是逐点积分

一般 L^2 函数未必可积，其 Fourier 变换通过范数极限定义。反演和乘子公式首先是 L^2 等式；若要逐点表示，还须增加可积性或正则性。

乘子有界不表示核处处有限

$(1 + |\xi|^2)^{-1}$ 给出有界的 L^2 乘子。对应核可能在原点无界，但仍可积；不能把一个点的奇性误当作表示公式失效。

色散与耗散的范数表现不同

热乘子模长小于一，会压低高频；Schrödinger 乘子模长为一，保持 L^2。后者的点态衰减来自振荡抵消，不来自 L^2 质量损失。

These are the overdamped, critical and oscillatory regimes. The common limiting formula at $q = 0$ is essential.

Theorem 4.6 (Asymptotic energy equipartition; source §4.3.1)
If $g \in H^1(\mathbb{R}^n)$ and $h \in L^2(\mathbb{R}^n)$, the wave solution has

$$\|Du(t)\|_2^2 \longrightarrow E_0, \qquad \|u_t(t)\|_2^2 \longrightarrow E_0, \qquad E_0 = \tfrac{1}{2}(\|Dg\|_2^2 + \|h\|_2^2).$$

Expanded proof.

Write $A = |\xi|\hat{g}$ and $B = \hat{h}$ in L^2. Plancherel and (WV4) give

$$\|Du\|_2^2 = \int |A|^2 \cos^2(t|\xi|) + |B|^2 \sin^2(t|\xi|) + 2\,\mathrm{Re}(A\overline{B}) \sin(t|\xi|)\cos(t|\xi|).$$

All three coefficient functions are L^1, the cross term by Cauchy–Schwarz. For any $F \in L^1(\mathbb{R}^n)$, polar coordinates give $\int e^{2it|\xi|}F(\xi)\,d\xi = \int_0^\infty e^{2itr}H(r)\,dr$ with $\|H\|_1 \le \|F\|_1$. The latter tends to zero: approximate H in L^1 by smooth functions supported inside $(0,\infty)$ and integrate those by parts. Now use the double-angle identities to obtain the first limit. The analogous formula for u_t gives energy conservation and the second limit. This also handles $n = 1$ without discarding a possible endpoint term in a radial integration by parts. □

4.3.3 Radon transform and plane-wave decomposition

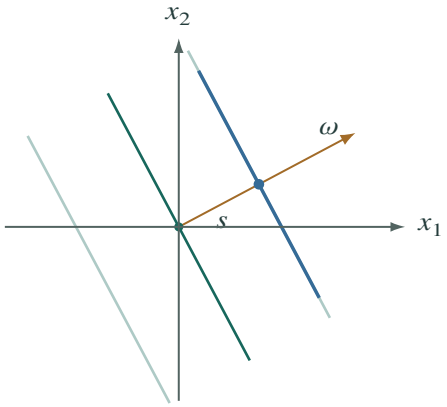
For $f \in C_c^\infty(\mathbb{R}^n)$, set

$$Rf(s,\omega) = \int_{x\cdot\omega = s} f(x)\,dS_x, \qquad \omega \in S^{n-1}.$$

Coordinates $x = s\omega + \sum y_j b_j$, where b_j are an orthonormal basis of $\omega^\perp$, make the surface measure dy. They immediately give

$$Rf(-s,-\omega) = Rf(s,\omega), \quad R(\partial_j f) = \omega_j \partial_s Rf, \quad R(\Delta f) = \partial_s^2 Rf.$$

For the middle identity decompose ∂_j into a normal and tangential derivative. Tangential integrals vanish by compact support. Induction gives $R(D^\gamma f) = \omega^\gamma \partial_s^{|\gamma|} Rf$. If $\mathrm{supp} f \subset B_R$, then $Rf = 0$ for $|s| \ge R$.



$$Rf(s,\omega) = \int_{x\cdot\omega=s} f\, dS$$

fix the unit normal ω;
vary s to scan parallel lines

Diagram T4.3. Radon data integrate along level hyperplanes. In dimension two the integration sets are lines perpendicular to ω. The scalar s is signed distance in the normal direction, not distance traveled along a line. The blue line is one member of the family; higher dimensions replace it by an $(n-1)$-plane.

Theorem 4.7 (Slice theorem and inversion; source Theorems 4–5)
With the unnormalized one-dimensional transform $B(r,\omega) = \int_{\mathbb{R}} e^{-irs} Rf(s,\omega)\, ds$,

$$B(r,\omega) = (2\pi)^{n/2}\hat{f}(r\omega).$$

For every dimension,

$$f(x) = \frac{1}{2(2\pi)^n}\int_{S^{n-1}}\int_{\mathbb{R}} B(r,\omega)|r|^{n-1}e^{irx\cdot\omega}\, dr\, dS_\omega.$$

If $n = 2k+1$ *is odd, this becomes the local formula*

$$f(x) = C_n\int_{S^{n-1}} \partial_s^{2k} Rf(x\cdot\omega,\omega)\, dS_\omega, \qquad C_n = \frac{(-1)^k}{2(2\pi)^{2k}}. \quad \text{(RI4)}$$

Expanded proof.

Fubini in the orthonormal coordinates gives

$$B(r,\omega) = \int f(x)e^{-ir\omega\cdot x}\, dx.$$

In Fourier inversion, polar coordinates with signed r double-count each nonzero frequency, giving the factor $1/2$ and $|r|^{n-1}$. All integrals converge since $\hat{f}$ is rapidly decreasing. For $n-1 = 2k$, one-dimensional inversion gives

$$\partial_s^{2k} Rf(s,\omega) = \frac{(-1)^k}{2\pi}\int_{\mathbb{R}} r^{2k}B(r,\omega)e^{irs}\, dr.$$

Substituting this in the general inversion formula proves (RI4). □

For odd n, define $\rho_f = C_n \partial_s^{n-1} Rf$. Then the plane-wave integral

$$u(x,t) = \int_{S^{n-1}} \rho_g(x\cdot\omega - t, \omega)\, dS_\omega$$

solves $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = 0$. The PDE holds because $|\omega| = 1$; the first initial condition is (RI4). The integrand for $u_t(0)$ is odd under $\omega \mapsto -\omega$, since an odd s derivative reverses the symmetry of Rg.

For the constant-coefficient system $\mathbf{u}_t + \sum B_j \mathbf{u}_{x_j} = 0$, suppose $B(\omega) = \sum \omega_j B_j$ is diagonalizable with real eigenvalues $\lambda_l(\omega)$. In Radon coordinates the diagonal components satisfy one-dimensional transport and equal

$$v_l(s,\omega,t) = h_l(s - \lambda_l(\omega)t, \omega).$$

After applying (RI4), use $\rho_l(s,\omega) = C_n \partial_s^{n-1} h_l(s,\omega)$ exactly once:

$$\mathbf{u}(x,t) = \int_{S^{n-1}} \sum_l a_l(\omega)\, \rho_l(x\cdot\omega - \lambda_l(\omega)t, \omega)\, dS_\omega.$$

Here a_l are the diagonalizing columns; the shift is not inserted again in the definition of ρ_l. For compact data in B_R, if $\min_{\omega,l} |\lambda_l(\omega)| = \sigma > 0$, then $\mathbf{u} = 0$ for $|x| < \sigma t - R$. Indeed each shifted argument has magnitude at least $\sigma t - |x| > R$. This is a conditional odd-dimensional Huygens statement; the diagonalization and nonzero-speed hypotheses cannot be removed.

奇偶维差别藏在绝对值里

带符号的极径在一般维数必须用 $|r|^{n-1}$；只有奇维时才等于多项式 r^{2k}，从而变成有限阶局部导数。偶维的反演通常带非局部算子。

4.3.4 Laplace transform, resolvents and the heat–wave link

If v and v_t have suitable exponential bounds, integration by parts gives

$$\mathscr{L}(v_t)(s) = s\mathscr{L}v(s) - v(0), \qquad \mathscr{L}v(s) = \int_0^\infty e^{-st} v(t)\, dt.$$

Thus for $v_t - \Delta v = 0$, $v(0) = f$,

$$(s - \Delta)\mathscr{L}v = f.$$

Use the same spatial boundary condition before and after transformation; for example a zero Dirichlet condition stays zero. In the whole space this recovers $(s-\Delta)^{-1} f = \int_0^\infty e^{-st}(G_t * f)\, dt$. For L^2 data the Bochner integral is justified by $\|G_t * f\|_2 \le \|f\|_2$.

平面支集条件怎样变成内部空区

若 $|x| < \sigma t - R$，所有方向的平面参数都移出初始支集区间，整个反演积分为零。零特征速度会留下不移动的分量，因此必须单独排除。

Theorem 4.8 (Gaussian averaging of the wave group; source Example 9)
Let g be smooth compactly supported, and let $U(x,s)$ be the wave solution with displacement g and zero initial velocity, extended evenly in s. Then

$$V(x,t) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-s^2/(4t)} U(x,s)\, ds$$

is the heat solution with datum g.

Expanded proof.

For smooth data the derivatives of U grow at most polynomially, as can also be checked from its Fourier representation. Gaussian weights justify differentiation and kill the boundary terms. Writing $q_t(s) = (4\pi t)^{-1/2} e^{-s^2/(4t)}$, observe $\partial_t q_t = \partial_{ss} q_t$. Hence

$$V_t = \int q_{ss} U = \int q U_{ss} = \int q \Delta U = \Delta V.$$

The Gaussian approximate identity in s gives $V(0) = g$. Alternatively the frequency multiplier is $\int q_t(s) \cos(s|\xi|)\, ds = e^{-t|\xi|^2}$, which proves the same statement in L^2. □

For $n = 2k+1 \geq 3$, let $M_g(x,r)$ be the spherical average of g. Equating the two Gaussian representations gives, with $\lambda = 1/(4t)$,

$$\int_0^\infty e^{-\lambda s^2} U(x,s)\, ds = \frac{|S^{n-1}|}{2\pi^k} \lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g(x,r)\, dr.$$

Use $\lambda e^{-\lambda r^2} = -(2r)^{-1} \partial_r e^{-\lambda r^2}$ and integrate by parts k times. The result is

$$\lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g\, dr = 2^{-k} \int_0^\infty e^{-\lambda r^2} r (r^{-1}\partial_r)^k (r^{2k-1} M_g)\, dr.$$

There are no boundary terms at infinity by compact support in r. At zero, M_g is smooth and even; before the last differentiation the relevant odd power has not fallen below one, so the boundary terms are zero. After $s^2 = \tau$, uniqueness of the Laplace transform implies

$$U(x,t) = \frac{1}{(n-2)!!} \partial_t (t^{-1}\partial_t)^{(n-3)/2} (t^{n-2} M_g(x,t)).$$

Here $|S^{n-1}|/(2^{k+1}\pi^k) = 1/(n-2)!!$ follows from the Gaussian sphere area formula. The one-dimensional case is $U = (g(x+t) + g(x-t))/2$. One can justify Laplace uniqueness without a new PDE theorem: after multiplying by $e^{-a\tau}$, substitute $z = e^{-\tau}$; vanishing at all $s = a + j$ gives zero moments on $[0,1]$. Polynomial density in continuous functions makes the finite signed measure zero. This applies to the continuous, exponentially integrable functions above.

初值正是 Laplace 分部积分的边界项

时间导数并非只替换成乘以 s。零时刻留下 $-v(0)$，它成为椭圆 resolvent 方程的右端；漏掉这一项就把非零初值问题变成了齐次问题。

4.4 Converting Nonlinear into Linear PDE

4.4.1 Cole–Hopf: choose the change of unknown by cancellation

Consider $u_t - a\Delta u + b|Du|^2 = 0$, $a > 0$. For $w = \phi(u)$, the chain rule yields

$$w_t - a\Delta w = -(b\phi'(u) + a\phi''(u))|Du|^2.$$

To cancel the nonlinear term solve $a\phi'' + b\phi' = 0$. For $b \neq 0$ choose $\phi(z) = e^{-bz/a}$.

Theorem 4.9 (Cole–Hopf; source §4.4.1)
If g is bounded and continuous, define

$$w(x,t) = \int_{\mathbb{R}^n} G_{at}(x-y)e^{-bg(y)/a}\,dy, \qquad u = -\frac{a}{b}\log w. \tag{CH4}$$

Then u is smooth for $t > 0$, solves the nonlinear PDE, and has initial value g at continuity points. It is the unique bounded classical solution with these data in the corresponding heat uniqueness class.

Expanded proof.

Boundedness of g gives positive constants $0 < c \le w \le C$, since the heat kernel is positive and has mass one. Heat smoothing, differentiation under Gaussian integrals and $w_t = a\Delta w$ hold for $t > 0$. Thus the logarithm is well defined and differentiable; substitution or the preceding chain rule proves the nonlinear equation. The approximate-identity limit gives $w \to e^{-bg/a}$ and hence $u \to g$. Conversely any bounded classical solution transforms into a bounded heat solution. Heat uniqueness and injectivity of the exponential prove the assertion. If $b = 0$, the original equation is already the heat equation; division by b in (CH4) is not allowed. □

For viscous Burgers, $u_t + uu_x - au_{xx} = 0$, let $P' = g$. A potential W with $W_x = u$ satisfies $W_t + \frac{1}{2}W_x^2 - aW_{xx} = c(t)$. Subtracting $\int c$ removes this purely time-dependent term. With $Z = e^{-W/(2a)}$, $Z_t = aZ_{xx}$, so

$$W = -2a\log\int_{\mathbb{R}} G_{at}(x-y)e^{-P(y)/(2a)}\,dy,$$

$$u(x,t) = \frac{\int_{\mathbb{R}}(x-y)t^{-1}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}{\int_{\mathbb{R}}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}. \tag{VB4}$$

For bounded g, choose $P(x) = \int_0^x g$; it is Lipschitz and need not be bounded. The Gaussian dominates its at-most-linear growth, so the integrals and their derivatives on compact positive-time sets converge. Integrating the y derivative of the exponential shows also u is its weighted

正性同时保证变换可逆

指数变换总是正的，而热核保持严格正性。反变换要取对数，所以不能把任意带符号的热解代入；有界初值还给出远离零的统一下界。

average of $g(y)$, a.e.:

$$\int_{\mathbb{R}} \big((x-y)/t - g(y)\big) e^{-K(y)/(2a)}\, dy = 0.$$

The endpoint terms vanish by quadratic Gaussian decay. In particular $|u| \le \|g\|_\infty$.

4.4.2 Potential flow and Bernoulli's law

For incompressible Euler, $\mathbf{u}_t + (\mathbf{u}\cdot D)\mathbf{u} = -Dp + Dh$ and $\operatorname{div}\mathbf{u} = 0$, impose the additional ansatz $\mathbf{u} = Dv$. Then $\Delta v = 0$, and componentwise symmetry of $D^2 v$ gives

$$[(\mathbf{u}\cdot D)\mathbf{u}]_i = \sum_j v_{x_j} v_{x_i x_j} = \partial_i \left(\tfrac{1}{2}|Dv|^2\right).$$

Therefore

$$D(v_t + \tfrac{1}{2}|Dv|^2 + p - h) = 0, \qquad p = h - v_t - \tfrac{1}{2}|Dv|^2 + c(t)$$

on each connected spatial component. To match initial data one needs $\mathbf{g} = Dv_0$ as well as $\operatorname{div}\mathbf{g} = 0$. Locally curl-free data admit a potential; globally a topological condition or a prescribed global potential is required.

4.4.3 Hodograph: invert dependent and independent variables

Write the two velocity components as $p = u^1$, $q = u^2$. For steady irrotational compressible flow the equations are

$$(c^2 - p^2)p_{x_1} - pq(p_{x_2} + q_{x_1}) + (c^2 - q^2)q_{x_2} = 0, \qquad p_{x_2} = q_{x_1},$$

where $c = c(p,q) > 0$ is given. If $J = \det D_x(p,q) \neq 0$, invert to $x = X(p,q)$. The inverse matrix identity is

$$\begin{pmatrix} p_{x_1} & p_{x_2} \\ q_{x_1} & q_{x_2} \end{pmatrix} = J \begin{pmatrix} X^2_q & -X^1_q \\ -X^2_p & X^1_p \end{pmatrix}.$$

Substitution and division by J give the linear system

$$(c^2 - p^2)X^2_q + pq(X^1_q + X^2_p) + (c^2 - q^2)X^1_p = 0, \qquad X^1_q = X^2_p.$$

Locally $X = DV$ for a scalar potential $V(p,q)$, and

$$(c^2 - q^2)V_{pp} + 2pqV_{pq} + (c^2 - p^2)V_{qq} = 0.$$

The coefficient determinant is $c^2(c^2 - p^2 - q^2)$: the equation is elliptic in the subsonic region and hyperbolic in the supersonic region, degenerate at sonic speed.

原函数的常数与时间规范

给 P 加常数只会把热解 Z 乘一个正数，不改变 $u = -2aZ_x/Z$。从 Burgers 积分得到的时间函数也能吸收进势函数，不能无说明地直接丢掉。

无散初值还不足以成为势流

$\operatorname{div}\mathbf{g} = 0$ 是不可压条件，不能推出 $\mathbf{g} = Dv_0$。还需无旋及相应的全局条件；这里线性化的是势流子类，并未解出一般 Euler 方程。

4.4.4 A local Legendre transform for the minimal surface equation

In two dimensions the minimal surface equation is

$$(1+u_{x_2}^2)u_{x_1x_1} - 2u_{x_1}u_{x_2}u_{x_1x_2} + (1+u_{x_1}^2)u_{x_2x_2} = 0.$$

Assume $\det D^2u = J \neq 0$ locally. Use $p = Du(x)$ as independent variable, let $x = X(p)$, and set $v(p) = X(p)\cdot p - u(X(p))$.

Expanded proof (the Hessian inverse and the linear equation).

The chain rule gives

$$\partial_{p_i} v = X_i + \sum_j p_j \partial_{p_i} X_j - \sum_j u_{x_j}\partial_{p_i}X_j = X_i.$$

Hence $D^2v = D_pX = (D^2u)^{-1}$. For a symmetric 2×2 matrix the inverse formula implies

$$u_{x_1x_1} = Jv_{p_2p_2}, \quad u_{x_1x_2} = -Jv_{p_1p_2}, \quad u_{x_2x_2} = Jv_{p_1p_1}.$$

Substitute and divide by J to obtain

$$(1+p_1^2)v_{p_1p_1} + 2p_1p_2v_{p_1p_2} + (1+p_2^2)v_{p_2p_2} = 0.$$

Its coefficient matrix $I + p\otimes p$ is positive definite. Conversely, a solution with invertible Hessian yields a local inverse $x = Dv(p)$ and reconstructs $u = x\cdot p - v$ satisfying the original PDE. Boundary data need a separate transformation; a fixed boundary in x generally becomes an unknown surface in p. □

变线性没有消除退化

系数现在依赖独立变量，因而方程线性，但声速边界仍使主部退化。还要把 $X(p,q)$ 反解回去；Jacobian 变零处不能继续套用变换。

4.5 Asymptotics

The source presents several asymptotic constructions formally. We distinguish exact algebraic identities, proved local integral asymptotics, and conditional PDE expansions. No convergence theorem is inferred merely by matching powers.

局部 Legendre 变换不要求凸性

这里凭 $\det D^2u \neq 0$ 局部反演，Hessian 可以不定。第三章的凸共轭是全局上确界构造；两者在严格凸、可逆情形相接，但假设与用途不同。

4.5.1 A small-diffusion plume: geometry and transverse scale

Consider $\operatorname{div}(\mathbf{b}\,\mu) = \delta_0$ in two dimensions. Let $\dot X = \mathbf{b}(X)$, $X(0) = 0$, and suppose $|\mathbf{b}(X)| > 0$. If the forward trajectory leaves compact sets, the locally finite measure

$$\int \phi\, d\mu = \int_0^\infty \phi(X(t))\, dt$$

solves this equation, since

$$-\int \mathbf{b}\cdot D\phi\, d\mu = -\int_0^\infty \frac{d}{dt}\phi(X(t))\, dt = \phi(0).$$

This is an outgoing solution; an extra background homogeneous solution is not excluded by this calculation. With speed $\sigma = |\mathbf{b}(X)|$, tangent τ and unit normal ν, use $x = X(t) + y\nu(t)$ in a nonoverlapping tubular neighborhood. Frenet identities give $\nu' = -\sigma\kappa\tau$ and

$$x_t = \sigma(1-\kappa y)\tau, \quad x_y = \nu, \quad dx = H\, dt\, dy, \qquad H = \sigma(1-\kappa y) > 0.$$

Thus the same measure has density $\delta(y)/\sigma(t)$ with respect to spatial area.

Now perturb to $-\varepsilon\Delta u^\varepsilon + \operatorname{div}(\mathbf{b}u^\varepsilon) = \delta_0$. Away from the injection point, set $z = y/\sqrt{\varepsilon}$, $u^\varepsilon = \varepsilon^{-1/2}v^\varepsilon(t,z)$. The coordinate metric gives the exact identity

$$\Delta w = \frac{1}{H}\partial_t(H^{-1}w_t) + \frac{1}{H}\partial_y(Hw_y).$$

Also write $\mathbf{b} = (\sigma + \alpha y + O(y^2))\tau + (\beta y + O(y^2))\nu$. Then on compact $t > 0$, bounded z, provided the needed derivatives of v^ε are uniformly bounded,

$$\mathbf{b}\cdot Dv^\varepsilon = v_t^\varepsilon + \beta z v_z^\varepsilon + O(\sqrt{\varepsilon}), \qquad \varepsilon\Delta v^\varepsilon = v_{zz}^\varepsilon + O(\sqrt{\varepsilon}),$$

$$\operatorname{div}\mathbf{b} = \sigma'/\sigma + \beta + O(\sqrt{\varepsilon}).$$

For example the transverse first-derivative term in the Laplacian is $\varepsilon(H_y/H)\partial_y v^\varepsilon = O(\sqrt{\varepsilon})$; the tangential second-derivative terms are $O(\varepsilon)$. Thus the conditional leading equation is

$$v_t - v_{zz} + (\beta z v)_z + (\sigma'/\sigma)v = 0, \qquad v(0) = \delta_0/\sigma(0). \tag{PL4}$$

Its exact Gaussian solution, derived again in Problem 13, is

$$v(t,z) = \frac{e^{-z^2/\Gamma(t)}}{\sigma(t)\sqrt{\pi\Gamma(t)}}, \qquad \Gamma' = 4 + 2\beta\Gamma, \quad \Gamma(0) = 0.$$

Indeed $A = \sigma v$ satisfies $A_t = A_{zz} - (\beta z A)_z$; substituting the Gaussian gives the displayed ODE and mass $\int A = 1$. The transverse width is $\sqrt{\varepsilon\Gamma(t)}$. This does not by itself prove approximation of the original elliptic Green function, nor control the injection region or large times.

线上的密度必须带速度因子

注入量按行进时间计量，而面积元在曲线处含 $\sigma(t)$。因此横向 Dirac 前要乘 $1/\sigma$。若轨道封闭或反复回返，时间积分可能不是局部有限测度。

4.5.2 Laplace concentration and vanishing viscosity

Lemma 4.10 (Source asymptotics lemma, with tail control)
Let k, l be continuous on $\mathbb{R}$, $|l(y)| \le C(1+|y|)$, and $k(y) \ge cy^2 - C$, $c > 0$. If k has a unique minimum at y_0, then

$$\frac{\int l(y) e^{-k(y)/\varepsilon}\, dy}{\int e^{-k(y)/\varepsilon}\, dy} \longrightarrow l(y_0). \tag{LC4}$$

Expanded proof.

Subtract the minimum k_0 and put $h = k - k_0 \ge 0$. Fix a small neighborhood N of y_0 on which $|l - l(y_0)| < \eta$. On a large compact interval outside N, $h \ge d > 0$ by uniqueness. Increase the interval so that outside it $h(y) \ge d/2 + (c/2)y^2$, decreasing d if necessary. By continuity there is an interval of length $2r > 0$ near y_0 where $h < d/4$. The denominator is at least $2re^{-d/(4\varepsilon)}$. The numerator of the absolute error over the compact complement is at most $C_1 e^{-d/\varepsilon}$. The tail numerator is bounded by

$$C_2 e^{-d/(2\varepsilon)} \int (1+|y|) e^{-cy^2/(2\varepsilon)}\, dy,$$

whose ratio to the denominator tends to zero. Inside N the normalized error is at most η. Let $\varepsilon \downarrow 0$ and then $\eta \downarrow 0$. This proves concentration together with the required control of the unbounded observable l. □

For bounded initial g, let $P' = g$ and $K(y) = (x-y)^2/(2t) + P(y)$. The quadratic term dominates the Lipschitz primitive. Formula (VB4) and (LC4), with parameter 2ε, imply

$$u^\varepsilon(x,t) \longrightarrow \frac{x - y(x,t)}{t}$$

whenever the minimizing point is unique. By Chapter 3, at each $t > 0$ only countably many x have multiple minimizers, and the right side is the Lax–Oleinik entropy solution. Since $|u^\varepsilon| \le \|g\|_\infty$, this convergence is also L^1_{loc} in x for each fixed t. Joint a.e. convergence and the same bound give local space-time L^1 convergence away from zero. The multiple-minimizer set is measurable, for instance by expressing separation of two minimizers using rational compact neighborhoods; Fubini therefore applies to the countable time slices.

形式余项依赖尚未证明的统一界

把 y 换成 $\sqrt{\varepsilon} z$ 后，系数误差很小，但它仍乘着解的导数。只有这些导数在所考察区域有统一界，才能写成统一的 $O(\sqrt{\varepsilon})$。

逐点密度趋零不足以推出集中

归一化密度在每个非极小点趋零，还不能控制积分或无穷远的质量。证明需要分母下界、远离极小点的能量间隙，以及带权尾积分。

If k is smooth with finitely many nondegenerate global minima y_j, local Taylor expansion $y = y_j + \sqrt{\varepsilon} z$ and Gaussian domination give

$$\int l e^{-k/\varepsilon} = e^{-k_0/\varepsilon} \sqrt{2\pi\varepsilon} \left(\sum_j \frac{l(y_j)}{\sqrt{k''(y_j)}} + o(1) \right).$$

To justify domination, restrict to a small neighborhood where $k - k_0 \geq c_j |y - y_j|^2$, and use the energy gap outside. Thus at a tie the limit is a weighted average, not necessarily the velocity at one selected minimizing point. Values on shock curves do not affect the weak solution.

4.5.3 Geometric optics and stationary phase

For $u^\varepsilon = a e^{ip/\varepsilon}$ the exact wave-operator identity is

$$\begin{aligned} & e^{-ip/\varepsilon} (u^\varepsilon_{tt} - \Delta u^\varepsilon) \\ & \quad = \varepsilon^{-2} a (|Dp|^2 - p_t^2) \\ & \qquad + \frac{i}{\varepsilon} \{ 2 p_t a_t - 2 Dp \cdot Da + a(p_{tt} - \Delta p) \} + (a_{tt} - \Delta a). \end{aligned}$$

Matching the first two orders gives the eikonal and amplitude equations

$$p_t^2 = |Dp|^2, \qquad 2 p_t a_t - 2 Dp \cdot Da + (p_{tt} - \Delta p) a = 0.$$

On a smooth nonzero-frequency branch choose $p_t = \pm |Dp|$. The sign cannot be chosen independently at every point. For $u_{tt} - \sum a_{ij}(x) u_{x_i x_j} = 0$ the first equation becomes $p_t^2 = \sum a_{ij} p_{x_i} p_{x_j}$, with the corresponding amplitude expression obtained by the same differentiation. These identities define a candidate expansion. A theorem about exact solutions needs control of the residual and the evolution, especially near caustics.

Lemma 4.11 (Nonstationary phase; strengthening source Lemma 1)
If $a \in C_c^\infty$ and $|D\phi| > 0$ on its support, then for every integer $N \geq 1$, $\int e^{i\phi/\varepsilon} a = O(\varepsilon^N)$.

Expanded proof.

Near the support let $V = D\phi / |D\phi|^2$. Then $V \cdot D e^{i\phi/\varepsilon} = (i/\varepsilon) e^{i\phi/\varepsilon}$. Integration by parts gives

$$\int e^{i\phi/\varepsilon} a = -\frac{\varepsilon}{i} \int e^{i\phi/\varepsilon} \operatorname{div}(aV).$$

Iterate N times; the final smooth compactly supported amplitude has finite L^1 norm independent of ε. This also proves the source's linear-phase case. □

与第三章的最小化公式接上了

黏性解是最优速度的指数加权平均。黏性趋零时，权重集中到作用量最低点；最低点的分支竞争正对应守恒律的激波。

相位误差除以小参数会放大

即使 $p^\varepsilon - p = o(1)$，也不能直接推出两个指数相近；指数里的差是 $(p^\varepsilon - p)/\varepsilon$。振荡展开需要比普通逐点收敛更精确的误差尺度。

Lemma 4.12 (Quadratic phase; source Lemma 2)
For real symmetric invertible H and $a \in C_c^\infty(\mathbb{R}^d)$,

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = (2\pi\varepsilon)^{d/2} \frac{e^{i\pi\,\operatorname{sgn} H/4}}{|\det H|^{1/2}} [a(0) + O(\varepsilon)]. \quad \text{(QP4)}$$

Expanded proof (regularization, branch and error).

For $\operatorname{Re} q > 0$,

$$\int_{\mathbb{R}} e^{-qs^2+ibs}\,ds = \sqrt{\pi}\, q^{-1/2} e^{-b^2/(4q)}.$$

Choose the square root with positive real part. For real positive q this is the Gaussian identity; analytic continuation in the right half-plane proves it for complex q, since compact subsets admit an integrable Gaussian majorant. Equivalently the contour argument of Problem 15 proves it directly.

Diagonalize H orthogonally, multiply the phase by $e^{-\eta|z|^2}$, and apply this identity in each coordinate. As $\eta \downarrow 0$, an eigenvalue λ contributes the factor $|\lambda|^{-1/2} e^{i\pi\,\operatorname{sgn}\lambda/4}$ after the common $(2\pi\varepsilon)^{1/2}$ factor. Insert Fourier inversion for a before this calculation. Its transform is rapidly decreasing. The regularized Gaussian integral is uniformly bounded in frequency, so dominated convergence yields the exact oscillatory identity

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = C_{\varepsilon,H}(2\pi)^{-d/2} \cdot \int \hat{a}(\xi) e^{-i\varepsilon\,\xi\cdot H^{-1}\xi/2}\,d\xi,$$

where $C_{\varepsilon,H}$ is the prefactor in (QP4). Since $|e^{ir} - 1| \le |r|$, the bracket differs from $a(0)$ by at most

$$C\varepsilon \int |\xi|^2 |\hat{a}(\xi)|\,d\xi < \infty.$$

This proves both the constant and the error in (QP4). □

Lemma 4.13 (Morse coordinates; source Lemma 3(ii))
If $D\phi(0) = 0$ and $H = D^2\phi(0)$ is invertible, there is a local smooth diffeomorphism Φ with $\Phi(0) = 0$, $D\Phi(0) = I$, and $\phi(\Phi(z)) = \phi(0) + z\cdot Hz/2$.

Expanded proof (an editorial matrix proof).

Taylor's formula with integral remainder gives $\phi(x) = \phi(0) + x^T A(x) x/2$, where $A(x) = 2\int_0^1 (1-s) D^2\phi(sx)\,ds$ and $A(0) = H$. On symmetric matrices solve

$$H + 2S + SH^{-1}S = A(x).$$

The derivative in S at zero is $2I$, so the implicit function theorem gives a smooth symmetric $S(x)$ with $S(0) = 0$. Set $B = I + H^{-1}S$; then $B^T H B = A$. The map $z = B(x)x$ has derivative I at zero. Its local inverse Φ has the required properties. This works for indefinite H as well as positive definite H. □

For completeness the source's noncritical coordinate lemma is also elementary. Choose rows $r_1, \dots, r_{d-1}$ complementary to $D\phi(0)$, impose $r_j \cdot (y - x) = 0$ and $\phi(y) - \phi(0) = D\phi(0) \cdot x$. The y Jacobian is invertible; the implicit function theorem gives $y = \Phi(x)$ and differentiation gives $D\Phi(0) = I$.

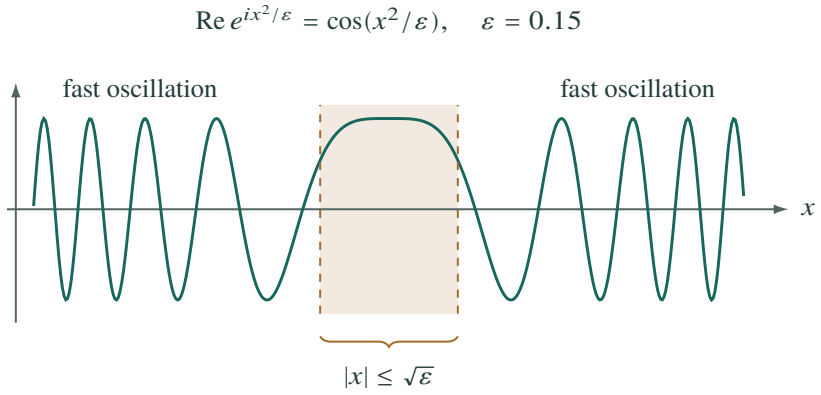


Diagram T4.4. Near a nondegenerate stationary point the scale is square-root. For phase x^2, oscillation is slow near its stationary point at zero and rapid farther away. The highlighted interval has width $2\sqrt{\varepsilon}$. Rescaling $x = \sqrt{\varepsilon}\, z$ explains the leading $\varepsilon^{1/2}$ factor in one dimension; cancellation, not pointwise smallness of the integrand, suppresses the outer region.

Theorem 4.14 (Nondegenerate stationary phase)
Suppose ϕ has just finitely many critical points z_j on supp a, *and all $H_j = D^2\phi(z_j)$ are invertible. Then*

$$\int e^{i\phi(z)/\varepsilon} a(z)\, dz = (2\pi\varepsilon)^{d/2} \sum_j \frac{e^{i\phi(z_j)/\varepsilon + i\pi\, \operatorname{sgn} H_j/4}}{|\det H_j|^{1/2}} [a(z_j) + O(\varepsilon)]. \tag{SP4}$$

Expanded proof.

Choose a partition of unity separating small neighborhoods of the critical points from the noncritical compact remainder. The latter is $O(\varepsilon^N)$ for arbitrary N by the first lemma. At each critical point change to Morse coordinates. The transformed amplitude is $a(\Phi(z))|\det D\Phi(z)|$ times its cutoff and equals $a(z_j)$ at the origin. Apply (QP4) to each localized integral and add them. Taking $N > d/2 + 1$ absorbs the noncritical remain-

der. □

Recovering the rays of Chapter 3. For high-frequency displacement $g^\varepsilon(z) = a(z)e^{ip_0(z)/\varepsilon}$ with $|Dp_0| > 0$ on $\operatorname{supp} a$ and zero initial velocity, Fourier representation splits the wave into two equal half-waves. After frequency rescaling their phases are

$$\phi_\pm(x, y, z, t) = (x - z) \cdot y \pm t|y| + p_0(z).$$

The critical equations are

$$y = Dp_0(z), \qquad x = z \mp t\,\frac{Dp_0(z)}{|Dp_0(z)|}. \tag{RAY4}$$

These are exactly the projected HJ rays for $p_t \mp |Dp| = 0$. The joint Hessian at a critical point is

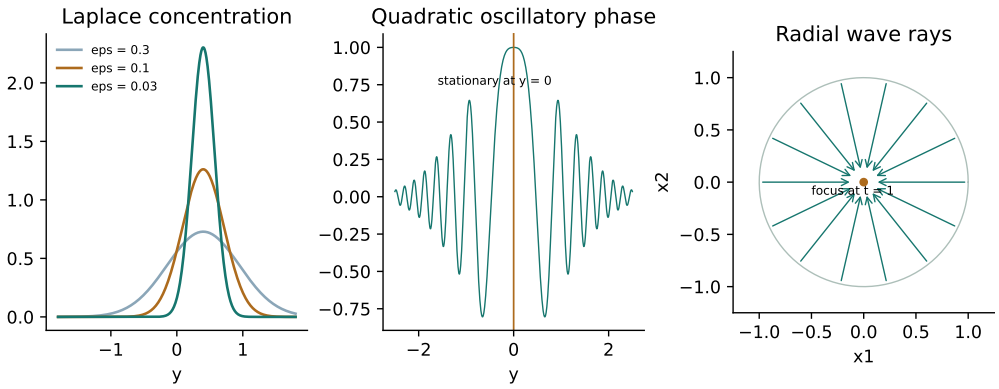
$$\begin{pmatrix} \pm\dfrac{t}{|y|}(I - \hat{y} \otimes \hat{y}) & -I \\ -I & D^2 p_0(z) \end{pmatrix}, \qquad \hat{y} = y/|y|.$$

Its determinant equals $(-1)^n$ times the Jacobian determinant of the ray map in (RAY4), by block elimination. For small time that map is locally invertible. The critical value is $p_0(z)$, since $(x-z)\cdot y \pm t|y| = 0$. Applying (SP4) in $2n$ variables cancels the half-wave prefactor $(2\pi\varepsilon)^{-n}$ and gives the amplitude $a(z)/|\det D_z x|^{1/2}$ and the Hessian signature phase. The two half-waves are then added with weight $1/2$.

The amplitude in y, z is initially not compact in y. To justify this application, first cut frequencies off to a compact neighborhood of $Dp_0(\operatorname{supp} a)$ disjoint from zero. Outside that neighborhood $D_z\phi = Dp_0 - y$ is nonzero. Repeated integration by parts in z makes the tail integrable in y, with arbitrarily high powers of ε; for large y each integration also supplies inverse powers of $|y|$. This reduces to the compactly supported theorem. The formula is local before caustics or summed over the finitely many nondegenerate preimages; it asserts no uniform approximation at a caustic.

行列式为零时不能继续用驻相公式

$|\det H|^{-1/2}$ 描述非退化临界点的集中强度。它发散时，表示这一阶近似失效；需研究更高阶相位，不能把发散系数直接当作精确解的无穷大。



Editorial figure. Laplace concentration is compared with oscillatory cancellation; the ray diagram displays a focusing map whose Jacobian eventually vanishes.

4.5.4 Periodic homogenization: expansion, cell problem, and a proof

We use the symmetric uniformly elliptic case, sufficient also for Problem 18: $A(y) = A(y)^T$ is smooth, Q-periodic, $Q = (0,1)^n$, and $\theta I \le A \le \Theta I$. Consider $-\operatorname{div}(A(x/\varepsilon)Du^\varepsilon) = f$ on a bounded Lipschitz domain with zero trace and $f \in L^2$. The source uses a formal expansion; we first derive it, then give a weak-convergence proof under these stated hypotheses.

Two-scale bookkeeping. Write $u^\varepsilon = u_0(x,y) + \varepsilon u_1(x,y) + \cdots$, $y = x/\varepsilon$, with all u_j periodic in y. Every spatial derivative becomes $D_x + \varepsilon^{-1} D_y$. Thus the three operator pieces are

$$L_1 = -D_y \cdot AD_y, \quad L_2 = -D_y \cdot AD_x - D_x \cdot AD_y, \quad L_3 = -D_x \cdot AD_x.$$

The coefficients of $\varepsilon^{-2}, \varepsilon^{-1}, 1$ give

$$L_1 u_0 = 0, \quad L_1 u_1 + L_2 u_0 = 0, \quad L_1 u_2 + L_2 u_1 + L_3 u_0 = f.$$

Testing the first identity by u_0 and integrating over Q gives $D_y u_0 = 0$; opposite periodic faces cancel. For zero-mean correctors χ_i, define

$$\operatorname{div}_y\{A(y)(e_i - D_y\chi_i)\} = 0, \qquad \int_Q \chi_i = 0. \tag{CELL4}$$

Then $u_1 = -\sum_i \chi_i(y) u_{x_i}$ solves the next equation. Averaging the order-one equation kills $L_1 u_2$ and yields

$$-\operatorname{div}(\overline{A}Du) = f, \qquad \overline{A}e_i = \int_Q A(y)(e_i - D\chi_i)\,dy. \tag{HOM4}$$

The minus sign in u_1 matches the source's corrector convention. Many other texts use $\varphi_i = -\chi_i$ and a plus sign instead.

Lemma 4.15 (Cell solvability and effective ellipticity)
(CELL4) has a unique zero-mean periodic weak solution. The matrix $\overline{A}$ is symmetric and satisfies $\theta I \le \overline{A} \le \Theta I$.

Expanded proof.

On the zero-mean periodic H^1 space minimize $\int_Q (e_i - Dv)\cdot A(e_i - Dv)$. Periodic Poincaré and ellipticity bound every minimizing sequence in H^1; weak compactness and weak lower semicontinuity give a minimizer. The first variation is $\int_Q A(D\chi_i - e_i)\cdot D\psi = 0$. Strict convexity in the gradient makes the difference of two minimizers constant, hence zero after normalization. These are standard Hilbert-space compactness and periodic Poincaré facts; the cell equation is not solved merely by asserting zero mean.

For $\xi \in \mathbb{R}^n$, let $\chi_\xi = \sum_i \xi_i \chi_i$. The cell identity allows any gradient $D\chi_\eta$ to be subtracted from the other factor, so

$$\eta \cdot \overline{A}\xi = \int_Q (\eta - D\chi_\eta)\cdot A(\xi - D\chi_\xi).$$

This proves symmetry. The minimum over all periodic v equals $\xi\cdot\overline{A}\xi$ by orthogonality. Its lower bound is $\theta\int_Q |\xi - Dv|^2 \ge \theta|\xi|^2$, since $\int_Q Dv = 0$ and $|Q| = 1$. The competitor $v = 0$ gives the upper bound. □

Theorem 4.16 (An editorial rigorous homogenization statement)
Under the symmetric hypotheses above, $u^\varepsilon \rightharpoonup u$ in H_0^1 and $u^\varepsilon \to u$ in L^2, where u is the unique weak solution of (HOM4).

Expanded proof (oscillating tests replace an unproved expansion).

Testing the original PDE by u^ε gives a uniform H_0^1 bound by ellipticity and Poincaré. Take a weakly convergent subsequence and use Rellich for strong L^2 convergence. The fluxes $\eta^\varepsilon = A(x/\varepsilon)Du^\varepsilon$ are bounded in L^2; take a weak limit η. The PDE gives $-\operatorname{div}\eta = f$.

For each i put

$$w_i^\varepsilon = x_i - \varepsilon\chi_i(x/\varepsilon), \qquad q_i^\varepsilon = A(x/\varepsilon)Dw_i^\varepsilon.$$

Periodic averaging implies $w_i^\varepsilon \to x_i$ in L^2_{loc} and $q_i^\varepsilon \rightharpoonup \overline{A}e_i$ in L^2_{loc}; also $\operatorname{div} q_i^\varepsilon = 0$. The averaging fact follows first by summing integrals on full ε-cells for continuous periodic functions and compact tests, then by $L^2(Q)$ approximation and uniform local cell bounds.

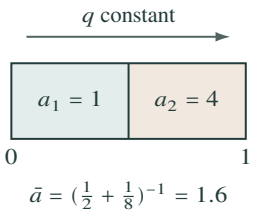
For $\phi \in C_c^\infty(U)$ symmetry gives $\eta^\varepsilon \cdot Dw_i^\varepsilon = Du^\varepsilon \cdot q_i^\varepsilon$. Integrate the two sides by parts in different ways:

$$\int \phi\, \eta^\varepsilon \cdot Dw_i^\varepsilon = \int f\phi w_i^\varepsilon - \int w_i^\varepsilon \eta^\varepsilon \cdot D\phi,$$

平均的是修正后的通量

有效系数通常不是直接对 A 求平均。微观梯度要先调整成 $e_i - D\chi_i$，使通量在单元中无散，再取平均；一维时结果是~~调和~~平均而非算术平均。

图 T4.M：一维有效系数



两个半格串联时通量相同，梯度与系数成反比。平均的是阻力 $1/a$，再取倒数。分段常系数可按弱形式理解。

$$\int \phi\, Du^\varepsilon \cdot q_i^\varepsilon = -\int u^\varepsilon q_i^\varepsilon \cdot D\phi.$$

Every product in these two right sides has a strong L^2 factor and a weak L^2 factor. The limits give $\eta_i = Du \cdot \overline{A}e_i$. By symmetry this says $\eta = \overline{A}Du$, proving (HOM4). Effective ellipticity gives uniqueness, so every subsequence has the same limit; the entire family converges. □

The expansion need not satisfy the original boundary condition termwise; boundary layers may be necessary for a quantitative error estimate. We have proved weak H^1 convergence, not strong convergence of the gradients or an $O(\varepsilon)$ global error. For nonsymmetric coefficients the source's formal cell calculation still applies with consistent index orientation; the rigorous testing argument uses adjoint correctors. That extension is not asserted here.

两个弱收敛量不能直接相乘

$A(x/\varepsilon)$ 与 Du^ε 都会振荡，不能分别取弱极限再相乘。振荡测试把乘积改写为一个强收敛因子和一个弱收敛因子的配对，才识别出有效通量。

4.6 Power Series

4.6.1 Noncharacteristic surfaces and determination of the full jet

For a quasilinear equation of order k, write

$$\sum_{|\gamma|=k} a_\gamma (D^{k-1}u, \dots, u, x) D^\gamma u + a_0(D^{k-1}u, \dots, u, x) = 0.$$

Its principal polynomial is $P(x, \zeta) = \sum_{|\gamma|=k} a_\gamma \zeta^\gamma$, with coefficients evaluated on the chosen lower-order jet. Cauchy data on a hypersurface Γ prescribe $u, \partial_\nu u, \dots, \partial_\nu^{k-1} u$. Here ∂_ν^j is the contraction of $D^j u$ with $\nu^{\otimes j}$ at the point, as in the source; iterating a variable normal field would also create lower-order curvature terms.

Theorem 4.17 (Source Theorem 1)
If $P(x, \nu) \neq 0$ along the specified data, then the PDE and the Cauchy data uniquely determine every derivative of any smooth solution along Γ. This is a determination statement, not an existence theorem.

Expanded proof.

On $\Gamma = \{x_n = 0\}$ the derivatives of the Cauchy functions in $x' = (x_1, \dots, x_{n-1})$ determine all derivatives of total order at most k, except $\partial_n^k u$. Its coefficient in the PDE is $a_{ke_n} = P(x, e_n) \neq 0$. Solving the PDE gives this missing derivative. Its tangential derivatives then give every derivative with exactly k normal differentiations. Differentiate the solved PDE once in x_n to find $\partial_n^{k+1} u$; all right-side normal orders are now at most k. Repeat, inducting on normal order and allowing arbitrary tangential differentiation at each stage.

For a curved surface use coordinates $y = \Psi(x)$ with $\Gamma = \{y_n = 0\}$. The chain rule shows that the coefficient of $\partial_{y_n}^k v$ in $u = v \circ \Psi$ is

$$\sum_{|\gamma|=k} a_\gamma (D\Psi_n)^\gamma .$$

On Γ, $D\Psi_n$ is a nonzero multiple of ν, so homogeneity and the noncharacteristic condition make this coefficient nonzero. The prescribed normal derivatives convert to y_n derivatives by a triangular system: at each order its leading coefficient is a nonzero power of $\partial_\nu \Psi_n$, while lower-order terms and tangential derivatives are already known. Apply the flat argument and transform the resulting jets back. □

For a surface $\{\rho = 0\}$, use $P(x, D\rho) \neq 0$. For the wave equation the principal polynomial is $\tau^2 - |\xi|^2$, giving characteristic light cones. For the heat equation, considered as a total-order-two PDE, the principal polynomial is $-|\xi|^2$: the time-zero surface has $(\xi, \tau) = (0, 1)$ and is characteristic. This convention does not negate forward parabolic well-posedness.

知道所有导数不等于知道函数

光滑函数可在一点具有全零 Taylor 系数而在邻域不为零，例如 e^{-1/x^2} 的零延拓。要由 jet 恢复解，需要额外的解析性或其他唯一性理论。

4.6.2 Analytic functions and coefficientwise majorants

An analytic function near zero is represented there by a convergent power series $f(x) = \sum_\gamma f_\gamma x^\gamma, f_\gamma = D^\gamma f(0)/\gamma!$. A formal series $F = \sum F_\gamma x^\gamma$ majorizes f if $F_\gamma \geq |f_\gamma|$ for every γ; write $F \gg f$.

Lemma 4.18 (Source majorant lemma, with the operations made explicit)
If F has nonnegative coefficients and converges near zero, then $F \gg f$ implies absolute convergence of f in a smaller neighborhood. Every analytic series has a majorant $C/(1 - (x_1 + \cdots + x_d)/r)$ after shrinking r. Majorization is preserved by addition, multiplication, partial differentiation, and composition with zero-constant-term series.

Expanded proof.

Coefficientwise comparison gives $\sum |f_\gamma x^\gamma| \leq F(|x_1|, \ldots, |x_d|)$. Analytic convergence on a smaller polydisc implies $|f_\gamma| \leq Cr^{-|\gamma|}$. But the coefficient of x^γ in the displayed rational series is $Cr^{-|\gamma|}|\gamma|!/\gamma! \geq Cr^{-|\gamma|}$. The multinomial expansion therefore proves the comparison. For products apply the triangle inequality to the finite coefficient convolution; for derivatives multiply each coefficient by its nonnegative integer exponent. Composition is a sum of such products. The zero constant term ensures that each fixed total-degree coefficient receives only finitely many composition contributions. These facts apply componentwise to vector and matrix series. □

4.6.3 Cauchy–Kovalevskaya: recurrence and actual convergence

After flattening an analytic noncharacteristic surface and subtracting analytic extensions of the Cauchy data, a quasilinear equation reduces to a first-order system for the derivatives of order below k. Write the normal coordinate as t and the $d = n - 1$ tangential coordinates as x':

$$U_t = \sum_{j=1}^{d} B_j(U, x')U_{x_j} + c(U, x'), \qquad U(x', 0) = 0. \tag{CK4}$$

Here U has m components. If coefficients originally depend on t, add a component $T_t = 1, T(x', 0) = 0$, and replace explicit t by T; this reduces to the displayed time-independent coefficient form.

Theorem 4.19 (Source Theorem 2)
For analytic B_j, c near $(0, 0)$, (CK4) has a unique analytic solution in a neighborhood of zero.

Expanded proof (the explicit majorant, fully verified).

1. Recursive formal solution. Write $U = \sum_{\ell \geq 1} t^\ell U_\ell(x')$. Equating the coefficient of t^ℓ gives

$$(\ell + 1)U_{\ell+1} = [t^\ell]\left\{\sum_j B_j(U, x')U_{x_j} + c(U, x')\right\}.$$

The right side only uses $U_1, \dots, U_\ell$ (and their tangential derivatives); $U_1 = c(0, x')$. Equivalently, each joint Taylor coefficient is determined recursively by smaller normal order. Composition is legitimate as a formal joint series since $U(0, 0) = 0$. No negative combinatorial coefficients occur in the operations on the right. This proves formal uniqueness, but not yet convergence.

2. Dominate the coefficient functions. Choose $r, C > 0$ so that every entry of every B_j and c is majorized by

$$F(Z, x') = \frac{Cr}{r - \sum_{j=1}^{d} x_j - \sum_{\ell=1}^{m} Z_\ell}.$$

The preceding lemma supplies such a common choice for the finite collection of analytic entries. The comparison system has all its components equal to a scalar v. With $s = \sum_j x_j$, its equation is

$$v_t = \frac{Cr}{r - s - mv}(md\, v_s + 1), \qquad v(s, 0) = 0. \tag{MAJ4}$$

控制函数值不等于控制系数

majorant 比较的是每个 Taylor 系数的绝对值。它之所以能控制递推解，是因为求导、乘积和复合只通过非负组合数来组合这些系数。

Each of the d matrix rows sums m identical derivatives, explaining the factor md.

3. Solve and check the majorant. Let $N = m(d+1)$ and set

$$v(s,t) = \frac{r - s - \sqrt{(r-s)^2 - 2NCr\,t}}{N}.$$

Use the square-root branch equal to $r - s$ near $(0,0)$. It is analytic near zero, is zero at $t = 0$, and obeys

$$(r-s)v - \frac{N}{2}v^2 = Cr\,t.$$

Implicit differentiation gives $v_t = Cr/(r-s-Nv)$ and $v_s = v/(r-s-Nv)$. Thus

$$\frac{Cr}{r-s-mv}(md\,v_s + 1) = \frac{Cr}{r-s-mv}\frac{r-s-Nv+md\,v}{r-s-Nv} = v_t,$$

because $N = md + m$. The expansion of $1 - \sqrt{1-z}$ has nonnegative coefficients; expanding $(r-s)^{-1}$ also does. Hence v is a nonnegative coefficient majorant in (x', t).

4. Compare recursively and obtain a PDE solution. At normal order zero both series vanish. Assume all coefficients up to normal order ℓ of U are dominated by those of $(v, \dots, v)$. The positive-operation rules and coefficient majorization of B_j, c give the same domination for the right side of the recurrence. Division by $\ell + 1 > 0$ preserves it. Thus the domination holds at every normal order. The convergent analytic series v bounds the absolute joint series of every component of U on a smaller polydisc. On still smaller polydiscs derivatives also converge normally. The analytic compositions are therefore genuine functions; their Taylor coefficients satisfy (CK4), so the analytic identity holds near zero. Analytic uniqueness follows from the coefficient recurrence and equality with the Taylor series. □

Returning to the scalar higher-order equation. Include every derivative $D^\gamma u$, $|\gamma| < k$, as a component. Normal derivatives of lower-order components are other components; a derivative of a top-order component either is a tangential derivative of another top-order component or, for the pure normal term, is obtained from the solved original PDE. This yields the quasilinear system above. The formal scalar jet determined by the noncharacteristic recursion satisfies these derivative identities and hence

优级数的作用是证明收敛

形式递推只确定候选解的系数。显式平方根解控制这些系数的绝对值，才把形式级数变成可求导的解析函数；数据解析本身还没有完成收敛证明。

the formal system. Its formally unique system solution is exactly the analytic series constructed above. Consequently all compatibility identities between the components hold as analytic identities. The zeroth component really is a scalar solution with the required derivatives; introducing a vector of independent unknowns did not silently discard compatibility.

For nonzero analytic initial data subtract a polynomial in t with analytic x' coefficients matching the prescribed normal jets, then incorporate its known terms into the coefficients. Analytic changes of coordinates preserve analyticity. This proves the quasilinear analytic noncharacteristic Cauchy theorem. A fully nonlinear version additionally solves its highest normal derivative by the analytic implicit function theorem and needs the corresponding nonzero derivative condition; that broader theorem is a further result, not proved by the quasilinear reduction alone.

解析存在不保证低正则范数下稳定

CK 给出非特征解析数据的局部解析解，未给出 C^0 或 Sobolev 数据到解的连续依赖。习题 2 的 Laplace Cauchy 问题正好说明存在与稳定必须分别检查。

4.7 Problems

Source: Evans §4.7, Problems 1–19. Statements are restated concisely; all solutions and domain clarifications below are editorial.

4.7.1 Separation, scaling and traveling profiles

1. Aronsson's infinity-harmonic example.

Solution. For $u = f(x) + g(y)$, the equation is $(f')^2 f'' + (g')^2 g'' = 0$. Set these terms equal to $64/81$ and $-64/81$ respectively. A resulting profile is

$$u(x, y) = |x|^{4/3} - |y|^{4/3}.$$

Off the axes, $(f')^2 f'' = 64/81$ and $(g')^2 g'' = -64/81$, so the PDE holds classically there. The function is C^1 on $\mathbb{R}^2$ but not C^2 on the axes. It is a global viscosity solution of $F(Du, D^2u) = -Du \cdot D^2u\, Du = 0$, as follows.

At $x = 0$, $y \neq 0$, no C^2 function touches u from above: the positive $|x|^{4/3}$ increment is larger than every quadratic with zero first derivative at zero. For a test ϕ touching from below, $D\phi = Du$ and $\phi_{yy} \leq u_{yy} < 0$, so $D\phi \cdot D^2\phi D\phi \leq 0$, giving $F \geq 0$. At $y = 0$, $x \neq 0$ the lower tests are absent, while an upper test has $\phi_{xx} \geq u_{xx} > 0$ and hence $F \leq 0$. At the origin every touching test has gradient zero, and $F = 0$. These are exactly the viscosity inequalities. An affine function supplies a global classical solution if that solution class is required; the nonaffine Aronsson example needs the explicit weaker interpretation. □

经典解与黏性解要分清

$|x|^{4/3} - |y|^{4/3}$ 不能在坐标轴上按经典二阶导数代入。本题用接触测试说明它的全局解意义；黏性解的一般理论留待第十章。

2. Hadamard instability of Laplace Cauchy data.

Solution. The separated ansatz $\sin(nx_1)T(x_2)$ gives $T'' = n^2 T$. The data $T(0) = 0, T'(0) = 1/n$ yield

$$u_n(x_1, x_2) = n^{-2} \sin(nx_1) \sinh(nx_2).$$

The Cauchy data tend uniformly to zero, but on any line $x_2 = h > 0$, $\sup_{x_1} |u_n| = n^{-2} \sinh(nh) \to \infty$. Thus the data-to-solution map is not continuous in these uniform data and solution norms. One must not claim pointwise divergence at every fixed x_1, since $\sin(nx_1)$ can vanish. Existence of analytic solutions does not restore this stability. □

3. A balanced bistable front.

Solution. For $f(v) = -2v^3 + 3v^2 - v$, $\int_0^1 f = 0$. The energy identity (BF4) forces speed $c = 0$ for a nonconstant front. Here $F(v) = -v^2(1 - v)^2/2$, so $(v')^2 = v^2(1 - v)^2$ and the increasing branch obeys $v' = v(1 - v)$. Hence $v(s) = (1 + e^{-(s-s_0)})^{-1}$. It has the required endpoints and vanishing derivatives. Its second derivative is $v(1 - v)(1 - 2v) = -f(v)$, verifying the ODE directly. □

4. Emden–Fowler coordinates.

Solution. Put $a = 2/(p - 1)$, $t = \log r$, and $X(t) = e^{at} v(e^t)$. Then

$$v' = r^{-a-1}(X' - aX), \quad v'' = r^{-a-2}\{X'' - (2a + 1)X' + a(a + 1)X\}.$$

Since $ap = a + 2$, the radial equation becomes the autonomous ODE

$$X'' + (n - 2 - 2a)X' + a(a - n + 2)X + X^p = 0.$$

This change treats $r > 0$; regularity at the center is an additional condition corresponding to $t \to -\infty$. □

5. Fast-diffusion scaling profile.

Solution. In the standard parabolic range $\max\{0, (n-2)/n\} < m < 1$, use $\beta = 1/(n(m-1)+2) > 0$, $\alpha = n\beta$ as in §4.2. Integrating $(W^m)' = -\beta r W$ gives

$$u(x, t) = t^{-\alpha} \left(C + \frac{(1 - m)\beta}{2m} |x|^2 t^{-2\beta} \right)^{-1/(1-m)}, \qquad C > 0.$$

It is positive and smooth at every $t > 0$. Its spatial tail is $|x|^{-2/(1-m)}$; integrability is equivalent to $2/(1-m) > n$, the stated mass-critical lower bound. Directly differentiating $W^{m-1} = C + (1-m)\beta r^2/(2m)$ verifies the profile ODE. Unlike the porous-medium profile it has no compact support. If $n = 1$, the source's written bound $(n-2)/n < m < 1$ alone also permits $m \le 0$. The above forward diffusion construction uses $m > 0$; it does not assert a decaying positive profile for those nonparabolic exponents. At $m = 0$ the positive equation is $u_t = 0$, a separate degeneracy. □

快扩散仍需正的扩散指数

扩散系数为 mu^{m-1}。$0 < m < 1$ 时在低密度处很大，产生代数尾；若 $m < 0$，最高阶符号反转，已不是同一类前向抛物扩散。

6. A boundary blowup solution.

Solution. Assume $n \ge 3$ and set $p = (n+2)/(n-2)$. For $u = A(1-r^2)^{-b}$, radial differentiation gives

$$\Delta u = 2bA(1 - r^2)^{-b-2}\{n + (2b + 2 - n)r^2\}.$$

Choose $b = (n-2)/2$ so the brace is n and $bp = b+2$. Then $-\Delta u + u^p = 0$ if

$$A = [n(n-2)]^{(n-2)/4}.$$

This is smooth and finite in the open ball but diverges at every boundary point, as required. □

7. Viscous shock profiles and the entropy direction.

Solution. For $u_t + F(u)_x - au_{xx} = 0$ put $u = v(x - ct)$. Integration yields

$$av' = F(v) - cv + b =: Q(v).$$

Separation gives $s = \int_{v(0)}^{v(s)} a/Q(z)\,dz$ wherever $Q \ne 0$. A connection from u_l to u_r forces both endpoint values of Q to vanish; otherwise v' stays away from zero near that endpoint and v cannot have the asserted limit. Subtracting the endpoint identities gives $c = (F(u_l) - F(u_r))/(u_l - u_r)$. Conversely, for $u_l > u_r$, strict convexity makes $Q < 0$ on (u_r, u_l) when b is chosen to make the endpoints zero. The zeros are simple: $Q'(u_r) < 0 < Q'(u_l)$. The integral therefore diverges to $+\infty$ as $v \downarrow u_r$ and to $-\infty$ as $v \uparrow u_l$. Its inverse is the decreasing heteroclinic profile, unique up to shift.

Normalize $v^\varepsilon(0) = (u_l + u_r)/2$. Then $v^\varepsilon(s) = v^1(s/\varepsilon)$. For $x - ct < 0$ its limit is u_l, for $x - ct > 0$ it is u_r, and on $x = ct$ it is the chosen midpoint. Boundedness gives local L^1 convergence to the Rankine–Hugoniot downward shock. The viscous connection thus selects the entropy direction. □

4.7.2 Transforms and linearization

8. Schrödinger dispersive bound.

Solution. For $g \in L^1$, the modulus of the kernel in (SC4) is $(4\pi|t|)^{-n/2}$. The triangle inequality gives $\|u(t)\|_\infty \le (4\pi|t|)^{-n/2}\|g\|_1$. The bound holds for both signs of $t \neq 0$. It is consistent with conservation of the L^2 norm, since the norms measure different features. □

9. Inverting a nonlinear heat equation.

Solution. For $y = u(x,t)$, $x = v(y,t)$ and $u_x > 0$, differentiate $u(v(y,t),t) = y$:

$$u_x = 1/v_y, \quad u_t = -v_t/v_y, \quad u_{xx} = -v_{yy}/v_y^3.$$

Substitution into $u_t = u_{xx}/u_x^2$ gives $v_t = v_{yy}$. This is local in the range of u; the inverse's spatial interval and boundary conditions may move with time. □

10. A local transformation to Liouville's equation.

Solution. Taken literally for every solution u with a smooth real $f : \mathbb{R}^3 \to \mathbb{R}$, the request is impossible: constant u would give constant $w = f(u,0,0)$, while $w_{xt} = e^w > 0$. The intended transformation must be local on a nondegenerate branch.

For $u_{xt} = 0$, write $u = A(x) + B(t)$. On $u \neq 0$, $u_x u_t > 0$, choose

$$w = f(u,u_x,u_t) = \log(2u_x u_t/u^2) = \log 2 + \log|u_x| + \log|u_t| - 2\log|u|.$$

The first two derivative terms have zero mixed derivative, while

$$(-2\log|u|)_{xt} = 2u_x u_t/u^2 = e^w.$$

This verifies the transformation on its real domain. More generally, for an additive ansatz $f = a(z) + b(p) + c(q)$, the mixed derivative equals $a''(u)u_x u_t$, which explains the logarithmic separation. □

变换的定义域不能隐去

本题对数公式只在 $u \neq 0$ 且 $u_x u_t > 0$ 的分支上成立。常数解给出一个直接检查：不存在对所有三元输入光滑、又能转换所有解的实函数。

11. Lax pairs preserve a smooth eigenvalue branch.

Solution. Differentiate $Lw = \lambda w$ and take the real Hilbert inner product with $w \neq 0$. Symmetry of L cancels the two w_t terms, leaving $\lambda' \|w\|^2 = (L_t w, w)$. If $L_t = BL - LB$, then

$$(L_t w, w) = \lambda (Bw, w) - (Bw, Lw) = 0.$$

Thus $\lambda' = 0$. For unbounded operators all these pairings require a common suitable domain and the differentiability stipulated by the exercise; formal operator symbols alone do not supply these hypotheses. □

12. The KdV commutator.

Solution. Let $D = \partial_x$, $L = -D^2 + u$, $B = -4D^3 + 6uD + 3u_x$. Using $D(av) = aDv + a_x v$ gives

$$\begin{aligned} [-4D^3, u] &= -12u_x D^2 - 12u_{xx} D - 4u_{xxx}, \\ [6uD, -D^2] &= 12u_x D^2 + 6u_{xx} D, \\ [3u_x, -D^2] &= 6u_{xx} D + 3u_{xxx}, \\ [6uD, u] &= 6uu_x. \end{aligned}$$

All derivative terms cancel, leaving $[B, L] = -u_{xxx} + 6uu_x$. Since L_t is multiplication by u_t, $(L_t - [B, L])v = (u_t + u_{xxx} - 6uu_x)v$. For a KdV solution, every eigenvalue branch meeting Problem 11's operator hypotheses is constant. This is a conditional spectral conclusion, not a domain-free proof about every possible realization of the differential expression. □

4.7.3 Asymptotics, homogenization and analytic obstructions

13. The plume's Gaussian width.

Solution. Set $A = \sigma v$ in (PL4). Then $A_t - A_{zz} + (\beta z A)_z = 0$ and $A(0) = \delta_0$. For $A = (\pi\Gamma)^{-1/2} e^{-z^2/\Gamma}$,

$$A_t/A = -\Gamma'/(2\Gamma) + z^2\Gamma'/\Gamma^2, \quad A_{zz}/A = -2/\Gamma + 4z^2/\Gamma^2.$$

Also $(\beta z A)_z/A = \beta - 2\beta z^2/\Gamma$. Both coefficients vanish precisely when $\Gamma' = 4 + 2\beta\Gamma$, with $\Gamma(0) = 0$ for the point-mass trace. Hence

$$\Gamma(t) = 4e^{2\int_0^t \beta(r)\,dr} \int_0^t e^{-2\int_0^s \beta(r)\,dr}\,ds > 0.$$

The normalized Gaussian has mass one and converges weakly to δ_0, verifying the initial condition after division by σ. □

14. A relaxation system produces logarithmic diffusion.

Solution. Write the equations as $u_t^\varepsilon + \varepsilon^{-1}u_x^\varepsilon = \varepsilon^{-2}((v^\varepsilon)^2 - (u^\varepsilon)^2)$ and the opposite equation for v^ε. Multiplying by ε^2 and using the assumed bounded remainders yields $(u^0)^2 = (v^0)^2$. Positivity forces $u^0 = v^0 = w$.

Define the bounded corrections $U^\varepsilon = (u^\varepsilon - w)/\varepsilon$ and $V^\varepsilon = (v^\varepsilon - w)/\varepsilon$; the source's remainder is only assumed to be order ε. The first equation, multiplied by ε, gives

$$(V^\varepsilon - U^\varepsilon)(u^\varepsilon + v^\varepsilon) = u_x^\varepsilon + \varepsilon u_t^\varepsilon.$$

On compact sets $w > 0$, so the denominator stays positive and $V^\varepsilon - U^\varepsilon \to w_x/(2w)$ locally uniformly. Adding the two original equations gives

$$u_t^\varepsilon + v_t^\varepsilon + \partial_x(U^\varepsilon - V^\varepsilon) = 0.$$

Pass to the limit in distributions, using the uniform derivative bounds in the hypotheses. The result is $2w_t - \partial_x(w_x/(2w)) = 0$, or

$$w_t - \tfrac{1}{4}(\log w)_{xx} = 0.$$

No separate limit of the two first-order corrections is needed. □

只能按实际余项阶数做比较

若余项仍是 ε 乘一个有界函数，就不能把固定的 u_1, v_1 自动认作唯一一阶系数。证明直接控制两者的差，再用两方程之和取极限。

15. Gaussian contour deformation.

Solution. For $q = \delta - i\lambda$, $\delta > 0$, the completed-square variable puts the contour on a translated line $e^{i\theta}\mathbb{R} + c$, where $\theta = \arg\sqrt{q}$ satisfies $|\theta| < \pi/4$. First remove the translation: close the two parallel line segments at parameters $\pm R$ by segments of bounded length. On those connectors $\operatorname{Re} z^2 \geq R^2\cos(2\theta) - CR - C$, so their integrals of e^{-z^2} tend to zero. Cauchy's theorem equates the two infinite line integrals.

Then rotate $e^{i\theta}\mathbb{R}$ onto $\mathbb{R}$. On connecting arcs of radius R within the fixed sector, $|e^{-z^2}| \leq e^{-R^2\cos(2\theta)}$; their length is $O(R)$, so they also vanish. The integral is $\sqrt{\pi}$ with the inherited orientation. This argument holds for each fixed $\delta > 0$; the limiting Fresnel branch is obtained afterward, not by claiming uniform decay on a sector boundary where the cosine is zero. □

16. Schrödinger initial recovery.

Solution. For $g \in C_c^\infty$, apply (SP4) to $\phi(y) = |x - y|^2/4$ with parameter $t > 0$. Its sole critical point is $y = x$ and Hessian is $I/2$. The factor $(4\pi t)^{n/2}e^{i\pi n/4}$ from stationary phase cancels the prefactor in (SC4), so $u(x,t) = g(x) + O(t)$, uniformly for x in compact sets. For general $g \in L^2$,

the multiplier and dominated convergence give $u(t) \to g$ strongly in L^2, as proved in §4.3. One cannot replace this by absolute-integral approximate-identity theory for a positive kernel: the Schrödinger kernel is oscillatory and is not in L^1 as a function of space. □

17. One-dimensional effective coefficient.

Solution. Set $F(x) = \int_0^x f(s)\,ds$ and $b = 1/a$. Integration of the flux equation gives

$$(u^\varepsilon)' = b(x/\varepsilon)(C_\varepsilon - F(x)), \qquad C_\varepsilon = \frac{\int_0^1 b(x/\varepsilon)F(x)\,dx}{\int_0^1 b(x/\varepsilon)\,dx}.$$

The boundary condition $\int_0^1 (u^\varepsilon)' = 0$ determines this constant. Periodic averaging gives $b(x/\varepsilon) \rightharpoonup^* \langle b\rangle$, $C_\varepsilon \to C = \int_0^1 F$. The formula is uniformly bounded in L^2, and its weak limit is $u' = \langle b\rangle(C - F)$. The zero traces and uniqueness identify the weak H_0^1 limit, with $-\overline{a}\,u'' = f$, $\overline{a} = \langle 1/a\rangle^{-1}$. For the cell problem, $a(1 - \chi')$ is constant; integrating $\chi' = 1 - \overline{a}/a$ over a period yields exactly the same harmonic mean. □

18. Primal and dual variational principles.

Solution. All averages use the unit cube, of volume one. For $\chi_\xi = \sum \xi_i \chi_i$, the cell orthogonality proves

$$\xi \cdot \overline{A}\xi = \min_{v \text{ periodic}} \int_Q (\xi + Dv) \cdot A(\xi + Dv),$$

with minimizer $v = -\chi_\xi$. For the dual problem fix η and set $\xi = \overline{A}^{-1}\eta$, $\sigma_* = A(\xi - D\chi_\xi)$. It is periodic, divergence free and has mean η. For any admissible $\sigma = \sigma_* + \tau$, $\int_Q \tau = 0$ and $\operatorname{div}\tau = 0$, so periodic integration by parts gives

$$\int_Q \tau \cdot A^{-1}\sigma_* = \int_Q \tau \cdot (\xi - D\chi_\xi) = 0.$$

Expansion of the square therefore shows that σ_* minimizes $\int_Q \sigma \cdot A^{-1}\sigma$, with minimum $\xi \cdot \overline{A}\xi = \eta \cdot \overline{A}^{-1}\eta$.

The primal competitor $v = 0$ gives $\overline{A} \le \int_Q A$. The constant dual competitor $\sigma = \eta$ gives $\overline{A}^{-1} \le \int_Q A^{-1}$. Inverting positive definite matrices reverses this order (conjugate by a square root and diagonalize), so

$$\left(\int_Q A^{-1}\right)^{-1} \le \overline{A} \le \int_Q A.$$

These are quadratic-form inequalities, not entrywise inequalities. □

19. Analytic initial data need not give a time-analytic heat solution.

Solution. The total-order-two principal polynomial of $u_t - u_{xx}$ is $-\xi^2$. At the normal to $\{t = 0\}$, $(\xi, \tau) = (0, 1)$, it vanishes. Thus this is a characteristic surface for the CK theorem.

If a solution with $u(x, 0) = 1/(1 + x^2)$ were analytic near $(0, 0)$, the PDE would give

$$\partial_t^k u(0, 0) = \partial_x^{2k}(1 + x^2)^{-1}|_{x=0} = (-1)^k (2k)!.$$

Its time Taylor series along $x = 0$ would be $\sum_{k\geq 0}(-1)^k (2k)! t^k/k!$. For every $t \neq 0$, successive term magnitudes have ratio $|t|(2k+2)(2k+1)/(k+1) \to \infty$, so terms do not even tend to zero. There is no analytic solution in a neighborhood of $(0, 0)$, and therefore none on all $\mathbb{R}^2$. The bounded forward heat solution still exists and is smooth for $t > 0$; it is time analyticity across the initial surface that fails. □

数据解析与跨初始面的解析性不同

空间初值在原点附近解析，却可能因连续两次空间求导反复产生过快的阶乘增长，使时间 Taylor 级数发散。这与正时间的热方程光滑化没有矛盾。

4.8 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 4, printed pp. 159–236 (local PDF pp. 176–253). Section order and explicit source theorem numbers follow that edition; exposition, proof expansions, Chinese annotations, figures and exercise solutions are editorial reconstructions.

The source's §4.8 directs readers to Pinsky, Strauss, Thoe and Zachmanoglou, and Weinberger for separation; Grindrod and Nishiura for Turing instability; Linares–Ponce, Tao and Olver for dispersive equations and symmetries; Stein–Weiss, Stein, Hörmander, Rauch, Treves and Zworski for Fourier methods; Helgason, John and Lax for Radon methods; and Courant–Hilbert for hodograph and Legendre transforms. Its asymptotics discussion credits Neu, Ralston and Bensoussan–Lions–Papanicolaou; its analytic Cauchy theory refers to Folland, John and DiBenedetto. These are source reading directions, not claims of independent consultation of every listed work. The Aronsson and Hadamard exercise attributions are retained; the source credits Kurtz for Problem 14 and Kohn for Problem 18.

The Chinese chapter opening (PDF p. 129) lists the section starts at printed pp. 111, 116, 124, 136, 140, 154, 162 and 165. Selected scanned pages and the exact review range are recorded in the chapter log. A line-by-line comparison of the whole Chinese chapter has not been made.

Scope of the proofs. Spectral basis existence, periodic Poincaré, weak Hilbert compactness, Rellich and standard local hyperbolic ODE manifold theory are explicitly invoked prerequisites. Formal plume and geometric-optics expansions are kept distinct from convergence theorems. Stationary phase is proved for smooth compact amplitudes and nondegenerate critical points. The homogenization proof uses the symmetric uniformly elliptic periodic setting; it makes no global boundary-layer error claim. CK is proved in its analytic quasilinear noncharacteristic setting, not for arbitrary smooth data. Problems 1, 5 and 10 explicitly record the relevant solution classes or transformation domains instead of silently using formulas outside their domains. Numerical model checks and typesetting checks do not constitute independent mathematical review.

Part II

Theory for Linear Partial Differential Equations

Chapter 5

Sobolev Spaces

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 5; printed pp. 239–292; PDF pages 256–309.

Chinese reference: 中译本：书内第 169–207 页；PDF 第 187–225 页。

Editorial study edition. This is an original expanded exposition following Evans, Chapter 5, rather than a verbatim transcription. Source section and theorem identifiers are given in headings. Proofs, worked solutions, and Chinese annotations are editorial reconstructions. Our theorem numbers run throughout the chapter and differ from the source. The 21 source problems are concisely restated with editorial solutions in §5.10.

The model is the Dirichlet problem $-\Delta u = f$. Its weak formulation is

$$\int_U Du \cdot D\phi \, dx = \langle f, \phi \rangle, \qquad \phi \in H_0^1(U).$$

Only one derivative of u is required. This chapter makes each symbol precise and explains which estimates survive limits of approximate solutions.

先看整条逻辑链

弱导数容纳能量有限但不光滑的函数；逼近让经典计算可用；延拓处理边界；迹解释边界值；嵌入与紧性负责估计和取极限。

Reading route and prerequisites. Read §5.2–§5.7 in order; use §5.8 as a toolbox and §5.9 for evolution equations. Unless stated otherwise, $U \subset \mathbb{R}^n$ is open, derivatives are weak, and functions equal almost everywhere are identified. The notation $V \Subset U$ means $\overline{V}$ is compact and contained in U. Constants C are independent of the functions being estimated. Basic integration theory, Hölder and Minkowski inequalities, completeness of L^p, smooth partitions of unity, and explicitly named functional analysis theorems are prerequisites; they are not proved anew here.

5.1 Hölder Spaces

5.1.1 Definitions and a complete function space

Definition 5.1 (Evans §5.1)
For $0 < \gamma \le 1$, define

$$[u]_{C^{0,\gamma}(\overline{U})} = \sup_{x \neq y \in U} \frac{|u(x) - u(y)|}{|x - y|^{\gamma}}.$$

The space $C^{k,\gamma}(\overline{U})$ consists of C^k functions on U whose derivatives through order k extend continuously to $\overline{U}$ and have finite norm

$$\|u\|_{C^{k,\gamma}} = \sum_{|\alpha| \le k} \|D^{\alpha} u\|_{\infty} + \sum_{|\alpha| = k} [D^{\alpha} u]_{C^{0,\gamma}}.$$

Here $C^{0,1}$ means Lipschitz continuous.

For $0 < \gamma < 1$, $|x|^{\gamma}$ on $[-1, 1]$ is γ-Hölder but not Lipschitz at zero. Concavity gives $|a^{\gamma} - b^{\gamma}| \le |a - b|^{\gamma}$ for $a, b \ge 0$, whereas $|h|^{\gamma}/|h| \to \infty$. This distinguishes a quantitative modulus of continuity from differentiability.

半范数为何不够

常数函数的 Hölder 半范数为零，故必须加上 $\|u\|_{\infty}$ 才成为范数。γ 控制的是振荡速度，不是函数大小。

Theorem 5.2 (Evans §5.1, Theorem 1)
$C^{k,\gamma}(\overline{U})$ is a Banach space.

Expanded proof (editorial; also source Problem 1).

Let (u_j) be Cauchy. Each $D^{\alpha} u_j$, $|\alpha| \le k$, converges uniformly on $\overline{U}$ to a continuous bounded function g_{α}. Put $u = g_0$. We must verify that these limits are its derivatives.

Fix a closed ball contained in U. For a coordinate segment in the ball,

$$D^{\alpha} u_j(x + h e_i) - D^{\alpha} u_j(x) = \int_0^h D^{\alpha + e_i} u_j(x + t e_i)\, dt \quad (|\alpha| < k).$$

Uniform convergence allows passage through the finite integral. The fundamental theorem of calculus yields $\partial_i g_{\alpha} = g_{\alpha + e_i}$. Induction in the derivative order proves $u \in C^k(U)$ with the required extensions.

Given $\varepsilon > 0$, choose N with $\|u_j - u_l\|_{C^{k,\gamma}} \le \varepsilon$ for $j, l \ge N$. Pass $l \to \infty$ in the difference quotients for each fixed pair x, y, then take suprema. Lower semicontinuity of the finite sum of norm components gives $\|u_j - u\|_{C^{k,\gamma}} \le \varepsilon$. Thus u belongs to the space and the convergence is in norm. The norm axioms follow from the triangle inequality and the included supremum norm. □

Interpolation (source Problem 2). For $0 < \beta < \gamma < 1$, set $\theta = (\gamma - \beta)/(1 - \beta)$. For every $x \neq y$,

$$\frac{|u(x) - u(y)|}{|x - y|^{\gamma}} = \left(\frac{|u(x) - u(y)|}{|x - y|^{\beta}} \right)^{1-\theta} \left(\frac{|u(x) - u(y)|}{|x - y|} \right)^{\theta}.$$

Hence $[u]_{\gamma} \leq [u]_{\beta}^{1-\theta} [u]_{1}^{\theta}$. The two-term Hölder inequality on the supremum and seminorm components also gives $\|u\|_{C^{0,\gamma}} \leq \|u\|_{C^{0,\beta}}^{1-\theta} \|u\|_{C^{0,1}}^{\theta}$. For $\gamma = 1$ this is an identity.

5.2 Sobolev Spaces

5.2.1 Weak derivatives: transfer derivatives to test functions

Definition 5.3 (Evans §5.2.1)
For $u, v \in L^1_{\mathrm{loc}}(U)$, write $D^{\alpha} u = v$ if

$$\int_U u D^{\alpha} \phi \, dx = (-1)^{|\alpha|} \int_U v \phi \, dx \qquad (\phi \in C_c^{\infty}(U)).$$

The function v is a weak derivative. A distributional derivative always exists as a functional; it need not be represented by a locally integrable function.

For a classical $C^{|\alpha|}$ function this identity follows by $|\alpha|$ integrations by parts, each contributing one minus sign. Thus the two notions agree whenever classical derivatives exist.

Lemma 5.4 (Evans §5.2.1: uniqueness)
A weak derivative is unique up to equality almost everywhere.

Expanded proof.

If v, w both work, $g = v - w$ satisfies $\int g\phi = 0$ for every test function. Take a nonnegative smooth kernel ρ, supported in the unit ball with integral one, and set $\rho_{\varepsilon}(x) = \varepsilon^{-n} \rho(x/\varepsilon)$. On $V \Subset U$, for sufficiently small ε, testing with $\rho_{\varepsilon}(x - \cdot)$ gives $g * \rho_{\varepsilon} = 0$. Approximate identities converge to g in $L^1(V)$, so $g = 0$ there. A countable exhaustion of U proves the assertion. The approximate-identity fact follows from

$$\|\rho_{\varepsilon} * g - g\|_{L^1(V)} \leq \int \rho(z) \|g(\cdot - \varepsilon z) - g\|_{L^1(V)} \, dz \to 0,$$

using a cutoff on a larger compact set, translation continuity in L^1, and domination by twice that cutoff's L^1 norm. □

导数的极限要认领

一致收敛保证候选导数连续；线段积分恒等式保证它确实是极限函数的导数。仅凭 $u_j \to u$ 一致，不能推出 $Du_j \to Du$。

为何没有边界项

试验函数的支集紧含于 U，所以在边界附近恒为零。弱导数的定义不规定 u 的边界值；边界条件要靠 $W_0^{1,p}$ 和迹来表达。

Corners, jumps, and singularities: editorial examples. On $(-1,1)$, splitting the integral at zero gives

$$\int_{-1}^{1} |x|\phi'(x)\,dx = -\int_{-1}^{1} \operatorname{sgn}(x)\phi(x)\,dx.$$

The interface terms vanish because $|0| = 0$. A corner therefore has a bounded weak first derivative. Its second distributional derivative is $2\delta_0$, which is not a locally integrable function.

For $H = \mathbf{1}_{(0,1)}$ on $(-1,1)$, $-\int H\phi' = \phi(0)$. If some locally integrable g represented this, choose $|\phi_\varepsilon| \le 1$, supported in $(-\varepsilon,\varepsilon)$, with $\phi_\varepsilon(0) = 1$. Then $|\int g\phi_\varepsilon| \le \int_{-\varepsilon}^{\varepsilon} |g| \to 0$, a contradiction.

On $B_1 \subset \mathbb{R}^n$, $n \ge 2$, put $u(x) = |x|^{-a}$, $a > 0$. Away from zero, $|Du| = a|x|^{-a-1}$. For finite p,

$$\int_{B_1} |u|^p < \infty \iff ap < n, \qquad \int_{B_1} |Du|^p < \infty \iff (a+1)p < n.$$

Under the second condition, integration by parts outside B_ε leaves an inner boundary term at most $C\varepsilon^{n-1-a}\|\phi\|_\infty \to 0$. This identifies the weak gradient. Equality of the exponent gives the divergent integral $\int_0^1 dr/r$.

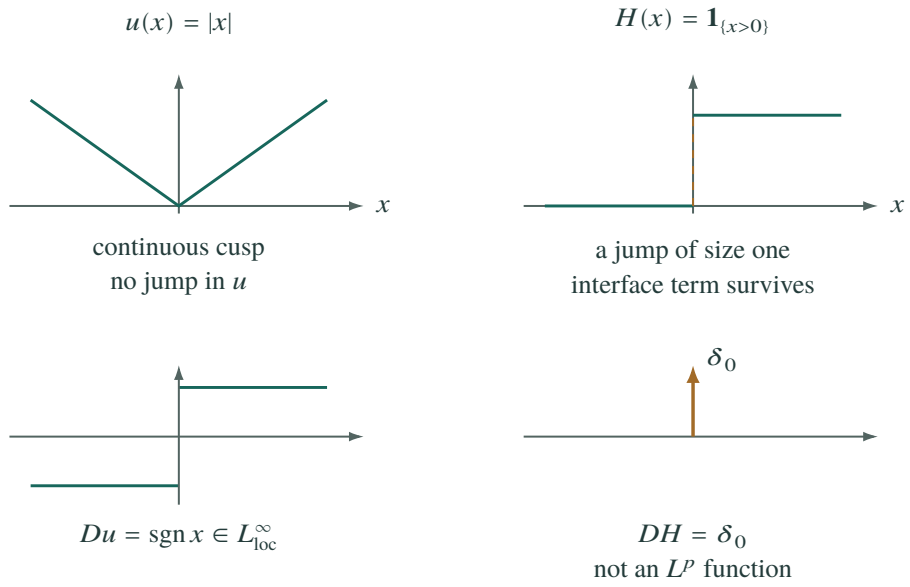


Diagram T5.1. A corner and a jump have different distributional derivatives. The continuous corner has a bounded weak first derivative. The step has a Dirac mass, so it has a distributional derivative but no weak first derivative representable by an L^p function. The vertical Dirac arrow denotes mass, not a graph of a finite-valued function.

尖角与跳跃

连续的分段光滑函数，其接口项可抵消；阶跃函数的跳跃留下 Dirac 质量。几乎处处有经典导数，不保证这个导数就是弱导数。

5.2.2 Spaces, norms, and convergence (Evans §5.2.2)

Definition 5.5

For an integer $k \geq 0$ and $1 \leq p \leq \infty$, $W^{k,p}(U)$ is the space of L^p functions whose weak derivatives through order k are all in $L^p(U)$. Use the norms

$$\|u\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_p^p \right)^{1/p} \qquad (p < \infty),$$

$$\|u\|_{W^{k,\infty}} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_\infty.$$

Write $H^k = W^{k,2}$. Define $W_0^{k,p}(U)$ as the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$. Local membership means membership on every $V \Subset U$. Functions equal almost everywhere are identified.

Our $p = \infty$ norm is equivalent to the source's convention. Strong convergence means L^p norm convergence of every derivative. For $1 < p < \infty$, weak convergence means weak L^p convergence of every component. The inner product on H^k is $\sum_{|\alpha| \leq k} \int_U D^\alpha u D^\alpha v$.

5.2.3 Calculus and completeness (Evans §5.2.3)

Theorem 5.6 (Source Theorem 1: weak calculus)

Weak derivatives are linear, restrict to open subsets, and commute: $D^\beta D^\alpha u = D^{\alpha+\beta} u$ if the total order is at most k. For $\zeta \in C_c^\infty(U)$,

$$D^\alpha(\zeta u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \zeta D^{\alpha-\beta} u.$$

Expanded proof.

Add the defining identities for linearity. A test function on an open subset extends by zero to one on U, proving locality. For commutation,

$$\int D^\alpha u D^\beta \phi = (-1)^{|\alpha|} \int u D^{\alpha+\beta} \phi = (-1)^{|\beta|} \int D^{\alpha+\beta} u \phi.$$

For the product, test with $\zeta\phi$:

$$\int \zeta u \partial_i \phi = \int u \partial_i(\zeta\phi) - \int u(\partial_i \zeta)\phi = -\int (\zeta \partial_i u + u \partial_i \zeta)\phi.$$

Iterate. The coefficients combine by the multiindex Pascal identity, giving the formula. All terms lie in L^p because all cutoff derivatives are bounded. □

零下标是闭包

先用紧支光滑逼近定义 $W_0^{1,p}$，不能提前把它解释成逐点边界值为零。有界 C^1 域、$p < \infty$ 时，迹定理才证明两种表述等价。

Theorem 5.7 (Source Theorem 2)
$W^{k,p}(U)$ is complete, including $p = 1$ and $p = \infty$.

Expanded proof.

The norm axioms follow from those of the finite product of L^p spaces. If (u_j) is Cauchy, completeness of L^p gives $D^\alpha u_j \to g_\alpha$ in L^p for every $|\alpha| \le k$. Put $u = g_0$. For $\phi \in C_c^\infty(U)$,

$$\int u D^\alpha \phi = \lim_j \int u_j D^\alpha \phi = (-1)^{|\alpha|} \lim_j \int D^\alpha u_j \phi = (-1)^{|\alpha|} \int g_\alpha \phi.$$

Hölder justifies each limit on the test function's compact support; for $p = 1$ use its L^∞ norm, and for $p = \infty$ its L^1 norm. Thus $g_\alpha = D^\alpha u$ and the finite sum of component errors tends to zero. □

For $1 < p < \infty$, the product of L^p spaces is reflexive, hence its closed Sobolev subspace is reflexive. Bounded sequences have weakly convergent subsequences. This uses reflexivity and weak compactness from functional analysis; strong convergence still requires additional work.

闭图思想

把 u 映到 $(u, Du, \dots, D^k u)$，所得像是有限个 L^p 的乘积中的闭子空间。闭性靠弱导数恒等式取极限，而不是对极限逐点求导。

5.3 Approximation

5.3.1 Interior smoothing (Evans §5.3.1, Theorem 1)

Use the kernel ρ above. If $u \in W^{k,p}(U)$, $V \Subset U$, and $\varepsilon < \operatorname{dist}(V, \partial U)$, then

$$D^\alpha(\rho_\varepsilon * u) = \rho_\varepsilon * D^\alpha u \quad \text{on } V,$$

$$\rho_\varepsilon * u \to u \quad \text{in } W^{k,p}(V) \quad (1 \le p < \infty).$$

卷积不能越界

卷积会读取半径 ε 内的数据。任意补零可能在边界制造跳跃，因此不能直接把补零后的 Du 当成补零函数的导数。

Expanded proof.

For fixed $x \in V$, differentiate the kernel under the integral; its smooth compact support and local integrability of u justify this. Since $D_x^\alpha \rho_\varepsilon(x - y) = (-1)^{|\alpha|} D_y^\alpha \rho_\varepsilon(x - y)$, the weak identity in y transfers derivatives to u and cancels the signs.

For $f = D^\alpha u$, multiply by a cutoff equal to one near $\overline{V}$ and extend as an L^p function to $\mathbb{R}^n$. Minkowski gives

$$\|\rho_\varepsilon * f - f\|_{L^p(V)} \le \int \rho(z) \|f(\cdot - \varepsilon z) - f\|_{L^p(V)} \, dz.$$

Translations are continuous in L^p for finite p: prove this first for continuous compactly supported functions by uniform continuity, then use their L^p density and the translation isometry. The integrand is dominated by $2\|f\|_p$, so dominated convergence applies. Sum the finitely many derivative errors. □

5.3.2 Smooth in the interior (Evans §5.3.2, Theorem 2)

Theorem 5.8 (Meyers–Serrin approximation)
For any open U and finite p, $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$.

Expanded proof.

Choose a locally finite smooth partition (ζ_j) with $\sum_j \zeta_j = 1$ on U, each supported compactly in a relatively compact patch A_j. Slight enlargements of these patches can also be chosen locally finite, by using layers of an exhaustion of U. Extend $f_j = \zeta_j u$ by zero. There are no interface derivatives: the cutoff vanishes in a neighborhood of the edge of its patch.

Given $\delta > 0$, choose radii ε_j so small that

$$\operatorname{supp}(\rho_{\varepsilon_j} * f_j) \subset A_j, \qquad \|\rho_{\varepsilon_j} * f_j - f_j\|_{W^{k,p}(\mathbb{R}^n)} < \delta 2^{-j}.$$

Then $v = \sum_j \rho_{\varepsilon_j} * f_j$ is locally a finite sum of smooth functions. The error series converges absolutely in the Banach space $W^{k,p}$, with norm at most δ. Locally it equals $v - u$, since $\sum_j f_j = u$ locally. Uniqueness of distributions identifies its global sum with $v - u$. Thus $\|v - u\|_{W^{k,p}(U)} \le \delta$. □

5.3.3 Smooth up to the boundary (Evans §5.3.3, Theorem 3)

Theorem 5.9
If U is bounded with C^1 boundary and $p < \infty$, restrictions of functions smooth in a neighborhood of $\overline{U}$ are dense in $W^{k,p}(U)$.

Expanded proof.

Use a finite partition on $\overline{U}$, subordinate to interior patches and boundary graph patches. In a rotated boundary patch write $U = \{x_n > g(x')\}$ with $g \in C^1$. On the compact base, g is Lipschitz with a constant L. Localize to $f = \zeta u$ supported away from the lateral edges. Translate inward: $f_\delta(x) = f(x + \delta e_n)$.

For $x_n \ge g(x')$ and $|z| < c\delta$,

$$x_n + \delta - z_n - g(x' - z') \ge \delta - (1 + L)|z| > 0 \quad \text{if } c < 1/(1 + L).$$

Thus a small convolution of f_δ is defined in a neighborhood of the closed patch. The translated weak derivatives are $D^\alpha f(x + \delta e_n)$, by a change of test variables. Extend each $D^\alpha f$ by zero only for estimating L^p translations. Translation continuity proves $f_\delta \to f$ in $W^{k,p}$ on the original patch intersected with U. For each fixed δ, choose a smaller convolution radius

每块半径不同

靠近边界的块需要更小的 ε_j。误差预算 $\delta 2^{-j}$ 控制无穷多块的总误差；局部有限性保证内点附近只相加有限项。

giving any desired Sobolev error. The result is smooth across the original boundary. Approximate interior patches by ordinary convolution and sum the finitely many pieces. □

Why the endpoint $p = \infty$ is excluded. The derivative of $|x|$ jumps from -1 to 1. No continuous derivative can approximate it in essential supremum norm with error less than one: the two one-sided requirements contradict continuity at zero. Smooth functions need not be norm dense in $W^{1,\infty}$. Local smoothing does converge in every finite $W^{1,q}$ norm on bounded interior sets and remains uniformly bounded in $W^{1,\infty}$. Later endpoint arguments use those facts instead.

两种稠密性不同

$C^\infty(\overline{U})$ 稠密不等于 $C_c^\infty(U)$ 稠密。常数 1 可光滑逼近，但在有界光滑域中不属于 $W_0^{1,p}$；它的迹不是零。

5.4 Extensions

5.4.1 Reflection and flattening (Evans §5.4, Theorem 1)

Theorem 5.10 (Extension)
Let U be bounded with C^1 boundary, and fix V with $U \Subset V$. For $1 \le p \le \infty$ there is a linear operator

$$E : W^{1,p}(U) \longrightarrow W^{1,p}(\mathbb{R}^n), \qquad Eu = u \text{ on } U,$$

with support in a fixed compact subset of V and $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \le C\|u\|_{W^{1,p}(U)}$. The constant depends on U, V, n, p.

Expanded proof.

先比较两种延拓

反射延拓在接口两侧保持同一函数值；补零只有在迹为零时才不会制造跳跃。延拓算子要同时控制函数和梯度，不能只保持 L^p 范数。

1. A flat boundary. On the upper half-space, for a smooth localized function define

$$\tilde{u}(x', t) = u(x', |t|).$$

For $t < 0$, its tangential derivatives are $(\partial_i u)(x', -t)$ and its normal derivative is $-(\partial_n u)(x', -t)$. Integrating against a test function separately in $t > 0$ and $t < 0$, the interface terms cancel because the values at $t = 0$ agree. Thus these piecewise derivatives are the weak derivatives. For finite p, reflection doubles both the p-th power of the L^p norm and the gradient integral; for $p = \infty$ it preserves their bounds.

2. Flatten a graph. In a boundary chart let $U = \{x_n > g(x')\}$ and set

$$\Phi(y', y_n) = (y', y_n + g(y')).$$

The map and its inverse have bounded first derivatives on the smaller chart, and $\det D\Phi = 1$. For smooth u, $D(u \circ \Phi) = (D\Phi)^T (Du) \circ \Phi$, so $\|u \circ \Phi\|_{W^{1,p}} \le C\|u\|_{W^{1,p}}$, with an analogous inverse bound. For Sobolev u, interior smooth approximation proves the same formula locally in the chart; the global integrability bounds then identify its weak derivatives on the entire chart. Reflect $u \circ \Phi$ and compose with Φ^{-1}.

3. Localize and assemble. Take finitely many smaller boundary charts and an interior patch covering $\overline{U}$, with a smooth partition (ζ_j). Extend each $\zeta_j u$ by the preceding construction, multiplying by a fixed exterior cutoff supported in its chart. The interior term extends by zero. Sum the pieces. On U the sum equals u, because $\sum_j \zeta_j = 1$. The product rule and finitely many chart estimates give the asserted norm bound. Choose all exterior cutoffs inside V. This construction is linear and uses fixed charts and cutoffs.

For finite p, one can also construct first on smooth functions and complete using Theorem 5.9. For $p = \infty$, apply the same reflection formula directly. Locally approximate in finite $W^{1,q}$ norms and pass to the weak identity; the candidate derivatives have the uniform L^∞ bounds already calculated. No $W^{1,\infty}$ norm density is needed. □

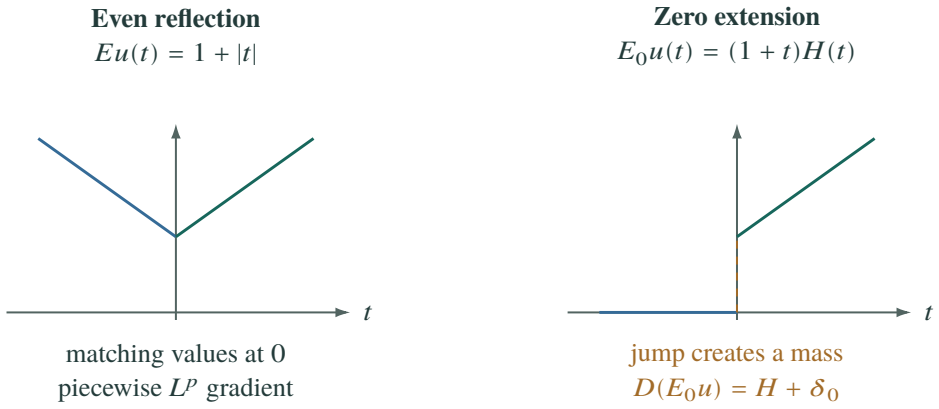


Diagram T5.2. Reflection and zero extension treat the interface differently. Take $u(t) = 1 + t$ for $0 < t < 1$ and inspect only the interface at zero. Reflection matches values and preserves first-order Sobolev regularity locally. Zero extension inserts a jump and a Dirac mass in the first derivative. Even reflection still has a corner, so higher-order extensions need additional matching.

边界正则性用在哪里

C^1 使边界可拉平，坐标变换及逆变换的一阶导数有界；这才保证 Sobolev 范数不被放大到无穷。尖裂缝域不能直接套用此结论。

5.4.2 Higher derivatives: match the jets

Ordinary even reflection need not preserve $W^{2,p}$: for $u(t) = t$ on $t > 0$ it gives $|t|$, whose second derivative has a Dirac mass. For an integer $k \ge 1$,

on $t < 0$ use instead

$$\tilde{u}(x', t) = \sum_{j=1}^{k} a_j u(x', -jt), \qquad \sum_{j=1}^{k} a_j (-j)^{\ell} = 1 \quad (0 \le \ell < k).$$

The Vandermonde matrix is invertible because $-1, \dots, -k$ are distinct. The derivatives of normal order $0, \dots, k-1$ match at zero, so repeated integration by parts has no interface distributions through order k. Each derivative on the lower side is bounded by a finite sum of scaled upper-side derivatives. This proves a $W^{k,p}$ bound on a sufficiently small strip. For $k = 2$ the formula is $3u(x', -t) - 2u(x', -2t)$.

If ∂U is C^k, flattening has bounded derivatives through order k on the chosen compact charts. The higher chain rule and the same partition construction produce an operator bounded simultaneously on $W^{j,p}$ for $0 \le j \le k$. For finite p, smooth boundary approximation passes the identities to the completion; for $p = \infty$, finite-exponent local convergence and the uniform derivative bounds give the same conclusion. We will use a single such operator on L^2 and H^2 in the time-dependent estimates.

高阶延拓不能照抄偶反射

一阶只需接口函数值匹配；二阶还需一阶导数匹配。系数 $3, -2$ 同时满足 $3 - 2 = 1$ 和 $-3 + 4 = 1$，正是为了消掉接口奇异项。

5.5 Traces

5.5.1 Boundary values as a continuous operator

Theorem 5.11 (Evans §5.5, Theorem 1)
If U is bounded with C^1 boundary and $1 \le p < \infty$, there is a unique bounded linear map

$$T : W^{1,p}(U) \longrightarrow L^p(\partial U)$$

agreeing with restriction for functions in $W^{1,p}(U) \cap C(\overline{U})$. In particular, $\|Tu\|_{L^p(\partial U)} \le C \|u\|_{W^{1,p}(U)}$.

Expanded proof.

边界不能随意赋值

改变 u 在零测集上的值不改变 Sobolev 元素；边界通常正是体积零测集。迹由域内逼近的极限决定，不能通过任意指定边界代表值来定义。

1. A flat strip estimate. For smooth v on a strip $Q' \times [0,a]$, the fundamental theorem gives

$$|v(x',0)|^p \le |v(x',t)|^p + p\int_0^a |v|^{p-1}|\partial_n v|(x',s)\,ds.$$

For $p = 1$, use the absolutely continuous function $|v|$ and its derivative bound $|(|v|)'| \le |v'|$. Average t over $(0,a)$ and integrate x'. For $p > 1$, Young's inequality bounds $p|v|^{p-1}|\partial_n v|$ by $C_p(|v|^p + |Dv|^p)$. Consequently

$$\int_{Q'} |v(x',0)|^p\,dx' \le C_{p,a}\int_{Q'\times(0,a)} (|v|^p + |Dv|^p)\,dx.$$

Boundary flattening introduces only bounded Jacobian and derivative factors, including $\sqrt{1+|Dg|^2}$ for surface measure. A finite partition yields the global estimate for smooth u.

2. Complete the restriction map. Choose u_j smooth near $\overline{U}$, with $u_j \to u$ in $W^{1,p}$. The estimate on $u_j - u_l$ shows that $u_j|_{\partial U}$ is Cauchy in $L^p(\partial U)$. Define Tu to be its limit. The same estimate on two approximating sequences proves independence of choice and boundedness. Linearity and uniqueness follow from density.

3. Agreement with continuous functions. For u continuous on $\overline{U}$, the inward-translation and smoothing construction in §5.3 can be chosen to converge uniformly as well as in $W^{1,p}$. Uniform continuity on the compact closure justifies this assertion for each localized piece. The boundary restrictions therefore converge to $u|_{\partial U}$ uniformly, identifying the trace. □

5.5.2 Zero trace and the closure definition

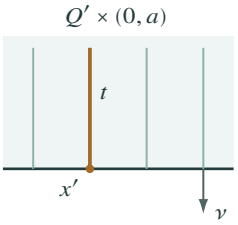
> **Theorem 5.12** (Evans §5.5, Theorem 2)
> *Under the same hypotheses,*
>
> $$W_0^{1,p}(U) = \{u \in W^{1,p}(U) : Tu = 0\}.$$

Expanded proof.

If $u_j \in C_c^\infty(U)$ tends to u in $W^{1,p}$, boundedness of T gives $Tu = \lim Tu_j = 0$.

For the reverse direction we supply the boundary-layer argument. In a flat chart the trace equals the value at $t = 0$ of the absolutely continuous

图 T5.M：沿法向控制边界值



先沿固定 x' 的线段比较内部值与边界值，再对高度和切向变量积分。图中外法向朝下。

迹估计的实际用处

只要 $u_j \to u$ 于 $W^{1,p}$，就自动有 $Tu_j \to Tu$ 于边界 L^p。边界条件因此可随强逼近传递，而不要求逐点追踪边界值。

representative on almost every vertical line. To justify this, use smooth approximations, Fubini, and the one-dimensional estimate

$$|w(0)| \le a^{-1}\|w\|_{L^1(0,a)} + \|w'\|_{L^1(0,a)}.$$

A subsequence converges in $W^{1,p}$ on almost every line, and this estimate identifies the endpoint limit with the trace limit.

Thus if $Tv = 0$, $v(x',t) = \int_0^t \partial_n v(x',s)\,ds$ on almost every such line. Hölder (or the integral triangle inequality for $p = 1$) gives

$$\int_{Q'}\int_0^{2\delta} |v(x',t)|^p\,dt\,dx' \le (2\delta)^p \int_{Q'}\int_0^{2\delta} |\partial_n v(x',t)|^p\,dt\,dx'.$$

Choose a smooth $\chi_\delta(t)$ equal to zero for $t < \delta$, one for $t > 2\delta$, with $|\chi_\delta'| \le C/\delta$. Then

$$D(\chi_\delta v) - Dv = (\chi_\delta - 1)Dv + vD\chi_\delta.$$

The first term tends to zero by absolute continuity of the integral. The p-th power of the second norm is bounded by $C\int_{0<t<2\delta}|Dv|^p$, hence also tends to zero. The function error tends to zero in the same way. Flattening and a finite partition transfer this argument to U. It gives functions supported a positive distance from ∂U converging to u in $W^{1,p}$. For each one, convolution with a smaller radius produces an element of $C_c^\infty(U)$ with arbitrarily small additional error. A diagonal choice completes the proof. □

截断产生的危险项

$D(\chi_\delta u)$ 中有 $uD\chi_\delta$，而 $D\chi_\delta$ 约为 $1/\delta$。零迹给出的法向积分公式补回一个 δ 因子，才使此项趋于零。

Zero extension. If $u \in W_0^{1,p}(U)$, extend compactly supported approximations by zero. They are Cauchy in $W^{1,p}(\mathbb{R}^n)$ with exactly the same norms, so completeness gives a zero extension $\tilde{u}$ satisfying $D\tilde{u} = \widetilde{Du}$. This argument works without regularity of U for finite p. A nonzero trace generally produces a boundary distribution instead.

5.6 Sobolev Inequalities

5.6.1 Scaling predicts the exponent

For $u_\lambda(x) = u(\lambda x)$ on $\mathbb{R}^n$,

$$\|u_\lambda\|_q = \lambda^{-n/q}\|u\|_q, \qquad \|Du_\lambda\|_p = \lambda^{1-n/p}\|Du\|_p.$$

An estimate $\|u\|_q \le C\|Du\|_p$ with a constant independent of scale must therefore have $1/q = 1/p - 1/n$. For $p < n$ put $p^* = np/(n-p)$.

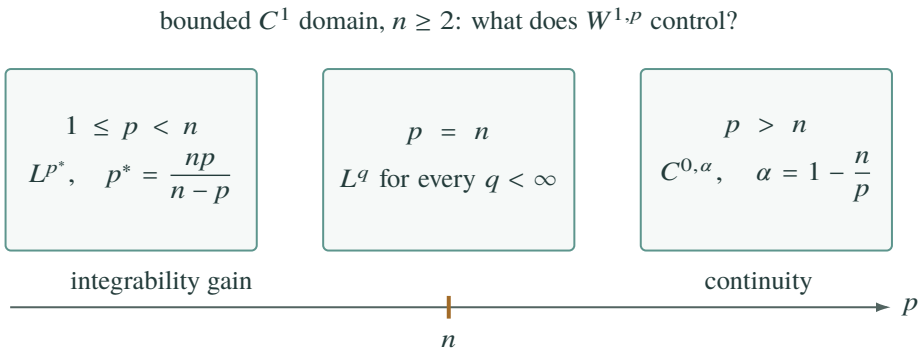


Diagram T5.3. The dimension separates three embedding regimes. The diagram records standard continuous embeddings on a bounded C^1 domain with $n \geq 2$. At $p = n$, the conclusion is valid for each finite q, with constants that may depend on q. It does not provide a uniform estimate as $q \to \infty$. Compactness is a further question addressed in the next section.

5.6.2 Gagliardo–Nirenberg–Sobolev (Evans §5.6.1)

We first give the integral inequality behind the endpoint $p = 1$. For $n \geq 2$ and nonnegative f_i on $\mathbb{R}^{n-1}$, with $\hat{x}_i$ denoting all coordinates except x_i,

$$\int_{\mathbb{R}^n} \prod_{i=1}^{n} f_i(\hat{x}_i)^{1/(n-1)}\, dx \leq \prod_{i=1}^{n} \left(\int_{\mathbb{R}^{n-1}} f_i \right)^{1/(n-1)}. \tag{LW}$$

Expanded proof (editorial integral lemma).

For $n = 2$ this is Fubini. Suppose it holds in dimension $n-1$. Integrate first in x_n and apply Hölder with $n-1$ equal exponents to the factors $i < n$. Put $F_i = \int_{\mathbb{R}} f_i(\hat{x}_i)\, dx_n$. What remains is

$$\int_{\mathbb{R}^{n-1}} f_n^{1/(n-1)} \prod_{i<n} F_i^{1/(n-1)}\, dx'.$$

Apply Hölder with exponents $n-1$ and $(n-1)/(n-2)$, obtaining

$$\left(\int f_n \right)^{1/(n-1)} \left(\int_{\mathbb{R}^{n-1}} \prod_{i<n} F_i^{1/(n-2)} \right)^{(n-2)/(n-1)}.$$

The induction hypothesis on the last integral gives (LW). Tonelli justifies the order changes, first for finite integrals and then by monotone truncation in general. □

Theorem 5.13 (Evans §5.6.1, Theorem 1)
For $1 \leq p < n$ and $u \in C_c^1(\mathbb{R}^n)$,

$$\|u\|_{p^*} \leq C_{n,p} \|Du\|_p.$$

Expanded proof.

先用缩放验算

$1/p^* = 1/p - 1/n$ 是量纲平衡，不是记忆技巧。若估计两侧缩放次数不同，就必须出现域尺度、低阶项或其他约束。

1. Start with $p = 1$. The fundamental theorem on the ith coordinate line gives

$$|u(x)| \le \int_{\mathbb{R}} |\partial_i u(\hat{x}_i, t)|\, dt =: f_i(\hat{x}_i).$$

Multiply the n inequalities and raise to the power $1/(n-1)$. By (LW),

$$\int |u|^{n/(n-1)} \le \prod_i \|\partial_i u\|_1^{1/(n-1)} \le \|Du\|_1^{n/(n-1)}.$$

Taking the power $(n-1)/n$ proves the endpoint.

2. Obtain $1 < p < n$ by a power substitution. Set $a = p(n-1)/(n-p) > 1$. Apply the endpoint to $|u|^a$, a compactly supported C^1 function. Hölder with $p' = p/(p-1)$ yields

$$\Big(\int |u|^{an/(n-1)}\Big)^{(n-1)/n} \le Ca \int |u|^{a-1}|Du| \le Ca\|Du\|_p \Big(\int |u|^{(a-1)p'}\Big)^{1/p'}.$$

Both exponents $an/(n-1)$ and $(a-1)p'$ equal p^*. The left side is $\|u\|_{p^*}^a$ and the last factor is $\|u\|_{p^*}^{a-1}$. Cancel it if nonzero; if zero the conclusion is immediate. □

Corollary 5.14 (Evans §5.6.1, Theorems 2–3)
On a bounded C^1 domain, $\|u\|_{p^*} \le C\|u\|_{W^{1,p}(U)}$. On any bounded open set, for $u \in W_0^{1,p}(U)$,

$$\|u\|_{p^*} \le C_{n,p}\|Du\|_p, \qquad \|u\|_p \le C_{n,p}|U|^{1/n}\|Du\|_p.$$

Expanded proof.

For the first statement extend by Theorem 5.10, smooth on $\mathbb{R}^n$ with fixed compact support, and apply Theorem 5.13. Applying it to differences shows the smooth sequence is Cauchy in L^{p^*}. Its L^p limit identifies the L^{p^*} limit with the extension, so the inequality passes to u. For zero boundary data use zero extensions of the defining compactly supported sequence instead. Finally Hölder gives $\|u\|_p \le |U|^{1/p-1/p^*}\|u\|_{p^*}$ and $1/p - 1/p^* = 1/n$. □

5.6.3 The critical case $p = n$

For $n \ge 2$, bounded C^1 domains satisfy $W^{1,n}(U) \hookrightarrow L^q(U)$ for every finite q. Indeed choose $1 \le r < n$ so close to n that $r^* \ge q$. Finite measure gives $W^{1,n} \subset W^{1,r}$, and the preceding inequality followed by Hölder proves the claim. The constant depends on q and need not stay bounded as $q \to \infty$.

幂次为何这样选

a 同时满足 $an/(n-1) = p^*$ 与 $(a-1)p' = p^*$，所以最后能约去同一个函数范数。随便取幂次，两个积分指数就对不上。

为何一般域估计有低阶项

常数 $u = 1$ 满足 $Du = 0$，所以对任意 $W^{1,p}$ 函数不能只有梯度控制。零迹排除非零常数，才可去掉右侧的 $\|u\|_p$。

A useful counterexample, equivalent in behavior to source Problem 14, is

$$u(x) = \log\log(e/|x|) \quad (0 < |x| < 1).$$

It is unbounded, but $|Du| = 1/(|x|\log(e/|x|))$ and

$$\int_{B_1} |Du|^n = C_n \int_0^1 \frac{dr}{r\log(e/r)^n} < \infty \quad (n > 1).$$

The function itself has finite L^n norm. Its weak derivative is identified by integration outside B_ε; the interface term is $O(\varepsilon^{n-1}\log\log(e/\varepsilon)) \to 0$. In dimension one, by contrast, $W^{1,1}$ on an interval has an absolutely continuous, hence bounded, representative.

5.6.4 Morrey's inequality (Evans §5.6.2, Theorems 4–5)

Theorem 5.15

If $n < p < \infty$, then $W^{1,p}(\mathbb{R}^n)$ has a representative satisfying

$$|u(x) - u(y)| \le C_{n,p}|x-y|^{1-n/p}\|Du\|_{L^p(\mathbb{R}^n)}, \qquad \|u\|_\infty \le C_{n,p}\|u\|_{W^{1,p}}.$$

On a bounded C^1 domain, $\|u\|_{C^{0,1-n/p}(\overline{U})} \le C\|u\|_{W^{1,p}(U)}$. For $p = \infty$, the corresponding exponent is 1.

Expanded proof.

1. A local kernel estimate. Let D be a ball of radius R and $x \in \overline{D}$. For smooth u, integrate the line identity $|u(x) - u(z)| \le \int_0^{|z-x|} |Du(x + t\omega)|\,dt$ over $z = x + s\omega \in D$. Convexity keeps each segment inside D. The s integration after changing order contributes at most CR^n. Dividing by $|D|$ gives

$$|u(x) - u_D| \le C_n \int_D \frac{|Du(z)|}{|x-z|^{n-1}}\,dz, \qquad u_D = |D|^{-1}\int_D u.$$

Hölder bounds the integral by

$$C\|Du\|_{L^p(D)} \left(\int_0^{2R} t^{n-1-(n-1)p'}\,dt\right)^{1/p'} \le CR^{1-n/p}\|Du\|_{L^p(D)}.$$

The radial integral is finite exactly when $p > n$ (for $n = 1$, the same computation is simply the integral triangle inequality and Hölder).

2. Compare two points. For $r = |x-y|$, choose $D = B(x, 2r)$ containing both. Bound $|u(x) - u_D|$ and $|u(y) - u_D|$ by step 1. This also yields the local estimate

$$|u(x) - u(y)| \le Cr^{1-n/p}\|Du\|_{L^p(B(x,2r))}$$

临界点不是无穷大嵌入

$p \uparrow n$ 时公式中的 p^* 趋于无穷，但常数也可能发散，不能把极限直接代入。$\log\log(e/|x|)$ 给出明确反例。

whenever that ball lies in the domain. With a unit ball in step 1, $|u(x)| \le |u_D| + C\|Du\|_p \le C(\|u\|_p + \|Du\|_p)$.

3. Pass to Sobolev functions. Smooth approximation on $\mathbb{R}^n$ and the inequalities on differences give uniform convergence to a Hölder representative, agreeing almost everywhere with the L^p limit. On U, use the extension operator. For $p = \infty$ on $\mathbb{R}^n$, local smoothing has gradient bounded by $\|Du\|_\infty$; the line identity gives a common Lipschitz constant. A locally uniform subsequential limit, identified by finite L^q convergence, proves the assertion. Extend first for the domain version. □

5.6.5 Higher-order embeddings (Evans §5.6.3, Theorem 6)

Let U be bounded with C^1 boundary, $k \ge 1$, and $1 \le p < \infty$. If $kp < n$,

$$W^{k,p}(U) \hookrightarrow L^q(U), \qquad \frac{1}{q} = \frac{1}{p} - \frac{k}{n}.$$

If $k > n/p$ and n/p is not an integer, put $l = \lfloor n/p \rfloor$, $m = k - l - 1$, and $\gamma = l + 1 - n/p$. Then $W^{k,p} \hookrightarrow C^{m,\gamma}(\overline{U})$. If $n/p = j$ is an integer and $k > j$, the conclusion is $C^{k-j-1,\gamma}$ for every $0 < \gamma < 1$. Each embedding has a norm bound depending on U, n, k, p and, where applicable, γ.

Expanded proof.

Apply the first-order estimate to each derivative of order at most $k-1$. This improves $W^{k,p}$ to W^{k-1,p_1} with $1/p_1 = 1/p - 1/n$. Iterate while the input exponent is below n. If $kp < n$, exactly k steps yield L^q. If n/p is nonintegral, after l steps the exponent r satisfies $1/r = 1/p - l/n$ and $r > n$. Apply Morrey to each derivative through order $k - l - 1$; $1 - n/r = l + 1 - n/p$. If $n/p = j$ is integral, after $j-1$ steps the exponent is n. The critical finite-q estimate gives $W^{k-j,q}$ for arbitrary finite q. Choose $q > n$ and apply Morrey, obtaining $C^{k-j-1,1-n/q}$. Any $\gamma < 1$ follows by choosing q sufficiently large. Multiplying the finitely many estimate constants gives the bounds. The continuous weak derivatives obtained here are classical derivatives: mollify locally and pass to the coordinate-segment integral identity, as in the completeness proof in §5.1. For $n = p = 1$, interval absolute continuity gives the needed bounded derivatives directly and hence the stated weaker Hölder conclusions. □

连续代表元的含义

原先 u 是几乎处处等价类。Morrey 证明其中存在连续函数；两个连续函数若几乎处处相同，便处处相同。因此此时谈逐点值才有唯一含义。

5.7 Compactness

5.7.1 Rellich–Kondrachov (Evans §5.7, Theorem 1)

Theorem 5.16
Let U be bounded with C^1 boundary. If $1 \le p < n$ and $1 \le q < p^*$, every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$. We write $W^{1,p}(U) \Subset L^q(U)$. For $p = n$ the same conclusion holds for every finite q. For $p > n$, including $p = \infty$, bounded sequences have subsequences converging uniformly on $\overline{U}$.

Expanded proof.

1. Put every function on the same compact set. Extend a bounded sequence (u_j) by the fixed operator in Theorem 5.10. Call the extensions v_j. They have uniformly bounded $W^{1,p}(\mathbb{R}^n)$ norm and support in a fixed compact set K.

2. Uniform smoothing error. For smooth v, the line integral formula and Minkowski give

$$\|v(\cdot + h) - v\|_p \le |h| \int_0^1 \|Dv(\cdot + th)\|_p \, dt = |h| \|Dv\|_p.$$

Smooth approximation passes this to every finite-p Sobolev function. Consequently

$$\|\rho_\varepsilon * v_j - v_j\|_p \le C\varepsilon \|Dv_j\|_p \le C\varepsilon$$

uniformly in j. For $q \le p$, finite support gives the corresponding L^q bound. For $p < q < p^*$ choose $0 < \theta < 1$ with $1/q = \theta/p + (1-\theta)/p^*$. Hölder interpolation gives

$$\|\rho_\varepsilon * v_j - v_j\|_q \le \|\rho_\varepsilon * v_j - v_j\|_p^\theta \|\rho_\varepsilon * v_j - v_j\|_{p^*}^{1-\theta} \le C\varepsilon^\theta.$$

The second factor is uniformly bounded by the Sobolev inequality and convolution's L^{p^*} bound.

3. Compactness for a fixed smoothing radius. For fixed ε, the functions $v_{j,\varepsilon} = \rho_\varepsilon * v_j$ have support in $K + \overline{B}_\varepsilon$. Moreover

$$\|v_{j,\varepsilon}\|_\infty \le \|\rho_\varepsilon\|_\infty \|v_j\|_1, \qquad \|Dv_{j,\varepsilon}\|_\infty \le \|D\rho_\varepsilon\|_\infty \|v_j\|_1.$$

The right sides are independent of j by bounded support and Hölder. Arzelà–Ascoli, invoked here as a prerequisite, supplies a uniformly convergent subsequence on that compact set.

高阶指数逐层算

每使用一次一阶嵌入，导数阶数减一，$1/p$ 减去 $1/n$。到达临界指数时先转到任意有限 q，所以整数临界处不能擅自宣称 Lipschitz。

连续嵌入与紧嵌入

连续嵌入只把有界集送到有界集；紧嵌入还能从任意有界序列抽出强收敛子列。强收敛常用于通过非线性项的极限。

4. Make one subsequence work at every scale. Choose $\varepsilon_l \downarrow 0$ so the L^q smoothing error is at most 2^{-l}. Extract nested subsequences for $l = 1, 2, \ldots$, and take the diagonal sequence (v_{j_a}). For fixed l, its smoothed sequence is Cauchy, so

$$\limsup_{a,b\to\infty} \|v_{j_a} - v_{j_b}\|_q \le 2^{1-l}.$$

Let $l \to \infty$. Completeness of L^q gives convergence, and restriction to U proves the result.

5. The remaining exponents. If $p = n \ge 2$, choose $1 \le r < n$ with $r^* > q$ and apply the preceding result to the bounded $W^{1,r}$ sequence. If $p > n$, Morrey gives uniform boundedness and a common Hölder modulus on $\overline{U}$; Arzelà–Ascoli gives uniform convergence. For the exceptional case $n = p = 1$, extension to a fixed bounded interval and the translation/smoothing argument above prove compactness in L^1. The interval fundamental theorem gives a uniform L^∞ bound; interpolation then gives compactness in every finite L^q. □

5.7.2 Why the critical exponent is excluded

Fix nonzero $\psi \in C_c^\infty(B_1)$ and an interior point x_0. For $p < n$ define, for sufficiently large λ,

$$u_\lambda(x) = \lambda^{(n-p)/p}\psi(\lambda(x - x_0)).$$

Then $\|Du_\lambda\|_p = \|D\psi\|_p$, $\|u_\lambda\|_p = \lambda^{-1}\|\psi\|_p$, but $\|u_\lambda\|_{p^*} = \|\psi\|_{p^*} > 0$. The sequence tends to zero at every $x \ne x_0$. A strongly L^{p^*} convergent subsequence would have a further almost-everywhere convergent subsequence, so its limit would be zero, contradicting its constant nonzero norm.

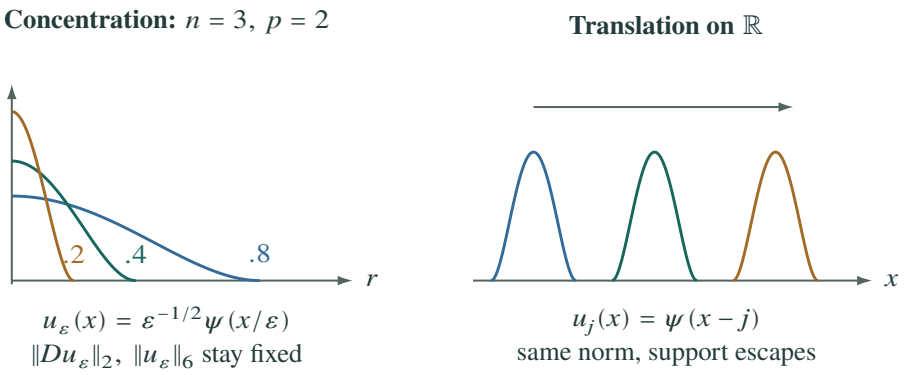


Diagram T5.4. Bounded norms do not prevent concentration or escape. The left panel uses radial bump profiles $\psi(x) = (1 - |x|^2)_+^2$ in $\mathbb{R}^3$, scaled with $\varepsilon = 0.8, 0.4, 0.2$; their gradient L^2 and critical L^6 norms stay fixed. The right panel illustrates translations of a compact bump on $\mathbb{R}$. Each mechanism obstructs strong compactness for a different reason.

对角线到底做什么

固定 ε 得到的子列可能随 ε 改变。先抽嵌套子列再取对角线，保证每个固定尺度最终都收敛；最后才令尺度趋零。

集中导致临界失紧

尖峰越来越窄、越来越高，梯度范数和临界范数都不变。次临界范数会衰减，因此 $q < p^*$ 留出的余量正好压住集中。

Zero boundary data on rough bounded domains. For finite p, zero extensions of a bounded sequence in $W_0^{1,p}(U)$ have a common compact support and bounded whole-space Sobolev norm. The proof above therefore gives compactness in $L^p(U)$ on any bounded open U, without regular boundary. On unbounded domains, translations of a fixed bump can escape to infinity and destroy global compactness.

5.8 Additional Topics

5.8.1 Poincaré inequalities (Evans §5.8.1, Theorems 1–2)

Lemma 5.17 (Zero gradient means constant)
If U is connected and $u \in W^{1,1}_{\mathrm{loc}}(U)$ satisfies $Du = 0$, then u is constant almost everywhere.

Expanded proof (also source Problem 11).

On a ball compactly contained in U, mollification has zero classical gradient on each smaller concentric ball and is constant there. As $\varepsilon \to 0$ it converges in L^1; the constants converge to the average of u on the smaller ball. Hence u is constant there almost everywhere. Constants agree on overlapping balls, because the overlap has positive measure. Any two points in an open connected subset of $\mathbb{R}^n$ can be joined by a finite chain of such balls: the set of points reachable from a fixed ball is both relatively open and relatively closed. Connectedness completes the proof. □

Theorem 5.18
On a bounded connected C^1 domain, for $1 \le p \le \infty$,

$$\|u - u_U\|_{L^p(U)} \le C_{U,n,p}\|Du\|_{L^p(U)}, \qquad u_U = |U|^{-1}\int_U u.$$

On a ball B_r the constant is $C_{n,p}r$.

连通性不可删

在两个不相交的球上分别取常数 0 和 1，梯度仍处处为零，却不是同一个常数。均值型 Poincaré 必须排除这种情况。

Expanded proof.

If the inequality failed, normalization would give

$$(v_j)_U = 0, \quad \|v_j\|_p = 1, \quad \|Dv_j\|_p < 1/j.$$

For finite p, compactness gives a subsequence converging strongly in L^p to v; for $p = \infty$, use the uniform compactness just proved. The mean and norm pass to the limit, giving $v_U = 0$ and $\|v\|_p = 1$. Testing the weak derivative identity and passing to the limit gives $Dv = 0$. Lemma 5.17 makes v constant, and its zero mean makes it zero, a contradiction. For

$B_r(x_0)$ use $w(y) = u(x_0 + ry)$: the function norm scales by $r^{-n/p}$ and the gradient norm by $r^{1-n/p}$, leaving precisely one factor r. □

For $u \in W^{1,n}(\mathbb{R}^n)$, $n \geq 2$, the $p = 1$ ball estimate and Hölder give

$$\frac{1}{|B_r|}\int_{B_r} |u - u_{B_r}| \leq Cr\frac{1}{|B_r|}\int_{B_r} |Du| \leq C\|Du\|_{L^n(\mathbb{R}^n)}.$$

Thus $W^{1,n}$ controls bounded mean oscillation (BMO), whose seminorm is the supremum of the left side over balls. BMO controls average oscillation rather than pointwise size.

5.8.2 Difference quotients (Evans §5.8.2, Theorem 3)

For $h \neq 0$ define

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}.$$

Theorem 5.19

If $u \in W^{1,p}(U)$, $V \Subset U$, and $|h| < \frac{1}{2}\operatorname{dist}(V, \partial U)$, then $\|D_i^h u\|_{L^p(V)} \leq \|\partial_i u\|_{L^p(U)}$ for finite p. Conversely, if $1 < p < \infty$, $u \in L^p(U)$, and all these quotients are uniformly bounded in $L^p(V)$ for every coordinate, then $u \in W^{1,p}(V)$, with the corresponding derivative bounds.

Expanded proof.

For smooth u, $D_i^h u(x) = \int_0^1 \partial_i u(x + the_i)\, dt$. Jensen and Fubini give

$$\int_V |D_i^h u|^p \leq \int_0^1 \int_V |\partial_i u(x + the_i)|^p \, dx\, dt \leq \int_U |\partial_i u|^p.$$

Interior approximation on a slightly larger set passes this to u.

For the converse, choose $h_j \to 0$. Reflexivity of $L^p(V)$ gives, after a finite diagonal extraction over coordinates, $D_i^{h_j} u \rightharpoonup g_i$ in $L^p(V)$. For a test function ϕ and small h a change of variables yields

$$\int D_i^h u\, \phi = -\int u D_i^{-h}\phi.$$

There are no edge terms because the test support and its translates stay inside V. Since $D_i^{-h}\phi \to \partial_i \phi$ uniformly on a common compact support, the right side tends to $-\int u\partial_i\phi$. The left side tends weakly to $\int g_i\phi$. Thus $g_i = \partial_i u$. Weak lower semicontinuity proves the norm bound. □

消掉常数的两种方式

零迹型要求 $Tu = 0$；均值型减去 u_U。二者都排除梯度算子的常数核，但适用条件和边界信息不同。

For finite p, one also has $D_i^h u \to \partial_i u$ strongly on interior sets: subtract the derivative in the line-average formula and use L^p translation continuity. This gives a practical way to recognize weak derivatives in elliptic regularity estimates.

5.8.3 Lipschitz functions (Evans §5.8.2, Theorem 4)

Locally, $W^{1,\infty}$ functions have Lipschitz representatives, by smoothing on balls and using the gradient bound on segments. Conversely an L-Lipschitz function has bounded difference quotients. On a relatively compact ball, extract weak limits in L^2. The weak identity above gives weak derivatives; the limit still satisfies $|g_i| \le L$ almost everywhere, since the closed convex set $\{g : |g| \le L\}$ is weakly closed in L^2. This identifies locally bounded weak derivatives without appealing to Rademacher's theorem.

On bounded C^1 domains the extension theorem gives a global Lipschitz representative for $W^{1,\infty}(U)$, with constant controlled by the full $W^{1,\infty}$ norm. Global Lipschitz continuity need not follow on arbitrary domains: points across a narrow slit can be very close in Euclidean distance but far apart along paths inside the domain.

5.8.4 Differentiability a.e. (Evans §5.8.3, Theorems 5–6)

Theorem 5.20
For $p > n$, a function $u \in W^{1,p}_{\mathrm{loc}}(U)$, identified with its continuous representative, is differentiable almost everywhere, and its classical gradient equals its weak gradient there. Every locally Lipschitz function is therefore differentiable a.e.

Expanded proof.

First let $n < p < \infty$. The L^p version of the Lebesgue differentiation theorem, invoked here, says that for almost every x,

$$\frac{1}{|B_r|}\int_{B_r(x)} |Du(z) - Du(x)|^p \, dz \to 0.$$

Fix such an x and put $v(z) = u(z) - u(x) - Du(x)\cdot(z-x)$. For $r = |y-x|$ small enough that $B(x,2r) \Subset U$, the local Morrey estimate gives

$$\begin{aligned} |v(y) - v(x)| &\le C r^{1-n/p} \left(\int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} \\ &= C r \left(\frac{1}{|B_{2r}|} \int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} = o(r). \end{aligned}$$

$p = 1$ 的反向结论失效

阶跃函数的差商高度为 $1/|h|$、支集宽约 $|h|$，故 L^1 范数有界；极限却是 Dirac 质量。L^1 有界不保证弱子列极限仍是函数。

避免循环证明

先由差商得到 Lipschitz 函数的弱导数，再由 Morrey 和微分定理证明几乎处处可微。若先引用 Rademacher 来构造弱导数，又用它证明 Rademacher，就循环了。

This is precisely differentiability with derivative $Du(x)$. For $p = \infty$, work locally with any finite exponent greater than n. A locally Lipschitz function belongs locally to $W^{1,\infty}$ by the preceding argument, proving Rademacher's theorem as a consequence. □

5.8.5 Hardy's inequality (Evans §5.8.4, Theorem 7)

Theorem 5.21
If $n \geq 3$ and $u \in H^1(B_r)$, then

$$\int_{B_r} \frac{|u|^2}{|x|^2}\,dx \leq C_n \int_{B_r} \left(|Du|^2 + r^{-2}|u|^2\right)\,dx.$$

Expanded proof.

Start with u smooth on $\overline{B}_r$ and set $X = x/|x|^2$, so $\operatorname{div} X = (n-2)/|x|^2$. Apply the divergence theorem on $B_r \setminus \overline{B}_\varepsilon$. The inner boundary term is $O(\varepsilon^{n-2})$ and tends to zero. With $I = \int_{B_r} u^2/|x|^2$ and $A = \|Du\|_2$, the result is

$$(n-2)I = r^{-1}\int_{\partial B_r} u^2\,dS - 2\int_{B_r} u\,Du \cdot \frac{x}{|x|^2}.$$

Cauchy–Schwarz and Young yield $I \leq C_n A^2 + C_n r^{-1}\int_{\partial B_r} u^2$. To control the last term, apply divergence to xu^2:

$$r\int_{\partial B_r} u^2 = \int_{B_r} (nu^2 + 2u\,x\cdot Du) \leq C_n \int_{B_r} (u^2 + r^2|Du|^2).$$

Combine the estimates. For general u, use smooth boundary approximation. Applying the estimate to differences shows $u_j/|x|$ is Cauchy in L^2. An a.e. convergent subsequence identifies its limit with $u/|x|$, and the bound passes to the limit. □

5.8.6 Fourier characterization (Evans §5.8.5, Theorem 8)

Use the unitary convention $\hat{u}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi}u(x)\,dx$. For integer $k \geq 0$,

$$u \in H^k(\mathbb{R}^n) \iff (1+|\xi|^2)^{k/2}\hat{u} \in L^2(\mathbb{R}^n),$$

$$\|u\|_{H^k} \asymp \|(1+|\xi|^2)^{k/2}\hat{u}\|_2.$$

Expanded proof.

For compactly supported smooth functions integration by parts gives $\widehat{D^\alpha u} = (i\xi)^\alpha \hat{u}$. Whole-space cutoff and convolution approximate every

可微比 Hölder 更强

Morrey 单独给出 $O(r^{1-n/p})$。减去候选线性项后，梯度的局部平均误差趋零，才把余项改善为 $o(r)$。

奇点与边界都要处理

不能直接在含原点的球上对奇异向量场用散度定理；先挖去小球再估内边界项。外边界项不为零，所以一般球域估计保留 $r^{-2}\|u\|_2^2$。

H^k function by such functions. Plancherel and passage to the L^2 limit preserve the identity. Also

$$c_{n,k}(1+|\xi|^2)^k \le \sum_{|\alpha|\le k} |\xi^\alpha|^2 \le C_{n,k}(1+|\xi|^2)^k.$$

This follows by the multinomial expansion of $(1+\xi_1^2+\cdots+\xi_n^2)^k$ and comparison of its finitely many positive coefficients. It proves the forward bound. Conversely, the weighted condition makes $g_\alpha = \mathscr{F}^{-1}((i\xi)^\alpha \hat{u})$ an L^2 function. Plancherel against test functions gives the weak derivative identity, hence $g_\alpha = D^\alpha u$ and $u \in H^k$. The same polynomial comparison proves both norm bounds. □

For real $s > 0$ this suggests the definition

$$\|u\|_{H^s(\mathbb{R}^n)} = \|(1+|\xi|^2)^{s/2}\hat{u}\|_2.$$

It is equivalent to the source weight $1+|\xi|^s$; equivalent norms describe the same space.

频率中的光滑度

求导变成乘以 $i\xi$。高频尾部越小，可承受的频率权重越高，意味着可积的导数越多；分数阶空间沿用同一机制。

5.9 Other Spaces of Functions

5.9.1 The dual space H^{-1} (Evans §5.9.1, Theorem 1)

Define $H^{-1}(U) = (H_0^1(U))^*$ with

$$\|f\|_{H^{-1}} = \sup_{\|v\|_{H^1}\le 1} |\langle f, v\rangle|.$$

Here we use the full H^1 norm, rather than the equivalent gradient norm available on bounded domains.

Theorem 5.22

For every $f \in H^{-1}(U)$ there are $f^0, \dots, f^n \in L^2(U)$ with

$$\langle f, v\rangle = \int_U f^0 v + \sum_{i=1}^n f^i \partial_i v \, dx, \qquad v \in H_0^1(U).$$

Moreover

$$\|f\|_{H^{-1}} = \min\left\{ \left(\sum_{i=0}^n \|f^i\|_2^2 \right)^{1/2} : (f^0, \dots, f^n) \text{ represents } f \right\}.$$

In distributions, $f = f^0 - \sum_i \partial_i f^i$.

Expanded proof.

Apply the Hilbert-space Riesz representation theorem, an external functional analysis result, to $H_0^1(U)$ with inner product $\int (wv + Dw \cdot Dv)$. There is a unique $w \in H_0^1(U)$ such that

$$\langle f, v\rangle = \int_U (wv + Dw \cdot Dv), \qquad \|f\|_{H^{-1}} = \|w\|_{H^1}.$$

Set $f^0 = w$ and $f^i = \partial_i w$. Their product L^2 norm equals $\|f\|_{H^{-1}}$. Any other representation (g^i) satisfies $|\langle f, v\rangle| \le (\sum_i \|g^i\|_2^2)^{1/2}\|v\|_{H^1}$ by Cauchy–Schwarz. Take the supremum over unit v to prove minimality. The distributional sign follows from $\langle \partial_i g, v\rangle = -\int g\partial_i v$. □

The embedding $L^2(U) \to H^{-1}(U)$ is $\langle f, v\rangle = \int_U fv$, with norm at most $\|f\|_2$. It is injective because test functions separate locally integrable functions. Thus the useful pivot-space chain is

$$H_0^1(U) \subset L^2(U) \subset H^{-1}(U).$$

负一阶不是负次求导

H^{-1} 的元素可以是导数型分布。若 $F \in L^2(U;\mathbb{R}^n)$，则 $-\operatorname{div} F$ 作用在 v 上是 $\int F \cdot Dv$，由 Cauchy 不等式连续。

5.9.2 Banach-valued time functions (Evans §5.9.2, Theorem 2)

For a real Banach space X, $L^p(0,T;X)$ consists of strongly measurable functions with integrable pth power of their X norm (or essentially bounded norm when $p = \infty$). Integration below is Bochner integration. Its construction, norm inequality, and Fubini properties are prerequisites from measure theory. We say $u' = g$ weakly if

$$\int_0^T u(t)\phi'(t)\,dt = -\int_0^T g(t)\phi(t)\,dt \quad (\phi \in C_c^\infty(0,T)),$$

as an equality in X. The space $W^{1,p}(0,T;X)$ requires both u and g in $L^p(0,T;X)$.

不要抹去中间的 L^2

Riesz 可把 Hilbert 空间与其对偶抽象同构，但演化方程要保留 L^2 配对。u_t 可只在 H^{-1} 中，$\langle u_t, u\rangle$ 仍有意义。

Theorem 5.23 (Fundamental theorem in X)
Every $u \in W^{1,p}(0,T;X)$ has a continuous representative with

$$u(t) - u(s) = \int_s^t u'(\tau)\,d\tau.$$

$$\max_{[0,T]} \|u(t)\|_X \le T^{-1/p}\|u\|_{L^p(X)} + T^{1-1/p}\|u'\|_{L^p(X)}.$$

Interpret $1/p = 0$ for $p = \infty$.

Expanded proof.

Set $G(t) = \int_0^t u'(\tau)\,d\tau$. The Bochner integral norm inequality makes G absolutely continuous. Fubini against scalar test functions shows $G' = u'$ weakly, hence $w = u - G$ has weak derivative zero.

To show w is constant without making an uncountable exceptional-set argument over X^*, smooth w in time on interior intervals. Its smooth derivative is zero, so w_ε is an X-valued constant there. The convergence $w_\varepsilon \to w$ in local $L^1(0,T;X)$ forces these constants to converge in X on each fixed subinterval. Overlapping subintervals give the same constant c. Thus $u = c + G$ almost everywhere; take $c + G$ as the representative. The formula holds for all s, t, including endpoints. Its norm inequality gives

$$\|u(t)\|_X \le \|u(s)\|_X + \int_0^T \|u'(\tau)\|_X \, d\tau.$$

Average in s and apply Hölder, obtaining the stated bound. □

5.9.3 Energy-space continuity (Evans §5.9.2, Theorem 3)

Write $V = H_0^1(U)$, $H = L^2(U)$, and $V^* = H^{-1}(U)$.

Theorem 5.24
If $u \in L^2(0,T;V)$ and $u' \in L^2(0,T;V^*)$, then u has a representative in $C([0,T];H)$ and

$$\|u(t)\|_H^2 - \|u(s)\|_H^2 = 2\int_s^t \langle u'(\tau), u(\tau)\rangle \, d\tau.$$

The squared norm is absolutely continuous and its derivative is the integrand almost everywhere. Also

$$\max_t \|u(t)\|_H \le C_T\big(\|u\|_{L^2(V)} + \|u'\|_{L^2(V^*)}\big).$$

Expanded proof.

1. Construct time-smooth approximations without endpoint jumps. Since $V \subset V^*$ continuously, $u \in W^{1,2}(0,T;V^*)$. Theorem 5.23 gives a continuous V^* representative. Reflect it across 0 and T to a slightly larger interval. The reflected function remains in $L^2(V)$, with weak derivative in $L^2(V^*)$ and the appropriate reflected signs. There is no endpoint Dirac term because the V^* values match. Convolve in time. Translation continuity of Bochner L^2 gives

$$u_\varepsilon \to u \text{ in } L^2(V), \qquad u_\varepsilon' \to u' \text{ in } L^2(V^*).$$

Each smoothed function is smooth with values in V.

2. A uniform-in-time Cauchy estimate. For any smooth $w : [0,T] \to V$, $\frac{d}{dt}\|w\|_H^2 = 2\langle w', w\rangle$. Integrate between s and t, take absolute values of

初值需要时间代表元

L^p 时间函数可以在单个时刻任意修改，因此 $u(0)$ 原先没有确定意义。时间基本定理选出连续代表元，才允许严谨地指定初值。

the integral, and average s over $(0,T)$. It follows that

$$\sup_t \|w(t)\|_H^2 \le T^{-1}\|w\|_{L^2(H)}^2 + 2\|w'\|_{L^2(V^*)}\|w\|_{L^2(V)}.$$

Apply this to $w = u_\varepsilon - u_\delta$. Both graph-norm errors tend to zero, so (u_ε) is Cauchy in $C([0,T];H)$. Its uniform limit agrees with u almost everywhere by the $L^2(H)$ convergence. This proves continuity, including at both endpoints.

3. Pass the energy identity to the limit. The endpoint terms converge uniformly. The integrands converge in L^1 because

$$\begin{aligned}\|\langle u_\varepsilon', u_\varepsilon\rangle - \langle u', u\rangle\|_{L^1} &\le \|u_\varepsilon' - u'\|_{L^2(V^*)}\|u_\varepsilon\|_{L^2(V)} \\ &\quad + \|u'\|_{L^2(V^*)}\|u_\varepsilon - u\|_{L^2(V)} \to 0.\end{aligned}$$

The limiting integral identity gives absolute continuity and its a.e. derivative. Applying the uniform estimate to u_ε and then Young's inequality proves the norm bound. □

5.9.4 Continuity in a better spatial space (source Theorem 4)

Let U be bounded with smooth boundary and $m \ge 0$ an integer. If

$$u \in L^2(0,T;H^{m+2}(U)), \qquad u' \in L^2(0,T;H^m(U)),$$

then $u \in C([0,T];H^{m+1}(U))$, with the corresponding bound

$$\max_t \|u(t)\|_{H^{m+1}} \le C_{U,T,m}(\|u\|_{L^2(H^{m+2})} + \|u'\|_{L^2(H^m)}).$$

Expanded proof.

First $m = 0$. Choose the same compact-support extension operator E bounded both on L^2 and H^2, as constructed in §5.4. Set $v = Eu$. Bounded linear maps commute with Bochner integration, so testing in time gives $v' = Eu'$. For time-smooth v, integrate by parts in space on $\mathbb{R}^n$:

$$\frac{d}{dt}\|v\|_{H^1(\mathbb{R}^n)}^2 = 2(v - \Delta v, v')_{L^2}.$$

This is valid for H^2 functions by spatial approximation, with no boundary term on the whole space. Thus its absolute value is at most $C\|v\|_{H^2}\|v'\|_2$. Average an integrated identity in the initial time to obtain

$$\sup_t \|v(t)\|_{H^1}^2 \le C_T(\|v\|_{L^2(H^2)}^2 + \|v'\|_{L^2(L^2)}^2).$$

Reflect and convolve in time as in the preceding proof; now use continuity in L^2, which follows from $u \in W^{1,2}(0,T;L^2)$. The estimate on differences gives convergence in $C(H^1)$. Restrict to U and use the bounds for E.

能量计算的合法化

形式上把方程乘 u 会得到 $\frac{1}{2}(d/dt)\|u\|_2^2$。当 u_t 只有 H^{-1} 正则性时，必须先时间平滑、建立一致估计，再通过配对的 L^1 收敛取极限。

For general m, apply this result to every $D^\alpha u$, $|\alpha| \le m$. Spatial derivatives commute with the time weak derivative, as one checks against separated space-time test functions. Summing the finitely many H^1 estimates yields an equivalent H^{m+1} norm and the claimed continuity. □

空间与时间交换一阶

空间有 $m+2$ 阶、时间导数有 m 阶，得到时间连续的中间空间 H^{m+1}。空间分部积分把两个导数移到同一侧，正是这个阶数配平的来源。

5.10 Problems

Evans §5.10, Problems 1–21, printed pp. 289–292 (PDF pp. 306–309). Statements are concisely restated; solutions are editorial. Boundary smoothness is assumed where needed.

5.10.1 Problems 1–6: the basic language

1. Completeness of Hölder spaces. Prove $C^{k,\gamma}(\overline{U})$ is Banach.

Solution. The full proof in §5.1 identifies uniform limits of derivative components through coordinate-segment integration, then passes the Cauchy bound to the Hölder seminorm. Derivative compatibility and seminorm convergence are both needed, beyond uniform convergence of u_j. □

2. Hölder interpolation. For $0 < \beta < \gamma \le 1$, prove

$$\|u\|_{C^{0,\gamma}} \le \|u\|_{C^{0,\beta}}^{(1-\gamma)/(1-\beta)} \|u\|_{C^{0,1}}^{(\gamma-\beta)/(1-\beta)}.$$

Solution. The quotient factorization and two-term Hölder inequality in §5.1 give this formula. The weights sum to one and satisfy $\gamma = (1-\theta)\beta + \theta$ for $\theta = (\gamma-\beta)/(1-\beta)$. □

3. A pyramid on a square. On $U = (-1,1)^2$ the source's piecewise function is $u(x) = 1 - \max(|x_1|, |x_2|)$. Determine its $W^{1,p}$ membership.

Solution. On the four sectors where one signed coordinate is largest, Du is respectively $(-1,0), (1,0), (0,-1), (0,1)$. Integrate against a test function separately on the sectors. At each shared edge, values of u agree and outward normals are opposite, so the interface terms cancel. This bounded vector field is the weak gradient. Both u and Du are bounded on a finite-measure domain, so $u \in W^{1,p}(U)$ for all $1 \le p \le \infty$. □

4. One-dimensional Sobolev functions. Show $u \in W^{1,p}(0,1)$, $p < \infty$, has an absolutely continuous representative, and derive a Hölder estimate for $p > 1$.

Solution. Let g be its weak derivative and $G(x) = \int_0^x g(t)\,dt$. Fubini shows $G' = g$ weakly. Since $(u - G)' = 0$, Lemma 5.17 makes $u - G$ constant. The representative $c + G$ is absolutely continuous, with classical derivative g a.e. For every $x, y \in [0,1]$, Hölder gives

$$|u(x) - u(y)| \le \int_{\min(x,y)}^{\max(x,y)} |g| \le |x - y|^{1-1/p} \|g\|_{L^p(0,1)}.$$

For $p = 1$, absolute continuity persists but this argument supplies no fixed positive Hölder exponent. □

5. A smooth cutoff. For $V \Subset U$, construct $\zeta \in C_c^\infty(U)$ with $\zeta = 1$ on V.

Solution. Put $d = \min\{1, \operatorname{dist}(\overline{V}, \mathbb{R}^n \setminus U)\} > 0$ and $W = \{x : \operatorname{dist}(x, \overline{V}) < d/3\}$. For $\varepsilon < d/6$, $\zeta = \rho_\varepsilon * \mathbf{1}_W$ equals one on V, lies in $[0,1]$, and has compact support within distance $d/2$ of $\overline{V}$, hence inside U. □

6. A finite partition of unity. If $\overline{U}$ is compact and contained in $\bigcup_{i=1}^N V_i$, construct smooth ζ_i supported in V_i, with $0 \le \zeta_i \le 1$ and $\sum_i \zeta_i = 1$ on U.

Solution. Cover $\overline{U}$ by finitely many smaller balls with closures in the given cover. Choose nonnegative smooth bumps positive on these balls and group them by V_i, obtaining $\psi_i \in C_c^\infty(V_i)$. Their sum S is positive on a neighborhood O of $\overline{U}$. Choose $\chi \in C_c^\infty(O)$ equal to one near $\overline{U}$, with $0 \le \chi \le 1$. Set $\zeta_i = \chi\psi_i/S$ on O, zero elsewhere. The cutoff makes this smooth globally, and $\psi_i \le S$ gives $\zeta_i \le 1$. Their sum is χ, as required. □

5.10.2 Problems 7–10: boundary and interpolation estimates

7. Trace from a transverse vector field. If α is smooth and $\alpha \cdot \nu \ge 1$ on ∂U, prove the trace estimate for $1 \le p < \infty$.

分片求导还要查接口

每块导数有界仍不够，必须检查接口函数值相同。若有跳跃，弱导数就会多出沿接口集中的测度。

分母先保证为正

S 只在目标集邻域中确定为正；乘上 χ 后，在离开该邻域前商已为零，才可安全延拓。

Solution. For smooth u and $p > 1$, divergence and Young's inequality give

$$\begin{aligned}\int_{\partial U} |u|^p &\le \int_{\partial U} |u|^p \alpha \cdot \nu \\ &= \int_U |u|^p \operatorname{div} \alpha + p|u|^{p-2} u\, \alpha \cdot Du \\ &\le C \int_U (|u|^p + |Du|^p).\end{aligned}$$

The coefficients $\alpha, D\alpha$ are bounded on the compact closure. For $p = 1$, replace $|u|$ by $\sqrt{u^2 + \varepsilon^2}$; its gradient has magnitude at most $|Du|$. Let $\varepsilon \to 0$. Smooth boundary approximation gives the Sobolev estimate. □

8. No bounded trace on L^p alone. On a nonempty bounded C^1 domain, show no bounded linear map $L^p(U) \to L^p(\partial U)$, $p < \infty$, can agree with continuous boundary restriction.

Solution. Write $d(x) = \operatorname{dist}(x, \partial U)$. Choose continuous $\eta : [0, \infty) \to [0, 1]$ with $\eta(0) = 1$ and $\eta = 0$ on $[1, \infty)$. The functions $v_\delta(x) = \eta(d(x)/\delta)$ are continuous on $\overline{U}$, equal one on ∂U, and tend pointwise to zero in U. Dominated convergence gives $\|v_\delta\|_{L^p(U)} \to 0$. A bounded trace would force $0 < |\partial U|^{1/p} = \|T v_\delta\|_p \to 0$, a contradiction. □

边界薄层反例

函数支撑在越来越薄的边界层中，体积范数趋零，边界值却始终为 1。其梯度变大，说明迹估计为何需要导数信息。

9. An L^2 interpolation estimate. Prove $\|Du\|_2 \le C\|u\|_2^{1/2}\|D^2u\|_2^{1/2}$ for $C_c^\infty(U)$ and $H^2(U) \cap H_0^1(U)$.

Solution. For compactly supported smooth u,

$$\|Du\|_2^2 = -\int_U u\Delta u \le \|u\|_2 \|\Delta u\|_2 \le C_n \|u\|_2 \|D^2 u\|_2.$$

For $u \in H^2 \cap H_0^1$, the distributional identity $\int Du \cdot D\phi = -\int (\Delta u)\phi$ extends from tests to $\phi \in H_0^1$ by density, since both sides are continuous in that norm. Take $\phi = u$. No compactly supported H^2 approximation is needed. □

不能假设更强的逼近

紧支函数在 H^2 中的闭包还会限制法向导数，通常比 $H^2 \cap H_0^1$ 小。此处只在 H_0^1 中延伸测试恒等式，避免加上多余边界条件。

10. Nonlinear interpolation. For compactly supported smooth u, prove

$$\|Du\|_p \le C\|u\|_p^{1/2}\|D^2u\|_p^{1/2} \quad (2 \le p < \infty),$$

$$\|Du\|_{2p} \le C\|u\|_\infty^{1/2}\|D^2u\|_p^{1/2} \quad (1 \le p < \infty).$$

Solution. For $s \ge 2$, integrate by parts and differentiate $|Du|^{s-2}Du$ to get

$$\int |Du|^s \le C_s \int |u|\, |Du|^{s-2} |D^2 u|.$$

The vector map $z \mapsto |z|^{s-2}z$ is C^1 for $s \geq 2$, including the linear case $s = 2$. With $s = p > 2$, Hölder with exponents $p, p/(p-2), p$ gives $\|Du\|_p^p \leq C_p \|u\|_p \|Du\|_p^{p-2} \|D^2 u\|_p$. At $p = 2$ use Cauchy directly. Cancel the gradient factor unless zero, then take a square root. With $s = 2p$, use $\|u\|_\infty$ and exponents $p, p/(p-1)$ to get

$$\|Du\|_{2p}^{2p} \leq C_p \|u\|_\infty \|D^2 u\|_p \|Du\|_{2p}^{2p-2}.$$

For $p = 1$ the last factor is one. Cancellation gives the second result. □

5.10.3 Problems 11–16: the limits of the theory

11. Zero weak gradient. Prove on connected U that $Du = 0$ implies u constant a.e.

Solution. Lemma 5.17 gives the full argument: mollify on balls, identify constant limits, and propagate the common constant along a finite chain of overlapping balls. Connectedness enters the last step. □

12. Failure at $p = 1$. Show bounded L^1 difference quotients do not imply $W^{1,1}$ membership.

Solution. Take $U = (-2, 2)$, $V = (-1, 1)$, and $u = \mathbf{1}_{(0,2)}$ on U. For small h, $|D^h u| = 1/|h|$ on an interval of length $|h|$, and is zero elsewhere in V. Thus $\|D^h u\|_{L^1(V)} = 1$. But its distributional derivative is δ_0, so it is not in $W^{1,1}(V)$. □

13. Bounded gradient without global Lipschitz continuity.

Solution. Let U be the slit annulus $\{(r\cos\theta, r\sin\theta) : 1/2 < r < 1,\ 0 < \theta < 2\pi\}$ and put $u(r, \theta) = \theta$. This is smooth on U, with $|u| \leq 2\pi$ and $|Du| = 1/r \leq 2$. At $r = 3/4$, points of angles ε and $2\pi - \varepsilon$ approach each other, but their values differ by $2\pi - 2\varepsilon$. The function is in $W^{1,\infty}(U)$ and is not globally Lipschitz. The domain is connected but fails the regular boundary hypothesis at the slit. □

14. The source's unbounded critical example. For $n > 1$ verify $u(x) = \log\log(1 + 1/|x|) \in W^{1,n}(B_1)$.

Solution. For $r = |x|$,

$$|u'(r)| = \frac{1}{r(r+1)\log(1+1/r)} \asymp \frac{1}{r\log(1/r)} \quad (r \downarrow 0).$$

梯度控制路径长度

梯度界控制沿域内路径的变化；路径长度未必能由直线距离统一控制。裂缝反例不违背球内的局部 Lipschitz 结论。

The gradient integral near zero is comparable to $\int_0^{1/2} dr/(r\log(1/r)^n)$, finite for $n > 1$. The function integral has an exponentially decaying weight after $t = \log(1/r)$, and is also finite. Away from zero all quantities are bounded. The inner integration-by-parts term is bounded by $C\varepsilon^{n-1}|u(\varepsilon)| \to 0$, proving that the classical gradient off zero is its weak gradient. The function itself diverges as $r \downarrow 0$. □

15. A zero set of positive measure. If $u \in H^1(B_1)$ and $|\{u = 0\}| \geq \alpha > 0$, prove $\|u\|_2 \leq C_{n,\alpha}\|Du\|_2$.

Solution. Let $c = u_{B_1}$ and $Z = \{u = 0\}$. Poincaré gives

$$\alpha|c|^2 \leq \int_Z |u - c|^2 \leq \|u - c\|_2^2 \leq C_n\|Du\|_2^2.$$

Then $\|u\|_2 \leq \|u - c\|_2 + |B_1|^{1/2}|c|$ proves the estimate, with constant $C_n(1 + |B_1|^{1/2}\alpha^{-1/2})$. □

16. Whole-space Hardy inequality. For $n \geq 3$ prove

$$\int_{\mathbb{R}^n} \frac{|u|^2}{|x|^2} \leq \frac{4}{(n-2)^2}\int_{\mathbb{R}^n} |Du|^2, \qquad u \in H^1(\mathbb{R}^n).$$

Solution. First take compactly supported smooth u. Integrate the identity

$$\operatorname{div}\left(\frac{x}{|x|^2}\right) = \frac{n-2}{|x|^2}$$

outside B_ε. The inner boundary term is $O(\varepsilon^{n-2})$ and vanishes; there is no outer term. Thus

$$(n-2)\int \frac{u^2}{|x|^2} = -2\int u\,Du \cdot \frac{x}{|x|^2} \leq 2\left(\int \frac{u^2}{|x|^2}\right)^{1/2}\|Du\|_2.$$

Cancel and square. For general u, compactly supported smooth H^1 approximations are available by cutoff and convolution. The estimate on differences makes $u_j/|x|$ Cauchy in L^2; an a.e. convergent subsequence identifies its limit with $u/|x|$. □

5.10.4 Problems 17–19: nonlinear operations

17. Chain rule. On bounded U, let $F \in C^1(\mathbb{R})$ have bounded derivative. Prove $F(u) \in W^{1,p}$ and $D(F(u)) = F'(u)Du$.

Solution. The bound $|F(s)| \le |F(0)| + \|F'\|_\infty |s|$ gives integrability. For finite p, smooth interior approximations converge in local $W^{1,p}$ and, along a subsequence, a.e. The Lipschitz bound gives $F(u_j) \to F(u)$ in local L^p. Split the gradients:

$$F'(u_j)Du_j - F'(u)Du = F'(u_j)(Du_j - Du) + (F'(u_j) - F'(u))Du.$$

The first term tends to zero by boundedness of F'; the second by dominated convergence with majorant $2\|F'\|_\infty |Du|$. Pass to the weak identity locally. The candidate derivative is globally L^p, so this proves membership on U. For $p = \infty$, use finite local exponents and retain the resulting L^∞ bounds on function and gradient. □

18. Positive parts and level sets. Prove $|u|, u^+, u^- \in W^{1,p}(U)$ and

$$Du^+ = \mathbf{1}_{\{u>0\}}Du, \qquad Du^- = -\mathbf{1}_{\{u<0\}}Du, \qquad Du = 0 \text{ a.e. on } \{u = 0\}.$$

Solution. The function

$$F_\varepsilon(s) = \begin{cases} \sqrt{s^2 + \varepsilon^2} - \varepsilon, & s \ge 0, \\ 0, & s < 0 \end{cases}$$

is C^1, with $|F_\varepsilon'| \le 1$, $F_\varepsilon' \to \mathbf{1}_{\{s>0\}}$ pointwise, and $|F_\varepsilon(s) - s^+| \le \varepsilon$. Problem 17 and dominated convergence show for finite p that

$$F_\varepsilon(u) \to u^+ \text{ in } L^p, \qquad D(F_\varepsilon(u)) \to \mathbf{1}_{\{u>0\}}Du \text{ in } L^p.$$

The weak identity yields the positive-part formula. Apply it to $-u$ for the negative part. Differentiate $u = u^+ - u^-$ to obtain $Du = \mathbf{1}_{\{u\neq 0\}}Du$, proving the level-set assertion. Use $|u| = u^+ + u^-$ for the absolute value. For $p = \infty$ use a finite exponent on bounded U and the derivative bounds to recover $W^{1,\infty}$ membership. □

用常数检验边界项

球域 Hardy 需要低阶项，因为常数在球上属于 H^1；全空间非零常数不在 L^2，所以不反驳纯梯度型估计。

链式法则如何取极限

$F'(u_j) \to F'(u)$ 只是逐点信息；第二项需要由 $2\|F'\|_\infty |Du|$ 支配，第一项则依赖梯度的强收敛。

19. A second level-set argument for H^1. Let ϕ be smooth, bounded, nondecreasing, with bounded derivative, and $\phi(s) = s$ for $|s| \le 1$. Set $u_\varepsilon = \varepsilon\phi(u/\varepsilon)$.

Solution. Since $\phi(0) = 0$ and ϕ' is bounded, $|u_\varepsilon| \le \|\phi'\|_\infty |u|$ while $|u_\varepsilon| \le \varepsilon\|\phi\|_\infty \to 0$ pointwise. Dominated convergence gives $u_\varepsilon \to 0$ in L^2, even when U has infinite measure. Also $Du_\varepsilon = \phi'(u/\varepsilon)Du$ is bounded in L^2. Every sequence $\varepsilon_j \to 0$ has a weak H^1 subsequence by reflexivity; its limit is zero by strong L^2 convergence. Thus the whole family converges weakly to zero. Pairing its gradient with Du gives

$$0 \le \int_{\{u=0\}} |Du|^2 \le \int_U \phi'(u/\varepsilon)|Du|^2 = \int_U Du_\varepsilon \cdot Du \to 0.$$

We used $\phi' \ge 0$ and $\phi'(0) = 1$, without needing any pointwise limit of $\phi'(u/\varepsilon)$ away from $\{u = 0\}$. □

不能漏掉零水平集

$\{u = 0\}$ 可能有正测度，不能只在正负区域求导就结束。先证明正负部公式，再由 $u = u^+ - u^-$ 推出该集合上梯度为零。

5.10.5 Problems 20–21: Fourier estimates

20. Boundedness above half the dimension. Show $H^s(\mathbb{R}^n) \hookrightarrow L^\infty$ for $s > n/2$.

Solution. Put $w_s(\xi) = (1 + |\xi|^2)^{s/2}$. Cauchy–Schwarz gives

$$\|\hat{u}\|_1 \le \|w_s\hat{u}\|_2 \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s}\, d\xi \right)^{1/2} \le C_{n,s}\|u\|_{H^s}.$$

The radial integral at infinity is $\int_1^\infty r^{n-1-2s}\, dr$, finite exactly when $2s > n$. The inverse Fourier integral is bounded and continuous, with supremum at most $(2\pi)^{-n/2}\|\hat{u}\|_1$, and agrees with u in L^2 by Plancherel/inversion. □

半维数的来源

要让逆变换绝对可积，频率权重的倒数需属于 L^2；径向积分恰好给出条件 $2s > n$，而不是随意选的正则性门槛。

21. The algebra property. For $s > n/2$, prove $\|uv\|_{H^s} \le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}$.

Solution. For Schwartz functions the triangle inequality gives $w_s(\xi) \le C_s(w_s(\eta) + w_s(\xi - \eta))$. The product-convolution formula implies

$$w_s|\widehat{uv}| \le C_{n,s}\big[(w_s|\hat{u}|) * |\hat{v}| + |\hat{u}| * (w_s|\hat{v}|)\big].$$

Young's $L^2 * L^1 \to L^2$ inequality and Problem 20 give

$$\begin{aligned} \|uv\|_{H^s} &\le C_{n,s}\big(\|u\|_{H^s}\|\hat{v}\|_1 + \|\hat{u}\|_1\|v\|_{H^s}\big) \\ &\le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}. \end{aligned}$$

For general u, v, approximate their transforms by smooth compactly supported functions in weighted L^2. Their inverse transforms are Schwartz. The bilinear estimate makes their products Cauchy in H^s. Problem 20 gives uniform convergence of the factors, and

$$\|u_j v_j - uv\|_2 \le \|u_j\|_\infty \|v_j - v\|_2 + \|u_j - u\|_\infty \|v\|_2 \to 0.$$

Thus the H^s limit is the actual product uv. □

乘积为何不丢阶

$s > n/2$ 提供频率的 L^1 控制，使卷积可用 $L^2 * L^1$ 估计。因此多项式非线性可以在同一个 H^s 空间内处理。

5.11 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 5, printed pp. 239–292 (local English PDF pp. 256–309). The chapter order and source theorem identifiers follow that edition. The Chinese scan has Chapter 5 at printed pp. 169–207 (PDF pp. 187–225); the chapter opening and its list of section starts were visually checked. It is a terminology reference, not the source of the English prose written here.

For Sobolev theory, Evans's §5.11 points to Gilbarg–Trudinger (Chapter 7), Lieb–Loss, Ziemer, and Evans–Gariepy. For BMO the source points to Stein, Chapter IV; for time-dependent spaces to Temam, pp. 248–273; and for whole-space Hardy to Tartar, Chapter 17. These are source pointers, not claims that those books have been independently consulted for this draft.

Proof dependencies and review boundary. This chapter invokes the measure-theoretic and functional-analytic prerequisites listed at its opening, including Lebesgue differentiation, Plancherel, Riesz representation, reflexivity, and Arzel–Ascoli. These foundational theorems are explicitly invoked rather than reproved. Other displayed chapter theorems have reconstructed proofs here. All 21 exercises have editorial solutions or exact pointers to full arguments earlier in the chapter. Source comparison, mathematical self-checking, compilation, and visual inspection have separate scopes. This study draft has not received independent mathematical review.

Chapter 6

Second-Order Elliptic Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 6; printed pp. 293–348; PDF pages 310–365.

Chinese reference: 中译本：书内第 208–246 页；PDF 第 226–264 页。

Editorial study edition. This chapter reconstructs Evans's exposition in original English, with expanded proofs and Chinese mathematical annotations. It is not a verbatim transcription. Headings identify the source results; our theorem numbers run throughout this chapter. All 17 source problems are restated concisely and solved. Foundational functional analysis is invoked explicitly; the Schaefer fixed-point theorem and Weyl's asymptotic law are identified where used or stated, rather than presented as proved here.

The central example is

$$-\operatorname{div}(ADu) + b \cdot Du + cu = f \quad \text{in } U, \qquad u = 0 \quad \text{on } \partial U.$$

There are four distinct questions: does a weak solution exist, how smooth is it, what controls its sign, and when does the homogeneous equation have a nonzero solution? The answers require different hypotheses.

弱解到经典解

存在性先在能量空间中完成，不要求二阶导数。正则性随后补足导数，才能使用经典最大值原理；特征值问题再把紧性与正性结合起来。

Conventions and reading route. Repeated indices are summed from 1 to n. Unless specified otherwise, U is a bounded open subset of $\mathbb{R}^n$, $A = (a^{ij})$ is real symmetric, and $a^{ij}\xi_i\xi_j \geq \theta|\xi|^2$ for some $\theta > 0$. The norm $\|\cdot\|_2$ is over the indicated domain, or over U if omitted. Begin with §6.1–§6.2, then follow the difference quotient argument in §6.3. Read §6.4 before the positivity arguments in §6.5. Chapter 5 supplies approximation, traces, Poincaré, difference quotients, and Rellich compactness. Riesz representation, weak subsequential compactness in Hilbert spaces, and basic finite-dimensional linear algebra are prerequisites.

6.1 Definitions

6.1.1 Two forms of an elliptic operator

The divergence form and nondivergence form are, respectively,

$$Lu = -\partial_j(a^{ij}\partial_i u) + b^i\partial_i u + cu, \tag{6.1}$$

$$Ku = -a^{ij}\partial_{ij}u + \tilde{b}^i\partial_i u + cu. \tag{6.2}$$

Uniform ellipticity means

$$\theta|\xi|^2 \le a^{ij}(x)\xi_i\xi_j \quad (x \in U,\ \xi \in \mathbb{R}^n).$$

For measurable coefficients this is required almost everywhere. Boundedness of the entries gives an upper bound $\Lambda|\xi|^2$. It is the principal part, not the lower-order terms, that defines ellipticity.

If A is C^1, the two formulas agree precisely when $\tilde{b}^i = b^i - \partial_j a^{ij}$. If A is merely bounded measurable, this conversion is not a pointwise identity between ordinary functions. Moreover, symmetry of A does not imply symmetry of the full operator: the drift $b \cdot Du$ generally destroys self-adjointness.

散度型与非散度型

散度型可把一个导数移到测试函数上，因而适合弱解。只有系数足够可微时，才能展开散度并把系数导数并入漂移项。

For example, $A = \operatorname{diag}(1,4)$ has $\theta = 1$ and $\Lambda = 4$; the principal quadratic form has elliptical level sets. In contrast, $u_{xx} - u_{yy}$ has a principal form of both signs and is not elliptic. If the smallest eigenvalue of $A(x)$ approaches zero, pointwise positive definiteness alone gives no uniform θ.

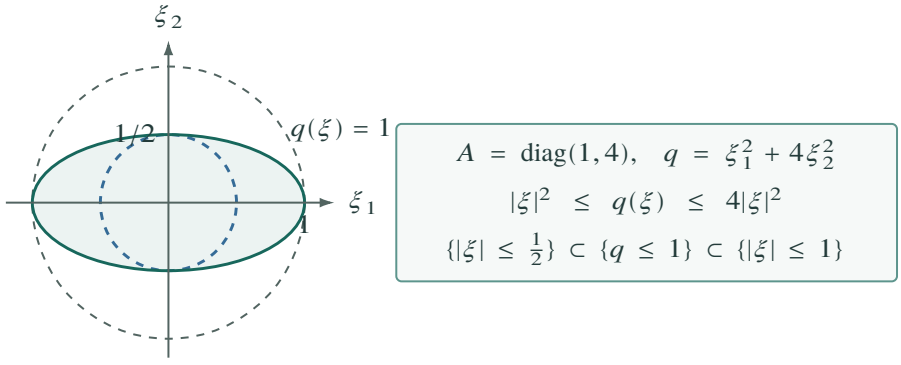


Diagram T6.1. Ellipticity bounds the shape of the principal quadratic form. Here $\theta = 1$ and $\Lambda = 4$. The sublevel ellipse contains the circle of radius $\Lambda^{-1/2}$ and lies inside the circle of radius $\theta^{-1/2}$. A uniform positive lower bound prevents this principal geometry from degenerating as the base point varies.

6.1.2 Weak solutions and the boundary condition

Assume $A, b, c \in L^\infty(U)$ and set

$$B[u,v] = \int_U (a^{ij}u_i v_j + b^i u_i v + cuv)\,dx, \qquad u, v \in H_0^1(U). \tag{6.3}$$

A weak solution of $Lu = f$, with zero Dirichlet boundary values, is $u \in H_0^1(U)$ such that

$$B[u, v] = \langle f, v \rangle \quad (v \in H_0^1(U)). \tag{6.4}$$

Here $f \in H^{-1}(U)$; in particular $f \in L^2(U)$ is allowed.

For smooth u with $u = 0$ on the boundary, integration by parts in the principal term gives (6.4). Conversely, testing against $C_c^\infty(U)$ shows that (6.4) implies $Lu = f$ as a distribution. If u subsequently belongs to $C^2(U)$ and the coefficients and f are continuous in the relevant form, the distributional equation becomes the classical equation.

A useful description of rough data is $f = f_0 - \operatorname{div} F$, with $f_0, F_i \in L^2(U)$. Its action is

$$\langle f, v \rangle = \int_U (f_0 v + F \cdot Dv)\, dx, \qquad |\langle f, v \rangle| \le C(\|f_0\|_2 + \|F\|_2)\|v\|_{H^1}.$$

The plus sign in the second term comes from distributional differentiation.

For prescribed trace $g = Tw$ with $w \in H^1(U)$, solve

$$B[z, v] = \langle f, v \rangle - B[w, v] \quad (v \in H_0^1(U)), \qquad z = u - w \in H_0^1(U).$$

Thus the homogeneous theory applies whenever a lift is available. The resulting u is independent of the chosen lift whenever the homogeneous problem is unique.

Adjoint and sign conventions. Define $B^*[v, u] = B[u, v]$. For smooth coefficients, the formal adjoint is

$$L^* v = -\partial_i(a^{ij}\partial_j v) - \partial_i(b^i v) + cv. \tag{6.5}$$

The adjoint drift differentiates the product $b^i v$. In particular, even if $c \ge 0$, the expanded zeroth-order coefficient of L^* can be negative. This matters in the nonsymmetric spectral theory.

零边界的准确含义

$H_0^1(U)$ 是紧支撑光滑函数在 H^1 中的闭包。光滑边界上它等价于零迹；一般弱解没有逐点边界值，更不能先假设法向导数存在。

非齐次边界先做提升

若边界数据是某个 $w \in H^1(U)$ 的迹，就写 $u = w + z$，再解 $z \in H_0^1(U)$。边界数据仅属于 $L^2(\partial U)$，并不自动保证存在这样的提升。

6.2 Existence of Weak Solutions

6.2.1 Lax–Milgram (Evans §6.2.1, Theorem 1)

Theorem 6.1

Let H be a real Hilbert space. Suppose a bilinear form B satisfies

$$|B[u,v]| \le M\|u\|_H\|v\|_H, \qquad B[u,u] \ge \beta\|u\|_H^2, \quad \beta > 0.$$

For each bounded linear functional ℓ on H there is a unique $u \in H$ with $B[u,v] = \ell(v)$ for all $v \in H$. Moreover, $\|u\|_H \le \beta^{-1}\|\ell\|_{H^*}$.

Expanded proof.

By Riesz representation there is a bounded linear operator $T : H \to H$ such that $B[u,v] = (Tu, v)_H$. Coercivity and Cauchy–Schwarz give

$$\beta\|u\|_H^2 \le (Tu,u)_H \le \|Tu\|_H\|u\|_H, \qquad \|Tu\|_H \ge \beta\|u\|_H.$$

Thus T is injective. Its range is closed: if Tu_j converges, the last inequality applied to $u_j - u_k$ makes (u_j) Cauchy, and its limit maps to the given range limit.

The range is also dense. If w is orthogonal to it, then $0 = (Tw, w)_H = B[w,w] \ge \beta\|w\|_H^2$, so $w = 0$. A closed dense subspace is all of H. Represent $\ell(v) = (g,v)_H$ and solve $Tu = g$. Uniqueness follows from injectivity; testing with $v = u$ gives the asserted bound. □

强制性排除失控方向

有界性保证双线性型连续；强制性则使 $B[u,u]$ 控制整个空间范数。这里不要求 B 对称，所以不能一律把证明改写成能量最小化。

6.2.2 Energy estimates and a shifted inverse (Evans §6.2.2)

Theorem 6.2 (Boundedness and Gårding inequality)

Under the hypotheses of §6.1, there are constants M, C_0 such that

$$|B[u,v]| \le M\|u\|_{H^1}\|v\|_{H^1}, \qquad B[u,u] \ge \tfrac{\theta}{2}\|Du\|_2^2 - C_0\|u\|_2^2.$$

Consequently, for sufficiently large $\kappa > 0$, $B_\kappa[u,v] = B[u,v] + \kappa(u,v)_{L^2}$ is coercive on $H_0^1(U)$. For every $f \in H^{-1}(U)$, $(L+\kappa)u = f$ has a unique weak solution, with $\|u\|_{H^1} \le C\|f\|_{H^{-1}}$.

Expanded proof.

Hölder bounds each integral defining B. For the diagonal,

$$\left|\int_U b\cdot Du\, u\right| \le \|b\|_\infty\|Du\|_2\|u\|_2 \le \tfrac{\theta}{2}\|Du\|_2^2 + \tfrac{\|b\|_\infty^2}{2\theta}\|u\|_2^2.$$

The potential contributes at worst $-\|c_-\|_\infty\|u\|_2^2$. Choose $C_0 = \|b\|_\infty^2/(2\theta) + \|c_-\|_\infty$ and $\kappa \ge C_0 + 1$. Then

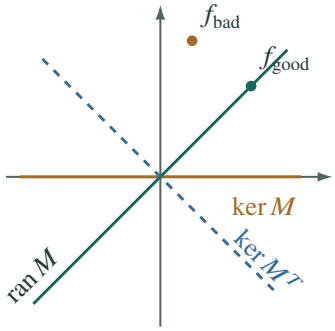
$$B_\kappa[u,u] \ge \min(\theta/2, 1)\|u\|_{H^1}^2.$$

Apply Theorem 6.1. All constants are independent of f. □

If $b = 0$ and $c \geq 0$, Poincaré gives coercivity without a shift. For general drift, $c \geq 0$ alone does not make this particular energy estimate coercive. Uniqueness may nevertheless follow from a maximum principle once the needed regularity is established.

6.2.3 The compact Fredholm mechanism (Evans §6.2.3)

Write $R = (L + \kappa)^{-1}$. As a map $L^2(U) \to H_0^1(U)$ it is bounded. The inclusion $H_0^1(U) \Subset L^2(U)$ is compact, so R is a compact operator on $L^2(U)$. For arbitrary bounded open U, this last compactness also follows by extending zero to a fixed enclosing ball and applying Chapter 5 there.



$$M = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad Mu = (u_2, u_2)$$

$$Mu = f \text{ is solvable iff } f_1 = f_2$$

$$\operatorname{ran} M = (\ker M^T)^{\perp}$$

$$\ker M \neq \ker M^T$$

Diagram T6.2. Compatibility uses the adjoint kernel. In this explicit nonsymmetric matrix example, the range is the diagonal line. A right side with $f_1 \neq f_2$ is unsolvable. Its obstruction is its component along $\ker M^T$, while the horizontal $\ker M$ describes nonuniqueness of solutions. The elliptic Fredholm alternative has the same distinction.

Lemma 6.3 (Compact Fredholm alternative)
For a compact operator T on a Hilbert space, $I - T$ has closed range, finite-dimensional kernel and cokernel of equal dimension, and

$$\operatorname{ran}(I - T) = \ker(I - T^*)^{\perp}.$$

In particular, injectivity is equivalent to surjectivity.

Expanded proof.

Compactness makes T a norm limit of finite-rank operators: choose a finite ε-net for the image of the unit ball, project onto the span of its centers, and set $F = PT$. Then $\|T - F\| \leq \varepsilon$. Choose $\varepsilon < 1$, put $E = T - F$, and invert $Q = I - E$ by its norm-convergent Neumann series. We have $I - T = Q(I - G)$, where $G = Q^{-1}F$ has finite rank.

平移算子而非平移解

加上 κu 是把算子换成 $L + \kappa I$，从而恢复强制性。原问题仍需另行研究；这个可逆的平移算子将成为 Fredholm 方法的起点。

紧性来自空间之间的往返

求解方程先把 L^2 数据送入更强的 H_0^1 空间，再忘掉导数回到 L^2。这第二步是紧嵌入；有界性本身不足以推出逆算子紧。

无限维问题的有限维障碍

紧性不能保证可逆，但能把不可逆的障碍压缩到有限维核中。右端必须与伴随核正交；非对称情形不能把伴随核随意换成原算子的核。

Let $M = \operatorname{ran} G$. Relative to $H = M \oplus M^{\perp}$,

$$I - G = \begin{pmatrix} D & -G|_{M^{\perp}} \\ 0 & I \end{pmatrix}, \qquad D = I_M - G|_M .$$

Its kernel is $\ker D \oplus \{0\}$. Its range consists of pairs (x, y) with $x + Gy \in \operatorname{ran} D$. This condition defines a closed subspace of codimension $\dim(M/\operatorname{ran} D) = \dim \ker D$. Multiplication by the isomorphism Q preserves closedness and both dimensions. Finally, for every bounded operator S, $(\operatorname{ran} S)^{\perp} = \ker S^{*}$; closedness now gives the formula. □

Theorem 6.4 (Solvability and resonance)
For each real λ, the weak Dirichlet equation $(L-\lambda)u = f, f \in L^2(U)$, has the following alternative. Either the homogeneous equation has only the zero solution and every f has a unique solution, or its kernel is nonzero and finite dimensional. In the latter case solvability is exactly

$$(f, v)_{L^2} = 0 \quad \text{for every weak solution } (L^{*} - \lambda)v = 0.$$

Whenever a solution exists, all solutions differ by a homogeneous solution.

Expanded proof.

For $\lambda \neq -\kappa$ the equation is equivalent to

$$[I - (\lambda + \kappa)R]u = Rf \quad \text{in } L^2(U). \tag{6.6}$$

Conversely, any L^2 solution of this identity satisfies $u = R((\lambda + \kappa)u + f)$, so lies in $H_0^1(U)$ and solves the original weak equation.

The adjoint inverse is $R^{*} = (L^{*} + \kappa)^{-1}$: if $u = Rf$ and $v = (L^{*} + \kappa)^{-1} g$, then $(u, g) = B_{\kappa}[u, v] = (f, v)$. Thus the kernel of $I - (\lambda + \kappa)R^{*}$ is precisely $\ker(L^{*} - \lambda)$. For v in this kernel,

$$(Rf, v) = (f, R^{*}v) = \frac{(f, v)}{\lambda + \kappa}.$$

Lemma 6.3 gives the required compatibility condition and the finite dimensional alternative. At $\lambda = -\kappa$ the shifted problem is already uniquely solvable. Subtraction describes all solutions. □

相容条件来自伴随恒等式

若伴随齐次解为 v，把原方程与它配对就得到 $(f, v) = 0$。Fredholm 定理补上反方向：所有这些正交条件满足时，解确实存在。

6.2.4 The exceptional set and bounds for the inverse

Theorem 6.5 (Evans §6.2.3, Theorems 5–6)
Let Σ be the set of real Dirichlet eigenvalues of L. It is bounded below and has only finitely many elements in any bounded interval. If infinite, its elements tend to $+\infty$. If $\lambda \notin \Sigma$, the solution of $(L-\lambda)u = f$ satisfies

$$\|u\|_{H^1(U)} \le C_\lambda \|f\|_2.$$

Expanded proof.

For an eigenfunction, Gårding gives $\lambda\|u\|_2^2 = B[u,u] \ge -C_0\|u\|_2^2$, hence $\lambda \ge -C_0$. Nonzero eigenvalues μ of the compact operator R have finite multiplicity and cannot accumulate away from zero. Here is the compactness argument for the latter assertion. If infinitely many distinct μ_j satisfy $|\mu_j| \ge \delta > 0$, let E_j be the span of eigenvectors for $\mu_1, \dots, \mu_j$, and choose $x_j \in E_j \cap E_{j-1}^{\perp}$ with $\|x_j\| = 1$. Then $(R-\mu_j I)x_j \in E_{j-1}$, whereas $Rx_i \in E_{j-1}$ for $i < j$. Consequently $\|Rx_j - Rx_i\| \ge |\mu_j| \ge \delta$, contradicting compactness. An infinite-dimensional eigenspace is excluded by the same argument with an orthonormal sequence. Since $\mu = (\lambda+\kappa)^{-1}$, the assertions about real Σ follow.

If an L^2 inverse bound failed at a fixed $\lambda \notin \Sigma$, there would be solutions u_j with $\|u_j\|_2 = 1$ and right sides $f_j \to 0$ in L^2. Gårding, applied to $B[u_j,u_j] = \lambda\|u_j\|_2^2 + (f_j,u_j)$, bounds (u_j) in H_0^1. A subsequence converges weakly in H_0^1 and strongly in L^2 to u of norm one. Passing to each fixed test function yields $(L-\lambda)u = 0$, a contradiction. The same energy inequality then upgrades the L^2 bound to the displayed H^1 estimate. □

For a nonsymmetric operator, complex eigenvalues can also occur after complexification, and the real exceptional set need not be infinite. An orthogonal eigenbasis or an exact inverse norm equal to a reciprocal spectral distance requires additional self-adjoint structure.

逼近共振时常数会变大

若 $Lw = \lambda_* w$ 且 $\|w\|_2 = 1$，则用右端 $(\lambda_* - \lambda)w$ 得到解 w。因此逆算子范数至少为 $|\lambda - \lambda_*|^{-1}$，不能给出跨越共振点的一致估计。

6.3 Regularity

6.3.1 Interior energy localization

Throughout the H^2 argument, $A \in C^1(U)$ is uniformly elliptic, $b, c \in L^\infty_{\mathrm{loc}}(U)$, $f \in L^2_{\mathrm{loc}}(U)$, and $u \in H^1(U)$ solves $Lu = f$ against compactly supported tests. All coefficient bounds below are on a fixed relatively compact region.

Lemma 6.6 (Caccioppoli estimate)
For a smooth cutoff $0 \le \eta \le 1$ with compact support in U,

$$\int_U \eta^2 |Du|^2 \le C \int_{\operatorname{supp}\eta} (|f|^2 + (1 + |D\eta|^2)|u|^2).$$

Expanded proof.

The admissible test $\eta^2 u$ gives

$$\int \eta^2 ADu \cdot Du = \int f \eta^2 u - 2 \int \eta u\, ADu \cdot D\eta - \int b \cdot Du\, \eta^2 u - \int c\eta^2 u^2.$$

The second and third terms are bounded, for any $\varepsilon > 0$, by

$$\varepsilon \int \eta^2 |Du|^2 + C_\varepsilon \int (|D\eta|^2 + \eta^2) u^2.$$

Use $|fu| \le (f^2 + u^2)/2$, bound the potential term, and choose the two ε's so their sum is less than $\theta/2$. Move these gradient terms to the left. □

6.3.2 Interior H^2 regularity (Evans §6.3.1, Theorem 1)

Theorem 6.7
For $V \Subset U$, a weak solution as above belongs to $H^2(V)$ and

$$\|u\|_{H^2(V)} \le C\big(\|f\|_{L^2(W)} + \|u\|_{L^2(W)}\big), \qquad V \Subset W \Subset U.$$

Here C depends on the nested regions, ellipticity and the local C^1 norm of A and L^∞ norms of b, c. Local Lipschitz bounds on A suffice for this proof.

Expanded proof.

1. Choose a legal discrete substitute for a second derivative. Fix a coordinate direction e_k and set $\delta_h z(x) = [z(x + he_k) - z(x)]/h$ for either sign of h. Choose $\eta = 1$ on V, compactly supported in an intermediate region, and $|h|$ small enough that all translated supports remain in W. For fixed h,

$$q = \eta^2 \delta_h u \in H_0^1(W), \qquad v = -\delta_{-h} q \in H_0^1(W).$$

The difference quotient is in H^1 for fixed h even though its H^1 norm has not yet been bounded uniformly. Thus v is a valid test.

截断函数的代价

局部估计中出现 $D\eta$，所以常数依赖内外区域的距离。把内区推向边界会使常数变大，不能把内部估计直接当成全局边界估计。

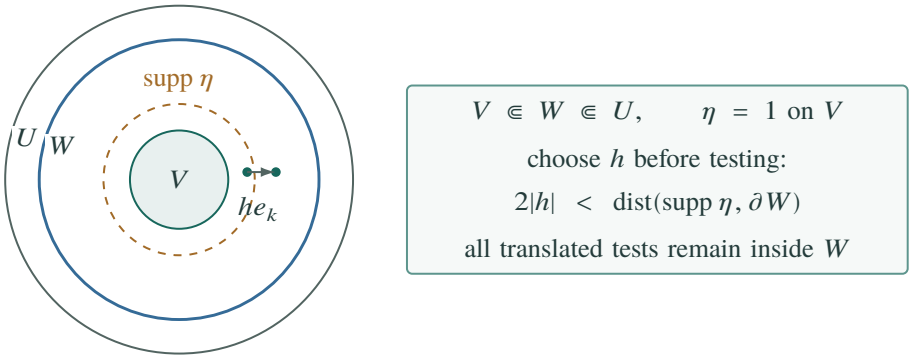


Diagram T6.3. Localize before taking differences. The cutoff equals one on V and is supported strictly inside W. A fixed geometric gap permits all sufficiently small positive and negative steps. The support condition is settled before the estimate; the estimate's constant must then stay independent of h as $h \to 0$.

2. Discrete integration by parts. A change of variables, with no support crossing the boundary, gives

$$\int g\, \delta_{-h} q = -\int (\delta_h g) q.$$

Absorb lower-order terms into $\tilde{f} = f - b \cdot Du - cu \in L^2(W)$. The weak equation becomes

$$\int \delta_h(a^{ij} u_i)\, \partial_j(\eta^2 \delta_h u) = -\int \tilde{f}\, \delta_{-h}(\eta^2 \delta_h u).$$

The exact product rule is

$$\delta_h(a^{ij} u_i) = a^{ij}(x + he_k)\delta_h u_i + (\delta_h a^{ij}) u_i.$$

The coefficient in the first term is translated and remains elliptic.

3. Estimate each error before absorbing it. Put $X = \|\eta\, \delta_h Du\|_2$ and $Y = \|Du\|_{L^2(W)} + \|u\|_{L^2(W)}$. The main term is at least θX^2. Since $\|\delta_h u\|_2 \le \|\partial_k u\|_{L^2(W)}$ on the relevant supports and $|\delta_h A| \le \|DA\|_\infty$, the other terms on the left are bounded by

$$CXY + CY^2 \le \varepsilon X^2 + C_\varepsilon Y^2.$$

For example the cutoff term is bounded by $C\|\eta\, \delta_h Du\|_2 \|D\eta\, \delta_h u\|_2$; the coefficient commutator is bounded by $C\|Du\|_2(X + \|\delta_h u\|_2)$.

For the right side, the difference quotient inequality applied to q gives

$$\|\delta_{-h} q\|_2 \le \|\partial_k q\|_2 \le C(X + Y).$$

Therefore it is at most $\varepsilon X^2 + C_\varepsilon(\|\tilde{f}\|_2^2 + Y^2)$. Since $\|\tilde{f}\|_2 \le C(\|f\|_2 + Y)$, absorption yields

$$\|\delta_h Du\|_{L^2(V)}^2 \le C\big(\|f\|_{L^2(W)}^2 + \|u\|_{H^1(W)}^2\big), \tag{6.7}$$

with C independent of h.

不能先假定二阶导数

直接测试 $-\partial_{kk} u$ 会预先要求尚未证明的正则性。固定步长的差商仍在 H^1 中，先合法计算，再用一致估计令步长趋零。

一致估计才能识别弱导数

差商判别准则作用于每个一阶导数 u_i：各方向的 $\delta_h u_i$ 一致有界，就得到 $u_{ik} \in L^2$。只证明每个固定步长有限并没有用。

4. Pass to weak derivatives and remove the first-order norm. The difference quotient criterion of Chapter 5 applied to each u_i gives $u_{ik} \in L^2(V)$. Repeating in every direction proves H^2. Apply (6.7) on slightly larger nested regions and use Lemma 6.6 there to bound $\|Du\|_2$ by f and u. This gives the stated estimate with only the L^2 norm of u. □

6.3.3 Higher interior regularity (Evans §6.3.1, Theorems 2–3)

Theorem 6.8

Let $m \geq 0$ be an integer, $A, b, c \in C^{m+1}(U)$, and $f \in H^m_{\mathrm{loc}}(U)$. Then $u \in H^{m+2}_{\mathrm{loc}}(U)$, with

$$\|u\|_{H^{m+2}(V)} \leq C\big(\|f\|_{H^m(W)} + \|u\|_{L^2(W)}\big), \qquad V \Subset W \Subset U.$$

Smooth coefficients and smooth f give $u \in C^\infty(U)$.

Expanded proof.

The case $m = 0$ is proved. Suppose $u \in H^{m+2}$ on intermediate regions and $f \in H^{m+1}$. For $|\alpha| = m+1$, differentiate the equation in distributions. With $z = D^\alpha u \in H^1$,

$$\begin{aligned} Lz = D^\alpha f &+ \sum_{0<\beta\leq\alpha} \binom{\alpha}{\beta} \partial_j\big((D^\beta a^{ij})\,\partial_i D^{\alpha-\beta} u\big) \\ &- \sum_{0<\beta\leq\alpha} \binom{\alpha}{\beta} \big((D^\beta b^i)\,\partial_i D^{\alpha-\beta} u + (D^\beta c) D^{\alpha-\beta} u\big). \end{aligned}$$

Expand the divergence in the first line. Its highest derivative of u has order $|\alpha| - |\beta| + 2 \leq m+2$, already known in L^2; coefficient derivatives have order at most $m+2$. Thus the entire right side lies in L^2. Interior H^2 regularity for z gives derivatives of u through order $m+3$. A finite chain of nested regions gives the norm bound inductively. For any fixed k, choose a Sobolev order $s > k + n/2$ and use the embedding $H^s_{\mathrm{loc}} \hookrightarrow C^k_{\mathrm{loc}}$. Since k is arbitrary, smoothness follows. □

6.3.4 Boundary H^2 regularity (Evans §6.3.2, Theorem 4)

Theorem 6.9
Suppose ∂U is C^2, $A \in C^1(\overline{U})$, $b, c \in L^\infty(U)$ and $f \in L^2(U)$. A weak Dirichlet solution $u \in H_0^1(U)$ belongs to $H^2(U)$ and satisfies

$$\|u\|_{H^2(U)} \le C(\|f\|_2 + \|u\|_2).$$

If the Dirichlet operator is invertible, the last term can be removed.

Expanded proof.

1. A flat boundary permits tangential differences. Work first in a half-ball $B_r^+ = \{x \in B_r : x_n > 0\}$, where the trace of u is zero on $x_n = 0$. Choose a cutoff vanishing near the curved boundary. For $k < n$, tangential translation preserves the half-space and the zero trace, so the test $-\delta_{-h}(\eta^2 \delta_h u)$ remains admissible. The preceding calculation bounds every u_{ik} with $k < n$ in L^2 on a smaller half-ball. This includes the mixed normal–tangential derivatives, since distributional derivatives commute.

2. Ellipticity recovers the missing derivative. Expand the divergence equation in distributions:

$$a^{nn} u_{nn} = - \sum_{(i,j) \neq (n,n)} a^{ij} u_{ij} + (b^i - \partial_j a^{ij}) u_i + cu - f. \tag{6.8}$$

Every term on the right is in L^2 by Step 1. Because $a^{nn} \ge \theta$, division bounds u_{nn}. One rigorous way to avoid assuming H^2 up to the boundary at this step is to use the already proved interior H^2 theorem: the identity holds almost everywhere away from $x_n = 0$, and its right side has a uniform L^2 bound on the whole smaller half-ball. The resulting L^2 function is the distributional derivative u_{nn} on this open half-ball. No boundary distribution is introduced because test functions are supported inside it.

3. Flatten a curved boundary without losing coefficients. After a rigid motion, write the boundary as $x_n = \gamma(x')$ and set

$$y = \Phi(x) = (x', x_n - \gamma(x')), \qquad v(y) = u(\Phi^{-1}(y)).$$

This particular map has determinant one. Writing $J = D\Phi$, the weak change of variables gives

$$\widehat{A}(y) = JAJ^T, \quad \hat{b}(y) = Jb, \quad \hat{c}(y) = c, \quad \hat{f}(y) = f,$$

提升正则性不是循环论证

先由已知阶数判断交换子右端属于 L^2，再把二阶正则性定理用于导函数。光滑系数得到光滑解，并不自动得到解析解。

正则性估计不等于可解性

估计以解已经存在为前提。右边保留 $\|u\|_2$ 是为容纳有限维齐次核；只有先证明算子可逆，才能用右端独自控制整个解。

法向差商为何失效

沿边界平移不会越出半空间，也保留零迹；法向平移会穿过边界，测试函数不再合法。因此先求所有带切向指标的导数，最后用方程解出纯法向导数。

where all right sides are evaluated at $\Phi^{-1}(y)$. Indeed $D_x u = J^T D_y v$ and $dx = dy$; the same gradient formula holds for each test function. Further,

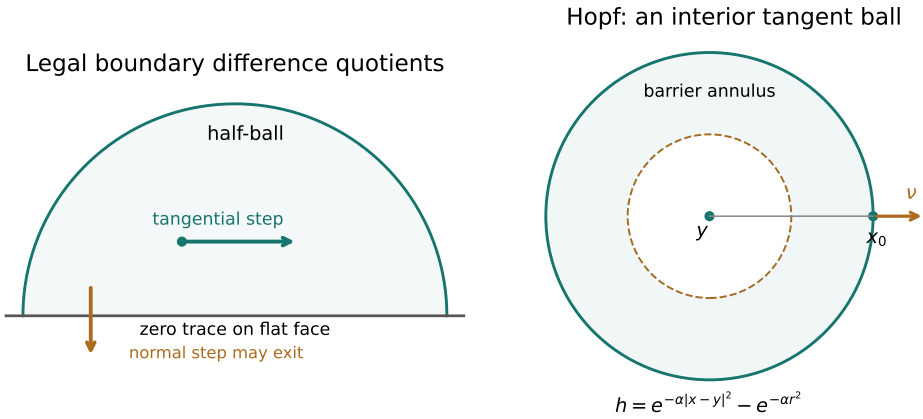
$$\xi^T \widehat{A}\xi \geq \theta |J^T \xi|^2 \geq \theta \|J^{-T}\|^{-2} |\xi|^2.$$

Bounded C^2 coordinate maps preserve the relevant H^2 estimates by the chain rule, including its lower-order terms. Since $\gamma \in C^2$ and $A \in C^1$, $\widehat{A} \in C^1$. The flat argument therefore applies.

4. Patch the local estimates. A finite boundary cover, together with interior patches, yields $\|u\|_{H^2} \leq C(\|f\|_2 + \|u\|_{H^1})$. For a solution, Gårding and $B[u,u] = (f,u)$ imply $\|Du\|_2^2 \leq C(\|f\|_2^2 + \|u\|_2^2)$. This removes the first-order term. At a nonresonant parameter, Theorem 6.5 removes $\|u\|_2$ as well. □

拉平边界也要变换算子

坐标变换后主系数是 JAJ^T。此处选用行列式为一的图形变换；一般坐标变换还会出现体积雅可比因子，不能把它漏掉。



Two boundary mechanisms. Tangential translations preserve the flat zero-trace face; normal translations need not. Hopf's exponential barrier is used only in the shaded annulus of an interior tangent ball. Diagrams are schematic.

6.3.5 Higher boundary regularity (Evans §6.3.2, Theorems 5–6)

Theorem 6.10

If ∂U is C^{m+2}, $A, b, c \in C^{m+1}(\overline{U})$ and $f \in H^m(U)$, then a weak zero-Dirichlet solution satisfies

$$u \in H^{m+2}(U), \qquad \|u\|_{H^{m+2}(U)} \leq C(\|f\|_{H^m(U)} + \|u\|_2).$$

Smooth boundary, coefficients and data give $u \in C^\infty(\overline{U})$.

Expanded proof.

The case $m = 0$ is Theorem 6.9. For the induction step assume $u \in H^{m+2}$ and $f \in H^{m+1}$, with coefficients C^{m+2} and boundary C^{m+3}.

In a flattened chart, differentiate $m+1$ times only tangentially. The resulting derivative has zero trace on the flat face, because tangential differentiation commutes with the trace at these orders. Its localized equation has an L^2 right side: expand the commutators exactly as in the interior proof. Cutoffs keep tests away from the curved part of the chart. Applying boundary H^2 gives all derivatives of total order $N = m+3$ having at most two normal differentiations.

Recover the others from (6.8). To estimate an order-N derivative with $q \geq 3$ normal indices, apply an order-$N-2$ derivative with $q-2$ normal indices to that identity. The desired term is multiplied by a^{nn}; the other order-N terms have at most $q-1$ normal indices, and coefficient commutators contain derivatives of u of order at most $N-1$, already controlled. Induct on q and divide by $a^{nn} \geq \theta$. These differentiated identities can be justified first by interior regularity and then by the uniform estimates up to the flat face, as in the H^2 case. Patching proves the global estimate. Finally extend Sobolev functions across the smooth boundary and use embedding at arbitrarily high orders to obtain $C^\infty(\overline{U})$. □

What the estimates actually say. The two-derivative gain is an a priori estimate for solutions, not an existence theorem by itself. At an eigenvalue, the term $\|u\|_2$ cannot be removed: a nonzero eigenfunction solves a homogeneous shifted equation. Nonzero boundary data require a sufficiently regular lift before the zero-boundary theorem can be used.

边界光滑度不能省略

内部正则性只看系数和右端，边界正则性还看区域形状。角点可能产生奇性；零边界条件本身并不能保证解在角点附近有平方可积的二阶导数。

6.4 Maximum Principles

Here we use nondivergence notation

$$Lu = -a^{ij}u_{ij} + b^i u_i + cu.$$

Assume U bounded, A uniformly elliptic, and coefficients bounded and continuous unless a stronger hypothesis is stated. Solutions are classical, $u \in C^2(U) \cap C(\overline{U})$. These are different assumptions from the measurable-coefficient weak theory.

6.4.1 Weak maximum principles (Evans §6.4.1, Theorems 1–2)

Theorem 6.11

If $c = 0$ and $Lu \leq 0$, then $\max_{\overline{U}} u = \max_{\partial U} u$. If $c \geq 0$ and $Lu \leq 0$, then

$$\max_{\overline{U}} u \leq \max_{\partial U} u^+.$$

In particular, $Lu \le Lv$ and $u \le v$ on ∂U imply $u \le v$ in U when $c \ge 0$.

Expanded proof.

First let $c = 0$. A strict subsolution $Lu < 0$ cannot have an interior maximum: at such a point $Du = 0$, D^2u is negative semidefinite, and $-A : D^2u \ge 0$, a contradiction. For a non-strict subsolution use $u_\varepsilon = u + \varepsilon e^{\tau x_1}$. Since

$$L(e^{\tau x_1}) = (-a^{11}\tau^2 + b^1\tau)e^{\tau x_1} < 0$$

for a fixed sufficiently large τ, u_ε has its maximum on the boundary. Let $\varepsilon \downarrow 0$.

For $c \ge 0$, on the open set $D = \{u > 0\}$ the operator $K = L - c$ satisfies $Ku = Lu - cu \le 0$. Apply the zero-potential result on D; boundary values on $\partial D \cap U$ are zero, while the remaining values come from ∂U. The proof uses compact maxima and does not require a smooth level-set boundary. If D is empty the conclusion is immediate. Apply this assertion to $u - v$ for comparison. □

Applying the theorem to $-u$ gives the corresponding minimum principles. The restriction on c is substantive: on $(0, \pi)$, $u = \sin x$ solves $-u'' - u = 0$ with zero boundary values but is positive inside.

For comparison, a direct weak energy argument is available in divergence form when $b \in C^1(\overline{U})$ and $c - \frac{1}{2}\operatorname{div} b \ge 0$. For a weak subsolution with $u^+ \in H_0^1(U)$, testing with u^+ gives

$$0 \ge B[u, u^+] = \int ADu^+ \cdot Du^+ + \int (c - \tfrac{1}{2}\operatorname{div} b)(u^+)^2 \ge \theta \|Du^+\|_2^2.$$

Thus $u \le 0$. The drift integral was integrated by parts after writing $u^+Du^+ = D((u^+)^2)/2$. This sufficient energy condition is distinct from $c \ge 0$ in the classical nondivergence argument.

6.4.2 Hopf's boundary lemma (Evans §6.4.2)

Lemma 6.12

Suppose an interior ball $B_r(y) \subset U$ touches ∂U at x_0, $u(x) < u(x_0) = M$ for all $x \in U$, and $Lu \le 0$. Assume either $c = 0$, or $c \ge 0$ and $M \ge 0$. If u is C^1 at x_0 in the inward-ball neighborhood, then

$$\partial_\nu u(x_0) > 0, \qquad \nu = (x_0 - y)/r.$$

At a smooth tangency this is the outward normal to U.

先固定算子符号

本章主部是负号 $-a^{ij}u_{ij}$。因此 $Lu \le 0$ 控制上界；在内部极大点，$Du = 0$、$D^2u \le 0$，主部反而非负。换符号时不等式也要一起换。

弱最大值原理中的“弱”

这里“弱”表示结论只控制边界极值，并非指 Sobolev 弱解。对弱解也有相应原理，但必须说明测试函数和系数条件，不能混用两个含义。

Expanded proof.

On $r/2 < |x-y| < r$, put $h(x) = e^{-\alpha|x-y|^2} - e^{-\alpha r^2}$. It is positive inside and zero on the outer sphere. A direct calculation, with $s = |x-y|$, gives

$$Lh \le e^{-\alpha s^2}\left(-4\theta\alpha^2 s^2 + 2\alpha \operatorname{tr} A + 2\alpha|b|s + c\right) \le 0$$

for sufficiently large α; for $c = 0$ the same bound applies. On the inner sphere $u - M$ has a strictly negative maximum, so choose $\varepsilon > 0$ with $u - M + \varepsilon h \le 0$ there. It is also nonpositive on the outer sphere. Moreover $L(u - M + \varepsilon h) \le -cM \le 0$. Comparison on the annulus gives $u - M \le -\varepsilon h$. Along the radius $x_0 - t\nu$ divide by $t > 0$ and let $t \downarrow 0$:

$$\partial_\nu u(x_0) \ge -\varepsilon \partial_\nu h(x_0) = 2\varepsilon\alpha r e^{-\alpha r^2} > 0.$$

□

6.4.3 Strong maximum principle (Evans §6.4.2, Theorems 3–4)

Theorem 6.13

Let U be connected. If $Lu \le 0$ and $c = 0$, attainment of a maximum at an interior point forces u to be constant. For $c \ge 0$ the same holds for a nonnegative interior maximum. Dually, a nonpositive interior minimum of a supersolution forces constancy.

Expanded proof.

Let $M = \max_U u$, attained inside, and suppose both sets $C = \{u = M\}$ and $V = \{u < M\}$ are nonempty. A path inside U from V to C shows that their common relative boundary contains a point in U. Choose $y \in V$ sufficiently close to this point so that the ball with radius $\operatorname{dist}(y, C)$ is contained in U. Its interior is in V and its boundary touches C at some $x_0 \in U$. Hopf's lemma on this ball gives a strictly positive radial derivative at x_0. But x_0 is an interior maximum of a differentiable function, so $Du(x_0) = 0$. This contradiction proves the assertion. The minimum statement follows by replacing u by $-u$. □

In particular, for $c \ge 0$, a nonzero nonnegative supersolution $Lu \ge 0$ is strictly positive in a connected domain.

Hopf 引理提供严格斜率

最大值原理只说内部不能超过边界。Hopf 引理进一步说：若内部严格更小，函数沿外法向到达边界极大值时必须有正斜率；内球条件保证可放入显式屏障。

6.4.4 Harnack's inequality (Evans §6.4.3, Theorem 5)

Theorem 6.14 (Smooth principal-part case proved here)
Suppose A is smooth and uniformly elliptic and $-a^{ij}u_{ij} = 0$, $u \geq 0$ in U. For connected $V \Subset U$,

$$\sup_V u \leq C \inf_V u.$$

The constant depends on the geometry of V inside U, ellipticity and local coefficient bounds, including the derivatives used below, but not on u.

Expanded proof.

1. Take a logarithm to estimate relative oscillation. First take $u > 0$, and put $v = \log u$ and $w = a^{ij}v_i v_j$. Then

$$-a^{ij}v_{ij} = w, \qquad \theta|Dv|^2 \leq w \leq \Lambda|Dv|^2. \tag{6.9}$$

Interior smoothness justifies the following differentiations: rewrite the smooth nondivergence equation in divergence form and apply §6.3. On a fixed compact region,

$$w_{kl} = 2a^{ij}v_{ikl}v_j + 2a^{ij}v_{ik}v_{jl} + R_{kl}, \quad |R_{kl}| \leq \varepsilon|D^2v|^2 + C_\varepsilon|Dv|^2.$$

The remainder consists of $D^2A\,Dv\,Dv$ and $DA\,D^2v\,Dv$. Differentiate (6.9) once to obtain $-a^{kl}v_{ikl} = w_i + (\partial_i a^{kl})v_{kl}$. Also $a^{ij}a^{kl}v_{ik}v_{jl} \geq \theta^2|D^2v|^2$; this is seen by diagonalizing the positive matrix A at the point. Absorbing coefficient remainders gives, for $\mathscr{M} = -a^{kl}\partial_{kl} - 2a^{ij}v_j\partial_i$,

$$\mathscr{M}w \leq -\gamma|D^2v|^2 + C|Dv|^2 \leq -\gamma_1 w^2 + Cw.$$

The second inequality uses $w = -A : D^2v \leq C|D^2v|$.

2. Localize and retain the drift–cutoff error. Take $\zeta \in C_c^\infty(U)$ with $0 \leq \zeta \leq 1$ and $\zeta = 1$ on a chosen ball V. Let $z = \zeta^4 w$. If its maximum is positive, it occurs where $\zeta > 0$, and $\zeta Dw = -4wD\zeta$. At this point $Dz = 0$ and $\mathscr{M}z = -A : D^2z \geq 0$. Expanding the product yields

$$0 \leq -\gamma_1\zeta^4 w^2 + C\zeta^2 w + C\zeta^3 w^{3/2}.$$

The last term comes from $-2ADv \cdot D(\zeta^4)\,w$ and uses $|Dv| \leq C\sqrt{w}$. It must be included because this drift depends on v. Set $Y = \zeta^2 w$ at this maximum point. Then

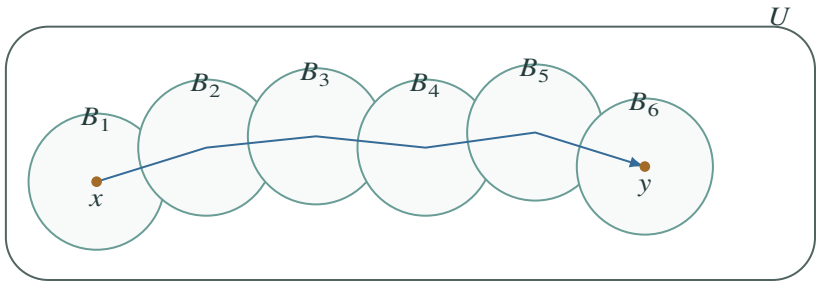
$$\gamma_1 Y^2 \leq C(Y + Y^{3/2}).$$

For $Y > 0$, divide by Y and solve the resulting quadratic inequality in $\sqrt{Y}$ to get $Y \leq C$. Hence $z \leq C$ at its maximum, and $w \leq C$, $|Dv| \leq C$ on V.

Harnack 是定量正性

强最大值原理说非负非零解处处为正；Harnack 还控制内区任意两点的比值。它需要方程等式，不能随意用于只有单边不等式的函数。

3. Integrate the logarithmic gradient. For x, y in the ball, integrate along their segment: $|v(x)-v(y)| \le C|x-y| \le C \operatorname{diam} V$. Exponentiation bounds $u(x)/u(y)$. For a general connected $V \Subset U$, cover it by finitely many balls whose slightly larger concentric balls remain inside U. A finite chain of overlapping balls connects their values; multiplying the finitely many comparison constants proves the result. Finally apply the estimate to $u+\delta$, which solves the same equation because there are no lower-order terms, and let $\delta \downarrow 0$. □



$$\sup_{B_i} u \le C_i \inf_{B_i} u \quad \Longrightarrow \quad u(x) \le (C_1 \cdots C_6)\, u(y)$$

Diagram T6.4. Local positivity comparisons propagate along a chain. Choose overlapping balls whose slightly larger balls still lie inside U. Comparing at overlap points transfers a local Harnack bound from x to y. The resulting constant depends on the finite chain and its distance from the boundary, which is why connected compact interior regions occur in the theorem.

Scope of the source theorem. Evans states Harnack for the full operator with lower-order terms and indicates extensions to bounded measurable coefficients. The proof supplied in the source, and reconstructed above, is restricted to smooth A and $b = c = 0$. The broader theorem is a further regularity result, cited there to Gilbarg–Trudinger; it is not proved by the logarithmic computation above. In the general case $u+\delta$ need not solve $Lu = 0$, and differentiating measurable coefficients is unavailable.

对数变换后的漂移项

新算子的漂移含有 $D\log u$，不能预先当成已有一致界的系数。截断计算保留三次项，再用四次项吸收，才能保证最终常数不依赖解。

6.5 Eigenvalues and Eigenfunctions

Assume in this section that U is bounded, connected and has smooth boundary, and that all coefficients are smooth on $\overline{U}$. This keeps the spectral arguments separate from technical boundary regularity issues. Eigenfunctions have zero Dirichlet trace.

6.5.1 Symmetric operators (Evans §6.5.1, Theorem 1)

First let $Lu = -\partial_j(a^{ij}u_i)$ with A symmetric uniformly elliptic. Then $B[u,v] = \int ADu \cdot Dv$ is a symmetric positive form.

Theorem 6.15

There are real eigenvalues, repeated according to multiplicity,

$$0 < \lambda_1 \le \lambda_2 \le \cdots, \qquad \lambda_j \longrightarrow \infty,$$

and a complete orthonormal basis (w_j) of $L^2(U)$ such that $Lw_j = \lambda_j w_j$. Each eigenspace is finite dimensional, and $w_j \in C^\infty(\overline{U})$.

Expanded proof.

Let $S = L^{-1} : L^2 \to L^2$. Lax–Milgram, Poincaré and Rellich show that S is bounded and compact. If $u = Sf$ and $v = Sg$, then

$$(Sf, g) = B[u,v] = (f, Sg), \qquad (Sf, f) = B[u,u] > 0 \quad (f \neq 0).$$

Thus S is self-adjoint, positive definite, and injective.

For completeness we give the compact spectral argument. Let $\mu_1 = \sup_{\|f\|_2=1}(Sf,f) > 0$. For a maximizing sequence f_k, pass to a weakly convergent subsequence $f_k \rightharpoonup f$ with $\|f\|_2 \le 1$. Compactness gives $Sf_k \to Sf$, so $(Sf,f) = \mu_1$. If $\|f\|_2 < 1$, scaling f to norm one increases this value, a contradiction. First variation on the unit sphere gives $Sf = \mu_1 f$.

Its orthogonal complement is invariant under S, by self-adjointness. Repeat the argument there, recording multiplicities, to obtain $\mu_1 \ge \mu_2 \ge \cdots > 0$ and orthonormal eigenvectors w_j. There are infinitely many, since an injective operator on infinite dimensional $L^2(U)$ cannot have finite-dimensional range. Compactness forces $\mu_j \to 0$: otherwise (Sw_j) has pairwise distances bounded below. If the closed span of the w_j had a nonzero orthogonal complement, the restriction of S to that complement would still be compact and positive definite and hence would have a positive maximal eigenvalue. That value would be at most every μ_j, contradicting $\mu_j \to 0$. This proves completeness. Set $\lambda_j = \mu_j^{-1}$. Finite multiplicity follows from compactness, and regularity bootstrapping applied to $Lw_j = \lambda_j w_j$ gives smoothness. □

6.5.2 Rayleigh quotient and positivity (Evans §6.5.1, Theorem 2)

Theorem 6.16

The first eigenvalue satisfies

$$\lambda_1 = \min_{\substack{u \in H_0^1(U) \\ \|u\|_2 = 1}} B[u,u].$$

A minimizer solves $Lu = \lambda_1 u$. There is a strictly positive first eigenfunction in U, and λ_1 is simple. Every eigenfunction for a larger eigenvalue changes sign.

Expanded proof.

The energy inner product is B, equivalent to the H_0^1 norm. The functions $w_j/\sqrt{\lambda_j}$ are orthonormal for this inner product, since $B[w_i, w_j] = \lambda_i (w_i, w_j)$. They are complete in the energy space: if $B[u, w_j] = 0$ for all j, then $(u, w_j) = 0$ for all j, hence $u = 0$ by the L^2 basis. Hilbert-space Parseval therefore gives

$$B[u,u] = \sum_j \lambda_j |(u, w_j)|^2, \qquad \|u\|_2^2 = \sum_j |(u, w_j)|^2. \tag{6.10}$$

The Rayleigh minimum follows, with equality at w_1. Alternatively, the direct method gives a minimizer: bounded energy gives weak H_0^1 convergence, Rellich preserves the L^2 normalization, and weak lower semicontinuity preserves minimality.

Differentiating the quotient $B[u + tv, u + tv]/\|u + tv\|_2^2$ at $t = 0$ yields $B[u, v] = \lambda_1 (u, v)$ for every v. If u is a minimizer, $|u| \in H_0^1$ has the same norm and energy by the Sobolev chain rule, so it is also a minimizer and hence smooth. It satisfies $L|u| = \lambda_1 |u| \geq 0$. The strong minimum principle, applied after rewriting L in nondivergence form with zero potential, shows $|u| > 0$ in U. Continuity and connectedness force u to have one fixed sign.

If two first eigenfunctions were independent, take a linear combination that vanishes at one interior point but is not identically zero. It is again a first eigenfunction, contradicting the preceding strict sign property. Thus the eigenspace is one dimensional. An eigenfunction for $\lambda > \lambda_1$ is orthogonal to the positive w_1, and consequently must take both signs. □

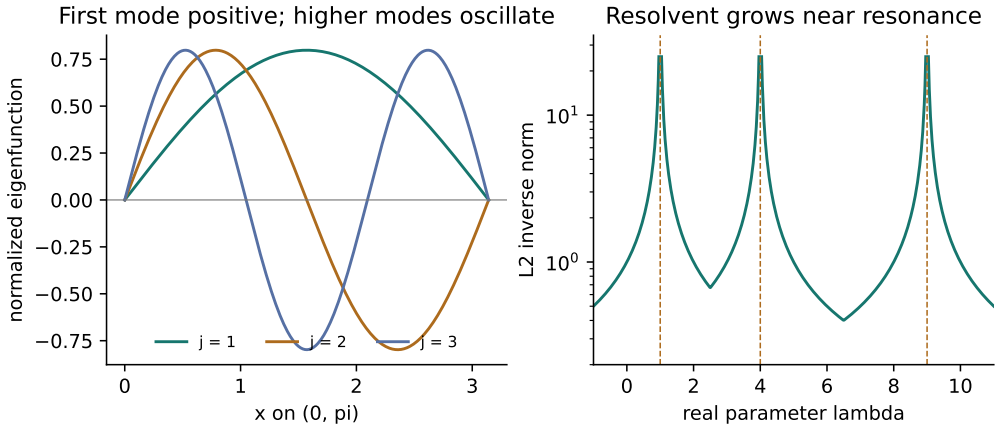
先研究紧的逆算子

L 在 L^2 上是无界算子，不宜直接套紧算子的谱定理。其逆先提升到 H_0^1，再由紧嵌入回到 L^2；逆的特征值趋零，对应 L 的特征值趋无穷。

能量空间中的归一化不同

w_j 在 L^2 中长度为一，但能量长度是 $\sqrt{\lambda_j}$。因此能量空间的正交归一基是 $w_j/\sqrt{\lambda_j}$，这决定了能量展开中的权重。

For $-\partial_{xx}$ on $(0,\ell)$, $w_j = \sqrt{2/\ell}\sin(j\pi x/\ell)$ and $\lambda_j = (j\pi/\ell)^2$. Their oscillations illustrate positivity of the first mode and sign changes of the higher modes.



Explicit interval model. On $(0,\pi)$ the Dirichlet eigenvalues are j^2. For real λ off the spectrum, the self-adjoint inverse norm is $\|(-\partial_{xx} - \lambda)^{-1}\|_{L^2\to L^2} = 1/\min_j |j^2 - \lambda|$. The poles mark loss of uniqueness.

Weyl's law: a source remark, not a proved theorem here. For the Dirichlet Laplacian on a smooth bounded domain,

$$N(\lambda) := \#\{j : \lambda_j \le \lambda\} \sim \frac{\omega_n |U|}{(2\pi)^n}\lambda^{n/2}.$$

Equivalently, $\lambda_j \sim 4\pi^2 (j/(\omega_n|U|))^{2/n}$. This asymptotic statement appears as a remark in Evans; its proof requires additional spectral estimates and is not part of the compactness argument above. Here ω_n is the volume of the unit ball.

6.5.3 A positive eigenfunction without symmetry (Evans §6.5.2)

Now write $L = -a^{ij}\partial_{ij} + b^i\partial_i + c$ and assume $c \ge 0$. The matrix A is symmetric, but the full operator need not be self-adjoint. For weak formulations and adjoints, first rewrite it in divergence form: the drift in B is $b^i + \partial_j a^{ij}$, rather than b^i itself.

> **Theorem 6.17**
> *There is a real eigenvalue $\lambda_1 > 0$ and an eigenfunction $\phi > 0$ in U, $\phi = 0$ on ∂U. Its eigenspace is one dimensional. Every complex eigenvalue λ satisfies* $\operatorname{Re}\lambda \ge \lambda_1$.

绝对值仍要通过变分方程

不能直接对 $|u|$ 做经典二阶求导。先用 Sobolev 链式法则证明能量不变，因而它仍是极小元；再由 Euler 方程与正则性得到光滑性和严格正性。

Expanded proof.

1. Construct a positive compact inverse. The homogeneous zero-Dirichlet problem has only the zero solution: regularity makes a weak homogeneous solution classical, and comparison applied to both signs gives zero. Fredholm therefore supplies $T = L^{-1}$. Fix an integer $m > n/2 + 2$ and let $X = H^m(U) \cap H_0^1(U)$ with its H^m norm. Boundary regularity makes $T : X \to H^{m+2}(U) \cap H_0^1(U)$ bounded, so $T : X \to X$ is compact. The closed convex cone $\mathscr{C} = \{u \in X : u \geq 0\}$ is preserved by T.

Fix $0 \neq w \in C_c^\infty(U)$, $w \geq 0$. The strong minimum principle and Hopf's lemma imply $Tw > 0$ inside and $\partial_\nu(Tw) < 0$ on the boundary. Since w has compact support, some $\mu > 0$ satisfies $\mu Tw \geq w$.

2. A compact fixed-point argument yields an eigenvector. We invoke the following *external* form of Schaefer's theorem: if a continuous compact map $F : \mathscr{C} \to \mathscr{C}$ has a bounded set $\{u : u = tF(u),\ 0 \leq t \leq 1\}$, then F has a fixed point. This topological theorem is a prerequisite here, not proved in this chapter.

For $\varepsilon > 0$, consider

$$u = \eta T(u + \varepsilon w), \qquad u \in \mathscr{C}.$$

No solution can have $\eta > \mu$. Indeed positivity gives $u \geq \varepsilon(\eta/\mu)w$; iterating $u \geq \eta Tu$ gives $u \geq \varepsilon(\eta/\mu)^k w$ for every k, impossible at a point where $w > 0$. If the solutions for $0 \leq \eta \leq 2\mu$ were bounded, Schaefer applied to $F(u) = 2\mu T(u + \varepsilon w)$ would give a solution at $\eta = 2\mu$, a contradiction. Hence that solution set is unbounded. Choose $\varepsilon_j \downarrow 0$ and solutions u_j, η_j with $\|u_j\|_X \geq \varepsilon_j^{-1}$ and $0 \leq \eta_j \leq 2\mu$. Normalize $v_j = u_j/\|u_j\|_X$. Then

$$v_j = \eta_j T\left(v_j + \frac{\varepsilon_j w}{\|u_j\|_X}\right).$$

Compactness gives a subsequence converging in X to v with $\|v\|_X = 1$, $v = \eta Tv$, and $\eta > 0$ (otherwise the right side would converge to zero). Thus $Lv = \eta v$, $v \geq 0$. Regularity, the strong minimum principle and Hopf give a smooth $\phi = v > 0$, $\partial_\nu \phi < 0$, with $\lambda_1 = \eta > 0$.

非对称时换一种结构

非对称算子通常没有正交特征基，Rayleigh 商也不再直接给出全部谱。主特征值的存在依赖紧逆算子保持非负性，极小实部则由正特征函数的商变换证明。

3. Exclude eigenvalues to the left of λ_1. For a complex eigenfunction z, smoothness and zero boundary values imply $|z(x)| \le Cd(x)$ near the boundary, with $d(x) = \operatorname{dist}(x, \partial U)$. Hopf and compactness of ∂U give $\phi(x) \ge ad(x)$. For $0 < \delta < 1$, put $h = \phi^{1-\delta}$ and $v = z/h$. Then v extends continuously as zero on ∂U.

A product calculation gives

$$L(hv) = hKv + vLh, \qquad K = -A : D^2 + (b - 2AD \log h) \cdot D,$$

and the chain rule gives

$$\frac{Lh}{h} = (1-\delta)\lambda_1 + \delta c + \delta(1-\delta)\frac{AD\phi \cdot D\phi}{\phi^2} \ge (1-\delta)\lambda_1.$$

Consequently, for $q = |v|^2$,

$$Kq = 2\operatorname{Re}(\bar{v}Kv) - 2ADv \cdot D\bar{v} \le 2\left(\operatorname{Re}\lambda - \frac{Lh}{h}\right)q.$$

If $\operatorname{Re}\lambda < \lambda_1$, choose δ small so that the coefficient on the right is strictly negative. The nonzero continuous function q, zero on ∂U, has a positive interior maximum, where $Dq = 0$ and $-A : D^2 q \ge 0$, a contradiction. This argument is pointwise at that maximum; it does not assume the drift $D \log h$ is bounded near the boundary.

4. Prove geometric simplicity by sliding positive functions. Let z be a real eigenfunction for λ_1, changing its sign if necessary so it is positive somewhere. The estimates $\phi \ge ad$ and $|z| \le Cd$, also valid after adjusting constants on the interior compact region, show that

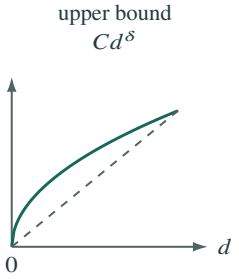
$$t_* = \sup\{t \ge 0 : \phi - tz \ge 0 \text{ in } U\}$$

is positive and finite. At t_* the residual $r = \phi - t_* z$ is nonnegative and satisfies $Lr = \lambda_1 r \ge 0$. If it were nonzero, the strong minimum principle and Hopf would give $r \ge a'd$. Since $|z| \le Cd$, a small further increase of t_* would preserve nonnegativity, a contradiction. Hence $r = 0$. Apply this separately to real and imaginary parts for complex eigenfunctions at the real value λ_1. □

归一化保住非零极限

紧性本身可能只给出零极限。把无界解列除以自身范数后，范数固定为一；扰动项同时趋零，极限才成为真正的非零特征函数。

图 T6.M：商为何在边界趋零



由 $|z| \le Cd$ 与 $\phi \ge ad$，得到 $|z|/\phi^{1-\delta} \le C'd^\delta \to 0$。图示上界取 $\delta = 1/2$。

6.5.4 The positive adjoint eigenfunction

Lemma 6.18
The formal Dirichlet adjoint has a positive eigenfunction ϕ^* for the same λ_1.

Expanded proof.

The inverse T^* is positive on L^2: for $f, g \geq 0$, $(T^*f, g) = (f, Tg) \geq 0$. Regularity makes its restriction to X compact. If $f \in \mathscr{C}$ is nonzero, $v = T^*f \geq 0$ is nonzero. Although the expanded potential c^* of L^* may be negative, choose M with $c^* + M \geq 0$; then $(L^* + M)v = f + Mv \geq 0$. The strong minimum principle and Hopf applied to $L^* + M$ give $v > 0$ and $\partial_\nu v < 0$. The compact-cone construction in Steps 1–2, with T^* in place of T, therefore gives $\phi^* > 0$ and $L^*\phi^* = \lambda^*\phi^*$, $\lambda^* > 0$. Both eigenfunctions have zero trace, so weak adjointness yields

$$\lambda_1(\phi, \phi^*) = B[\phi, \phi^*] = (\phi, L^*\phi^*) = \lambda^*(\phi, \phi^*).$$

The inner product is positive, hence $\lambda^* = \lambda_1$. □

If c has arbitrary sign but is bounded, apply the theorem to $L + \kappa$ with $c + \kappa \geq 0$. Translating its eigenvalues back preserves the positive principal eigenfunction, simplicity and the least-real-part property; the resulting λ_1 need not be positive. The word *simple* above describes the eigenspace. In fact there is also no generalized eigenvector at λ_1: an equation $(L - \lambda_1)z = \phi$ contradicts pairing with ϕ^*, since $(\phi, \phi^*) > 0$.

伴随的零阶项可能变号

原算子的 $c \geq 0$ 不推出伴随算子的零阶项非负。先由逆算子的对偶关系得到非负性，再加常数使用强最大值原理，才能严谨得到伴随正特征函数。

6.6 Problems and Worked Solutions

These are concise restatements of the 17 source problems, with editorial solutions. Unless a problem says otherwise, U is bounded with smooth boundary, coefficients are smooth and bounded, and ellipticity is uniform. Connectedness is specified where constants or positivity are involved. The same assumptions apply to each weak formulation below.

1. Removing a potential by a positive solution. Relate $-\Delta u + cu = 0$ to $-\operatorname{div}(aDv) = 0$.

Solution. If $w > 0$ solves the first equation, set $u = wv$. The product rule gives

$$-\Delta(wv) + cwv = -w\Delta v - 2Dw \cdot Dv + v(-\Delta w + cw) = -w^{-1} \operatorname{div}(w^2 Dv).$$

Thus u solves the first equation exactly when $v = u/w$ solves the second with $a = w^2$. Conversely, given a smooth positive a, set $w = \sqrt{a}$ and $c = \Delta w/w$. Then $-\Delta w + cw = 0$, so the same identity proves that $u = \sqrt{a}\, v$ solves the potential equation. These are local identities wherever $w > 0$; uniform bounds on a are additional requirements for a uniform elliptic theory. □

正解作为新的权重

除以一个正解后，零阶项被吸收到权重 w^2 中。关键是展开 $-\Delta(wv)$ 时保留两倍交叉项；反向构造的势是 $\Delta\sqrt{a}/\sqrt{a}$。

2. A small negative potential. For $L = -\partial_j(a^{ij}\partial_i) + c$, find $\mu > 0$ so that $c \geq -\mu$ ensures Lax–Milgram.

Solution. Let $\|u\|_2 \leq C_P\|Du\|_2$ on $H_0^1(U)$ and choose $\mu = \theta/(2C_P^2)$. Then

$$B[u,u] \geq \theta\|Du\|_2^2 - \mu\|u\|_2^2 \geq \tfrac{\theta}{2}\|Du\|_2^2 \geq C\|u\|_{H^1}^2.$$

Bounded coefficients give boundedness of B. Apply Theorem 6.1. The permitted negative part depends on both ellipticity and the domain's Poincaré constant. □

负势可以由梯度能量吸收

零阶系数略微为负并不立即破坏可解性。Poincaré 不等式把函数平方项转回梯度项；负势大小的阈值因此与区域几何有关。

3. The clamped biharmonic problem. For $f \in L^2(U)$, solve $\Delta^2 u = f$, with $u = \partial_\nu u = 0$, weakly in $H_0^2(U)$.

Solution. Use $B_2[u,v] = \int \Delta u\, \Delta v$ and $\ell(v) = \int fv$. For $u \in C_c^\infty(U)$, integration by parts twice gives

$$\int (\Delta u)^2 = \sum_{i,j} \int u_{ii}u_{jj} = \sum_{i,j} \int u_{ij}^2.$$

Approximation extends this to $H_0^2(U)$. Since u and each u_i belong to $H_0^1(U)$, Poincaré applied twice gives $\|u\|_{H^2} \leq C\|D^2u\|_2$. Thus B_2 is bounded and coercive on H_0^2 and ℓ is bounded. Lax–Milgram gives exactly one weak solution. On smooth boundaries the closure definition of H_0^2 encodes both zero traces. □

四阶方程用二阶能量空间

双调和方程分部积分两次后只需二阶弱导数。H_0^2 同时编码函数及一阶迹为零，不能用只有函数零迹的 $H^2 \cap H_0^1$ 随意替代。

4. Neumann compatibility. For connected U, prove that $\int Du \cdot Dv = \int fv$ for every $v \in H^1(U)$ is solvable exactly when $\int_U f = 0$.

Solution. Testing with $v = 1$ proves necessity. On $V = \{u \in H^1(U) : \int_U u = 0\}$, the mean-zero Poincaré inequality holds. To see this, a counterexample can be normalized to have $\|u_j\|_2 = 1$, mean zero and $\|Du_j\|_2 \to 0$. Rellich gives a strong L^2 limit, whose weak gradient is zero. It is constant by connectedness and zero by its mean, contradicting norm one. Lax–Milgram on V gives a unique mean-zero solution. For arbitrary v, subtract its mean; the removed constant contributes zero on both sides because $\int f = 0$. Thus the full weak equation holds. The difference of any two solutions has zero gradient, so all solutions differ by a constant. □

5. Robin boundary conditions. Formulate and solve $-\Delta u = f$, $u + \partial_\nu u = 0$.

Solution. Use $H^1(U)$ and its trace T, with

$$B_R[u,v] = \int_U Du \cdot Dv + \int_{\partial U} Tu\, Tv, \qquad \ell(v) = \int_U fv.$$

The plus boundary term follows from $\partial_\nu u = -u$ in Green's formula. Trace boundedness makes B_R bounded. Also

$$\|u\|_{H^1}^2 \le C\left(\|Du\|_2^2 + \|Tu\|_{L^2(\partial U)}^2\right).$$

Otherwise normalize a violating sequence to norm one. Its gradients and traces tend to zero; Rellich and weak H^1 convergence yield a constant on each component, with zero trace, hence zero. Strong L^2 convergence and vanishing gradients contradict norm one. This proves coercivity; Lax–Milgram yields a unique solution for every f. □

6. Mixed Dirichlet–Neumann data. Let connected U have $\partial U = \Gamma_1 \cup \Gamma_2$ with disjoint closed pieces. Impose $u = 0$ on Γ_1 and $\partial_\nu u = 0$ on Γ_2.

Solution. Set $V = \{u \in H^1(U) : Tu = 0 \text{ on } \Gamma_1\}$ and require $\int Du \cdot Dv = \int fv$ for every $v \in V$. The Dirichlet condition is built into V; the Neumann condition appears naturally after integration by parts. If Γ_1 has positive surface measure, Poincaré holds on V: a normalized counterexample converges to a constant by the argument in Problem 4, and its zero trace on Γ_1 forces that constant to vanish. Lax–Milgram gives existence and uniqueness for every f. Under the stated smoothness and disjoint closed partition, a nonempty Γ_1 is relatively open in ∂U and has positive measure. If Γ_1 is empty, Problem 4 gives the Neumann compatibility and constant ambiguity. If Γ_2 is empty, this is the usual Dirichlet problem. For more general partitions that meet along an interface, the weak formulation still makes sense, but global H^2 regularity is a separate issue. □

7. A monotone semilinear estimate. Let $u \in H^1(\mathbb{R}^n)$ have compact support and solve $-\Delta u + c(u) = f$ weakly, where c is smooth, $c(0) = 0$, $c' \ge 0$, and $c(u), f \in L^2(\mathbb{R}^n)$. Prove $\|D^2 u\|_2 \le C\|f\|_2$.

Solution. For fixed h in direction e_k, use $v = -\delta_{-h}\delta_h u \in H^1(\mathbb{R}^n)$, an admissible compactly supported weak test by density. Discrete integration by parts gives

$$\|\delta_h Du\|_2^2 + \int_{\mathbb{R}^n} \delta_h(c(u))\, \delta_h u = -\int_{\mathbb{R}^n} f\, \delta_{-h}\delta_h u.$$

边界项也能控制常数

Neumann 能量看不见常数，所以需要相容条件并允许加常数。Robin 能量多了边界平方项，常数方向也受到控制，于是无需零均值条件且解唯一。

自然边界与本质边界

Dirichlet 条件先限制解和测试函数所在空间，称为本质条件；Neumann 条件由分部积分后的边界项体现，称为自然条件。更换边界类型也就更换了变分空间。

The second term is nonnegative because $(c(s) - c(t))(s - t) \geq 0$. Moreover, $\|\delta_{-h}\delta_h u\|_2 \leq \|\partial_k \delta_h u\|_2 \leq \|\delta_h Du\|_2$. Hence $\|\delta_h Du\|_2 \leq \|f\|_2$ uniformly in h. The difference quotient criterion in each direction yields $u \in H^2$ and, for the Frobenius norm, $\|D^2 u\|_2 \leq \sqrt{n}\,\|f\|_2$. No upper bound on c' is required in this monotonicity argument. □

单调性替代导数上界

非线性项的差商乘积非负，因此可以留在左边后直接舍去。这里使用的是 c 的单调性，不需要把 $c'(u)$ 当作有界系数来估计。

8. Bernstein's gradient estimate. For $Lu = -a^{ij}u_{ij} = 0$, show that $v = |Du|^2 + Mu^2$ is a subsolution for large M and deduce a boundary estimate.

Solution. Differentiate the equation to get $Lu_k = (\partial_k a^{ij})u_{ij}$. Thus

$$L|Du|^2 = -2a^{ij}u_{ki}u_{kj} + 2u_k(\partial_k a^{ij})u_{ij} \leq -\theta|D^2u|^2 + C|Du|^2.$$

Also $L(u^2) = -2a^{ij}u_i u_j \leq -2\theta|Du|^2$. Choose $2M\theta \geq C$. Then $Lv \leq 0$. The weak maximum principle gives $\sup_U |Du|^2 \leq \sup_{\partial U}(|Du|^2 + Mu^2)$, hence

$$\|Du\|_{L^\infty(U)} \leq C\big(\|Du\|_{L^\infty(\partial U)} + \|u\|_{L^\infty(\partial U)}\big).$$

□

9. A boundary barrier estimate. For $Lu = -a^{ij}u_{ij} = f$, $u = 0$ on ∂U, suppose $Lw \geq 1$, $w \geq 0$ on ∂U, and $w(x_0) = 0$.

Solution. The minimum principle gives $w \geq 0$ in U. Let $F = \|f\|_\infty$. Since $L(\pm u - Fw) \leq 0$ and $\pm u - Fw \leq 0$ on the boundary, comparison gives $|u| \leq Fw$. Assuming the indicated boundary derivatives exist, evaluate at $x_0 - t\nu$ and divide by t to obtain $|\partial_\nu u(x_0)| \leq F|\partial_\nu w(x_0)|$. All tangential derivatives of the smooth zero-boundary function u vanish at x_0, so $|Du(x_0)| = |\partial_\nu u(x_0)|$. This is the desired estimate, with constant F. □

屏障把函数值变成法向斜率

先比较得到整个区域的 $|u| \leq \|f\|_\infty w$，再沿内法线取差商。边界上 u 恒为零，所以梯度只剩法向分量。

同一唯一性可有两种证明

能量法用积分把整个梯度消掉；最大值原理通过极值位置与边界斜率制造矛盾。前者更适合弱解，后者直接揭示正性与边界几何的作用。

10. Harmonic Neumann solutions are constant. Give energy and maximum-principle proofs on connected U.

Solution. Green's identity gives $\int_U |Du|^2 = -\int_U u\Delta u + \int_{\partial U} u\partial_\nu u = 0$, so $Du = 0$ and connectedness gives constancy. Alternatively, if u were nonconstant, its maximum would be on the boundary by the strong maximum principle and all interior values would be strictly smaller. Hopf's lemma at a boundary maximum would imply $\partial_\nu u > 0$, contradicting the homogeneous Neumann condition. The latter proof uses a smooth boundary, or at least the needed interior ball. □

11. Convex functions of weak solutions. If bounded $u \in H^1(U)$ satisfies $\int ADu \cdot Dv = 0$ for $v \in H_0^1(U)$, prove that $w = \varphi(u)$ is a weak subsolution for smooth convex φ.

Solution. The bounded range of u makes φ', φ'' bounded on that range, so $w \in H^1(U)$ and $Dw = \varphi'(u)Du$. First take bounded $0 \le v \in H_0^1(U)$. The product $\varphi'(u)v \in H_0^1(U)$ is an admissible test and gives

$$\int ADw \cdot Dv = -\int \varphi''(u)v\, ADu \cdot Du \le 0.$$

For arbitrary $0 \le v \in H_0^1(U)$, use $v_M = \min(v, M)$ and pass in the left side by $v_M \to v$ in H^1. This proves the required inequality for every nonnegative weak test. The function w need not have zero trace. □

测试函数的乘积要先检查

若 v 无界，$\varphi''(u)vDu$ 未必属于 L^2，不能直接宣布 $\varphi'(u)v$ 合法。先截断 v，证明不等式后再用 H^1 收敛去掉截断。

12. A positive supersolution implies comparison. Assume $h \in C^2(U) \cap C(\overline{U})$, $h > 0$ on *all of* $\overline{U}$, and $Lh \ge 0$. Prove the zero-boundary weak maximum principle without assuming $c \ge 0$.

Solution. For $Lu \le 0$ and $u \le 0$ on ∂U, set $z = u/h$. The product formula gives

$$Lu = hMz + zLh, \qquad M = -A : D^2 + (b - 2ADh/h) \cdot D.$$

On $\{z > 0\}$, $Mz \le 0$ and M has no potential. For each $\varepsilon > 0$, the closure of $\{z > \varepsilon\}$ is compact in U, because $z \le 0$ on the boundary and $\min_{\overline{U}} h > 0$. The drift is bounded there, so the zero-potential maximum principle gives $z \le \varepsilon$ on this set, which must therefore be empty. Let $\varepsilon \downarrow 0$. Positivity only in U would not suffice: $h = \sin x$ for $L = -\partial_{xx} - 1$ on $(0, \pi)$ is an interior positive solution but vanishes at the endpoints, and comparison fails. □

13. Courant's max–min principle. For the symmetric operator of §6.5, show

$$\lambda_k = \max_{\dim S = k-1} \min_{\substack{u \in H_0^1(U) \cap S^\perp \\ \|u\|_2 = 1}} B[u,u],$$

where $S \subset H_0^1(U)$ and orthogonality is in $L^2(U)$.

Solution. For any such S, linear algebra supplies a nonzero vector in $\operatorname{span}(w_1, \dots, w_k) \cap S^\perp$. Normalize it. Formula (6.10) gives its energy at most λ_k, so the inner minimum is at most λ_k. For $S = \operatorname{span}(w_1, \dots, w_{k-1})$, every admissible u has only modes $j \geq k$, and the same formula gives $B[u,u] \geq \lambda_k$. Equality occurs at w_k. For any fixed S, the minimum exists by the direct method; its orthogonality constraints are preserved by strong L^2 convergence. □

极小极大中的正交空间

这里去掉的是前若干模态在 L^2 中的方向。任意少于 k 个线性约束都不可能消灭前 k 个模态的全部组合，这就是一半不等式的有限维理由。

14. A principal-eigenvalue max–min formula. For the nonsymmetric operator, prove

$$\lambda_1 = \sup_{\substack{u \in C^\infty(\overline{U}),\ u>0 \text{ in } U \\ u=0 \text{ on } \partial U}} \inf_{x \in U} \frac{Lu(x)}{u(x)}.$$

Solution. Let $\phi^* > 0$ be the adjoint eigenfunction from Lemma 6.18. Both functions have zero trace, so Green's identity, or the weak adjoint identity, gives

$$\int_U \phi^* L u = \int_U u L^* \phi^* = \lambda_1 \int_U u \phi^*.$$

Thus the weighted average of Lu/u with positive weight $u\phi^*$ is λ_1. Its infimum cannot exceed this average. If the infimum is $-\infty$ the inequality is immediate; otherwise integrating its pointwise lower bound proves it directly. Taking $u = \phi$ achieves equality everywhere. A constant shift handles potentials of arbitrary bounded sign. □

非对称公式中的权重来自伴随

不能只对 Lu/u 做普通空间平均。与正的伴随特征函数配对后，正确权重是 $u\phi^*$；分部积分才把这个加权平均固定为主特征值。

15. Hadamard's domain-variation formula. Let $-\Delta w = \lambda w$ on a smoothly moving domain $U(\tau)$, $w = 0$ on its boundary and $\int_{U(\tau)} w^2 = 1$. Assume a smooth eigenvalue/eigenfunction branch, and denote boundary velocity by $\mathbf{v}$.

Solution. Use the Eulerian derivative $\dot{w}$ at a fixed spatial point. Differentiating $w(x(\tau), \tau) = 0$ on the moving boundary gives

$$\dot{w} = -Dw \cdot \mathbf{v} = -\partial_\nu w \, (\mathbf{v} \cdot \nu).$$

Reynolds' formula for normalization yields $2\int_U w\dot w + \int_{\partial U} w^2 \mathbf{v}\cdot\nu = 0$, hence $\int_U w\dot w = 0$. Since $\lambda = \int_U |Dw|^2$, differentiate once more:

$$\begin{aligned}\dot\lambda &= 2\int_U Dw\cdot D\dot w + \int_{\partial U} |Dw|^2 \mathbf{v}\cdot\nu \\ &= 2\lambda\int_U w\dot w + 2\int_{\partial U} \partial_\nu w\,\dot w + \int_{\partial U} (\partial_\nu w)^2 \mathbf{v}\cdot\nu \\ &= -\int_{\partial U} (\partial_\nu w)^2 \mathbf{v}\cdot\nu.\end{aligned}$$

Tangential derivatives vanish because w is zero on the boundary. The smooth-branch hypothesis is essential; a multiple eigenvalue need not admit a single arbitrarily chosen differentiable eigenfunction branch. □

扩张区域使特征值下降

若边界向外移动，$\mathbf{v}\cdot\nu > 0$，公式给出 $\dot\lambda < 0$。负号来自移动边界条件 $\dot w = -\partial_\nu w\,\mathbf{v}\cdot\nu$，不是凭直觉添加。

16. Helmholtz waves and radiation. Let real $\sigma \neq 0$ and $\lambda = \sigma^2$.

Solution. (a) For $|\omega| = 1$, $-\Delta e^{i\sigma\omega\cdot x} = \sigma^2 e^{i\sigma\omega\cdot x}$. Thus $e^{i\sigma(\omega\cdot x - t)}$ solves the wave equation.

(b) In dimension three put $\Phi(x) = e^{i\sigma r}/(4\pi r)$, $r = |x|$. The radial identity $\Delta g = g'' + 2g'/r$ shows $(-\Delta - \sigma^2)\Phi = 0$ for $r > 0$. At $r = \varepsilon$,

$$-\int_{\partial B_\varepsilon} \partial_r \Phi\, dS = e^{i\sigma\varepsilon}(1 - i\sigma\varepsilon) \longrightarrow 1.$$

Integration by parts against a smooth test function outside B_ε therefore gives $(-\Delta - \sigma^2)\Phi = \delta_0$; the other shrinking-sphere term and the volume integral of $\sigma^2\Phi$ tend to zero.

(c) For Φ, $r(\Phi_r - i\sigma\Phi) = -e^{i\sigma r}/(4\pi r) \to 0$ uniformly in angle. For the plane wave at $x = r\vartheta$,

$$r(w_r - i\sigma w) = i\sigma r(\omega\cdot\vartheta - 1)e^{i\sigma r\omega\cdot\vartheta},$$

whose modulus diverges on every ray with $\omega\cdot\vartheta \neq 1$. Thus the plane wave does not satisfy the Sommerfeld condition. □

辐射条件选择向外传播

时间因子取 $e^{-i\sigma t}$ 时，$e^{i\sigma r}/r$ 描述向外传播的球面波。条件中的减号与这一时间约定配套；平面波不会在所有方向都向外辐射。

17. Uniqueness of an entire radiating solution. Suppose w solves $-\Delta w = \sigma^2 w$ throughout $\mathbb{R}^3$ and satisfies $r(w_r - i\sigma w) \to 0$ *uniformly in angle*. Prove $w = 0$.

Solution. Green's identity and the real coefficient σ^2 imply

$$0 = \int_{\partial B_R} (\overline{w}\, w_r - w\,\overline{w_r})\, dS.$$

The imaginary flux is zero, so expansion of the square gives

$$\int_{\partial B_R} (|w_r|^2 + \sigma^2|w|^2)\, dS = \int_{\partial B_R} |w_r - i\sigma w|^2\, dS = o(1).$$

Uniform radiation justifies the last step, since the sphere has area $4\pi R^2$ and the integrand is $o(R^{-2})$ uniformly. For a fixed x_0 and $R > |x_0|$, Green's representation with the outgoing kernel $\Phi(x - x_0)$ gives

$$w(x_0) = \int_{\partial B_R} (\Phi\, w_r - w\, \partial_r \Phi)\, dS.$$

Here ∂_r on the kernel means the outward derivative on ∂B_R, not radial differentiation about x_0. For large R, both $\Phi(x-x_0)$ and this derivative are $O(R^{-1})$, so their $L^2(\partial B_R)$ norms remain bounded. Cauchy–Schwarz and the vanishing flux norm above imply $w(x_0) = 0$. Since x_0 was arbitrary, $w \equiv 0$. □

辐射条件还包含一致性

从逐方向极限跳到球面积分极限需要控制角变量。这里明确采用对方向一致的 Sommerfeld 条件；基本解在原点有奇性，所以并不违背“全空间光滑辐射解只能为零”。

6.7 References and Proof Dependencies

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 6, printed pp. 293–348 (local English PDF pp. 310–365). The Chinese scan corresponds to printed pp. 208–246 (PDF pp. 226–264); selected page comparisons are recorded in `docs/CHAPTER-06.md`. The English prose, proof expansions, solutions and Chinese annotations here are editorial reconstructions.

Evans's references point to Gilbarg–Trudinger for weak elliptic theory, regularity, maximum principles and Harnack; Krylov and Ladyzhenskaya–Uraltseva for regularity; Protter–Weinberger and Pucci–Serrin for maximum principles; and Smoller for spectral theory. For domain variations the source cites Garabedian; for radiation it cites Colton. These are source bibliography pointers, not claims that those books have been independently consulted for this draft.

Dependencies and boundaries. Riesz representation, Hilbert weak compactness and Hilbert Parseval are prerequisites. Chapter 5 supplies the Sobolev tools. Lax–Milgram, the compact Fredholm lemma, regularity, maximum principles, the smooth principal-part Harnack inequality, compact self-adjoint spectral decomposition and the principal-eigenvalue arguments are expanded above. The nonsymmetric existence proof explicitly invokes Schaefer's fixed-point theorem. Weyl's law is stated as an unproved source remark. The general measurable-coefficient Harnack theorem is described as an extension, not claimed as proved. All 17 source problems have editorial solutions. Compilation, model calculations and layout checks do not constitute independent mathematical review; this chapter remains an author-checked study draft.

Chapter 7

Linear Evolution Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 7; printed p. 349; PDF page 366.

Chinese reference: 中译本对应章：线性发展方程，书内第 248 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

7.1 Second-Order Parabolic Equations

Outline only. Text, annotations, and proof details are pending.

7.2 Second-Order Hyperbolic Equations

Outline only. Text, annotations, and proof details are pending.

7.3 Systems of Hyperbolic First-Order Equations

Outline only. Text, annotations, and proof details are pending.

7.4 Semigroup Theory

Outline only. Text, annotations, and proof details are pending.

7.5 Problems

Outline only. Text, annotations, and proof details are pending.

7.6 References

Outline only. Text, annotations, and proof details are pending.

Part III

Theory for Nonlinear Partial Differential Equations

Chapter 8

The Calculus of Variations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 8; printed p. 427; PDF page 444.

Chinese reference: 中译本对应章：变分法，书内第 303 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

8.1 Introduction

Outline only. Text, annotations, and proof details are pending.

8.2 Existence of Minimizers

Outline only. Text, annotations, and proof details are pending.

8.3 Regularity

Outline only. Text, annotations, and proof details are pending.

8.4 Constraints

Outline only. Text, annotations, and proof details are pending.

8.5 Critical Points

Outline only. Text, annotations, and proof details are pending.

8.6 Invariance, Noether's Theorem

Outline only. Text, annotations, and proof details are pending.

8.7 Problems

Outline only. Text, annotations, and proof details are pending.

8.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 9

Nonvariational Techniques

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 9; printed p. 495; PDF page 512.

Chinese reference: 中译本对应章：非变分技术，书内第 350 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

9.1 Monotonicity Methods

Outline only. Text, annotations, and proof details are pending.

9.2 Fixed Point Methods

Outline only. Text, annotations, and proof details are pending.

9.3 Method of Subsolutions and Supersolutions

Outline only. Text, annotations, and proof details are pending.

9.4 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

9.5 Geometric Properties of Solutions

Outline only. Text, annotations, and proof details are pending.

9.6 Gradient Flows

Outline only. Text, annotations, and proof details are pending.

9.7 Problems

Outline only. Text, annotations, and proof details are pending.

9.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 10

Hamilton–Jacobi Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 10; printed p. 543; PDF page 560.

Chinese reference: 中译本对应章：Hamilton–Jacobi 方程，书内第 384 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

10.1 Introduction, Viscosity Solutions

Outline only. Text, annotations, and proof details are pending.

10.2 Uniqueness

Outline only. Text, annotations, and proof details are pending.

10.3 Control Theory, Dynamic Programming

Outline only. Text, annotations, and proof details are pending.

10.4 Problems

Outline only. Text, annotations, and proof details are pending.

10.5 References

Outline only. Text, annotations, and proof details are pending.

Chapter 11

Systems of Conservation Laws

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 11; printed p. 571; PDF page 588.

Chinese reference: 中译本对应章：守恒律组，书内第 404 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

11.1 Introduction

Outline only. Text, annotations, and proof details are pending.

11.2 Riemann's Problem

Outline only. Text, annotations, and proof details are pending.

11.3 Systems of Two Conservation Laws

Outline only. Text, annotations, and proof details are pending.

11.4 Entropy Criteria

Outline only. Text, annotations, and proof details are pending.

11.5 Problems

Outline only. Text, annotations, and proof details are pending.

11.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 12

Nonlinear Wave Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 12; printed p. 617; PDF page 634.

Chinese reference: 中译本对应章：非线性波动方程，书内第 437 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

12.1 Introduction

Outline only. Text, annotations, and proof details are pending.

12.2 Existence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.3 Semilinear Wave Equations

Outline only. Text, annotations, and proof details are pending.

12.4 Critical Power Nonlinearity

Outline only. Text, annotations, and proof details are pending.

12.5 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.6 Problems

Outline only. Text, annotations, and proof details are pending.

12.7 References

Outline only. Text, annotations, and proof details are pending.

Appendices

Appendix

A

Notation

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix A; printed p. 653; PDF page 670.

Chinese reference: 中译本对应章：记号，书内第 463 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

A.1 Notation for matrices

Outline only. Text, annotations, and proof details are pending.

A.2 Geometric notation

Outline only. Text, annotations, and proof details are pending.

A.3 Notation for functions

Outline only. Text, annotations, and proof details are pending.

A.4 Vector-valued functions

Outline only. Text, annotations, and proof details are pending.

A.5 Notation for estimates

Outline only. Text, annotations, and proof details are pending.

A.6 Some comments about notation

Outline only. Text, annotations, and proof details are pending.

Appendix

B

Inequalities

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix B; printed p. 661; PDF page 678.

Chinese reference: 中译本对应章：不等式，书内第 471 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

B.1 Convex functions

Outline only. Text, annotations, and proof details are pending.

B.2 Useful inequalities

Outline only. Text, annotations, and proof details are pending.

Appendix

C

Calculus

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix C; printed p. 665; PDF page 682.

Chinese reference: 中译本对应章：微积分，书内第 475 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

C.1 Boundaries

Outline only. Text, annotations, and proof details are pending.

C.2 Gauss–Green Theorem

Outline only. Text, annotations, and proof details are pending.

C.3 Polar coordinates, coarea formula

Outline only. Text, annotations, and proof details are pending.

C.4 Moving regions

Outline only. Text, annotations, and proof details are pending.

C.5 Convolution and smoothing

Outline only. Text, annotations, and proof details are pending.

C.6 Inverse Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.7 Implicit Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.8 Uniform convergence

Outline only. Text, annotations, and proof details are pending.

Appendix

D

Functional Analysis

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix D; printed p. 674; PDF page 691.

Chinese reference: 中译本对应章：泛函分析，书内第 482 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

D.1 Banach spaces

Outline only. Text, annotations, and proof details are pending.

D.2 Hilbert spaces

Outline only. Text, annotations, and proof details are pending.

D.3 Bounded linear operators

Outline only. Text, annotations, and proof details are pending.

D.4 Weak convergence

Outline only. Text, annotations, and proof details are pending.

D.5 Compact operators, Fredholm theory

Outline only. Text, annotations, and proof details are pending.

D.6 Symmetric operators

Outline only. Text, annotations, and proof details are pending.

Appendix

E

Measure Theory

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix E; printed p. 684; PDF page 701.

Chinese reference: 中译本对应章：测度论，书内第 490 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

E.1 Lebesgue measure

Outline only. Text, annotations, and proof details are pending.

E.2 Measurable functions and integration

Outline only. Text, annotations, and proof details are pending.

E.3 Convergence theorems for integrals

Outline only. Text, annotations, and proof details are pending.

E.4 Differentiation

Outline only. Text, annotations, and proof details are pending.

E.5 Banach space-valued functions

Outline only. Text, annotations, and proof details are pending.

Bibliography

The source bibliography starts on English printed p. 689 (PDF page 706). Bibliographic entries and any additional references remain to be entered.

Outline only. Text, annotations, and proof details are pending.

Index

The source index starts on English printed p. 703 (PDF page 720). An index for these notes will be prepared as the text is written; page numbers must refer to this edition.

Outline only. Text, annotations, and proof details are pending.

Bilingual Glossary
英汉术语表

The entries below cover the annotated material in Chapters 1–6.

English	中文及正文位置
multiindex	多重指标；§1.1
principal part	主部；§1.1
linear / homogeneous	线性 / 齐次；§1.1
semilinear	半线性；§1.1
quasilinear	拟线性；§1.1
fully nonlinear	完全非线性；§1.1
divergence form	散度形式；§1.2
formal adjoint	形式伴随；§1.2
uniform ellipticity	一致椭圆性；§1.2
well-posedness	适定性；§1.3
continuous dependence	连续依赖；§1.3
classical solution	经典解；§1.3
weak solution / regularity	弱解 / 正则性；§1.3
shock / entropy condition	激波 / 熵条件；§1.3
gradient blow-up	梯度爆破；§1.3
viscosity solution	黏性解；§1.4
uniform remainder estimate	一致余项估计；§1.5

Classical model equations / 经典模型方程

English	中文及正文位置
characteristic	特征线；§2.1
fundamental solution	基本解；§2.2
harmonic function	调和函数；§2.2
spherical mean	球面平均；§2.2.2
ball mean	球体平均；§2.2.2
Poisson kernel	Poisson 核；§2.2
Green function	Green 函数；§2.2
heat ball	热球；§2.3.2
parabolic scaling	抛物缩放；§2.3.2
backward heat kernel	向后热核；§2.3.2
parabolic boundary	抛物边界；§2.3
subsolution	次解；§2.3
Duhamel’s principle	Duhamel 原理；§2.3
method of descent	降维法；§2.4
finite propagation speed	有限传播速度；§2.4

Nonlinear first-order equations / 非线性一阶方程

English	中文及正文位置
complete integral	完全积分；§3.1
envelope / singular integral	包络 / 奇异积分；§3.1
first jet	一阶 jet（函数值与梯度）；§3.1
noncharacteristic data	非特征数据；§3.2
characteristic projection	特征投影；§3.2
Legendre transform	Legendre 变换；§3.3
convex conjugate	凸共轭；§3.3
superlinear growth	超线性增长；§3.3
dynamic programming	动态规划；§3.3
semiconcavity	半凹性；§3.3.3
Rankine–Hugoniot condition	跳跃条件；§3.4
one-sided entropy bound	单边熵估计；§3.4.3
rarefaction wave	稀疏波；§3.4
Riemann problem	Riemann 问题；§3.4
N-wave	N 波；§3.4

Representation methods / 解的表示方法

English	中文及正文位置
separation of variables	分离变量法；§4.1
eigenfunction expansion	特征函数展开；§4.1
Turing instability	Turing 失稳；§4.1
phase / group velocity	相速度 / 群速度；§4.2
solitary wave	孤立波；§4.2
Barenblatt solution	Barenblatt 解；§4.2.4
Plancherel theorem	Plancherel 定理；§4.3
Radon transform	Radon 变换；§4.3.3
resolvent	预解算子；§4.3
Cole–Hopf transformation	Cole–Hopf 变换；§4.4
hodograph transform	速度图变换；§4.4
vanishing viscosity	消失黏性；§4.5.2
stationary phase	驻相法；§4.5.3
caustic	焦散；§4.5.3
homogenization	均匀化；§4.5.4
cell problem / corrector	单元问题 / 修正子；§4.5.4
noncharacteristic surface	非特征曲面；§4.6
coefficientwise majorant	优级数；§4.6.3

Sobolev theory / Sobolev 理论

English	中文及正文位置
Hölder seminorm	Hölder 半范数；§5.1
weak derivative	弱导数；§5.2
test function	试验函数；§5.2
Sobolev space	Sobolev 空间；§5.2
mollification	磨光 / 光滑化；§5.3
partition of unity	单位分解；§5.3
extension operator	延拓算子；§5.4
boundary flattening	边界拉平；§5.4
trace / zero trace	迹 / 零迹；§5.5
Sobolev conjugate	Sobolev 共轭指数；§5.6
critical exponent	临界指数；§5.6
continuous representative	连续代表元；§5.6
compact embedding	紧嵌入；§5.7
concentration	集中；§5.7
difference quotient	差商；§5.8
mean oscillation	平均振荡；§5.8
negative Sobolev space	负阶 Sobolev 空间；§5.9
Bochner integral	Bochner 积分；§5.9
duality pairing	对偶配对；§5.9
absolutely continuous	绝对连续；§5.9

Elliptic theory / 椭圆方程理论

English	中文及正文位置
uniform ellipticity	一致椭圆性；§6.1
divergence form	散度型；§6.1
nondivergence form	非散度型；§6.1
bilinear form	双线性型；§6.1
weak Dirichlet solution	弱 Dirichlet 解；§6.1
coercivity	强制性；§6.2
Gårding inequality	Gårding 不等式；§6.2
Fredholm alternative	Fredholm 二择一定理；§6.2
formal adjoint	形式伴随；§6.1
compatibility condition	相容条件；§6.2
Caccioppoli estimate	Caccioppoli 估计；§6.3
tangential difference quotient	切向差商；§6.3
boundary regularity	边界正则性；§6.3
Hopf boundary lemma	Hopf 边界引理；§6.4
strong maximum principle	强最大值原理；§6.4
Harnack inequality	Harnack 不等式；§6.4
Rayleigh quotient	Rayleigh 商；§6.5
principal eigenvalue	主特征值；§6.5
Hadamard variation formula	Hadamard 变分公式；§6.6
Sommerfeld radiation condition	Sommerfeld 辐射条件；§6.6